

FIRST RETURNS, SINGULAR EXITS, AND ACTION FINITE PARTS NEAR A REVERSIBLE HAMILTONIAN SADDLE-FOCUS

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ABSTRACT. We consider a four-dimensional reversible exact Hamiltonian flow whose saddle-focus germ is real analytic in the state variables and has a transverse homoclinic orbit. A fixed local flow tube has two noncompact forward exits of different types: an algebraic end reached in infinite physical time and a regular-singular pole reached in finite time. Assuming a clean limiting first-hit arrangement, normal expansion with order-three bunching at the algebraic boundary, and a fixed conjugacy class of indicial operators with simple positive-integer spectrum at the pole, we prove that every orbit with sufficiently long saddle residence in a fixed compact subcell of the prescribed source cell belongs to exactly one first-return domain or one first-exit stratum. The local exact symplectic coordinates used in the proof are constructed from these geometric data and need not exist globally over the parameter space.

For a compact C^r family $(H_p)_{p \in \mathcal{P}}$, $0 \leq r \leq 2$, the normalized return maps, flight times, and physical Hamiltonian actions have all mixed derivatives $D_z^i D_p^j$, $i + j \leq 2$, $j \leq r$, and converge with error $O((1+n)^3 e^{-\kappa n})$ for every $0 < \kappa < \inf_{p \in \mathcal{P}} 2\pi\alpha_p/\beta_p$. At the algebraic end, the time and action finite parts are obtained by comparing trajectories at the same compactifying coordinate. At the pole, physical flight time has an ordinary finite limit, while the action finite part is obtained at fixed remaining physical time by subtracting the Laurent-log terms determined by the boundary jet. The two action finite parts are compatible with exact composition along concatenated orbit segments and with changes of local coordinates.

The return domains consequently carry a countable Markov coding with summable-variation roof and action functions, yielding quantitative period, action, and entropy asymptotics. Exact calculations and interval enclosures verify the hypotheses for the normal form associated with a reversible folded saddle-node of type II. The conclusion persists in the relative coefficient topology of the fixed flow tube and end structures specified in the paper.

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1. INTRODUCTION

1.1. The geometric problem. A transverse homoclinic orbit to a Hamiltonian saddle-focus creates return strips with arbitrarily long residence near the equilibrium. The recurrent part of this picture is classical. We study the complementary first-hit problem in a fixed flow tube: which points return, which points leave through a prescribed boundary, and what remains of physical time and Hamiltonian action when a nonreturning orbit reaches a singular end? The two ends considered here have genuinely different clocks. An algebraic end is approached only as physical time tends to infinity, whereas a regular-singular pole is reached in finite physical time.

Our concrete application is the normal-form system associated with a reversible folded saddle-node of type II (RFSN-II),

$$(1.1) \quad U' = P, \quad P' = -U^2 - V, \quad V' = Q, \quad Q' = U.$$

With $\lambda = P dU - Q dV$ and $\omega = d\lambda$, it is exact Hamiltonian:

$$\iota_X \omega = dH, \quad H = \frac{1}{2}(Q^2 - P^2) - \frac{1}{3}U^3 - UV,$$

and it is reversed by $\mathcal{R}(U, P, V, Q) = (U, -P, V, -Q)$. The origin is a saddle-focus with eigenvalues $(\pm 1 \pm i)/\sqrt{2}$, and we work on the critical level $H = 0$ through the origin. Section 8 constructs a transverse symmetric homoclinic orbit and both noncompact forward channels for this system.

An orbit entering a small saddle-focus neighborhood may remain there for a long time, make an arbitrarily large number of turns, and then reach an outgoing section. Some outgoing points traverse the homoclinic tube and return to the incoming section. Others reach one of the two noncompact ends or cross a fixed boundary of the local flow-tube domain. The question is whether all sufficiently high-winding initial points can be divided, without gaps or overlaps, into return domains and first-exit sets, while retaining estimates uniform both in the winding number and under perturbation of the Hamiltonian vector field.

We use standard return-map terminology for this geometry. A *return strip* is a connected component of the domain of the local first-return map. A *first-exit component* is a connected component on which the first fixed event is arrival at the algebraic end, the pole end, the stable-manifold cut, or one of the local boundary faces. Their disjoint union is the *first-return/first-exit decomposition*. The technical sections equip this decomposition with fixed normalized charts, cross forms, and time/action data; these auxiliary choices are not additional dynamical objects. Figure 1 summarizes the physical alternatives and the resulting two-vertex return graph.

Contributions. The main theorem starts from an intrinsic analytic saddle-focus germ, a transverse physical homoclinic tube, a clean first-hit arrangement, and geometric compactifications of the two ends. From these data we construct local exact symplectic coordinates on a finite cover of parameter space; no global eigenframe, angle, or winding function is assumed.

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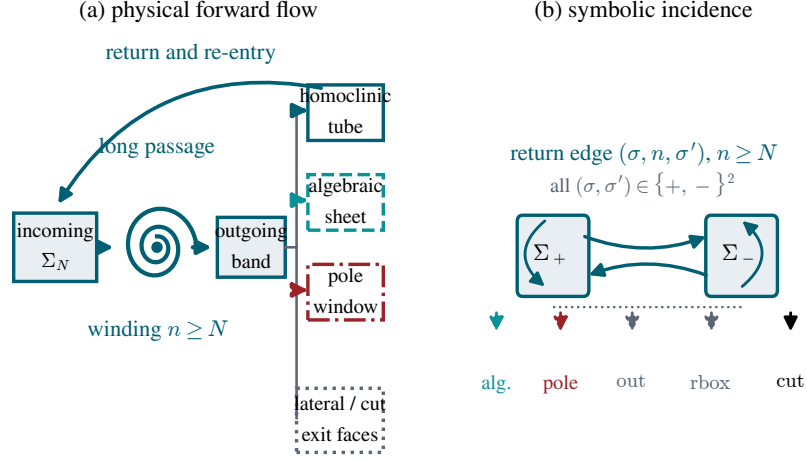


FIGURE 1. The fixed local return-exit geometry (schematic, not to scale). (a) A point in the punctured incoming section makes a long saddle-focus passage. It either traverses the homoclinic tube and re-enters the section, or first reaches the codimension-one algebraic sheet, the open pole window, the stable-manifold cut, or a fixed lateral face. Arrows stop at the first fixed event; no subsequent orbit after a lateral exit is represented. The algebraic end has infinite physical flight time, whereas the pole is reached in finite physical time. (b) Regular returns form a two-vertex countable graph whose edges are the triples (σ, n, σ') , $n \geq N$, for all four sign pairs. The first-exit labels are alg (algebraic end), p (pole), out (outgoing-box lateral face), rbox (return-box lateral face), and cut (the stable-manifold cut). In particular, cut is the zero-action boundary between the two target-sign return components, not an omitted return edge. The drawing represents the fixed flow-tube domain, not a global phase-space partition.

The main new conclusion is that the complete collection of labelled connected strata in the limiting first-hit arrangement persists in every sufficiently high-winding cell. On all of these components simultaneously, the return maps, physical time, and physical action have uniform C^2 asymptotics. The algebraic and pole actions admit finite parts defined by two different comparison procedures, and both remain compatible with exact composition. At the pole the finite boundary jet is allowed to vary; a regular-singular recursion determines the required Laurent-log subtraction.

The return domains also give a countable Markov shift with summable-variation roof and action functions. Periodic-orbit asymptotics, a Lambert- W entropy law, and an $O(L^{-1})$ action-rate estimate follow. Exact calculations and interval enclosures establish the geometric hypotheses for the RFSN-II normal form. The logarithmic saddle-focus passage, exactness of an individual Poincaré branch, and thermodynamic formalism for a prescribed countable shift are standard ingredients; the contribution is their uniform compatibility with the exhaustive first-hit geometry and both singular ends. The conclusion persists in the relative coefficient topology of the fixed flow tube and end structures in Corollary 8.7.

1.2. Relation with earlier homoclinic theory. Transverse Hamiltonian saddle-focus homoclinics already imply finite horseshoes of arbitrary cardinality [6], and reversibility yields blue-sky periodic families and multipulse homoclinic cascades [7, 10]; related reversible bifocus configurations also support chaotic invariant sets [3]. The critical countable admissibility relation and its set-level orbit realization are also classical [18, 19, 20]. These results describe recurrent homoclinic geometry, but they do not give a uniform return/first-exit decomposition that also resolves the algebraic and pole ends.

The RFSN-II blow-up, its explicit algebraic separatrix, and the associated canard mechanisms are likewise established [27]. Thus neither the saddle-focus horseshoes nor the RFSN-II ends are new in isolation. The point here is to couple them in one local theorem: the countably many return strips and all complementary first-exit components are constructed from the vector field, represented in fixed normalized charts, and controlled together with their flight times and exact actions. This joint control yields Corollary 3.5: for each fixed cyclic sign pattern it gives the full linear winding term, an exponentially small remainder, and a limiting closed-orbit action. Thus, within the prescribed physical flow tube, it refines the qualitative period-divergence mechanism in the reversible blue-sky setting [7] and adds an action asymptotic; it is not a global classification of blue-sky families.

Bolotin derives a Shilnikov separatrix map for slow-fast Hamiltonian systems whose frozen dynamics has a transverse saddle-focus homoclinic and uses it to prescribe the evolution of the slow variables [4]. Homburg, Lamb, and Turaev obtain positive topological entropy from strongly transverse symmetric homoclinic tangles associated with normally hyperbolic families of periodic orbits in smooth reversible systems [12]. These results have natural system-level hypotheses and conclusions extending beyond one fixed flow tube. The present theorem is narrower: it resolves every sufficiently high-winding return and first exit in a prescribed local domain, and retains the singular time/action data at two different ends.

Countable Markov shifts with summable-variation potentials have an established thermodynamic theory [24], while exact scattering maps for normally hyperbolic invariant manifolds carry action-based primitives and

coboundary laws [5]. These are important method precedents, but both begin after structures that must be constructed here have already been supplied. Sarig’s theory starts with the countable alphabet, transition law, and potentials. Exact scattering maps start from a normally hyperbolic invariant manifold. Our task is instead to construct the shift and its regular potentials from the singular return/exit geometry, and to prove that both end finite parts are compatible with the standard exact branch identity.

Rigorous computer-assisted verification of a transverse Hamiltonian saddle-focus homoclinic in a concrete model was carried out by Kepley and Mireles James [14]. Their result illustrates the distinction maintained here between a general forcing theorem and the finite validation needed to apply it. At a different singular scale, Baldomá, Giralt, and Guardia obtain an exponentially small homoclinic splitting formula by complex matching [1]. We do not study separatrix splitting; this comparison emphasizes that the new quantitative object claimed here is instead the compatibility of the return/first-exit maps with the two renormalized ends.

TABLE 1. Relation to the closest established frameworks. The last column states the additional object constructed here, not a priority claim beyond the cited scope.

Framework	Established starting point	Role in the present paper
Saddle-focus homoclinics	Finite horseshoes and countable set-level symbolic organization near a homoclinic saddle-focus	A return family uniform over the Hamiltonian class, together with all first exits inside one fixed local domain
RFSN-II blow-up	The algebraic separatrix, noncompact scaling, and associated canard geometry	Coupling of the algebraic and pole channels to the same return map and their time/action renormalizations
Countable shifts and scattering maps	Regularity theory for a given countable shift; exact primitives for scattering maps associated with an NHIM	Construction of the shift, roof, and exact branch primitives directly from the return–exit flow geometry

1.3. Main result and its scope. The next statement is the geometric form of the main result. It separates the dynamical hypotheses from the local coordinates used in the proof. The precise uniformity, boundary regularity, and parameter dependence are given in Definition 2.1; the fully normalized version is Theorems 3.1–3.3.

Theorem 1.1 (First returns and singular exits). *Let $\mathcal{P}$ be a compact C^r parameter manifold, $r \in \{0, 1, 2\}$. Assume that the family $(H_p)_{p \in \mathcal{P}}$, together with its exact primitive, reverser, fixed flow tube, and end data, satisfies the geometric hypotheses of Definition 2.1. In particular, the following conditions hold uniformly in p .*

- (i) *The critical energy surface contains a real-analytic saddle-focus with eigenvalues $\pm\alpha_p \pm i\beta_p$, $\alpha_p, \beta_p > 0$, and a transverse homoclinic orbit.*

- (ii) A fixed local flow tube has a finite clean first-hit stratification. Its nonreturn pieces consist of two specified noncompact exits, the stable-manifold cut, and fixed lateral faces; this list exhausts the boundary through which a nonreturning orbit can first leave the tube.
- (iii) At the infinite-time algebraic exit, a desingularized flow extends to a locally maximal normally expanding invariant hypersurface satisfying the order-three bunching inequalities. At the finite-time pole, the desingularized boundary is a regular-singular sink with a fixed indicial operator whose spectrum is simple and contained in the positive integers.

The same definition fixes the exact primitive and zero energy, the source cell and its proper angular arc, the algebraic clock and action scalings, the pole action order, and the regularity of every transition map. The saddle and homoclinic coefficients and the algebraic compactification are C^k , $k \geq 5$; the event data are C^3 ; the normally expanding algebraic hypersurface has the mixed-total C^3 regularity in (2.2); the saddle germ is analytic in the state variables; and the pole coefficients have C^{m_p+3} regularity. Fix

$$0 < \kappa < \inf_{p \in \mathcal{P}} \frac{2\pi\alpha_p}{\beta_p}.$$

Then a finite cover of $\mathcal{P}$ carries compatible local exact symplectic coordinates, and there is a threshold T_* for the time spent in the saddle block with the following properties.

- (a) After one restriction to a fixed compact source subcell, every point in that subcell whose saddle residence is at least T_* belongs, without a gap or overlap, either to a connected domain of the first-return map or to exactly one first-exit stratum. In each local angular coordinate, all sufficiently large winding cells have the same labelled connected strata and incidence relations as the limiting first-hit arrangement.
- (b) On every return component the Poincaré map has a uniformly contracting Shilnikov cross form, and its transverse return coordinate has scale $\epsilon_n \asymp \exp(-2\pi\alpha_p n/\beta_p)$. After pullback to fixed reference cells and application of the respective normalizations, the cross forms, first-hit maps, flight times, and physical action integrals have all mixed derivatives $D_z^i D_p^j$, $i + j \leq 2$, $j \leq r$, and converge with error $O((1+n)^3 e^{-\kappa n})$. The flight time is normalized by its linear winding term; the two singular ends use the finite parts in (c).
- (c) The algebraic time and action have finite parts when trajectories are compared at the same compactifying coordinate. The pole time has an ordinary finite limit. Its action has a finite part at fixed remaining physical time after subtraction of the finite Laurent-log expansion determined by the boundary jet. On every finite Poincaré branch

$$P_a : \Sigma_a^- \longrightarrow \Sigma_a^+, \quad B_a^{\text{phys}}(z) = \int_0^{\tau_a(z)} \lambda(X)(\Phi^t z) dt,$$

one has

$$P_a^*(\lambda|_{\Sigma_a^+}) - \lambda|_{\Sigma_a^-} = dB_a^{\text{phys}}.$$

These action primitives compose exactly, and both end finite parts obey the same composition and coordinate-change laws.

- (d) *In every local angular coordinate the return domains give a two-vertex countable Markov shift with summable-variation roof and action functions. On overlaps the first-hit relation, the trapped Poincaré system, closed periods, and closed actions agree.*

The first-hit arrangement in hypothesis (ii) is an input at the limiting physical tube. The persistence theorem proves something different: every sufficiently high-winding cell has precisely the same labelled strata and connected components, with no additional component born under the exponentially small passage perturbation. Likewise, the normally expanding hypersurface in (iii) is a standard geometric hypothesis; the product block, cone, and corridor inequalities used for the normal-form example are only sufficient criteria for checking it.

Corollary 1.2 (The reversible folded saddle-node normal form). *For the Hamiltonian system (1.1), the hypotheses of Theorem 1.1 hold on a fixed critical-energy flow tube. Consequently, for every $\kappa \in (0, 2\pi)$ the sufficiently long saddle-residence region within the fixed compact source subcell has the exhaustive first-return/first-exit decomposition, the two action finite parts, exact action composition, and the countable coding described above. The corresponding roof and action functions satisfy the period, action-rate, and entropy conclusions of Corollary 3.5 and Theorem 3.6.*

Moreover, the same conclusion holds for the relatively open family in Corollary 8.7; that topology fixes the physical block and tubes, the cooriented event germs, the algebraic boundary factorizations and invariant faces, the analytic saddle class, and the pole action order and indicial class. This is relative openness, not ambient openness among unrestricted Hamiltonian systems. The exact and interval arguments establishing the normal-form hypotheses are given in Section 8 and Appendix C.

Theorem 2.12 constructs the local exact symplectic coordinates used in the proof. Theorem 3.1 is the complete weighted and parameter-dependent statement, Theorem 3.2 treats the two end finite parts, and Theorem 5.7 proves independence of the compatible local descriptions. Period, action, and entropy asymptotics are stated separately because they are consequences rather than hypotheses of the return–exit theorem.

This is not an additional existence theorem for a horseshoe. Classical saddle-focus homoclinic theory provides finite horseshoes, and earlier work gives a set-level countable admissibility relation, but neither identifies the complement of the return domains nor controls flight time and Hamiltonian action at the two noncompact exits considered here. The logarithmic local passage and the transverse homoclinic map produce one return strip

for each sufficiently large winding and source–target sign pair; transverse first-hit hypersurfaces exhaust the complement; and the algebraic and pole asymptotics give finite C^2 time and action limits after the stated subtractions. Because the action is integrated along the physical orbit, its branch primitives compose exactly, and the same return domains give the countable Markov coding and summable-variation roof and action functions. The conclusion is local to the prescribed physical flow tube and its long-passage collar. The coordinate estimates exclude local windings $n_i < N_i$ and retain the fixed pole indicial conjugacy class.

The upper rate $\inf_{p \in \mathcal{P}} \frac{2\pi\alpha_p}{\beta_p}$ is geometric. In a local marking, a branch of winding n spends time

$$T_n \sim \frac{2\pi}{\beta_{\mathbf{X}}} n$$

near the saddle-focus, so its transverse scale is

$$e^{-\alpha_{\mathbf{X}} T_n} \sim \exp\left(-\frac{2\pi\alpha_{\mathbf{X}}}{\beta_{\mathbf{X}}} n\right).$$

Fixing $\kappa < \inf_{p \in \mathcal{P}} \frac{2\pi\alpha_p}{\beta_p}$ leaves the uniform strict rate gap needed for two-jet estimates on the finite parameter cover. For the normal-form system (1.1), $\alpha_0 = \beta_0$, and the upper exponent is 2π . We do not claim uniform constants as $\kappa \uparrow \inf_{p \in \mathcal{P}} \frac{2\pi\alpha_p}{\beta_p}$.

The result is local to the fixed flow tube and its long-passage region: a lateral exit makes no assertion about the subsequent orbit, and low windings are excluded. The local angular cuts and winding labels need not globalize, and a one-sided symbolic word determines a stable plaque rather than a single point. Section 10 records the remaining scope limits in one place.

1.4. Proof strategy. The proof separates local coordinate construction, the saddle-focus passage, the finite first-hit geometry, and the two end compactifications.

Local coordinates and long passages. Local spectral trivializations, equivariant exact Darboux coordinates, and the reversible analytic normal form produce transverse sections with exact action-angle coordinates and a flight-time function. Event-free orbit slides connect them to the physical faces. In these coordinates the long saddle passage is a mixed boundary-value problem: stable data are imposed at the incoming end and unstable data at the outgoing end. Volterra estimates give the orbit and its mixed two-jets. Comparing its action with the two limiting half-orbit integrals leaves an exponentially small interaction term.

Matching and the complete first-hit partition. The logarithmic phase of the local passage and the finite homoclinic map give a matching equation. The implicit-function theorem produces the high-winding return strips and their contracting Shilnikov cross forms on the scale $\exp(-2\pi\alpha n/\beta)$. Local implicit graphs alone do not give exhaustiveness. That step uses whole-cell C^2 convergence and a controlled isotopy of the clean, neat first-hit stratifications to

obtain a bijection of connected strata; strict first-hit margins preserve event order. Hence no additional high-winding component is hidden between the constructed return and exit sets.

Finite parts, exact composition, and coordinate changes. At the algebraic end, an escaping orbit and a fixed reference orbit are compared at the same compactifying coordinate. At the pole, physical remaining time is the asymptotic variable and a regular-singular recursion determines the Laurent–log subtraction. The remaining time and action terms have uniform mixed C^2 limits. Because action is always the physical integral $\int \lambda(X) dt$, it composes exactly before the limit. Changes of section or primitive add endpoint coboundaries; changes of angular cut give bounded winding recodings. A finite common refinement therefore identifies the physical first-hit relation, trapped system, closed periods, and closed actions on chart overlaps.

Coding and the normal-form application. The cross-form equations define a contraction on sequence space and hence a countable Markov coding; exponential dependence on the first differing symbol gives summable variations. The entropy and action-rate estimates are consequences of this coding. Exact calculations and interval enclosures establish the geometric hypotheses for (1.1); their strict margins yield the relative-open family of Corollary 8.7. The selected reversible branch and its restricted-Hamiltonian critical points are a separate consequence, not an input to the return–exit theorem.

Sections 2–3 formulate the geometric hypotheses, construct the local coordinates, and state the quantitative results. Sections 4–7 prove the passage, first-hit, finite-part, and coding statements; Section 8 treats the normal-form application. Appendix A gives the computable interval-block test for the algebraic hypersurface; weighted function spaces, validation records, uniform end limits, and the optional Jost calculation occupy the remaining appendices.

Dependency map. Table 2 separates the logical input/output layers of the main theorem from the optional modules. Its short package codes are expanded immediately below; Appendix C gives the exact certificate paths and quantitative margins.

TABLE 2. Hypotheses, theorem outputs, and computational dependencies. The optional rows are not in the logical dependency chain of Theorem 1.1 or Corollary 1.2.

Input layer	Mathematical output	RFSN-II validated evidence (supplementing exact identities; rows may overlap)
Analytic saddle and transverse homoclinic, (I1)–(I2)	Parameter-local exact symplectic charts, the long saddle passage, and the matching graph	M3
Clean event geometry, (I3)	Persistence of the prescribed limiting census, with no new component, on every sufficiently long-residence cell	Jointly M1–M4
Algebraic compactification, (I4)	Local graph representation and persistence of the supplied normally expanding hypersurface; algebraic time/action finite parts	M1, M2
Regular-singular pole, (I5)	Pole entry, finite remaining time, and the intrinsic Laurent–log action finite part	M4
The preceding four rows	Theorem 1.1, its exact cocycle, coding, and Corollary 1.2	M1–M4, plus the provenance audit during replay
Optional general selected-branch theorem	Corollary 3.9 and its analytic proof	No certificate dependency
Optional normal-form selected branch	Proposition 8.5, Corollary 8.9, and Appendix E	S1–S5; none is a main-theorem dependency
Optional restricted matching genericity	Section 9	Analytic perturbation argument; no certificate dependency

M1: `future-target-fold`; M2: `origin-algebraic-heteroclinic`; M3: `universal-core-symmetric-homoclinic`; M4: `origin-unstable-pole-entry`. The codes S1–S5 denote, in the order printed in Appendix C, the five additional claim-bearing packages for the optional normal-form selected branch. The complete `all` replay also runs two ancillary packages and the `toolchain-metadata-audit` node; these do not supply hypotheses of the main theorem.

Figure 2 summarizes this route and separates the analytic construction from the normal-form verification.

SCHEMATIC THEOREM ROUTE

TWO-END RETURN--EXIT THEOREM

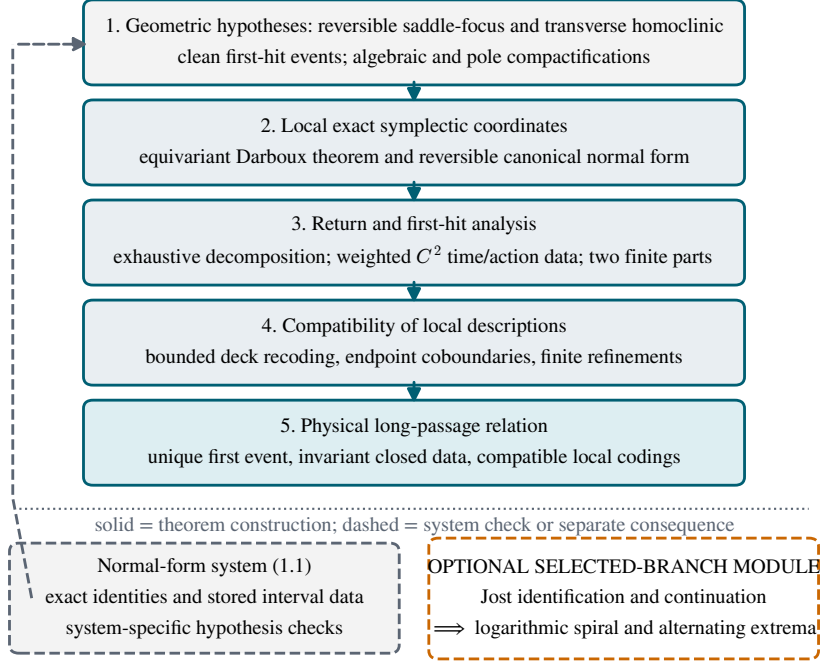


FIGURE 2. Route through the theorem. Intrinsic physical data produce a finite parameter-local family of reversible exact symplectic coordinates. The local analytic construction gives the return/first-exit, time, action, and end data; compatible coordinate changes identify these descriptions as one physical long-passage relation. Neither end is derived from the saddle-focus germ, and no global winding label is asserted. The dashed arrow records exact and interval verification of the intrinsic hypotheses for (1.1). The selected reversible branch is an optional consequence using the Jost identification and finite continuation. The diagram is schematic and is not an exhaustive dependency graph.

2. GEOMETRIC HYPOTHESES AND LOCAL EXACT SYMPLECTIC COORDINATES

2.1. Geometric data. The return–exit theorem is proved in coordinates, but its geometric input need not include those coordinates. We therefore distinguish a physical configuration from a *marked representation* of that configuration. The definition below deliberately retains the two compactified

ends and the event hypersurfaces as geometric hypotheses: they are not consequences of a saddle-focus homoclinic orbit.

Definition 2.1 (Intrinsic two-ended Hamiltonian configuration). Let $\mathcal{P}$ be a compact C^r parameter manifold, $r \in \{0, 1, 2\}$. An intrinsic two-ended configuration over $\mathcal{P}$ consists of a fixed exact symplectic four-manifold $(M, d\lambda)$ with an anti-symplectic involution $\mathcal{R}$ and a chosen physical primitive satisfying

$$\mathcal{R}^*\lambda = -\lambda,$$

an integer $m_p \geq 2$, a conjugacy class of real 3×3 matrices whose spectrum is simple and contained in the positive integers

$$(2.1) \quad \Lambda_p = \{\lambda_1 < \lambda_2 < \lambda_3\} \subset \mathbb{N},$$

and a C^r family of $\mathcal{R}$ -invariant Hamiltonians H_p , $p \in \mathcal{P}$, with the following data and hypotheses. All compactifications, bundles, submanifolds, boundary defining functions, fields, bundle atlases, splittings, metrics, and first-hit maps listed below form parameter-locally trivial C^r families in the displayed C^3 , C^k , or C^{m_p+3} state norms, except for the normally expanding hypersurface in (I4). Parameter-locally write that hypersurface as a normal section Γ_p over a fixed domain; its required regularity is the mixed-total scale

$$(2.2) \quad \sup_{\substack{i+j \leq 3 \\ j \leq r}} \|D_z^i D_p^j \Gamma_p\| < \infty.$$

All strict inequalities and compactness statements are uniform over $p \in \mathcal{P}$.

- (I1) *Analytic saddle-focus and physical saddle block.* There is a C^r family $O_p \in \text{Fix } \mathcal{R}$ of critical points on $H_p^{-1}(0)$ such that

$$\text{spec } DX_{H_p}(O_p) = \{\pm\alpha_p \pm i\beta_p\}, \quad \inf_{p \in \mathcal{P}} \min\{\alpha_p, \beta_p\} > 0.$$

The saddle neighborhoods carry parameter-locally trivial real-analytic structures in which $d\lambda$, a local representative of λ , and $\mathcal{R}$ are real analytic. The germs of H_p at O_p are real analytic in the state variables on parameter-locally uniform complex neighborhoods and, together with these analytic symplectic and reversible data, depend C^r on p in the corresponding analytic norms. A fixed physical saddle block $\mathcal{B}_{\text{sf},p}$ is supplied with transverse incoming and outgoing faces S_p^s, S_p^u . The first-hit time across this block is denoted by $\mathcal{T}_{\text{sf},p}$; the faces and the block vary C^r with p . Its punctured long-passage collar contains no event face, so auxiliary transverse sections inside the block may be joined to the physical faces by event-free orbit slides with uniformly bounded times and mixed derivatives through order three.

- (I2) *Transverse homoclinic tube.* There is a C^r family of selected homoclinic orbits γ_p and a compact C^k , $k \geq 5$, family of flow tubes containing a fundamental segment of each γ_p ; H_p and X_{H_p} are C^k in

the state variables on these tubes and C^r in p in the corresponding norms. Along that segment,

$$(2.3) \quad T_z W^s(O_p) + T_z W^u(O_p) = T_z H_p^{-1}(0).$$

The finite homoclinic flight from S_p^u to S_p^s has a uniform first-hit margin.

- (I3) *Physical event arrangement.* The incoming face carries a compact physical source cell whose boundary is transverse to the phase-circle fibers of the oriented blow-up and contains the homoclinic and the two designated end-anchor projections in its interior. The cell is local in the phase direction: on the oriented blow-up, the union of its two signed limiting residual-phase traces and the designated anchor traces is contained, parameter-locally, in an open arc bundle whose complementary arc has positive uniform length. This proper-arc condition is unchanged by a degree-one reparametrization of the phase-circle fibers. The theorem concerns this one local source cell; a larger phase cover is not asserted. The compact return–exit tube carries a finite C^3 family of cooriented event hypersurface germs. Together with the return aperture, these event faces exhaust the relevant outgoing boundary of the fixed tube: an orbit that leaves before returning has a prescribed first hit. The connected pure-dimensional strata on the outgoing band and, inside the homoclinic aperture, the strata obtained after pulling back the return-band arrangement form a finite exhaustive list. Each relative interior is assigned exactly one outcome: genuine return, pole or algebraic exit, outgoing- or return-box lateral exit, or the stable-manifold cut. There is no residual component, and a fixed priority assigns an outcome on common faces and corners without merging dimensions. On the oriented real blow-up of the stable and unstable axes, their limiting pullbacks and the homoclinic aperture form a finite clean stratification, neat at the boundary faces. Assigned first hits are transverse, and the inactive-face and strict event-order inequalities have positive uniform margins. The limiting algebraic and pole strata each contain a specified interior point with positive boundary, inactive-face, and event-order margins. Cleanliness here is independent of a local eigenframe: two such frames act on the boundary circle bundle by a degree-one diffeomorphism.

- (I4) *Algebraic compactification.* The algebraic channel has a C^k manifold-with-boundary compactification $\overline{M}_{a,p}$, a boundary defining function e , and a desingularized field $Y_{a,p}$ satisfying

$$(2.4) \quad \frac{dt}{ds} = e^{1/2} \widehat{\chi}_{a,p}, \quad Y_{a,p}e = e^2 v_{e,p}, \quad v_{e,p} < 0,$$

with uniform bounds, while $E = 2H_p$ and $e^3 \lambda(X_{H_p})$ extend with the stated C^k regularity. The extension of E is a submersion defining

the boundary energy face. Inside the boundary energy face there is a cooriented invariant flag with one-dimensional normal quotient whose desingularized linear rate is uniformly negative. Equivalently, a local defining function b for that flag satisfies, by Hadamard's lemma,

$$Y_{a,p}b = ef_{e,p} + Ef_{E,p} + bv_{b,p}, \quad v_{b,p} < 0,$$

on a sufficiently small collar. Along the common flag, (de, dE, db) is a clean defining system: its conormals are linearly independent with a positive uniform least singular value. On a fixed collar this system extends as a rank-three submersion. There is a compact C^3 locally invariant hypersurface with boundary $\mathcal{W}_{a,p}^{\text{tail}}$ in this collar. It is locally maximal on a compact relative isolating neighborhood: the points whose positive semiorbits remain in that neighborhood are precisely $\mathcal{W}_{a,p}^{\text{tail}}$. The restriction of (e, E, b) to this hypersurface is a C^3 diffeomorphism onto a compact base domain B_p with corners whose interior contains $(e_*, 0, 0)$ for a fixed $e_* > 0$. Every tangential boundary face is strictly inward or structurally invariant. For some $\delta_a > 0$, let Φ_p^s denote the flow of $Y_{a,p}$ and set

$$\begin{aligned} A_{p,z}^\delta &= D\Phi_p^{\delta_a}|_{T_z\mathcal{W}_{a,p}^{\text{tail}}}, \\ N_{p,z}^\delta : T_z\overline{M}_{a,p}/T_z\mathcal{W}_{a,p}^{\text{tail}} &\longrightarrow T_{\Phi_p^{\delta_a}z}\overline{M}_{a,p}/T_{\Phi_p^{\delta_a}z}\mathcal{W}_{a,p}^{\text{tail}}. \end{aligned}$$

With continuous adapted metrics, this hypersurface is normally expanding in forward desingularized time and is order-three bunched:

$$(2.5) \quad \vartheta_{a,j} := \sup_{p,z \in \mathcal{W}_{a,p}^{\text{tail}}} \|(N_{p,z}^\delta)^{-1}\| \|A_{p,z}^\delta\|^j < 1, \quad j = 0, 1, 2, 3.$$

Here $j = 0$ is normal expansion and $j = 1, 2, 3$ are the standard bunching conditions for persistence with three derivatives.

The interval-block hypotheses in Theorem A.1 give a computable sufficient criterion for the existence, local maximality, and bunching of this hypersurface. The product trivialization and its cone estimates are not part of the intrinsic hypothesis. A physical cut $e = e_*$ in the interior of the collar and a finite first-hit/saturation map from a fixed open neighborhood of the whole compact cut slice to the algebraic event face are part of the data, with compact closure and uniform first-hit and coorientation margins. Its domain contains a fixed neighborhood of $\mathcal{W}_{a,p}^{\text{tail}} \cap \{e = e_*\}$.

- (I5) *Pole compactification.* The pole channel has a C^{m_p+3} manifold-with-boundary compactification with boundary defining function ρ . The desingularized physical field ρX_{H_p} extends to a hyperbolic boundary sink $q_{\infty,p}$, and

$$d\rho(\rho X_{H_p}) = \rho a_p, \quad a_p(q_{\infty,p}) < 0.$$

On the intrinsic tangential quotient at the sink, the regular-singular endomorphism

$$(2.6) \quad \mathcal{I}_p = a_p(q_{\infty,p})^{-1} D(\rho X_{H_p}|_{\partial})_{q_{\infty,p}}$$

has the fixed conjugacy class specified above, with spectrum Λ_p in (2.1). The coefficient $\rho^{m_p} \lambda(X_{H_p})$ extends in C^{m_p+3} , and the pole entrance has compact image in the basin of the sink with uniform entry, radial, and spectral-gap margins.

All objects above are physical submanifolds, compactifications, invariant hypersurfaces, or finite-time tangent-cocycle inequalities. The definition contains no eigenframe, angle lift, sign or winding label, matching graph, return strip, or renormalized end potential. In a concrete application the future-staying hypersurface may instead be constructed from the interval-block criterion in Appendix A.

For later comparison, in a fixed product bundle chart with product connection, $DY = \begin{pmatrix} C & B \\ D & a \end{pmatrix}$ and $P = \text{graph } S$, $\|S\| \leq \alpha$. If $\mu_2(C) \leq c$ and $\|B\| \leq b$ in the adapted metrics, then

$$(2.7) \quad \begin{aligned} L(P) &= C + BS, & n(P) &= a - SB, \\ n(P) - j\mu(L(P)) &\geq a - jc - (j+1)b\alpha. \end{aligned}$$

If the right-hand side is bounded below by $\gamma_j > 0$ along every δ_g -admissible evolving plane, integration gives $\vartheta_{g,j} \leq e^{-\gamma_j \delta_g}$. Thus the block inequalities used in the RFSN-II verification are a computable chartwise sufficient criterion for (A.2); they are not part of the intrinsic definition.

2.2. Exact reversible fields and local coordinate descriptions. Let (M, λ) be an exact symplectic four-manifold, $\omega = d\lambda$, and let $\mathcal{R}$ be an anti-symplectic involution. We use the convention

$$(2.8) \quad \iota_X \omega = dH, \quad H \circ \mathcal{R} = H, \quad \mathcal{R}^* \lambda = -\lambda, \quad \mathcal{R}_* X = -X.$$

A *marked Hamiltonian presentation* is the datum

$$(2.9) \quad \mathbf{X} = (X, H, \lambda, \mathcal{R}; \Xi_{\text{sf}}, \Xi_{\text{fin}}, \Xi_{\text{a}}, \Xi_{\text{p}}),$$

with fixed chart domains and fixed transition maps on their overlaps. The equilibrium is marked by $O = 0$ and $H(O) = 0$.

In the saddle-focus chart the field has reversible cross form

$$(2.10) \quad \begin{aligned} \dot{\xi} &= A_{\mathbf{X}} \xi + F_{\mathbf{X}}(\xi, \eta), & \dot{\eta} &= -A_{\mathbf{X}} \eta + G_{\mathbf{X}}(\xi, \eta), & A_{\mathbf{X}} &= \alpha_{\mathbf{X}} I + \beta_{\mathbf{X}} \mathbb{J}, \end{aligned}$$

where $\alpha_{\mathbf{X}}, \beta_{\mathbf{X}} > 0$, $\mathbb{J}$ is the standard complex structure on $\mathbb{R}^2$, the reverser exchanges ξ and η , and

$$(2.11) \quad F(\xi, 0) = F(0, \eta) = G(\xi, 0) = G(0, \eta) = 0.$$

The four vanishing relations are recorded by Hadamard coefficients in the two variables. The compact chart contains a fixed finite union K_{fin} of flow-tube boxes: the primary homoclinic tube, the finite part of the algebraic

connection, the pole-entry tube, and the outgoing, return, and boundary collars used below.

The saddle-focus marking contains one further, genuinely Hamiltonian piece of data. On fixed exact-symplectic incoming and outgoing section charts, and on each component $\sigma\nu > 0$ of the critical energy, the exact local passage is required to be

$$(2.12) \quad \mathcal{L}_{\mathbf{X},\sigma}(\phi, \nu) = (\phi + \Delta_{\mathbf{X},\sigma}(\nu), \nu), \quad \Delta_{\mathbf{X},\sigma}(\nu) = \frac{\beta_{\mathbf{X}}}{\alpha_{\mathbf{X}}} \log |\nu| + b_{\mathbf{X},\sigma} + \rho_{\mathbf{X},\sigma}(\nu).$$

Thus equality of the two action coordinates is exact, not an asymptotic claim. With $D_{\log \nu} := \nu \partial_\nu$, the marked remainder obeys

$$(2.13) \quad \max_{0 \leq j \leq k-2} \sup_{0 < |\nu| \leq \nu_*} \frac{|D_{\log \nu}^j \rho_{\mathbf{X},\sigma}(\nu)|}{|\nu|(1 + |\log |\nu||)} < \infty.$$

The same actual local passage, in the same section charts, carries the physical flight-time law

$$(2.14) \quad T_{\mathbf{X},\sigma}(\nu) = -\frac{1}{\alpha_{\mathbf{X}}} \log |\nu| + t_{\mathbf{X},\sigma} + \tau_{\mathbf{X},\sigma}(\nu), \quad c_{\mathbf{X},\sigma} := e^{\alpha_{\mathbf{X}} t_{\mathbf{X},\sigma}} \geq c_* > 0,$$

where

$$(2.15) \quad \max_{0 \leq j \leq k-2} \sup_{0 < |\nu| \leq \nu_*} \frac{|D_{\log \nu}^j \tau_{\mathbf{X},\sigma}(\nu)|}{|\nu|(1 + |\log |\nu||)} < \infty.$$

Thus $c_{\mathbf{X},\sigma}$ is derived from the exact local flight time, not an independently chosen coefficient. The map (2.12) and the clock (2.14) are two components of the same marked local flow. They are local section data; no return orbit is used to define them.

At the algebraic end use coordinates $z_a = (e, E, b, \zeta)$, where $E = 2H$. The physical field is still denoted by X , but the regular compactified field is the desingularization

$$Y_a = \chi_a X, \quad \frac{dt}{ds} = \chi_a = e^{1/2} \hat{\chi}_a, \quad 0 < \chi_* \leq \hat{\chi}_a \leq \chi^*.$$

Thus $E_s = 0$, and a dot in the algebraic chart denotes differentiation with respect to s . The regular compactified physical action coefficient is

$$(2.16) \quad a_a^c = e^3 \lambda(X).$$

The raw coefficient conditions are

$$(2.17) \quad Y_a^e = e^2 v_e, \quad v_e \leq -a_* < 0, \quad Y_a^b = e f_e + E f_E + b v_b, \quad v_b \leq -\lambda_* < 0.$$

Fix a compact rectangular base corridor B_Γ in the $x = (e, E, b)$ coordinates and a number $\zeta_\Gamma > 0$ such that

$$(2.18) \quad Q_\Gamma = B_\Gamma \times [-\zeta_\Gamma, \zeta_\Gamma]$$

has a fixed collar in the algebraic chart. Write the compactified field and its derivative in base-normal blocks as

$$(2.19) \quad Y_{a,\mathbf{X}}(x, \zeta) = (F_{\mathbf{X}}(x, \zeta), G_{\mathbf{X}}(x, \zeta)), \quad DY_{a,\mathbf{X}} = \begin{pmatrix} C_{\mathbf{X}} & B_{\mathbf{X}} \\ D_{\mathbf{X}} & a_{\mathbf{X}} \end{pmatrix}.$$

Here $C_{\mathbf{X}}$ is a 3×3 matrix, $B_{\mathbf{X}}$ is a column, $D_{\mathbf{X}}$ is a row, and $a_{\mathbf{X}} = \partial_{\zeta} G_{\mathbf{X}}$. We use the Euclidean logarithmic norm

$$(2.20) \quad \mu_2(C) = \lambda_{\max} \left(\frac{C + C^T}{2} \right).$$

The face, cone, and finite-order rate conditions imposed on these raw coefficients are stated in Theorem A.1; they are only a sufficient test for (H3). In particular, f_e may be nonzero: no complete reference curve is fixed in the definition of the class.

At the pole put

$$(2.21) \quad \rho = x^{-1/2}, \quad A = \rho^3 y, \quad B = \rho q, \quad C = \rho^4 w, \quad \frac{d\tau}{dt} = \rho^{-1}.$$

For the core, the compactified field is

$$(2.22) \quad \dot{\rho} = -\frac{1}{2}A\rho, \quad \dot{A} = 1 - \frac{3}{2}A^2 - C, \quad \dot{B} = 1 - \frac{1}{2}AB, \quad \dot{C} = -2AC + \rho^4 B,$$

and has the hyperbolic sink

$$(2.23) \quad p_{\infty} = (0, \sqrt{2/3}, \sqrt{6}, 0).$$

For a fixed marked coefficient class, choose an integer $m_p \geq 2$ and a real matrix $\mathcal{I}$ with simple positive integral spectrum Λ_p as in (2.1), and set

$$(2.24) \quad a_p^c = \rho^{m_p} \lambda(X).$$

For the normal-form system, $m_p = 6$. For a general presentation, use a marked C^{m_p+3} boundary chart centered at its pole equilibrium and write

$$(2.25) \quad X_p = \rho a(\rho, u) \partial_{\rho} + b(\rho, u) \cdot \partial_u, \quad u \in \mathbb{R}^3, \quad b(0, 0) = 0.$$

The radial coefficient satisfies $a(0, 0) < 0$. After a marked linear change in u , the principal regular-singular operator is required to be

$$(2.26) \quad \mathcal{I}_X := a(0, 0)^{-1} D_u b(0, 0) = \mathcal{I}, \quad \text{spec } \mathcal{I} = \Lambda_p \subset \mathbb{N}, \quad \Lambda_p \text{ simple.}$$

For the normal-form system these fixed data are

$$(2.27) \quad m_p = 6, \quad \mathcal{I} = \begin{pmatrix} 6 & 0 & \sqrt{6} \\ 3 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix}, \quad \text{spec } \mathcal{I} = \{1, 4, 6\}.$$

For the core, take $u = (A - \sqrt{2/3}, B - \sqrt{6}, C)$ in (2.22); direct linearization gives (2.27). Fix a relatively compact neighborhood $U_p \subset \mathbb{R}^3$ of $u = 0$. For a scalar or vector coefficient R in this chart, put

$$(2.28) \quad \|R\|_{m_p+3;p} = \max_{j+|\gamma| \leq m_p+3} \sup_{(\rho, u) \in [0, \rho_*] \times U_p} |\partial_{\rho}^j \partial_u^{\gamma} R|.$$

Only m_p , the radial scaling, and the indicial operator are fixed. The non-linear boundary coefficients, every positive normal jet of the field, and the complete boundary jet of a_p^c are allowed to vary.

The positive-integer-spectrum assumption selects a standard resonance class for which the singular density has an ordinary finite Laurent-log expansion. The arguments below apply to every fixed $(m_p, \mathcal{I})$ in this class; the values $(6, \{1, 4, 6\})$ enter only in the normal-form specialization. A fixed nonintegral spectrum would instead generate a finite polyhomogeneous index set with nonintegral powers, which is not asserted here.

Definition 2.2 (Admissible coefficient class). Fix $k \geq 5$. The space $\mathfrak{H}_{\text{adm}}^k$ consists of marked Hamiltonian presentations satisfying (2.8)–(2.11), the exact marked passage and clock (2.12)–(2.15), the fixed overlap identities, (2.17), and the pole conditions

$$(2.29) \quad a(0, 0) < 0, \quad b(0, 0) = 0, \quad \mathcal{I}_X = \mathcal{I}, \quad \|(a, b, a_p^c)\|_{m_p+3; p} < \infty.$$

Equip this class with the product coefficient topology specified in Appendix B.1. In brief, it controls the marked saddle-focus, finite-flight, algebraic, and pole coefficients in the displayed regularity classes, together with all marked sections, the C^3 event faces and cell charts, a finite algebraic tubular atlas (including the interval-block atlas when that certificate is used), and the chart transitions in their respective regularity classes; it contains no return orbit or first-exit component. Exactness, reversibility, the boundary factorizations, the compatibility identity $Y_a = e^{1/2} \widehat{\chi}_a X$ on the algebraic overlap, and the fixed pole indicial type define the ambient coefficient subspace. All open sets below are relatively open in this space.

If $\varpi \mapsto \mathbf{X}_\varpi$ is a C^r family in this coefficient space, derivatives in ϖ are taken in every Banach norm specified in Appendix B.1. In particular, for $r \in \{1, 2\}$,

$$(2.30) \quad \sup_{\varpi} \max_{0 \leq j \leq r} \|D_{\varpi}^j(a_{\varpi}, b_{\varpi}, a_{p, \varpi}^c)\|_{m_p+3; p} < \infty.$$

Thus the pole coefficients and their finite jets may vary with the parameter. Only the compactified-action order m_p , the radial scaling convention, and the normalized indicial matrix (2.26) remain fixed throughout the family.

Remark 2.3 (Role of the marked coefficient class). The space $\mathfrak{H}_{\text{adm}}^k$ is the coordinate model in which Theorems 3.1–3.3 are proved. It is not the input space of the intrinsic Theorem 1.1. Theorem 2.12 constructs a finite parameter-local atlas of such representations from Definition 2.1. A global reversible symplectic eigenframe or angle lift is neither assumed nor concluded.

The analytic hypothesis at the saddle is essential for this construction. No finite-smooth normal-form theorem with the present weighted remainder and parameter budget is asserted. An independently supplied finitely smooth marking may still be used directly in the coordinate theorems. The coefficient topology supplies the local coordinate descriptions used in the proof,

while Definition 5.6 and Theorem 5.7 describe their coordinate changes and identify the resulting physical dynamical objects.

Lemma 2.4 (Parameter-local equivariant exact Darboux coordinates). *Fix $p_0 \in \mathcal{P}$ and assume (I1). After shrinking to a parameter neighborhood U of p_0 , there are $R > 0$ and a C^r family of real-analytic embeddings*

$$\Psi_p : B_R \subset \mathbb{R}^4 \longrightarrow M, \quad \Psi_p(0) = O_p.$$

Put

$$\omega_0 = \sum_{j=1}^2 dy_j \wedge dx_j, \quad \lambda_0 = \frac{1}{2} \sum_{j=1}^2 (y_j dx_j - x_j dy_j),$$

let $C = \text{diag}(1, -1)$, and set $\mathcal{R}_0(x, y) = (Cy, Cx)$. The embeddings may be chosen so that

$$(2.31) \quad \Psi_p^* d\lambda = \omega_0, \quad \Psi_p \circ \mathcal{R}_0 = \mathcal{R} \circ \Psi_p, \quad \Psi_p^* \lambda = \lambda_0 + df_p,$$

where $f_p(0) = 0$ and $f_p \circ \mathcal{R}_0 = -f_p$. On every smaller complex ball, $p \mapsto (\Psi_p, f_p)$ is C^r in analytic norms. The same coordinates satisfy

$$(2.32) \quad (H_p \circ \Psi_p)_2 = \alpha_p I_1 + \beta_p I_2.$$

Proof. Put

$$L_p = DX_{H_p}(O_p), \quad J_p = D\mathcal{R}(O_p), \quad \Omega_p = d\lambda|_{O_p}.$$

Then

$$J_p L_p = -L_p J_p, \quad J_p^* \Omega_p = -\Omega_p, \quad \Omega_p(L_p u, v) + \Omega_p(u, L_p v) = 0.$$

After shrinking U , four fixed disjoint contours enclose the simple eigenvalues $\pm\alpha_p \pm i\beta_p$. The corresponding Riesz projectors are C^r in p . Choose a C^r local section of the $(\alpha_p + i\beta_p)$ eigenline. The reverser and complex conjugation supply the other three eigenlines. The last displayed identity shows that opposite eigenlines pair nondegenerately under Ω_p , while all other spectral pairings vanish. Normalizing this pairing and taking a fixed real recombination gives a C^r symplectic frame in which

$$(2.33) \quad J_p = \mathcal{R}_0, \quad H_{2,p} = \alpha_p I_1 + \beta_p I_2.$$

This construction is parameter-local; no global eigenframe is used.

Choose a C^r family of analytic coordinate maps $c_p : V_p \rightarrow \mathbb{R}^4$, $c_p(O_p) = 0$, whose differentials realize the symplectic identifications just constructed, and define

$$\widehat{c}_p(q) = \frac{1}{2}(c_p(q) + \mathcal{R}_0 c_p(\mathcal{R}q)).$$

This is an identity between coordinate maps, and

$$\widehat{c}_p \circ \mathcal{R} = \mathcal{R}_0 \circ \widehat{c}_p, \quad D\widehat{c}_p(O_p) = Dc_p(O_p).$$

The analytic inverse-function theorem on a common smaller complex neighborhood gives equivariant embeddings $j_p = \widehat{c}_p^{-1}$. They satisfy

$$j_p^* d\lambda(0) = \omega_0, \quad (H_p \circ j_p)_2 = \alpha_p I_1 + \beta_p I_2.$$

Set $\delta_p = j_p^* d\lambda - \omega_0$ and let $\mathcal{E}$ be the radial vector field. The radial homotopy operator gives

$$a_p(z)[v] = \int_0^1 t \delta_p(tz)[z, v] dt, \quad da_p = \delta_p, \quad \mathcal{R}_0^* a_p = -a_p.$$

After one uniform shrinking, $\omega_{p,s} = \omega_0 + s\delta_p$ is symplectic for $0 \leq s \leq 1$. Define $Y_{p,s}$ by $\iota_{Y_{p,s}} \omega_{p,s} = -a_p$. The two anti-invariance identities imply $(\mathcal{R}_0)_* Y_{p,s} = Y_{p,s}$. If $\varphi_{p,s}$ denotes its flow, set $\Psi_p = j_p \circ \varphi_{p,1}$. The Moser identity and the commutation of $\varphi_{p,s}$ with $\mathcal{R}_0$ give the first two identities in (2.31). Analytic ODE dependence on a common complex ball gives the asserted C^r parameter dependence. Since $\delta_p(0) = 0$, the Moser field vanishes to second order; its flow is tangent to the identity and therefore preserves (2.33).

Finally, $\Psi_p^* \lambda - \lambda_0$ is closed on a ball and is anti-invariant under $\mathcal{R}_0$. Its radial primitive, normalized to vanish at the origin, is the odd function f_p in (2.31). This also fixes the exact gauge used below. $\square$

Lemma 2.5 (Second-order parameter majorant for complex-saddle normalization). *Let $r \in \{0, 1, 2\}$, let $U \Subset \mathbb{R}^d$ be a parameter coordinate neighborhood, and let $\Delta_R \subset \mathbb{C}^4$ be a polydisc. Suppose*

$$F_p = F_{2,p} + \sum_{q \geq 1} \widehat{F}_{q,p}, \quad F_{2,p} = \lambda_{1,p} z_1 w_1 + \lambda_{2,p} z_2 w_2,$$

where $\widehat{F}_{q,p}$ is homogeneous of degree $q + 2$ and

$$\lambda_{1,p} = \alpha_p + i\beta_p, \quad \lambda_{2,p} = \alpha_p - i\beta_p, \quad \inf_{p \in U} \min(\alpha_p, \beta_p) > 0.$$

Assume that $p \mapsto F_p$ extends as a C^r map from a neighborhood of $\overline{U}$ into the analytic functions on a neighborhood of $\overline{\Delta_R}$, with finite C^r norm there. For $G = \sum_{a,b} G_{ab} z^a w^b$, use the coefficient-majorant norm

$$(2.34) \quad \|G\|_{\theta R}^\# = \sum_{a,b} |G_{ab}| \theta^{|a|+|b|} R^{a+b}, \quad 0 < \theta \leq 1,$$

where $R = (R_1, R_2)$ and $R^{a+b} = \prod_{j=1}^2 R_j^{a_j+b_j}$, and let Π be the fixed projection onto the monomials with $a = b$.

Apply the normalized Lie recursion of [9, Sections 2–4], with $\Pi \chi_{q,p} = 0$, and denote its degree- $(q+2)$ generating and normal-form terms by $\chi_{q,p}$ and $Z_{q,p}$. To keep the source recursion visible, write the Hamiltonian after the first q normalized steps as

$$(2.35) \quad F_p^{(q)} = F_{2,p} + \sum_{j=1}^q Z_{j,p} + \sum_{s>q} H_{s,p}^{(q)}, \quad H_{s,p}^{(0)} = \widehat{F}_{s,p},$$

where $H_{s,p}^{(q)}$ is homogeneous of degree $s + 2$. Fix $0 < d < 1/2$ and put

$$(2.36) \quad d_q = \frac{6d}{\pi^2 q^2}, \quad \delta_0 = 0, \quad \delta_q = \sum_{j=1}^q d_j, \quad R_q = (1 - \delta_q)R.$$

There are constants $B, C_0, \dots, C_r$, independent of q and p , such that

$$(2.37) \quad \sup_{p \in U} \left(\|D_p^\ell \chi_{q,p}\|_{R_{q-1}}^\# + \|D_p^\ell Z_{q,p}\|_{R_{q-1}}^\# \right) \leq C_\ell q^{2\ell} B^q, \quad 0 \leq \ell \leq r.$$

Moreover, after replacing R by εR for one sufficiently small $\varepsilon > 0$, the finite normalizing transformations $\Theta_p^{(N)}$, finite normal forms $g_p^{(N)}$, and normalized exact primitives

$$(2.38) \quad B_p^{(N)}(z) = \int_0^1 \left[((\Theta_p^{(N)})^* \lambda_0 - \lambda_0)_{tz} \right] (z) dt$$

converge through parameter order r in analytic norms on every polydisc compactly contained in $\Delta_{(1-d)\varepsilon R}$. In particular, for $r = 2$ the convergence includes both parameter derivatives asserted in (2.37).

Proof. After one preliminary radius shrink, the analytic C^r parameter bound gives constants E_ℓ, H such that

$$(2.39) \quad \sup_{p \in U} \|D_p^\ell \widehat{F}_{q,p}\|_R^\# \leq E_\ell H^q, \quad 0 \leq \ell \leq r.$$

All derivatives are taken in the fixed parameter chart on U .

Put $\mathcal{L}_p = \{F_{2,p}, \cdot\}$. On the complement of Π it is diagonal. If $k = a - b \neq 0$, its divisor, up to the immaterial sign fixed by the Poisson-bracket convention, is $\Delta_{k,p} = \langle k, \lambda_p \rangle$. With $m_* = \inf_{p \in U} \min(\alpha_p, \beta_p)$,

$$(2.40) \quad \begin{aligned} |\Delta_{k,p}|^2 &= \alpha_p^2 (k_1 + k_2)^2 + \beta_p^2 (k_1 - k_2)^2 \\ &\geq 2m_*^2 (k_1^2 + k_2^2). \end{aligned}$$

Thus $|\Delta_{k,p}| \geq m_* |k|_1$. Since λ_p has the available C^r derivatives on a neighborhood of $\overline{U}$, for parameter directions v, w one has, whenever the displayed derivative order is at most r ,

$$|D_v \Delta_{k,p}| \leq M_1 |v| |k|_1, \quad |D_{v,w}^2 \Delta_{k,p}| \leq M_2 |v| |w| |k|_1.$$

The exact inverse-divisor identities are

$$(2.41) \quad D_v(\Delta_{k,p}^{-1}) = -\frac{D_v \Delta_{k,p}}{\Delta_{k,p}^2},$$

$$(2.42) \quad D_{v,w}^2(\Delta_{k,p}^{-1}) = 2 \frac{(D_v \Delta_{k,p})(D_w \Delta_{k,p})}{\Delta_{k,p}^3} - \frac{D_{v,w}^2 \Delta_{k,p}}{\Delta_{k,p}^2}.$$

Consequently the first two parameter derivatives of the inverse divisor have the same $|k|_1^{-1}$ degree gain as the inverse divisor itself.

Equivalently, define the normalized homological inverse by

$$(2.43) \quad \mathcal{G}_p = (\mathcal{L}_p|_{\ker \Pi})^{-1}(I - \Pi), \quad \Pi \mathcal{G}_p = 0.$$

We orient the Lie pullback by $\frac{d}{dt}(F \circ \Phi_\chi^t) = \{F, \chi\} \circ \Phi_\chi^t$. With this convention for the time-one maps below, the step- q homological equation is, more explicitly,

$$(2.44) \quad -\mathcal{L}_p \chi_{q,p} + Z_{q,p} = H_{q,p}^{(q-1)}, \quad Z_{q,p} = \Pi H_{q,p}^{(q-1)}, \quad qquad \chi_{q,p} = -\mathcal{G}_p H_{q,p}^{(q-1)}.$$

Its exact first and second parameter recurrences are

$$(2.45) \quad D_v \mathcal{G}_p = -\mathcal{G}_p(D_v \mathcal{L}_p) \mathcal{G}_p,$$

$$(2.46) \quad D_{v,w}^2 \mathcal{G}_p = \mathcal{G}_p(D_w \mathcal{L}_p) \mathcal{G}_p(D_v \mathcal{L}_p) \mathcal{G}_p + \mathcal{G}_p(D_v \mathcal{L}_p) \mathcal{G}_p(D_w \mathcal{L}_p) \mathcal{G}_p \\ - \mathcal{G}_p(D_{v,w}^2 \mathcal{L}_p) \mathcal{G}_p.$$

Equations (2.40)–(2.46) give

$$(2.47) \quad \|(D_p^\ell \mathcal{G}_p) G\|_{\theta R}^\# \leq G_\ell \|G\|_{\theta R}^\#, \quad \Pi G = 0, \quad 0 \leq \ell \leq r,$$

with constants independent of the homogeneous degree, p , and $0 < \theta \leq 1$.

The remaining parameter recurrences in the Lie algorithm are the exact Leibniz identities

$$(2.48) \quad D_v \{G_1, G_2\} = \{D_v G_1, G_2\} + \{G_1, D_v G_2\},$$

$$(2.49) \quad D_{v,w}^2 \{G_1, G_2\} = \{D_{v,w}^2 G_1, G_2\} + \{D_v G_1, D_w G_2\} \\ + \{D_w G_1, D_v G_2\} + \{G_1, D_{v,w}^2 G_2\}.$$

They also give the parameter derivatives of every iterated Lie derivative.

Table 3 records exactly what is imported from the fixed-Hamiltonian recursion and what is added here. In particular, after step $q-1$ the homological equation is driven by $H_{q,p}^{(q-1)}$, not by the original Taylor block $\widehat{F}_{q,p}$. Giorgilli uses $L_\phi f = \{f, \phi\}$. With our operator $\mathcal{L}_p = \{F_{2,p}, \cdot\}$, his normalized homological solution therefore uses $-\mathcal{G}_p$ under the forward Lie-pullback convention. This convention-dependent sign has no effect on any coefficient majorant below.

We now compare the differentiated recursion with the fixed-parameter majorant. For fixed p , the coefficientwise nonnegative recursion associated with formula (8) of [9] is, under Table 3, exactly the unmarked recursion for (2.35). In two degrees of freedom $\mathcal{P}^a = \mathcal{P}^b = 0$. Consequently (35), (37), and (39), together with the estimates of Section 3.4 there and the schedule (2.36), give geometric coefficient majorants for $\chi_{q,p}$ and $Z_{q,p}$ on $\Delta_{R_{q-1}}$ and for $H_{s,p}^{(q)}$ on Δ_{R_q} . Enlarging one constant packages the factors denoted $\mu_{q,s}$ and $T_{q,s}$ in that paper into $C_0 B^s$. This is the entire imported part of the argument; all parameter marking below is additional.

For the additional parameter bookkeeping, represent an iterated Poisson-bracket term by its rooted binary tree and assign to a homogeneous expression its excess degree, namely its degree minus two. At a bracket vertex excess degrees add. Since every nonlinear leaf and every normalizing generator has positive excess degree, a tree of total excess q has at most q nonlinear leaves and $q-1$ bracket vertices. Adding the projections and homological

TABLE 3. Crosswalk between the normalization used here and the fixed-Hamiltonian recursion of [9]. Equation numbers in the middle column refer to that paper.

Present notation	Source notation	Exact role and scope
$F_{2,p}$	H_0 in (1)	Quadratic Hamiltonian defining the homological operator. The subscript shift is essential: $F_{2,p}$ does not correspond to H_2 .
$\widehat{F}_{s,p} = H_{s,p}^{(0)}$	$H_s^{(0)}$	Initial homogeneous block of degree $s + 2$.
$\chi_{q,p}, Z_{q,p}$	χ_q, Z_q in (7)	Generator and normalized term at step q . Equation (7) agrees up to the sign imposed by the opposite order in our definition of $\mathcal{L}_p$.
$H_{s,p}^{(q)}$	$H_s^{(q)}$ in (6)–(8)	Remainder block after step q . Formula (8), with its Lie multiplicity index distinct from s , is the unmarked coefficient recursion used below.
Π onto $a = b$; $I - \Pi$	$\mathcal{Z} \oplus \mathcal{W}$ in (9)–(15)	In two degrees of freedom the classes $\mathcal{P}^{\mathfrak{a}}$ and $\mathcal{P}^{\mathfrak{b}}$ vanish; the resonant projection is exactly $a = b$.
$\ \cdot\ _{\theta R}^{\#}$	$\ \cdot\ _{\theta R}$ in (18)	The same coefficient norm on each homogeneous block; summing the blocks is the analytic-series extension used here.
d_q, δ_q, R_q in (2.36)	$d_q = 6d/(\pi^2 q^2)$, $\delta_q = \sum_{j \leq q} d_j$ in Section 3.4	The same cumulative radius loss: generators and normal terms are estimated on R_{q-1} , transformed remainders on R_q .
The unmarked geometric bounds	(35), (37), (39) and Section 3.4	In two degrees of freedom these are respectively the bounds for χ_q , Z_q , and $H_s^{(q)}$; the resulting bounds are B^q for the first two and B^s for the remainder, after packaging the μ -, T -, and radius majorants.
Marked trees, D_p^ℓ , common parameter domain, and $B_p^{(N)}$	no fixed-parameter counterpart	These are the present lemma's additions. The source supplies neither parameter derivatives nor convergence of the normalized exact primitive.

inverses leaves at most $4q + 1$ parameter-dependent slots: every normalized generator contributes only one projection and one inverse, and the number of generator occurrences is at most q by the same excess-degree count.

A parameter derivative marks either an input coefficient or one homological inverse. Equations (2.39), (2.47), and (2.48)–(2.49) show that replacing a slot by its first or second derivative changes its majorant by only a uniform factor. There are at most $(4q + 1)^\ell$ ordered placements of ℓ marks, allowing repetition when both derivatives hit the same slot, and the binomial multiplicities for $\ell \leq 2$ are uniform. The same simultaneous induction on normalization stage and excess degree as in the unmarked proof therefore gives

$$\sup_{p \in U} \|D_p^\ell \chi_{q,p}\|_{R_{q-1}}^\# \leq \tilde{C}_\ell q^\ell B^q \leq C_\ell q^{2\ell} B^q.$$

The induction for $Z_{q,p}$ and the normalized remainders is identical. This proves (2.37). A parameter mark creates no state derivative, so every Cauchy loss is already present in the fixed-parameter tree; in particular the cumulative radii R_q , rather than the single-step radii $(1 - d_q)R$, apply through every available parameter order $r \leq 2$.

It remains to pass from homogeneous coefficients to analytic maps. Choose $B_1 > B$ and then $\varepsilon > 0$ so small that $\vartheta = B_1 \varepsilon < 1$. If $T_{q,p}$ is the time-one Hamiltonian map generated by $\chi_{q,p}$, homogeneity, (2.37), and the variational equations of its Hamiltonian flow through order r give

$$(2.50) \quad \sup_{p \in U} \|D_p^\ell (T_{q,p} - \text{id})\|_{\varepsilon R_q} \leq \tilde{C}_\ell \varepsilon q^{2\ell+2} \vartheta^q, \quad 0 \leq \ell \leq r.$$

After decreasing ε once more, the case $\ell = 0$ is smaller than the available gap $\varepsilon d_q R$ for every q ; hence

$$T_{q,p}(\Delta_{\varepsilon R_q}) \subset \Delta_{\varepsilon R_{q-1}}.$$

Since $\sum_{q \geq 1} q^6 \vartheta^q < \infty$, the chain rule for the finite compositions gives summable recurrences for the available parameter derivatives. At orders one and two these have the form

$$\begin{aligned} u_q &\leq (1 + a_q)u_{q-1} + b_q, \\ v_q &\leq (1 + a_q)v_{q-1} + b_q(1 + u_{q-1})^2, \quad \sum_q (a_q + b_q) < \infty. \end{aligned}$$

The same estimates applied to differences show that $\Theta_p^{(N)}$ is Cauchy through parameter order r on every strictly smaller polydisc.

The normal coefficient series is Cauchy there by (2.37). The normalized remainder after step N has Taylor order $N + 3$ and obeys the same marked majorant, hence tends to zero through parameter order r . Consequently

$$\Theta_p^{(N)} \longrightarrow \Theta_p, \quad g_p^{(N)} \longrightarrow g_p, \quad F_p \circ \Theta_p = g_p$$

in the asserted analytic parameter norms.

Finally, every $\Theta_p^{(N)}$ is symplectic. Thus $(\Theta_p^{(N)})^* \lambda_0 - \lambda_0$ is closed, and (2.38) is its uniquely normalized radial primitive. The radial homotopy operator

is bounded on every strictly smaller analytic polydisc. Convergence of $\Theta_p^{(N)}$ and its state derivative therefore gives $B_p^{(N)} \rightarrow B_p$ through two parameter derivatives when $r = 2$ (and through the available lower order when $r < 2$), as claimed. $\square$

Lemma 2.6 (Uniform reversible complex-saddle normal form). *Let $K_p = H_p \circ \Psi_p$ be the family from Lemma 2.4. After reducing U and choosing $0 < \rho < R$, there are C^r families of real-analytic exact symplectic maps*

$$\Phi_p : B_\rho \longrightarrow B_R, \quad \Phi_p(0) = 0, \quad D\Phi_p(0) = I,$$

and real-analytic functions h_p such that

$$(2.51) \quad \Phi_p \circ \mathcal{R}_0 = \mathcal{R}_0 \circ \Phi_p, \quad K_p \circ \Phi_p = h_p(I_1, I_2),$$

where

$$I_1 = x_1 y_1 + x_2 y_2, \quad I_2 = x_1 y_2 - x_2 y_1, \quad dh_p(0) = (\alpha_p, \beta_p).$$

Moreover,

$$\Phi_p^* \lambda_0 - \lambda_0 = dA_p, \quad A_p(0) = 0, \quad A_p \circ \mathcal{R}_0 = -A_p,$$

and $p \mapsto (\Phi_p, h_p, A_p)$ is C^r in analytic norms on every smaller ball.

Proof. For fixed p , convergence of the canonical normalization is [9, Theorem 1, p. 856]; Moser's earlier analytic saddle construction [22] is the historical solution-family antecedent. The parameter-uniform statement needed here does not follow directly from either fixed-system theorem.

The spectral trivialization in Lemma 2.4 gives $K_{2,p} = \alpha_p I_1 + \beta_p I_2$ in one fixed real coordinate model over U . Upon complexification, the eigenvalues are $\lambda_{1,p} = \alpha_p + i\beta_p$ and $\lambda_{2,p} = \alpha_p - i\beta_p$. Hypothesis (I1), after one uniform radius shrink, gives the input bound (2.39). Lemma 2.5 therefore supplies a common analytic domain and the C^r normalizing transformation, normal form, and normalized exact primitive. Restricting the limits to the real domain, write them as Φ_p , h_p , and A_p , respectively. Because every $\chi_{q,p}$ has degree at least three, $\Phi_p(0) = 0$ and $D\Phi_p(0) = I$.

It remains to record the real and reversible structures. Let $\mathcal{C}_\mathcal{R} F = F \circ \mathcal{R}_0$ and $\mathcal{L}_p = \{K_{2,p}, \cdot\}$, and let Π and $\mathcal{G}_p$ be the normalized projection and inverse from Lemma 2.5. The projection is fixed by the spectral trivialization, and

$$\Pi \mathcal{C}_\mathcal{R} = \mathcal{C}_\mathcal{R} \Pi, \quad \mathcal{C}_\mathcal{R} \mathcal{L}_p = -\mathcal{L}_p \mathcal{C}_\mathcal{R}.$$

At a normalized step, if $F \circ \mathcal{R}_0 = F$, then $Z = \Pi F$ is invariant and the uniquely normalized solution $S = \mathcal{G}_p(I - \Pi)F$, $\Pi S = 0$, satisfies $S \circ \mathcal{R}_0 = -S$. Its Hamiltonian flow therefore commutes with $\mathcal{R}_0$. Induction shows that every finite normalizing map is reversible. The projection, homological inverse, and Lie recursion also commute with the real structure induced by complex conjugation, so all finite maps and normal forms restrict to real analytic objects. These properties pass to the common analytic limit.

In two degrees of freedom the resonant algebra is precisely the algebra of analytic functions of I_1, I_2 . Thus

$$K_p \circ \Phi_p = h_p(I_1, I_2), \quad dh_p(0) = (\alpha_p, \beta_p).$$

Finally, $\Phi_p^* \lambda_0 - \lambda_0$ is closed and anti-invariant under $\mathcal{R}_0$. Its normalized radial primitive is the limit supplied by Lemma 2.5; hence

$$\Phi_p^* \lambda_0 - \lambda_0 = dA_p, \quad A_p(0) = 0, \quad A_p \circ \mathcal{R}_0 = -A_p.$$

This proves all assertions through parameter order $r \leq 2$. $\square$

Proposition 2.7 (Parameter-local reversible saddle-focus coordinates). *Assume (I1) of Definition 2.1, and fix $p_0 \in \mathcal{P}$. After shrinking to a parameter neighborhood U of p_0 , there are C^r families of reversible exact symplectic coordinates and auxiliary radial incoming and outgoing sections inside the physical saddle block in which the following hold.*

The Hamiltonian has convergent saddle-focus normal form

$$(2.52) \quad H_p = h_p(I_1, I_2), \quad dh_p(0) = (\alpha_p, \beta_p),$$

for the two standard quadratic invariants of the chosen reversible symplectic eigenframe. On $H_p = 0$, the local passage preserves the transverse action $\nu = I_2$ exactly and has the form (2.12); its physical first-hit time is (2.14). For every fixed m and all $0 \leq \ell \leq r$, $0 \leq j \leq m$, after possibly shrinking the section width,

$$(2.53) \quad \sup_{\substack{p \in U \\ 0 < |\nu| \leq \nu_*}} \frac{|D_p^\ell D_{\log \nu}^j \rho_{p,\sigma}(\nu)| + |D_p^\ell D_{\log \nu}^j \tau_{p,\sigma}(\nu)|}{|\nu|(1 + |\log |\nu||)} < \infty.$$

The coordinate transition to any second chart produced this way extends to the oriented real blow-up and has degree one on the phase-circle boundary. The auxiliary radial faces are joined to the physical faces by the event-free orbit slides in (I1). Their slide times and endpoint maps are uniformly bounded, so the radial passage time and the physical block residence time differ by a bounded function with the required mixed state-parameter derivatives. Consequently, if n_i denotes the integer selected by any half-open local angle lift, then on every sufficiently long local passage

$$(2.54) \quad \left| n_i - \frac{\beta_p}{2\pi} \mathcal{T}_{\text{sf},p} \right| \leq C_i.$$

The constants are uniform on compact subsets of U .

Proof. Lemma 2.4 followed by Lemma 2.6 gives the composite parameter-local reversible exact symplectic chart and (2.52) on one common analytic ball. Its primitive differs from the standard primitive by the sum of the two odd exact gauges constructed in those lemmas. Choose the two auxiliary radial faces inside this ball, transverse to the normal-form flow. Their restrictions have exact section form $d\phi \wedge d\nu$, and the stable and unstable traces are the zero-action curves $\nu = 0$.

On the zero-energy surface, solve $h_p(I_1, I_2) = 0$ as $I_1 = q_p(I_2)$. The two invariants are constant along the normal-form flow. Quadrature between the two fixed radial faces gives

$$(2.55) \quad T_{p,\sigma}(\nu) = -\alpha_p^{-1} \log |\nu| + t_{p,\sigma} + O(|\nu|(1 + |\log |\nu||)),$$

$$(2.56) \quad \Delta_{p,\sigma}(\nu) = \frac{\beta_p}{\alpha_p} \log |\nu| + b_{p,\sigma} + O(|\nu|(1 + |\log |\nu||)).$$

The exact section forms are $d\phi \wedge d\nu$, so preservation of the second invariant is exact preservation of transverse action. These formulas and the exact section map are the classical saddle-focus passage calculation; see [15, Section 2.1]. Analyticity of q_p and of the normal-form frequencies shows that $D_{\log \nu} = \nu \partial_\nu$ preserves the displayed remainder class. The same argument after differentiating in p proves (2.53).

Two local normal-form charts preserve the stable and unstable axes. Their derivative on either complex normal line is multiplication by a nonzero complex scalar; the fixed choice $\beta_p > 0$ excludes complex conjugation. Hence their oriented-blow-up phase maps have degree +1 and bounded periodic displacement on a compact overlap. A winding label is the deck integer in (2.56); subtracting $(\beta_p/2\pi)$ times (2.55) leaves a bounded function. The two event-free slide times change the radial passage time to $\mathcal{T}_{\text{sf},p}$ by another uniformly bounded function. This proves (2.54). $\square$

2.3. Geometric hypotheses. The assumptions below are imposed only on a compact part of the flow and on the two end germs. In particular, they do not presuppose a winding partition, return strips, symbolic dynamics, or renormalized potentials.

We first define the two cut-face templates used in the assumptions. Fix a preliminary source-angle block I_s^0 , a target-action collar J_ν^0 , and a closed lift $\bar{J}_0 = [\theta_0, \theta_0 + 2\pi]$ in the marked section charts. For each source sign put

$$(2.57) \quad \tilde{b}_{\mathbf{X},\sigma} = b_{\mathbf{X},\sigma} + \beta_{\mathbf{X}} t_{\mathbf{X},\sigma}.$$

In the outgoing action–angle chart, the residual-phase cut-face template is

$$(2.58) \quad \pi_{\mathbf{X},\sigma,\infty}^{\text{loc}}(\phi, \theta) = (\phi - \theta + \tilde{b}_{\mathbf{X},\sigma}, 0), \quad (\phi, \theta) \in I_s^0 \times \bar{J}_0.$$

Let $\mathcal{T}_{\mathbf{X}}^u$ be the marked finite-flight section map from the local outgoing section to the compact outgoing band B_0^u encoded by Ξ_{fin} in (2.9). Its fixed domain contains the image of (2.58). Define

$$(2.59) \quad \Pi_{\mathbf{X},\sigma,\infty} := \mathcal{T}_{\mathbf{X}}^u \circ \pi_{\mathbf{X},\sigma,\infty}^{\text{loc}} : Z_{\text{exit}}^0 = I_s^0 \times \bar{J}_0 \longrightarrow B_0^u.$$

These are marked cut-face templates determined before any winding strip is constructed. In particular, (2.59) is not a first-exit conclusion and is not defined by assuming convergence of the finite-winding exit maps; that convergence is proved in Proposition 5.4.

Definition 2.8 (Hypotheses for the two-ended return problem). A presentation $\mathbf{X}_*$ in $\mathfrak{H}_{\text{adm}}^k$, $k \geq 5$, satisfies the geometric hypotheses if the following conditions hold.

(H1) *Saddle-focus passage and homoclinic transition.* The equilibrium has eigenvalues $\pm\alpha_* \pm i\beta_*$ with $\alpha_*, \beta_* > 0$, and its local passage and flight time satisfy (2.12)–(2.15). There is a transverse homoclinic orbit whose finite transition from the outgoing to the incoming section is defined on a compact lifted angular interval J_ψ . In exact action–angle coordinates write this transition as

$$(2.60) \quad (\bar{\phi}, \bar{\nu}) = \mathcal{H}_{\mathbf{X}_*}(\psi, \nu_u) = (F_{\mathbf{X}_*}^{\text{hom}}(\psi, \nu_u), G_{\mathbf{X}_*}^{\text{hom}}(\psi, \nu_u)).$$

Its target-action component satisfies

$$(2.61) \quad |\partial_\psi G_{\mathbf{X}_*}^{\text{hom}}(\psi, 0)| \geq d_* > 0$$

throughout the chosen window J_ψ . There is $\nu_* > 0$ such that

$$(2.62) \quad [-\nu_*, \nu_*] \Subset J_\psi^0 \cap G_{\mathbf{X}_*}^{\text{hom}}(J_\psi, 0),$$

with a positive distance from the two endpoints of $G_{\mathbf{X}_*}^{\text{hom}}(J_\psi, 0)$. For each source sign define the limiting matching relation by the zero set of

$$(2.63) \quad \mathcal{F}_{\mathbf{X}_*, \sigma, \infty}(\psi, \theta; \phi, \bar{\nu}) = \left(\begin{array}{c} G_{\mathbf{X}_*}^{\text{hom}}(\psi, 0) - \bar{\nu} \\ \psi - \phi + \theta - \tilde{b}_{\mathbf{X}_*, \sigma} \end{array} \right).$$

The derivative bound and the interval crossing imply that $G_{\mathbf{X}_*}^{\text{hom}}(\cdot, 0)$ is one-to-one on J_ψ and has a C^2 inverse on $[-\nu_*, \nu_*]$; denote this inverse by Ψ_* . The only additional range condition on the second matching row is the lifted-angle margin

$$(2.64) \quad \inf_{\substack{\phi \in I_s^0, |\bar{\nu}| \leq \nu_* \\ \sigma \in \{+, -\}}} \text{dist}(\phi - \Psi_*(\bar{\nu}) + \tilde{b}_{\mathbf{X}_*, \sigma}, \partial \bar{J}_0) > 0, \quad \phi - \Psi_*(\bar{\nu}) + \tilde{b}_{\mathbf{X}_*, \sigma} \in \text{int } \bar{J}_0.$$

(H2) *Clean first-event geometry.* The homoclinic aperture, the algebraic entrance hypersurface, the open pole window, and a finite collection of lateral faces lie in fixed compact flow boxes and have C^3 defining functions and cell charts. Every active intersection of their defining hypersurfaces is transverse: the active conormals are linearly independent with a uniform lower bound. The connected sign strata form a finite decomposition of each compact flow-box section. This decomposition is exhaustive in the following literal sense. The closures of the homoclinic aperture, algebraic entrance, pole window, and named lateral strata cover the entire relevant outgoing band. At the end of the homoclinic tube, the closures of the re-entry strata, the stable-manifold cut, and the named lateral strata cover the entire relevant return band. Every pure-dimensional connected outgoing sign stratum outside the homoclinic aperture, and every connected

stratum obtained inside that aperture after refinement by the pulled-back return-band functions described below, is assigned exactly one outcome in

$$\{\text{return}\} \sqcup \{\text{p}, \text{a}, \text{out}, \text{rbox}, \text{cut}\}.$$

Here `return` denotes genuine re-entry into the incoming section, whereas the terminal label `rbox` denotes a lateral exit from the return box. The assignment is total, so there is no unlabelled residual component. Every point in the relative interior of an assigned stratum has the stated finite return or first physical exit event. On a common face or corner, the fixed priority rule assigns the event without adjoining that lower-dimensional stratum to either neighboring interior. Thus the labelled connected strata, event existence, and first-event assignment are hypotheses, whereas their high-winding persistence is a conclusion. For each σ , pull back the active outgoing-face functions by $\Pi_{\mathbf{X}_*, \sigma, \infty}$. The prescribed finite homoclinic flight and the return-band defining functions extend to a fixed neighborhood of the compact closure of each homoclinic cell; pull those functions back on that closure. Every nonempty active pullback family has linearly independent differentials with a uniform lower bound. At an intersection with a boundary face of Z_{exit}^0 , adjoining the conormal of every active boundary face preserves linear independence with a uniform lower bound. Thus the finite pullback stratification is neat with respect to the boundary of the reference cell. Every assigned finite hit is transverse to the flow, and the event order and all inactive face inequalities have positive margins on the corresponding compact pre-event tubes; these tubes cover every assigned relative interior in the preceding exhaustive list. The three apertures have disjoint closures inside the chosen outgoing window. For each $\mathbf{t} \in \{\mathbf{a}, \mathbf{p}\}$, fix a source sign $\sigma_{\mathbf{t}}$, an interior anchor $z_{\mathbf{t}}^{\text{lim}} \in \text{int } Z_{\text{exit}}^0$ at a positive distance from $\partial Z_{\text{exit}}^0$, and a connected component $\mathcal{C}_{\mathbf{t}, \sigma_{\mathbf{t}}, \ell_{\mathbf{t}}}^{\text{lim}}$ of the corresponding pullback sign stratum containing that anchor. At the anchor the active differentials retain the stated rank, and the inactive-face and event-order inequalities have positive margins. No nonemptiness is required for every source sign or cell label.

- (H3) *Algebraic end.* The algebraic compactification satisfies (2.17) on a fixed collar and contains a compact locally maximal C^3 future-staying hypersurface $\mathcal{W}_{\mathbf{a},*}^{\text{tail}}$. The restriction of (e, E, b) to this hypersurface is a diffeomorphism onto a compact base with corners, and its tangential boundary faces are inward or structurally invariant. It is normally expanding in forward desingularized time and satisfies the order-three bunching inequalities (2.5). A fixed cut $e = e_*$ and a finite saturation tube are defined on a neighborhood of the whole cut slice of this hypersurface, with positive first-hit, coorientation,

and same- c face margins. These are the geometric conditions in (H3). The interval-block criterion in Theorem A.1 may be used beforehand to construct this hypersurface and verify these conditions. Only the resulting geometric data, not its product coordinates or cone estimates, enter the return–exit theorems.

- (H4) *Pole end.* The pole entrance has a strict first hit in the basin of the hyperbolic boundary sink of the compactified pole field. Its image is contained in a compact subset of that basin, and the radial decay, spectral gap, and entry transversality have positive margins.

Proposition 2.9 (Persistence of a normally expanding algebraic hypersurface). *Let Y_0 be a C^5 vector field and let $\mathcal{W}_{a,0}^{\text{tail}}$ be a compact locally maximal C^3 hypersurface with the relative boundary behavior and the order-three normal-expansion inequalities in (H3). In a sufficiently small relative C^5 neighborhood of Y_0 , there is a unique continuation $\mathcal{W}_{a,Y}^{\text{tail}}$ in a fixed tubular neighborhood. It is locally maximal there and satisfies the same strict normal-expansion and bunching inequalities after a uniform weakening of their margins.*

If $\varpi \mapsto Y_\varpi$ is a C^r family, $r \in \{0, 1, 2\}$, with compact image in that neighborhood, the continued hypersurfaces can be written in a fixed normal bundle as sections Γ_ϖ satisfying

$$(2.65) \quad \sup_{i+j \leq 3, j \leq r} \|D_z^i D_\varpi^j \Gamma_\varpi\| < \infty.$$

The assertion is relative to the subspace preserving every designated structurally invariant boundary face; if all such faces are strictly inward, the neighborhood is an ordinary C^5 neighborhood.

Proof. Use a fixed tubular neighborhood and the normal quotient as the fiber. The backward graph transform has homogeneous i th-jet part

$$(N^\delta)^{-1} \circ K \circ (A^\delta)^{\otimes i}, \quad 0 \leq i \leq 3,$$

so its norm is bounded by the corresponding strict factor in (2.5). The C^r section theorem and its parameter version therefore give persistence, uniqueness, and (2.65); see [11, Chapter 3] and the classical normal-hyperbolicity framework in [8]. At an inward boundary face the transform is formed in a fixed collar and restricted back to the relative domain. At a structurally invariant face it restricts to that face. These two constructions agree at corners by uniqueness. Local maximality persists after shrinking the fixed isolating neighborhood. This is the standard overflowing-invariant-manifold form of normal hyperbolicity; no product interval coordinate is used. $\square$

A coordinate criterion that constructs the normally expanding hypersurface from an interval block, invariant cones, and finite-time bunching estimates is stated and proved in Appendix A. Once the hypersurface and its strict geometric margins have been obtained, only Proposition 2.9 is used in the return–exit argument.

Let $\mathcal{W}_{a,*}^{\text{tail}}$ be the normally expanding hypersurface in (H3). Its finite saturation through the marked tube is the cooriented invariant entrance hypersurface $\mathcal{W}_{a,*}$. Since $Y_a^e = e^2 v_e < 0$ at $e = e_* > 0$, the cut meets the hypersurface transversely. After shrinking a coefficient neighborhood, Proposition 2.9 constructs its unique continuation and gives the scale regularity (2.65) for every compact parameter family. In an application based on an interval block, Theorem A.1 first supplies precisely this geometric input.

Lemma 2.10 (The limiting matching graph). *Under (H1), the part of the zero set of (2.63) over $I_s^0 \times [-\nu_*, \nu_*]$ is, for each source sign σ , the graph of the single C^2 map*

$$(2.66) \quad Y_{\mathbf{X}_*,\sigma,\infty}(\phi, \bar{\nu}) = (\Psi_*(\bar{\nu}), \phi - \Psi_*(\bar{\nu}) + \tilde{b}_{\mathbf{X}_*,\sigma}).$$

Its image has positive distance from the boundary of $J_\psi \times \bar{J}_0$, and its inverse-function bounds are uniform on the displayed compact rectangle.

Proof of Lemma 2.10. The continuous derivative $\partial_\psi G_{\mathbf{X}_*}^{\text{hom}}(\psi, 0)$ never vanishes on the interval J_ψ , so it has constant sign. Hence $G_{\mathbf{X}_*}^{\text{hom}}(\cdot, 0)$ is strictly monotone there. The interval inclusion (2.62) gives existence and uniqueness of $\psi = \Psi_*(\bar{\nu})$, and (2.61) gives the inverse derivative bound. Compactness and the positive endpoint margin in (2.62) keep $\Psi_*([-\nu_*, \nu_*])$ away from ∂J_ψ . Substitution in the second row gives (2.66); (2.64) supplies its remaining boundary margin. Along this graph the selector matrix is

$$(2.67) \quad D_{(\psi,\theta)} \mathcal{F}_{\mathbf{X}_*,\sigma,\infty} = \begin{pmatrix} \partial_\psi G_{\mathbf{X}_*}^{\text{hom}}(\psi, 0) & 0 \\ 1 & 1 \end{pmatrix}.$$

Its determinant is the first displayed entry, so its inverse is uniformly bounded; in the notation of (4.42), the Schur complement is identically 1. The claimed C^2 bounds now follow from the inverse-function formula on a compact interval. $\square$

Proposition 2.11 (Transverse homoclinics produce the matching hypothesis). *Assume (I1)–(I3) of Definition 2.1. For every $p_0 \in \mathcal{P}$, the reversible saddle-focus coordinates of Proposition 2.7 and exact action–angle coordinates on their auxiliary radial faces may be chosen, after shrinking their domains and the parameter neighborhood, so that (H1) holds uniformly. In particular, the inverse matching graph is a conclusion of transversality; it is not part of the intrinsic data.*

Proof. Absorb the event-free orbit slides between the radial and physical faces into the compact homoclinic flight. Its stable and unstable traces on the two-dimensional auxiliary section faces are the zero-action curves in the restrictions of the canonical normal-form coordinates from Proposition 2.7; those restrictions already have section forms $d\psi \wedge d\nu_u$ and $d\bar{\phi} \wedge d\bar{\nu}$. No second Darboux straightening is made, so the exact action-preserving passage and its clock are unchanged. Let $\mathcal{H}_p = (F_p^{\text{hom}}, G_p^{\text{hom}})$ be the physical finite-flight

map in these coordinates. At the selected homoclinic point, $G_p^{\text{hom}}(\psi_p, 0) = 0$. Modulo the common flow direction, (2.3) is precisely

$$(2.68) \quad \partial_\psi G_p^{\text{hom}}(\psi_p, 0) \neq 0.$$

Indeed, the derivative on the left is the target conormal of the stable trace evaluated on the transported tangent to the unstable trace.

Shrink one interval J_ψ about ψ_{p_0} . Continuity, compactness of a smaller parameter neighborhood, and (2.68) give the lower bound (2.61). The image of J_ψ then contains a fixed interval $[-\nu_*, \nu_*]$ with positive endpoint margin, giving (2.62). Finally shrink the source-angle and target-action intervals inside the physical source cell. The proper-arc clause of (I3) says that the union of the two signed residual-phase images has a complementary arc of positive length. Choose one cut in that complementary arc and lift its complement to a length- 2π interval; one further shrinking gives (2.64). Lemma 2.10 now supplies the unique matching graph and the uniform inverse bounds. $\square$

2.4. Local production of marked representations.

Theorem 2.12 (Local production of admissible marked charts). *Let an intrinsic two-ended Hamiltonian configuration over a compact C^r parameter manifold $\mathcal{P}$, $r \in \{0, 1, 2\}$, satisfy Definition 2.1. Then there are finitely many pairs*

$$(2.69) \quad V_i \Subset U_i \subset \mathcal{P}, \quad \mathcal{P} = \bigcup_{i=1}^m V_i,$$

and, on every U_i , a C^r family of marked representations

$$(2.70) \quad p \longmapsto \mathbf{X}_{i,p} \in \mathfrak{H}_{\text{adm}}^k$$

with the following properties.

- (a) *The marked data are constructed from the physical configuration. They comprise a reversible exact saddle-focus chart, auxiliary radial incoming and outgoing faces with exact action-angle coordinates and event-free orbit slides to the fixed physical faces, a finite coherent system of angular cuts and event charts, a finite good tubular atlas of the normally expanding algebraic hypersurface, and a pole boundary chart. They satisfy (H1)–(H4) uniformly on $\overline{V}_i$. No matching graph, return or first-exit component, winding label, or end counterterm is used to construct them. The normally expanding algebraic hypersurface is an intrinsic hypothesis. In a concrete application it may first be constructed by Theorem A.1.*
- (b) *On every nonempty overlap $U_i \cap U_j$, the two representations are related by an admissible change in the sense of Definition 5.6, after a finite clean refinement of their angular cuts and event cells. If one refined physical component has local winding labels n_i and n_j , then*

$$(2.71) \quad |n_i - n_j| \leq K$$

for one constant K independent of the overlap, parameter, and component.

(c) On every represented sufficiently long local passage component,

$$(2.72) \quad \left| n_i - \frac{\beta_p}{2\pi} \mathcal{T}_{\text{sf},p} \right| \leq C_i.$$

Consequently, for any finite collection of local marked winding thresholds N_i , there is one physical residence-time threshold T_* such that

$$(2.73) \quad \mathcal{T}_{\text{sf},p}(z) \geq T_* \implies n_i(z) \geq N_i$$

in every chart in which z is represented.

There need not be a global eigenframe, angle lift, sign labeling, or integer winding function on $\mathcal{P}$. The global objects are the physical first-event relation and, after applying the marked theorems, the overlap class of the corrected weighted data.

Proof. Fix $p_0 \in \mathcal{P}$.

Saddle-focus chart and clock. Proposition 2.7 gives a parameter-local reversible exact symplectic normal-form chart on one common analytic ball. It also gives the exact action-preserving section passage, the physical clock, their weighted remainders, and the extension to the oriented real blow-up. This constructs all saddle data in (2.9) without choosing a return orbit. The construction is parameter-local because the stable complex eigenline bundle need not be globally trivial.

Homoclinic matching. Use the auxiliary radial faces and absorb their event-free slides to the physical faces into the finite transition. Exact relative Darboux coordinates and Proposition 2.11 turn the transverse physical homoclinic into the derivative and image margins of (H1). Lemma 2.10 then derives the matching graph. Thus neither that graph nor its inverse bounds have been inserted into the intrinsic assumptions.

Event charts. Coorientation gives local defining functions for the physical event hypersurfaces. Clean and neat intersection is invariant under a change of defining functions. On compact pre-event tubes, the least singular value of every active conormal matrix and the first-hit, inactive-face, and event-order functions have positive minima. The two intrinsic interior points give the algebraic and pole anchors required in (H2). The exhaustion clause of (I3) is preserved by these flow-box diffeomorphisms and event-free orbit slides. Hence the images of the homoclinic aperture, two end entrances, and lateral sign strata cover the whole marked outgoing band; after the homoclinic flight, the re-entry, stable cut, and lateral strata cover the whole marked return band. Carry the physical return/first-exit assignment and its fixed face priority through these diffeomorphisms. This gives the exhaustive list of labelled connected strata and the finite return/first-hit assertion in (H2), not merely its local conormal-rank conditions. In particular, no unlabelled

marked component is introduced when defining functions or angular lifts are chosen.

The proper-arc clause of (I3) leaves a complementary phase arc with positive margin. We postpone the choice of a cut inside that arc until the preliminary finite parameter cover has been extracted below. The complement of the cut will then be the half-open angular lift. Thus the physical stratification, and not a finite-winding list of connected components, is the input to (H2).

Algebraic end. Choose local defining functions for the boundary energy face and the cooriented invariant flag in (I4). Hadamard's lemma gives (2.17); the negative flag-quotient rate gives the strict sign of v_b . The intrinsic hypersurface's local maximality, normal expansion, and order-three bunching give (H3) directly; Proposition 2.9 gives its parameter-local continuation and mixed regularity. The clock, compactified action coefficient, physical cut, and finite saturation tube give the remaining clauses. Theorem A.1 and the one-chart bounds (A.4)–(A.6) are used only to verify the intrinsic hypersurface in applications such as the normal-form system below.

Pole end. Take a C^{m_p+3} collar at the physical boundary sink. The radial defining function gives the first equation in (2.25). The indicial spectrum is simple and its conjugacy class is constant, so parameter-local C^r eigenframes conjugate (2.6) to (2.26). The C^{m_p+3} extension of $\rho^{m_p}\lambda(X_{H_p})$, together with the compact basin-entry, radial, and spectral-gap margins, gives (H4). No finite pole jet is frozen by this normalization.

Finite cover and coherent cuts. All constructions so far, except the phase cut, persist after shrinking a parameter neighborhood. Compactness of $\mathcal{P}$ therefore gives a finite preliminary cover; shrink it to pairs $V_i \Subset U_i$ as in (2.69). On each U_i choose a C^r cut section inside the complementary arc supplied by (I3). After one further finite shrinking, every such cut has positive distance from the closure of every represented residual-phase image and every anchor trace. Hence on an overlap no cut locus meets the common represented source or event cells. Those cells are restrictions of the same finite physical clean stratification, so their intersections give a finite clean common refinement. This is a simultaneous finite choice made after the cover is fixed; no claim that arbitrary independent cuts intersect cleanly is used.

Overlap maps. Two reversible symplectic saddle frames differ by a nonzero complex scalar; their blow-up transition therefore has degree one and bounded periodic phase displacement. The section maps describe the same physical faces, or event-preserving orbit slides between auxiliary faces. The algebraic and pole transitions are respectively C^3 and C^{m_p+3} b -diffeomorphisms, and the physical primitive is fixed. The preceding finite cut construction supplies the clean common refinement on every represented overlap. These are the clauses of Definition 5.6. Comparing the two degree-one deck decompositions gives (2.71); compactness of the finite overlap atlas makes K uniform.

Uniform physical cutoff. Equation (2.72) is (2.54). Taking the maximum of the finitely many constants and using the positive lower bound for β_p gives (2.73). Notice that the physical cutoff may intersect a local marked component; the assertion is only that every point in the long-passage region belongs, in each applicable marking, to a component above its marked threshold. It does not rename the intersections as new connected branches. $\square$

These are open conditions in the coefficient topology. Condition (H1) is the usual transverse homoclinic matching condition, (H2) is a finite clean-intersection and first-hit condition which also requires one nonempty limiting family for each noncompact end, and (H3)–(H4) are asymptotic stability conditions at the two ends. Section 8 verifies them for (1.1); the long-passage, winding, and return–exit conclusions are proved from them in the following sections.

The precise admissible transition class is stated next to the invariance theorem in Definition 5.6. It consists of smooth orbit slides, exact gauges, blow-up and b -coordinate changes, and parameter-uniform clean finite cut arrangements with the boundary regularity needed by the weighted estimates.

3. RETURN–EXIT GEOMETRY AND MAIN RESULTS

3.1. Incoming section, return strips, and exit faces. Fix incoming and outgoing sections S^s, S^u in the marked saddle-focus chart. For the normal-form system (1.1) they are the critical-energy sections at local radius .01. On one incoming rectangle use coordinates (ϕ, ν) , where ν is the signed transverse action. Choose $I_s \Subset I_s^0$ so that its interior contains the ϕ -projections of the two anchors z_t^{lim} in (H2), still at a positive distance from ∂I_s . Choose its endpoints so that every limiting pullback stratum either avoids the new boundary or meets it transversely; this is possible because the arrangement is finite and neat. For a number $\nu_N > 0$, set

$$(3.1) \quad \Sigma_N = \Sigma_+ \sqcup \Sigma_-, \quad \Sigma_\sigma = I_s^\circ \times \{\nu : 0 < \sigma\nu < \nu_N\}.$$

The zero-action boundary $\nu = 0$ is removed; its two sides are retained only as cut faces. The components Σ_+ and Σ_- are the two vertices of the return graph. After a fixed affine normalization of I_s to $I = [-1, 1]$, use the vertex-compatible charts

$$(3.2) \quad \chi_\sigma(x, y) = (x, \sigma\nu_N y), \quad (x, y) \in I \times (0, 1).$$

Thus every occurrence of y at a vertex means the positive normalized magnitude of the signed action; $y = 0$ is the removed zero-action face. For a return branch, $\sigma = \text{sgn } \nu$ denotes its source vertex and $\sigma' = \text{sgn } \bar{\nu}$ its target vertex.

If T is the local passage length, use the half-open lift $J_0 = [\theta_0, \theta_0 + 2\pi)$ associated with $\bar{J}_0$ and assign the unique pair (n, θ) such that

$$(3.3) \quad \beta_{\mathbf{X}} T = 2\pi n + \theta, \quad \theta \in J_0.$$

All finite outgoing cells are cut into angular lifts of length less than 2π . This convention labels the entire punctured collar without a gap or a double label.

The finite outgoing phase is not assumed monotone. The exact local saddle-focus passage (2.12)–(2.14), followed by the finite section shears, gives on each signed side

$$\begin{aligned}
 \Theta_{\mathbf{X}}(\phi, \nu) &= -\frac{\beta_{\mathbf{X}}}{\alpha_{\mathbf{X}}} \log |\nu| + \psi_{\mathbf{X},\sigma}(\phi) \\
 &\quad + O_{C^2_{\phi, \log |\nu|}}(|\nu|(1 + |\log |\nu||)), \\
 |\partial_{\nu} \Theta_{\mathbf{X}}| &\geq \frac{\beta_{\mathbf{X}}}{2\alpha_{\mathbf{X}}|\nu|}.
 \end{aligned}
 \tag{3.4}$$

after decreasing ν_N . Hence every lifted angular cell has at most one component of each winding and sign, and every compact subset away from the cut faces meets only finitely many such components.

The fixed local domain in (H2) contains three outgoing apertures: a homoclinic return tube, an open pole window, and a codimension-one algebraic sheet. After one compact interior restriction of those base data, write $B^u \subseteq B_0^u$ and B^r for the outgoing and return bands, and write h_j for the finite family of C^3 face functions. These are the marked data already required in (H2), not additional hypotheses; their uniform continuation is part of Theorem 3.1. Their zero sets contain the three aperture boundaries, the stable-manifold cut, and the lateral faces; all active intersections are transverse on the assigned cells. Their connected sign cells give a finite disjoint decomposition into two-dimensional interiors, one-dimensional faces, and corners. The formal sign-vector definition is given in Section 5, where it is used in the proof of the first-event assertion. A fixed priority assigns a first-exit *type* when several event faces meet, but it never absorbs a face or corner into a two-dimensional cell. The half-open convention is used only for the lifted angular interval in (3.3). The complement of the aperture closures in B^u , and the complement of the re-entry strip and stable-manifold cut in B^r , are consequently finite disjoint unions of named pure-dimensional lateral strata. At the end of the homoclinic tube, a point either re-enters Σ_N , hits the stable-manifold cut, or crosses its uniquely assigned return-box stratum. Thus the first-exit types are

$$\mathbf{t} \in \{\mathbf{p}, \mathbf{a}, \mathbf{out}, \mathbf{rbox}, \mathbf{cut}\}.
 \tag{3.5}$$

The type cut is a finite hit of the stable-manifold cut face; it is not a compactified saddle endpoint. Its winding-resolved preimages will be denoted $\mathcal{I}_{\sigma,n}$; they belong to the first-exit partition and never to either adjacent regular return edge. Figure 3 shows the two endpoint conditions, the transverse-action scale, and the target cut in their respective coordinates.

3.2. Main results.

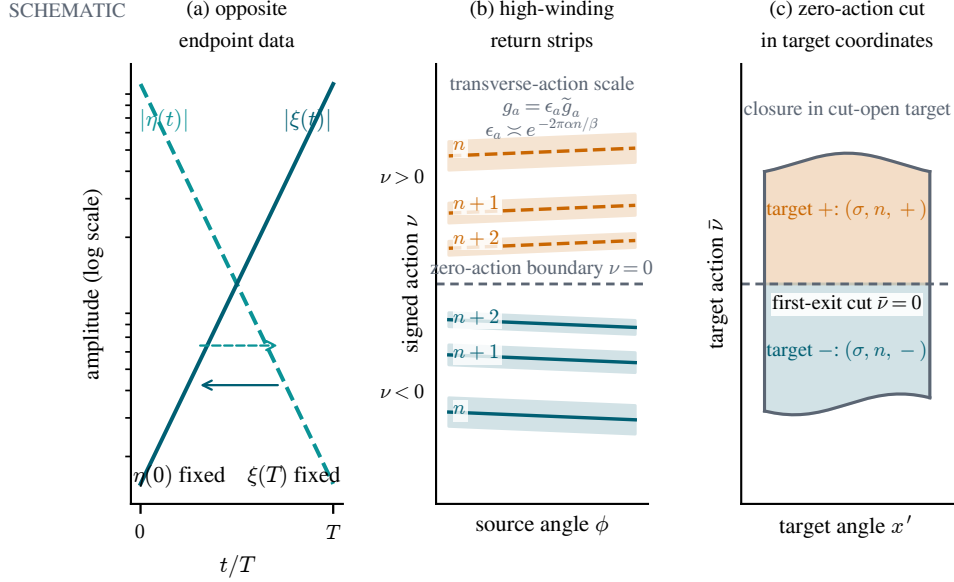


FIGURE 3. Opposite-endpoint boundary data and the transverse-action scale. (a) The stable coordinate is prescribed at $t = 0$ and the unstable coordinate at $t = T$; the curves are the leading linear decay profiles. (b) High-winding return strips on both sides accumulate at the zero-action boundary on the scale $g_a = \epsilon_a \tilde{g}_a$, $\epsilon_a \asymp e^{-2\pi\alpha n/\beta}$. (c) The target zero-action set cuts the closed return strip into its two target-sign components; their common boundary $\mathcal{I}_{\sigma,n}$ is a first-exit set, not a return branch. All three panels are schematic.

Theorem 3.1 (Marked high-winding returns and first exits near a Hamiltonian saddle-focus). *Fix $k \geq 5$. Let $\mathbf{X}_*$ satisfy the geometric hypotheses of Definition 2.8. For every chosen exponential rate*

$$(3.6) \quad 0 < \kappa < \frac{2\pi\alpha_*}{\beta_*}$$

there exist a relatively open neighborhood $\mathcal{U}_\kappa \subset \mathfrak{H}_{\text{adm}}^k$ of $\mathbf{X}_$, a fixed interval I_s , an integer N , a width $\nu_N > 0$, and constants*

$$(3.7) \quad 0 < \vartheta < 1, \quad \rho_{\text{in}} > 0, \quad \delta_{\text{event}} > 0, \quad C < \infty$$

such that the following statements hold for every $\mathbf{X} \in \mathcal{U}_\kappa$.

- (i) **Continued geometry and the two ends.** *The continued homoclinic and compact first-hit tubes, the algebraic normally expanding hypersurface, and the pole sink channel exist with common coorientations, bunching, and first-hit margins. At the algebraic end a reference orbit is selected at the fixed cut $e = e_*$, $E = b = 0$. Every*

algebraic first-exit orbit can be compared with it at the same value of e ; their difference is $O(e^m)$ in C^2 for every fixed m , uniformly on the compact algebraic entrance-label set.

- (ii) **Complete first-return/first-exit partition.** *The actual flow gives a disjoint and exhaustive first-return/first-exit partition of Σ_N into the nonempty connected return and first-exit strata produced by the fixed cell decomposition. These consist of high-winding return strips, stable-manifold cut curves, and the nonempty algebraic, pole, outgoing-boundary, and return-boundary components. Every disjoint union below ranges only over nonempty connected components. Every return or first-exit component has a unique winding $n \geq N$ and current sign. Every recurrent edge also has a unique next sign; the zero set separating the two next signs is the type-cut first-exit curve, not a return edge. Finite first-exit components have their unique cell label. The event speeds, strict auxiliary face-ordering inequalities, and least singular values of the active defining-function families are bounded below by δ_{event} ; the labelled sign pattern is preserved.*

For every source sign and every $n \geq N$, both target-sign return components are nonempty, and their common cut-face curve $\mathcal{I}_{\sigma,n}$ is nonempty; equivalently, $\mathcal{G}_N = \{+, -\}^2$. For each $\mathbf{t} \in \{\mathbf{a}, \mathbf{p}\}$, the interior anchor selected in (H2) continues into a nonempty $E_{\mathbf{t}, \sigma, n, \ell_{\mathbf{t}}}$ for every $n \geq N$. No nonemptiness is asserted for every source sign or for any other algebraic or pole cell label.

- (iii) **Uniform return maps.** *In the two fixed vertex charts (3.2) and after pullback to the fixed rectangle $I \times [0, 1]$ and fixed first-exit reference domains, the first-return map $P_a : V_a \rightarrow W_a$ for every edge $a = (\sigma, n, \sigma')$ has a Shilnikov cross form (called a cross form below)*

$$(3.8) \quad x' = f_a(x, y'), \quad y = g_a(x, y'),$$

and put

$$(3.9) \quad \epsilon_{\mathbf{X}, a} = \nu_N^{-1} \exp\left(-\frac{2\pi\alpha_{\mathbf{X}}}{\beta_{\mathbf{X}}} n(a)\right), \quad \tilde{g}_a = g_a / \epsilon_{\mathbf{X}, a}.$$

The cross form has contraction constant at most ϑ , $f_a(I \times [0, 1])$ is at distance at least ρ_{in} from ∂I , and

$$(3.10) \quad \rho_{\text{in}} \leq \tilde{g}_a \leq \rho_{\text{in}}^{-1}, \quad g_a \leq 1 - \rho_{\text{in}}, \quad \|Dg_a\|_{C^1} \leq C\epsilon_{\mathbf{X}, a}.$$

For each fixed sign pair (σ, σ') there are C^2 limit maps $f_{\mathbf{X}, \sigma, \sigma', \infty}$ and $\tilde{g}_{\mathbf{X}, \sigma, \sigma', \infty}$ such that

$$(3.11) \quad \|f_a - f_{\mathbf{X}, \sigma, \sigma', \infty}\|_{C^2} + \|\tilde{g}_a - \tilde{g}_{\mathbf{X}, \sigma, \sigma', \infty}\|_{C^2} \leq C(1+n)^3 e^{-\kappa n}.$$

Figure 4 contrasts the two finite-part constructions and shows where exact composition enters.

SCHEMATIC; THE TWO COUNTERTERMS ARE END-SPECIFIC

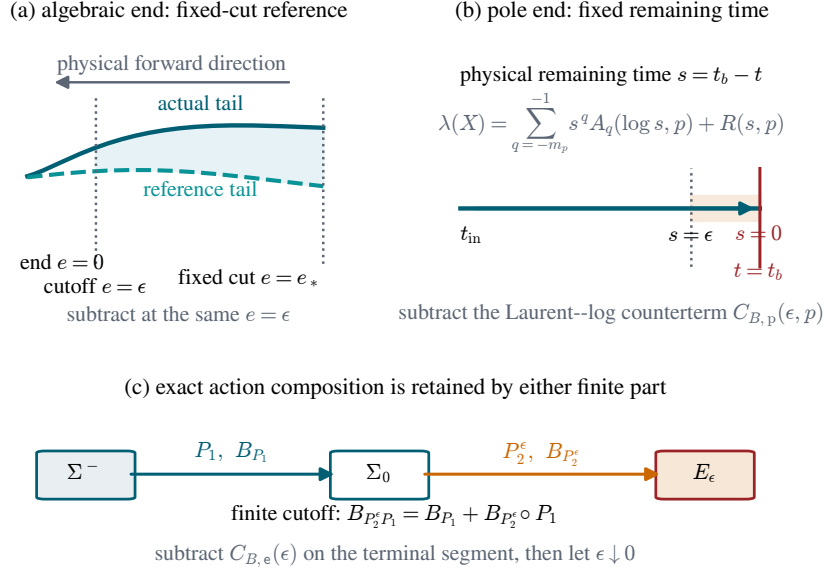


FIGURE 4. Finite parts at the two singular ends (schematic). In both cases the renormalized action has the form $B^{\text{ren}} = \lim_{\epsilon \downarrow 0} (B^\epsilon - C_B^\epsilon)$, but the counterterm C_B^ϵ is end-specific. (a) At the algebraic end, the actual and reference tails are distinct trajectories compared at the same cutoff $e = \epsilon$ after a fixed cut $e = e_*$; the difference of their action integrals defines the finite part. (b) At the pole, the asymptotic variable is the physical remaining time $s = t_b - t$; the end-specific Laurent-log counterterm is subtracted at $s = \epsilon$. (c) Exact action composition is first used at each finite cutoff. The appropriate counterterm is then subtracted only from the terminal segment before the cutoff tends to the end. The algebraic and pole counterterms are different objects; the diagram does not assert that they are equal.

Theorem 3.2 (Renormalized time and exact action). *Under the hypotheses and with the return and first-exit components of Theorem 3.1, the following statements hold uniformly for $\mathbf{X} \in \mathcal{U}_\kappa$.*

(i) **Renormalized return times and actions.** *For an edge a , let*

$$\iota_{\mathbf{X},a}(x, y') = (x, g_a(x, y'))$$

in the fixed source chart. The physical branch time and abbreviated action on V_a are

$$(3.12) \quad \tau_{\mathbf{X},a}^{\text{phys}} = \frac{2\pi}{\beta_{\mathbf{X}}} n(a) + \hat{\tau}_{\mathbf{X},a}^{\text{phys}}, \quad B_{\mathbf{X},a}^{\text{phys}}(z) = \int_0^{\tau_{\mathbf{X},a}^{\text{phys}}(z)} \lambda_{\mathbf{X}}(X_{\mathbf{X}})(\Phi_{\mathbf{X}}^t z) dt.$$

In every C^2 norm and evaluation below we use the fixed-rectangle pullbacks $\widehat{\tau}_{\mathbf{X},a} = \widehat{\tau}_{\mathbf{X},a}^{\text{phys}} \circ \iota_{\mathbf{X},a}$ and $B_{\mathbf{X},a} = B_{\mathbf{X},a}^{\text{phys}} \circ \iota_{\mathbf{X},a}$. For each sign pair (σ, σ') there are C^2 functions $\widehat{\tau}_{\mathbf{X},\sigma,\sigma',\infty}$ and $B_{\mathbf{X},\sigma,\sigma',\infty}$ such that every return edge with these signs satisfies

$$(3.13) \quad \|\widehat{\tau}_{\mathbf{X},a} - \widehat{\tau}_{\mathbf{X},\sigma,\sigma',\infty}\|_{C^2} + \|B_{\mathbf{X},a} - B_{\mathbf{X},\sigma,\sigma',\infty}\|_{C^2} \leq C(1+n)e^{-\kappa n}.$$

For each nonempty first-exit family with type $\mathbf{t}$, source sign σ , and fixed cell label ℓ , the corresponding renormalized time and action converge in C^2 to functions $\widehat{\tau}_{\mathbf{X},\mathbf{t},\sigma,\infty,\ell}^{\text{ren}}$ and $B_{\mathbf{X},\mathbf{t},\sigma,\infty,\ell}^{\text{ren}}$, with error at most $C(1+n)^3e^{-\kappa n}$. The algebraic branch uses fixed-cut reference counterterms for time and action. Pole flight time is already finite and uses zero end counterterm; only the pole action uses the physical-remaining-time Laurent-log counterterm. Finite lateral and cut-face events have zero end counterterm.

- (ii) **Exact branch primitives.** On every finite branch, writing Σ_a^- and Σ_a^+ for its marked source and target sections,

$$(3.14) \quad P_a^*(\lambda_{\mathbf{X}}|_{\Sigma_a^+}) - \lambda_{\mathbf{X}}|_{\Sigma_a^-} = d_{\Sigma_a^-} B_{\mathbf{X},a}^{\text{phys}}.$$

Composition is exact because the same orbit integral is split at intermediate sections. Orbit-following section changes and exact changes of primitive satisfying Proposition 5.5 alter the renormalized actions only by endpoint coboundaries.

Theorem 3.3 (Coding and parameter dependence). *Under the hypotheses and with the objects of Theorems 3.1 and 3.2, the following statements hold.*

- (i) **Countable Markov coding.** The return graph has the two vertices Σ_+ and Σ_- and all edges $a = (\sigma, n, \sigma')$ with source σ and target σ' . Let $\Sigma_{\mathcal{G}_N}$ be its two-sided edge shift, and let $K_{\mathbf{X}} \subset \Sigma_N$ be the set on which the first-return map is defined for all positive and negative iterates. The map $h_{\mathbf{X}} : \Sigma_{\mathcal{G}_N} \rightarrow K_{\mathbf{X}}$ is a homeomorphism for the product-discrete topology. Every one-sided infinite future word determines a stable plaque; every point outside the forward trapped set has a finite return word followed by a unique first-exit component. For each source sign, the closures of the uncut return strips in the cut-open section form a Hausdorff compact alphabet; the edge alphabet is its finite discrete next-sign extension. First-exit labels remain discrete and are not compactified. The renormalized roof and action on recurrent codes have summable variations.
- (ii) **Persistence and parameter dependence.** The construction is continuous from $\mathcal{U}_{\kappa}$ to a fixed weighted C^2 space of return and first-exit data. Let $\mathcal{P}$ be a compact finite-dimensional parameter manifold, let $r \in \{1, 2\}$, and let

$$\mathcal{P} \longrightarrow \mathcal{U}_{\kappa}, \quad \varpi \longmapsto \mathbf{X}_{\varpi},$$

be a C^r map whose image is compact in the coefficient topology of Definition 2.2. The state-only geometric constants work uniformly

on its image. The return and compact first-hit maps, the recurrent time/action data, and both renormalized end limits have all mixed derivatives $D_z^i D_\omega^j$, $i + j \leq 2$, $j \leq r$, with the same $e^{-\kappa n}$ weights. The multiplicative constants for derivatives with $j > 0$ may depend on the C^r coefficient norm of the chosen parameterization. The same mixed two-jet statement applies to the same- e algebraic comparison in Theorem 3.1(i). At the pole the finite jets may vary subject to the C^{m_p+3} family condition (2.30); Proposition D.9 extracts every nonintegrable Laurent-log term with the same mixed regularity.

Proof of Theorem 1.1. Apply Theorem 2.12. After a finite refinement of (2.69), the compact image of every $\bar{V}_i$ is covered by finitely many coefficient neighborhoods on which Theorems 3.1–3.3 apply. The chosen inequality for κ leaves a strict rate gap on all of them. Take the maximum of their marked winding thresholds and all finite first-event thresholds. The interior margins in the physical source cell allow one smaller physical source collar to be used throughout the finite cover.

Theorem 3.1(i)–(ii) gives the exhaustive marked return/first-event partition, including the two selected nonempty end families, and its part (iii) gives the cross forms. Theorem 3.2 gives the time, end finite-part, and exact action statements. Theorem 3.3(i) gives the local coding and summable variations.

Now use (2.73) to choose T_* above every marked threshold. The marked partitions therefore restrict to an exhaustive physical relation on the part of the smaller source collar whose saddle-block residence time is at least T_* . On overlaps, Theorem 5.7 identifies that relation, the trapped Poincaré system, and the corrected weighted data after a finite clean refinement. Its deck and endpoint-coboundary formulas preserve closed periods and closed actions. This proves all four parts, including the stated limitation on global symbolic labels. $\square$

Corollary 3.4 (Descent from a finite marked atlas). *Let $\mathcal{P}$ be compact, $r \in \{0, 1, 2\}$, and consider a C^r family of physical reversible exact Hamiltonian return–exit tubes. Suppose a finite cover $\mathcal{P} = \bigcup_i V_i$ carries C^r marked representations*

$$p \longmapsto \mathbf{X}_{i,p} \in \mathfrak{H}_{\text{adm}}^k$$

with compact local images to which Theorems 3.1–3.3 apply. Suppose that on every overlap the representations describe the same physical event hypersurfaces and are admissibly related in the sense of Definition 5.6. Then, after increasing the finitely many local winding thresholds, the following statements hold.

- (i) *The local exhaustive return/first-exit partitions descend to one physical first-event relation. The Poincaré-conjugacy class of its trapped return system, the periods of its closed orbits, and their actions for the fixed primitive are independent of the representation.*

- (ii) *The corrected return maps, roof and action functions, first-exit data, and the two renormalized end limits form the compatible family of weighted mixed- C^2 data defined by Theorem 5.7. Raw winding labels, individual limit maps, references, and end counterterms remain local representatives.*
- (iii) *Each marking has its local full countable edge shift. When two cuts cross, the two codings compare through the realized-itinerary factor on a finite common refinement; a global labeled full shift is not inferred. Event-preserving section slides and exact gauges change roof and action only by the endpoint coboundaries in Proposition 5.5. Closed periods and closed exact-gauge actions are unchanged.*

A single global marked lift is the special case of a one-chart atlas; it is not required.

Corollary 3.5 (High-winding period, action, and first-exit asymptotics). *Fix $\mathbf{X} \in \mathcal{U}_\kappa$ and a cyclic sign sequence $\sigma = (\sigma_0, \dots, \sigma_m)$, $m \geq 1$, with $\sigma_m = \sigma_0$. For $\mathbf{n} = (n_0, \dots, n_{m-1})$, $n_j \geq N$, put*

$$(3.15) \quad a_j = (\sigma_j, n_j, \sigma_{j+1}), \quad \Delta_\kappa(\mathbf{n}) = \max_{0 \leq j < m} (1 + n_j)^3 e^{-\kappa n_j}.$$

Assume that the cyclic edge word $a_0 \cdots a_{m-1}$ is primitive. Its repetition codes a unique periodic orbit $\gamma_{\mathbf{n}}$ with exactly m first returns. A nonprimitive word is covered by applying the statement to its primitive cyclic subword.

Let $(x_j^\infty)_{j \in \mathbb{Z}/m\mathbb{Z}}$ be the unique solution of

$$(3.16) \quad x_{j+1}^\infty = f_{\mathbf{X}, \sigma_j, \sigma_{j+1}, \infty}(x_j^\infty, 0).$$

Define

$$(3.17) \quad \mathcal{T}_{\sigma, \mathbf{X}} = \sum_{j=0}^{m-1} \widehat{\tau}_{\mathbf{X}, \sigma_j, \sigma_{j+1}, \infty}(x_j^\infty, 0),$$

$$(3.18) \quad \mathcal{A}_{\sigma, \mathbf{X}} = \sum_{j=0}^{m-1} B_{\mathbf{X}, \sigma_j, \sigma_{j+1}, \infty}(x_j^\infty, 0).$$

Then, as $\min_j n_j \rightarrow \infty$,

$$(3.19) \quad \text{per}(\gamma_{\mathbf{n}}) = \frac{2\pi}{\beta_{\mathbf{X}}} \sum_{j=0}^{m-1} n_j + \mathcal{T}_{\sigma, \mathbf{X}} + O(\Delta_\kappa(\mathbf{n})),$$

$$(3.20) \quad \oint_{\gamma_{\mathbf{n}}} \lambda_{\mathbf{X}} = \mathcal{A}_{\sigma, \mathbf{X}} + O(\Delta_\kappa(\mathbf{n})).$$

There is an analogous C^2 statement for first exits. Fix a finite sign path $\sigma_0, \dots, \sigma_m$, a first-exit type $\mathfrak{t}$, and a cell label ℓ . Restrict to winding vectors for which the full finite cylinder consisting of the prescribed return branches followed by that first-exit component is nonempty, and identify these cylinders by the fixed component chart. For return windings $n_0, \dots, n_{m-1}$ and

final winding n_m , set $a_j = (\sigma_j, n_j, \sigma_{j+1})$ for $j < m$ and define, with every term evaluated along the corresponding finite composition,

$$\begin{aligned}\tau_{\mathbf{n}}^{\text{end}} &= \sum_{j=0}^{m-1} \left(\frac{2\pi}{\beta_{\mathbf{X}}} n_j + \widehat{\tau}_{a_j} \right) + \frac{2\pi}{\beta_{\mathbf{X}}} n_m + \widehat{\tau}_{\mathbf{t}, \sigma_m, n_m, \ell}^{\text{ren}}, \\ B_{\mathbf{n}}^{\text{end}} &= \sum_{j=0}^{m-1} B_{a_j} + B_{\mathbf{t}, \sigma_m, n_m, \ell}^{\text{ren}}.\end{aligned}$$

Here the final renormalized terms are those of (5.60)–(5.61). At the algebraic end they are defined by a cutoff limit after subtraction of the end counterterm, never by subtracting two infinite quantities; the pole action is understood in the same cutoff-renormalized sense. On the fixed component chart there are C^2 limit functions $\mathcal{T}_{\sigma, \mathbf{t}, \ell, \mathbf{X}}^{\text{exit}}$ and $\mathcal{A}_{\sigma, \mathbf{t}, \ell, \mathbf{X}}^{\text{exit}}$ such that

$$\begin{aligned}(3.21) \quad & \left\| \tau_{\mathbf{n}}^{\text{end}} - \frac{2\pi}{\beta_{\mathbf{X}}} \sum_{j=0}^m n_j - \mathcal{T}_{\sigma, \mathbf{t}, \ell, \mathbf{X}}^{\text{exit}} \right\|_{C^2} \\ & + \left\| B_{\mathbf{n}}^{\text{end}} - \mathcal{A}_{\sigma, \mathbf{t}, \ell, \mathbf{X}}^{\text{exit}} \right\|_{C^2} \leq C \max_{0 \leq j \leq m} (1 + n_j)^3 e^{-\kappa n_j}.\end{aligned}$$

The two limit functions are the sums of the limiting return functions and the limiting first-exit function along the corresponding limiting composition.

The complete period and action asymptotics are unchanged under the section slides and exact gauges in Proposition 5.5. The individual constants in (3.17)–(3.18) use the fixed winding and endpoint normalizations; their transformation laws are (5.35)–(5.57).

Proof of Corollary 3.5. On the cyclic product of m section rectangles, the cross-form equations define a contraction with constant at most ϑ . By (3.9)–(3.11), this contraction differs from the limiting contraction in (3.16) by $O(\Delta_{\kappa}(\mathbf{n}))$. The inverse bound $(1 - \vartheta)^{-1}$ therefore gives the same estimate for their fixed points. Substitution into (3.13) and summation over the fixed number m of branches proves (3.19)–(3.20). The period is the sum of the actual branch flight times, and (5.32) identifies the action sum with the closed orbit integral. Injectivity of the coding identifies the fixed point with the periodic orbit of the primitive word.

On a nonempty fixed finite cylinder ending in a first exit, the same argument is a finite induction through the return maps, followed by the C^2 first-exit limit in Theorem 3.2. This gives (3.21). Section and primitive changes contribute endpoint coboundaries. They telescope on a periodic word; for a first-exit path they are the endpoint terms fixed by the chosen normalization. $\square$

For a quantitative dynamical consequence, discard every return whose winding is smaller than a second threshold $L \geq N$. The resulting subsystem is still countable: from either signed section there is one edge of winding

n to each signed section for every $n \geq L$. Completeness of the return–exit partition is important here. It identifies all trapped orbits satisfying this winding restriction, rather than only a subshift chosen inside the return map.

Theorem 3.6 (Entropy and action of the high-winding trapped sets). *Fix $\mathbf{X} \in \mathcal{U}_\kappa$ and put*

$$(3.22) \quad c_{\mathbf{X}} = \frac{2\pi}{\beta_{\mathbf{X}}}.$$

For $L \geq N$, let

$$(3.23) \quad \Sigma_L = \{\omega \in \Sigma_{\mathcal{G}_N} : n(\omega_j) \geq L \text{ for every } j \in \mathbb{Z}\},$$

and let Λ_L be the corresponding suspension of the trapped Poincaré system inside the fixed flow tube. On Σ_L denote the physical roof and branch action by

$$(3.24) \quad r_{\mathbf{X}}(\omega) = \tau_{\mathbf{X}, \omega_0}^{\text{phys}}(h_{\mathbf{X}}(\omega)),$$

$$(3.25) \quad b_{\mathbf{X}}(\omega) = B_{\mathbf{X}, \omega_0}^{\text{phys}}(h_{\mathbf{X}}(\omega)).$$

There are constants $M_\tau, M_B < \infty$, independent of L , such that

$$(3.26) \quad |r_{\mathbf{X}}(\omega) - c_{\mathbf{X}}n(\omega_0)| \leq M_\tau, \quad |b_{\mathbf{X}}(\omega)| \leq M_B.$$

In the pressure notation below, $P_{\Sigma_L}(-sr_{\mathbf{X}})$ means the pressure of the cohomologous one-sided representative $r_{\mathbf{X}}^+$ supplied by Lemma 3.7; bounded cohomology makes this choice immaterial.

For all sufficiently large L , the equation

$$(3.27) \quad P_{\Sigma_L}(-sr_{\mathbf{X}}) = 0$$

for Gurevich pressure has a unique positive root $h_L(\mathbf{X})$. This root is the variational entropy of the suspension flow on Λ_L , and

$$(3.28) \quad h_L(\mathbf{X}) = \frac{1}{c_{\mathbf{X}}L} \left\{ W(2L) + O\left(\frac{W(L)}{L}\right) \right\},$$

where W is the principal branch of the Lambert W -function. Equivalently,

$$(3.29) \quad h_L(\mathbf{X}) = \frac{\beta_{\mathbf{X}}}{2\pi L} (\log L - \log \log L + \log 2 + o(1)).$$

The error estimates are uniform when $\mathbf{X}$ ranges over a compact subset of $\mathcal{U}_\kappa$.

Every periodic orbit $\gamma \subset \Lambda_L$ satisfies

$$(3.30) \quad \frac{\left| \oint_\gamma \lambda_{\mathbf{X}} \right|}{\text{per}(\gamma)} \leq \frac{M_B}{c_{\mathbf{X}}L - M_\tau} = O(L^{-1}).$$

More generally, if μ is a shift-invariant probability measure on Σ_L and $\int r_{\mathbf{X}} d\mu < \infty$, then

$$(3.31) \quad \frac{|\int b_{\mathbf{X}} d\mu|}{\int r_{\mathbf{X}} d\mu} \leq \frac{M_B}{c_{\mathbf{X}}L - M_{\tau}}.$$

Thus the entropy of the complete high-winding tail has the explicit scale $(\log L)/L$, while its action per unit physical time tends uniformly to zero.

Lemma 3.7 (Thermodynamic hypotheses for the high-winding tail). *For all sufficiently large L , the one-sided presentation of Σ_L is a topologically mixing countable Markov shift, and the following facts hold.*

- (i) *There are bounded transfer functions, with bounds independent of L , which replace $r_{\mathbf{X}}$ and $b_{\mathbf{X}}$ by cohomologous one-sided functions $r_{\mathbf{X}}^+$ and $b_{\mathbf{X}}^+$. These functions have summable variations and*

$$(3.32) \quad r_{\mathbf{X}}^+ \geq c_{\mathbf{X}}L - C_{\tau} > 0, \quad |r_{\mathbf{X}}^+ - c_{\mathbf{X}}n(\omega_0)| \leq C_{\tau}$$

for a constant C_{τ} independent of L .

- (ii) *For every $s > 0$, the Gurevich pressure $P_{\Sigma_L}(-sr_{\mathbf{X}}^+)$ is finite. It is continuous and strictly decreasing in s , tends to $+\infty$ as $s \downarrow 0$, and tends to $-\infty$ as $s \rightarrow \infty$.*
- (iii) *The suspension variational principle applies and gives*

$$(3.33) \quad h_{\text{var}}(\Lambda_L) = \sup_{\substack{\mu \text{ shift-invariant} \\ \int r_{\mathbf{X}}^+ d\mu < \infty}} \frac{h_{\mu}(\sigma)}{\int r_{\mathbf{X}}^+ d\mu}.$$

Here h_{var} denotes the supremum of metric entropies over invariant probability measures of the suspension flow. No positive-recurrence assumption is needed for this assertion: it is a statement about the supremum and the pressure zero, not about existence of a maximizing probability measure.

Proof of Lemma 3.7. The full two-vertex incidence, including self-edges of every winding, makes the one-sided shift topologically mixing. The exponential two-sided variation estimates from Theorem 3.3 give the usual telescoping two-sided-to-one-sided cohomology construction. Its transfer functions are bounded by the sum of a geometric series, uniformly after symbols with winding below L are discarded. Together with (3.26), this proves summable variations and (3.32) after increasing L .

For $r_0(\omega) = c_{\mathbf{X}}n(\omega_0)$, the weighted transition matrix has spectral radius

$$2 \sum_{n=L}^{\infty} e^{-sc_{\mathbf{X}}n} = 2 \frac{e^{-sc_{\mathbf{X}}L}}{1 - e^{-sc_{\mathbf{X}}}}.$$

The bounded difference between $r_{\mathbf{X}}^+$ and r_0 changes pressure by at most sC_{τ} . Hence pressure is finite for $s > 0$ and has the stated endpoint limits. The positive lower roof bound makes it strictly decreasing; convexity makes it

continuous. Finally, the variational principle for finite-pressure summable-variation potentials, followed by the suspension variational principle and Abramov's formula, gives (3.33) [24, 2]. These results identify the supremum without requiring an equilibrium measure. $\square$

Proof of Theorem 3.6. Theorem 3.2 first gives (3.26). Apply Lemma 3.7 and use its one-sided representatives below; bounded cohomology preserves pressure, periodic sums, invariant-measure integrals, and the suspension up to a change of section. Increase M_τ , if necessary, so that $\|r_{\mathbf{X}}^+ - r_0\|_\infty \leq M_\tau$.

We use the convention that P_{Σ_L} denotes Gurevich pressure for the one-sided presentation of the shift. First replace the roof by the one-coordinate function $r_0(\omega) = c_{\mathbf{X}}n(\omega_0)$. The weighted transition matrix between the two signed vertices has all four entries equal to

$$(3.34) \quad R_L(s) = \sum_{n=L}^{\infty} e^{-sc_{\mathbf{X}}n} = \frac{e^{-sc_{\mathbf{X}}L}}{1 - e^{-sc_{\mathbf{X}}}}.$$

Its spectral radius is $2R_L(s)$, and hence

$$(3.35) \quad P_{\Sigma_L}(-sr_0) = \log \left(2 \frac{e^{-sc_{\mathbf{X}}L}}{1 - e^{-sc_{\mathbf{X}}}} \right).$$

Since Gurevich pressure changes by at most the uniform norm of a potential, the preceding one-sided bound gives

$$(3.36) \quad P_{\Sigma_L}(-sr_0) - sM_\tau \leq P_{\Sigma_L}(-sr_{\mathbf{X}}^+) \leq P_{\Sigma_L}(-sr_0) + sM_\tau.$$

Lemma 3.7 proves existence and uniqueness of the pressure zero and identifies it with

$$(3.37) \quad h_L(\mathbf{X}) = \sup_{\substack{\mu \text{ a shift-invariant probability measure} \\ \int r_{\mathbf{X}} d\mu < \infty}} \frac{h_\mu(\sigma)}{\int r_{\mathbf{X}} d\mu}.$$

Set $x = c_{\mathbf{X}}Lh_L$. Before expanding the denominator in (3.35), note that $x \rightarrow \infty$ and $x/L \rightarrow 0$. Indeed, bounded x would make the model pressure grow like $\log L$, contrary to $|P_{\Sigma_L}(-h_L r_0)| \leq h_L M_\tau$; while x/L bounded away from zero would make its negative linear term dominate the same right-hand side. Using $s = h_L$ in (3.35)–(3.36) together with the expansion $1 - e^{-x/L} = (x/L)(1 + O(x/L))$ gives

$$(3.38) \quad x + \log x = \log(2L) + O(x/L).$$

The inverse of $x \mapsto x + \log x$ is $W(e^{(\cdot)})$ and has derivative $x/(1+x)$. Hence

$$x = W(2L) + O(W(L)/L),$$

which proves (3.28). The standard expansion of W gives (3.29). The constants used in the comparison are uniform on compact field families, proving the uniform version.

If a periodic code has m returns, exact action composition and (3.26) give

$$\left| \oint_{\gamma} \lambda_{\mathbf{X}} \right| \leq m M_B, \quad \text{per}(\gamma) \geq m(c_{\mathbf{X}} L - M_{\tau}).$$

This proves (3.30). Integrating the same pointwise inequalities proves (3.31). $\square$

Remark 3.8 (Physical conclusions and auxiliary choices). The first-return/first-exit partition is defined by the physical flow in the fixed flow tube, and moving an intermediate section along that flow gives the Poincaré conjugacy and endpoint coboundaries proved in Proposition 5.5. Periods are independent of coordinates. For the fixed primitive, $\oint \lambda_{\mathbf{X}}$ is also coordinate-independent and is unchanged by an exact gauge $\lambda \mapsto \lambda + d\chi$; it need not be unchanged after adding a non-exact closed one-form. The integer winding labels, individual limiting cross forms, and chosen algebraic reference do change with the chosen presentation. Under the admissibility conditions of Definition 5.6, Theorem 5.7 gives their exact deck, coboundary, and end-correction laws and proves that, after a common finite refinement, the metric overlap maps are locally bi-Lipschitz. Thus the topology defined by these overlaps, rather than any one labeled representative, is intrinsic.

The following selected-branch consequence is an optional module. It is not used in the return–exit theorem, its RFSN-II realization, or the coding and action conclusions.

Corollary 3.9 (Optional module: critical points on a selected reversible branch). *Assume, in addition to the hypotheses of Theorem 3.1, that the algebraic entrance hypersurface $\mathcal{W}_{a,*}$ contains a selected critical-energy connection*

$$\Gamma_{a,*} \subset W^u(O_*) \cap \mathcal{W}_{a,*} \cap H_*^{-1}(0),$$

and that the two invariant manifolds meet transversely modulo their common flow direction. Equivalently, at one, hence every, regular point $q_ \in \Gamma_{a,*}$,*

$$T_{q_*} W^u(O_*) + T_{q_*} \mathcal{W}_{a,*} = T_{q_*} M, \quad T_{q_*} W^u(O_*) \cap T_{q_*} \mathcal{W}_{a,*} = \text{span}\{X_*(q_*)\}.$$

Then the neighborhood $\mathcal{U}_{\kappa}$ in Theorem 3.1 may be decreased so that the following holds for every $\mathbf{X} \in \mathcal{U}_{\kappa}$. Denote by $\Gamma_{a,\mathbf{X}}$ the transverse continuation of $\Gamma_{a,}$ and by $\mathcal{W}_{a,\mathbf{X}}$ the continued algebraic entrance hypersurface. On the fixed-point plane of the reverser, let (r, θ) be local polar coordinates defined by $\xi = \eta = r q(\theta)$, and denote by $\gamma_{\mathbf{X}}^a(r)$ the corresponding local branch selected by $\Gamma_{a,\mathbf{X}}$. Its lifted angle satisfies*

$$(3.39) \quad \theta_{\mathbf{X}}(r) - \frac{\beta_{\mathbf{X}}}{\alpha_{\mathbf{X}}} \log r = \theta_{\mathbf{X}}^* + O_{C_{\log r}^2}(r^{\eta})$$

for one uniform $\eta > 0$. All sufficiently small critical points of $H_{\mathbf{X}}$ restricted to this branch are nondegenerate, and they alternate between maxima and

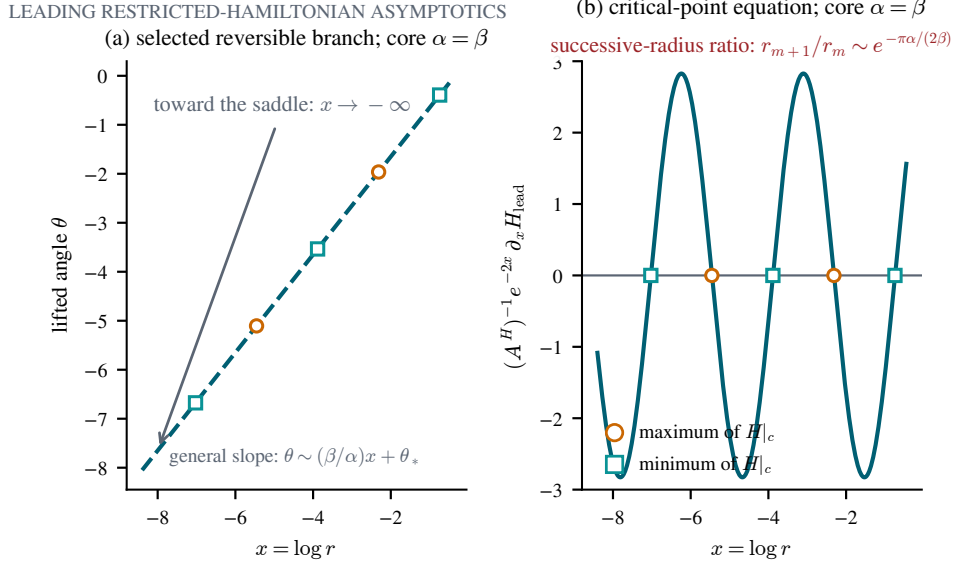


FIGURE 5. Leading-order logarithmic-spiral mechanism on the selected reversible-plane branch. The curves are plotted for the normal-form system $\alpha = \beta$; the displayed formulas give the general slope $\theta \sim (\beta/\alpha) \log r + \theta_*$ and the general ratio $r_{m+1}/r_m \sim \exp[-\pi\alpha/(2\beta)]$. Open circles and open squares denote, respectively, maxima and minima of the leading restriction of H . These are critical points of the scalar restriction, not zeros of H . The diagram is asymptotic and does not provide a rigorous enclosure of the finite continuation.

minima. The radii of consecutive critical points satisfy

$$(3.40) \quad \frac{r_{m+1}(\mathbf{X})}{r_m(\mathbf{X})} = \exp\left(-\frac{\pi\alpha\mathbf{X}}{2\beta\mathbf{X}}\right) (1 + O(r_m(\mathbf{X})^\eta)).$$

These are critical points of the scalar function $r \mapsto H_{\mathbf{X}}(\gamma_{\mathbf{X}}^{\mathbf{a}}(r))$, not zeros of H .

Figure 5 illustrates the leading logarithmic-spiral calculation and distinguishes the normal-form plot from the general ratio in (3.40).

The order of quantifiers in Theorem 3.1 is essential: the neighborhood $\mathcal{U}_\kappa$ and the threshold N are chosen after fixing the chosen rate κ . We do not assert one neighborhood that works uniformly as $\kappa \uparrow 2\pi\alpha_*/\beta_*$. For the normal-form system (1.1), this endpoint is 2π .

Remark 3.10 (Relative openness and the pole type). The perturbation conclusions are open in the marked coefficient topology of Definition 2.2, not in an unrestricted compact-open C^k topology. Condition (2.29) fixes the compactified-action order m_p , the principal radial scaling, and a simple

positive-integer indicial spectrum, but not the finite pole jet or the leading compactified action coefficient. The latter may vary and are handled by Proposition D.9. Allowing the indicial roots themselves to move or cross an integral resonance would require parameter-dependent powers and uniform divided-difference coordinates; that is outside the theorem.

4. LONG SADDLE-FOCUS PASSAGES AND RECURRENT STRIPS

4.1. Mixed boundary-value problem in Volterra form. The axis identities (2.11) imply by the double Hadamard lemma that, in a smaller bidisk,

$$(4.1) \quad |F(\xi, \eta)| + |G(\xi, \eta)| \leq C|\xi||\eta|.$$

For $T > 0$ prescribe the mixed boundary conditions

$$(4.2) \quad \eta(0) = \eta_-, \quad \xi(T) = \xi_+, \quad |\eta_-| = |\xi_+| = \delta.$$

For the parameter statement below, let $\mathcal{P}$ be compact, assume $\varpi \mapsto (A_\varpi, F_\varpi, G_\varpi)$ is C^2 in the preceding coefficient norms, and require the four axis identities in (2.11) for every $\varpi \in \mathcal{P}$. The boundary vectors are parametrized by finitely many fixed C^2 charts. Put

$$(4.3) \quad \underline{\alpha} = \inf_{\varpi \in \mathcal{P}} \alpha_\varpi, \quad 0 < \alpha_* < \underline{\alpha}.$$

At a fixed ϖ we abbreviate $A = A_\varpi$ and $\alpha = \alpha_\varpi$ in the formulas below.

Lemma 4.1 (Uniform mixed passage). *After decreasing δ , for every $T \geq T_0$ and every boundary pair in (4.2) there is a unique solution in the class*

$$(4.4) \quad |\xi(t)| \leq 2\delta e^{-\alpha(T-t)}, \quad |\eta(t)| \leq 2\delta e^{-\alpha t}.$$

It has the form

$$(4.5) \quad \xi(t) = e^{A(t-T)}\xi_+ + R_\xi(t), \quad \eta(t) = e^{-At}\eta_- + R_\eta(t),$$

where

$$(4.6) \quad \sup_{0 \leq t \leq T} (|R_\xi(t)| + |R_\eta(t)|) \leq C\delta^2 e^{-\alpha T}.$$

The same estimate holds through two boundary derivatives at fixed T . With $t = Tu$ and every boundary, T , and field-parameter multi-index γ of total order at most two, where T differentiation is taken at fixed u ,

$$(4.7) \quad \sup_{0 \leq u \leq 1} \left| D_{(\text{bdry}, T, \varpi)}^\gamma (R_\xi(Tu), R_\eta(Tu)) \right| \leq C_\gamma (1+T)^2 e^{-\alpha_* T}.$$

For every fixed $\alpha_ < \underline{\alpha}$ the two remote traces satisfy, for every mixed multi-index of total order at most two,*

$$(4.8) \quad \left| D_{\text{bdry}}^i D_{(T, \varpi)}^j (R_\xi(0), R_\eta(T)) \right| \leq C_{ij} (1+T)^{2j+i} e^{-\alpha_* T}, \quad i+j \leq 2.$$

The corresponding estimate for action applies to the renormalized gluing remainder in Lemma 4.4, not to the raw moving integral. Thus the present claim concerns endpoint traces, not an undefined T derivative of a function living on $[0, T]$.

Proof. The mixed problem is equivalent to Volterra integral equations with data imposed at opposite endpoints:

$$(4.9) \quad \xi(t) = e^{A(t-T)}\xi_+ - \int_t^T e^{A(t-s)}F(\xi(s), \eta(s)) ds,$$

$$(4.10) \quad \eta(t) = e^{-At}\eta_- + \int_0^t e^{-A(t-s)}G(\xi(s), \eta(s)) ds.$$

For existence at a fixed T , the usual endpoint-weighted norm makes the right-hand side a contraction: the double Hadamard factorization gives $|F| + |G| \leq C|\xi||\eta|$, hence the nonlinear source is $O(\delta^2 e^{-\alpha T})$ at every time. The two Volterra kernels have uniformly bounded L^1 norms, and the Lipschitz constant is $C\delta$. After decreasing δ , the contraction constant is some $\vartheta < 1$, independent of T and of $\varpi \in \mathcal{P}$. This proves existence, uniqueness, (4.4), and (4.6).

It remains to justify derivatives with respect to a moving interval. Put $t = Tu$ and work on the single Banach space

$$(4.11) \quad \mathcal{E} = C([0, 1]; \mathbb{R}^2 \times \mathbb{R}^2), \quad \|z\|_{\mathcal{E}} = \max_{0 \leq u \leq 1} \max\{|z_{\xi}(u)|, |z_{\eta}(u)|\}.$$

Define

$$(4.12) \quad \widehat{\xi}(u) = e^{\alpha_* T(1-u)}\xi(Tu), \quad \widehat{\eta}(u) = e^{\alpha_* Tu}\eta(Tu).$$

For the next display abbreviate

$$\xi_T(v) = e^{-\alpha_* T(1-v)}\widehat{\xi}(v), \quad \eta_T(v) = e^{-\alpha_* Tv}\widehat{\eta}(v).$$

In these variables the fixed-point map on $\mathcal{E}$ is

$$(4.13) \quad \begin{aligned} \widehat{\mathcal{T}}_{T, \varpi}^{\xi}(\widehat{\xi}, \widehat{\eta})(u) &= e^{\alpha_* T(1-u)}e^{A\varpi T(u-1)}\xi_+ \\ &\quad - T \int_u^1 e^{\alpha_* T(1-u)}e^{A\varpi T(u-v)}F_{\varpi}(\xi_T(v), \eta_T(v)) dv, \end{aligned}$$

$$(4.14) \quad \begin{aligned} \widehat{\mathcal{T}}_{T, \varpi}^{\eta}(\widehat{\xi}, \widehat{\eta})(u) &= e^{\alpha_* Tu}e^{-A\varpi Tu}\eta_- \\ &\quad + T \int_0^u e^{\alpha_* Tu}e^{-A\varpi T(u-v)}G_{\varpi}(\xi_T(v), \eta_T(v)) dv. \end{aligned}$$

The product of the two physical arguments in either nonlinear term is $e^{-\alpha_* T}|\widehat{\xi}(v)||\widehat{\eta}(v)|$. Moreover,

$$(4.15) \quad \sup_{T \geq T_0} \sup_{0 \leq u \leq 1} T \int_0^1 (1 + T|u - v|)^2 e^{-(\underline{\alpha} - \alpha_*)T|u - v|} dv < \infty.$$

This bound covers the kernels and their first two T - and ϖ -derivatives; derivatives of $e^{A\varpi s}$ contribute at most two powers of s . The homogeneous boundary terms obey the same bound because

$$\sup_{s \geq 0} (1 + s)^2 e^{-(\underline{\alpha} - \alpha_*)s} < \infty.$$

Thus (4.13)–(4.14) define a uniformly C^2 family, in the boundary-chart coordinates, T , and ϖ , on one fixed closed ball of $\mathcal{E}$. Its derivative in $(\widehat{\xi}, \widehat{\eta})$ has norm at most the same $\vartheta < 1$.

For completeness, the equations used for all mixed derivatives are as follows. Write $p = (\mathbf{d}, T, \varpi)$, where $\mathbf{d}$ is a boundary chart coordinate, let $\widehat{z}_p$ be the fixed point, and set $L_p = D_z \widehat{\mathcal{T}}_p(\widehat{z}_p)$. For any coordinate directions r, s in p ,

$$(4.16) \quad (I - L_p) \partial_r \widehat{z}_p = \partial_r \widehat{\mathcal{T}}_p(\widehat{z}_p),$$

$$(4.17) \quad (I - L_p) \partial_{rs} \widehat{z}_p = \partial_{rs} \widehat{\mathcal{T}}_p(\widehat{z}_p) + D_z \partial_r \widehat{\mathcal{T}}_p(\widehat{z}_p) \partial_s \widehat{z}_p \\ + D_z \partial_s \widehat{\mathcal{T}}_p(\widehat{z}_p) \partial_r \widehat{z}_p + D_z^2 \widehat{\mathcal{T}}_p(\widehat{z}_p) [\partial_r \widehat{z}_p, \partial_s \widehat{z}_p].$$

Since $\|(I - L_p)^{-1}\| \leq (1 - \vartheta)^{-1}$, these equations give uniform derivatives through total order two on the fixed Banach space.

Let $\widehat{R}$ be the difference between $\widehat{z}_p$ and the two homogeneous terms in (4.13)–(4.14). Endpoint evaluation has norm one on $\mathcal{E}$, and

$$R_\xi(0) = e^{-\alpha_* T} \widehat{R}_\xi(0), \quad R_\eta(T) = e^{-\alpha_* T} \widehat{R}_\eta(1).$$

Equations (4.16)–(4.17) therefore prove the stronger bound $C_{ij} e^{-\alpha_* T}$ in (4.8). Returning to the physical Volterra equations, the parameterwise axis identities give the double Hadamard factorization also after differentiating the coefficients. Every differentiated nonlinear source through order two still contains one incoming and one outgoing factor. More explicitly, retain the outgoing and incoming single-tail weights in the differentiated ξ and η estimates, respectively, and differentiate the unweighted Volterra equations for the remainders at $t = Tu$, with u fixed. The double-Hadamard source and all of its derivatives through order two are bounded by $C(1+T)^2 e^{-\alpha_* T}$. The Volterra kernels have the uniform L^1 bound in (4.15); induction in derivative order therefore gives (4.7).

Finally, the solutions are compatible under restriction. Construct the fixed point first for boundary vectors in a slightly larger fixed ball. For $0 < S < T$, uniqueness gives

$$(4.18) \quad z_T|_{[0,S]} = z_S(\eta_-, \xi_T(S)), \quad z_T(S + \cdot)|_{[0,T-S]} = z_{T-S}(\eta_T(S), \xi_+).$$

The intermediate values are traces of the same orbit, not independently chosen boundary data. The non-small endpoint density of the raw action integral is cancelled in Lemma 4.4. $\square$

The selected reversible branch uses a different local problem. It starts on the fixed-point plane of the reverser and stops at its first unstable exit; it is not an instance of the mixed boundary-value problem above.

Lemma 4.2 (Uniform reversible-plane first exit). *Let $q(\phi) = (\cos \phi, \sin \phi)$ and, for $r = e^x$, start the local orbit at*

$$(4.19) \quad \xi(0) = \eta(0) = rq(\phi).$$

There are $\delta > 0$, $x_0 < 0$, and $\eta_0 \in (0, 1)$, uniform on a sufficiently small coefficient ball, such that every $x < x_0$ has a unique first exit $\tau(x, \phi)$ through $|\xi| = \delta$. If ϑ_{out} is the lifted unstable angle there, then

$$(4.20) \quad \tau(x, \phi) = \alpha_{\mathbf{X}}^{-1}(\log \delta - x) + O_{C_{x, \phi}^2}(e^{\eta_0 x}),$$

$$(4.21) \quad \vartheta_{\text{out}}(x, \phi) = \phi + \frac{\beta_{\mathbf{X}}}{\alpha_{\mathbf{X}}}(\log \delta - x) + O_{C_{x, \phi}^2}(e^{\eta_0 x}),$$

$$(4.22) \quad \eta_{\text{out}}(x, \phi) = O_{C_{x, \phi}^2}(e^{(1+\eta_0)x}).$$

For a compact C^s family $\varpi \mapsto \mathbf{X}_{\varpi}$ in the raw coefficient space, $s \in \{1, 2\}$, the same estimates hold for mixed (x, ϕ, ϖ) derivatives of total order at most two and parameter order at most s after the displayed, parameter-dependent leading terms are subtracted.

Proof. Until the exit, variation of constants gives

$$(4.23) \quad e^{-At}\xi(t) = r q(\phi) + \int_0^t e^{-As} F(\xi(s), \eta(s)) ds,$$

$$(4.24) \quad e^{At}\eta(t) = r q(\phi) + \int_0^t e^{As} G(\xi(s), \eta(s)) ds.$$

The double Hadamard factorization writes $G(\xi, \eta) = \mathcal{B}_G(\xi, \eta)\eta$ with $\|\mathcal{B}_G\| \leq C|\xi|$. Choose δ so that $C\delta < \inf \alpha_{\mathbf{X}}/4$. On the standard exit bootstrap,

$$(4.25) \quad |\eta(t)| \leq C r e^{-\alpha t}, \quad \int_0^t |\eta(s)| ds \leq C r.$$

Moreover, since the rotational part of A is skew,

$$(4.26) \quad \frac{d}{dt} \log |\xi(t)| = \alpha_{\mathbf{X}} + O(|\eta(t)|) \geq \frac{1}{2} \inf \alpha_{\mathbf{X}}.$$

Thus the bootstrap closes, $|\xi|$ is strictly increasing, and the first exit is unique. Equations (4.23) and (4.25) give

$$(4.27) \quad r^{-1} e^{-At} \xi(t) = q(\phi) + O(r)$$

uniformly up to the exit. The exit equation and its angular part now give (4.20)–(4.21) with an $O(r)$ remainder, while (4.24) and $e^{-\alpha\tau} = O(r)$ give $\eta_{\text{out}} = O(r^2)$.

For the mixed-derivative statement use the normalized variables $\hat{\xi} = r^{-1} e^{-At} \xi$ and $\hat{\eta} = r^{-1} e^{At} \eta$. Differentiate their two Volterra equations through total order two. A field derivative of a semigroup may produce powers of t , and on the moving interval $t \leq \tau = O(1 + |x|)$. The differentiated sources still retain one complementary tail factor. Induction in

derivative order and Gronwall therefore give, for one fixed integer M ,

$$(4.28) \quad \max_{\substack{|\iota|+j \leq 2 \\ j \leq s}} |D_{x,\phi}^\iota D_\varpi^j (\widehat{\xi} - q(\phi))| \leq Cr(1+|x|)^M,$$

$$\max_{\substack{|\iota|+j \leq 2 \\ j \leq s}} |D_{x,\phi}^\iota D_\varpi^j \widehat{\eta}| \leq C(1+|x|)^M.$$

The derivative of the exit equation with respect to time is bounded below by (4.26); two implicit differentiations therefore give, after subtracting the displayed parameter-dependent leading terms, time and angle remainders whose mixed derivatives of orders $|\iota| + j \leq 2$, $j \leq s$, are bounded by $Cr(1+|x|)^M$; the corresponding derivatives of η_{out} are bounded by $Cr^2(1+|x|)^M$. Since $r(1+|\log r|)^M \leq Cr^{\eta_0}$ and $r^2(1+|\log r|)^M \leq Cr^{1+\eta_0}$, these are the asserted mixed endpoint two-jets. No uniform field-parameter jet is claimed for the interior profile at a common unscaled time on the moving interval. $\square$

Proposition 4.3 (Logarithmic approach along a transverse reversible branch). *Suppose a C^3 invariant target hypersurface meets the local unstable trace along a selected orbit and the intersection is transverse modulo the flow direction. On the fixed-point plane of the reverser, the end selected by that orbit is, for all sufficiently small radii, one embedded curve with exactly one point at each radius and*

$$(4.29) \quad \theta_{\mathbf{X}}(r) - \frac{\beta_{\mathbf{X}}}{\alpha_{\mathbf{X}}} \log r = \theta_{\mathbf{X}}^* + O_{C_{\log r}^2}(r^{\eta_0}).$$

For a compact C^s coefficient family, $s \in \{1, 2\}$, on which the transversality margin at the target hypersurface is uniform, the remainder in (4.29) has mixed derivatives $D_{\log r}^i D_\varpi^j$ for $i+j \leq 2$, $j \leq s$, with the same $O(r^{\eta_0})$ bound.

Proof. Propagate the transverse intersection to the exit section of Lemma 4.2. If g defines the target trace there, transversality is the nonzero angular derivative $\partial_\vartheta g(\vartheta_*, 0) \neq 0$. On one fixed target patch the implicit-function theorem writes it as $\vartheta = \Theta(\eta)$ with a uniform derivative margin. Substitute (4.21)–(4.22), put $\psi = \phi - (\beta_{\mathbf{X}}/\alpha_{\mathbf{X}})x$, and choose the one lift centered at the marked orbit. The target equation becomes

$$(4.30) \quad \psi - \psi_* + R_{\mathbf{X}}(x, \psi) = 0, \quad R_{\mathbf{X}} = O_{C_{x,\psi}^2}(e^{\eta_0 x}), \quad \partial_\psi(\psi + R) > \tfrac{1}{2}.$$

It has exactly one root on the fixed lift for every sufficiently negative x , and two implicit differentiations give the asserted logarithmic two-jet. Equivariance identifies the reversible plane with $\xi = \eta$. The radius and angle in the proposition are, by definition, those of this marked source gauge. This proves (4.29), embeddedness, and the parameter statement. $\square$

Let $a = \widetilde{\lambda}(X)$ in the marked local exact gauge with $\widetilde{\lambda}(0) = 0$. The marked primitive is part of the coefficient data, so along a C^2 field family the functions a_ϖ form a C^2 coefficient family. We suppress ϖ below. Then

$a(0) = Da(0) = 0$. Define the convergent half-tail actions

$$(4.31) \quad B_s(\eta_-) = \int_0^\infty a(0, e^{-At}\eta_-) dt, \quad B_u(\xi_+) = \int_{-\infty}^0 a(e^{At}\xi_+, 0) dt.$$

Lemma 4.4 (Action gluing). *For the orbit in Lemma 4.1,*

$$(4.32) \quad \int_0^T a(\xi(t), \eta(t)) dt = B_s(\eta_-) + B_u(\xi_+) + O_{C_{\text{bdry}}^2}((1+T)e^{-\alpha_* T}).$$

With passage-length or parameter derivatives through order two, the error is $O((1+T)^3 e^{-\alpha_ T})$.*

Proof. Put $x_0(t) = e^{A(t-T)}\xi_+$ and $y_0(t) = e^{-At}\eta_-$. The quadratic vanishing of a gives

$$(4.33) \quad |a(\xi, \eta) - a(\xi, 0) - a(0, \eta)| \leq C|\xi||\eta|.$$

The gluing remainder is exactly

$$(4.34) \quad \begin{aligned} \mathcal{E}_T = & \int_0^T \{a(\xi, \eta) - a(\xi, 0) - a(0, \eta)\} dt \\ & + \int_0^T \{a(\xi, 0) - a(x_0, 0)\} dt + \int_0^T \{a(0, \eta) - a(0, y_0)\} dt \\ & - \int_T^\infty a(0, y_0(t)) dt - \int_{-\infty}^{-T} a(e^{As}\xi_+, 0) ds. \end{aligned}$$

The first density is $O(e^{-\alpha_* T})$ throughout the interval. The next two integrals have the same total bound because $|Da(z)| \leq C|z|$, the orbit replacement errors are exponentially small, and the complementary single-tail factors are integrable. The last two tails are $O(e^{-2\alpha_* T})$.

The moving-endpoint cancellation is exact. Changing variables $s = t - T$ in the unstable term gives

$$(4.35) \quad \int_0^T a(x_0(t), 0) dt + \int_{-\infty}^{-T} a(e^{As}\xi_+, 0) ds = B_u(\xi_+),$$

$$(4.36) \quad \int_0^T a(0, y_0(t)) dt + \int_T^\infty a(0, e^{-At}\eta_-) dt = B_s(\eta_-).$$

The two sums are independent of T , so differentiating them produces no uncanceled endpoint density.

For each finite-interval integral in (4.34), first set $t = Tu$ and $dt = T du$; every T derivative below is taken at fixed u . Use the fixed-domain derivatives from Lemma 4.1 and let D^γ be any boundary, T , or field-parameter derivative of total order at most two. The three finite-interval densities in (4.34) satisfy

$$(4.37) \quad |D^\gamma \{a(\xi, \eta) - a(\xi, 0) - a(0, \eta)\}| \leq C(1+T)^2 e^{-\alpha_* T},$$

$$(4.38) \quad |D^\gamma \{a(\xi, 0) - a(x_0, 0)\}| \leq C(1+T)^2 e^{-\alpha_* T} e^{-\alpha_*(T-t)},$$

$$(4.39) \quad |D^\gamma \{a(0, \eta) - a(0, y_0)\}| \leq C(1+T)^2 e^{-\alpha_* T} e^{-\alpha_* t}.$$

The first estimate follows by differentiating the double Hadamard formula for the interaction term. The last two use $|Da(z)| \leq C|z|$ and the differentiated orbit remainders. The corresponding differentiated half-line densities are bounded by $C(1 + |t|)^2 e^{-2\alpha_*|t|}$. Integrating these majorants proves

$$(4.40) \quad \left| D_{\text{bdry}}^i D_{(T, \varpi)}^j \mathcal{E}_T \right| \leq C_{ij} (1 + T)^3 e^{-\alpha_* T}, \quad i + j \leq 2.$$

At fixed T , boundary differentiation does not differentiate a semigroup parameter, and the sharper factor $(1 + T)e^{-\alpha_* T}$ follows. This is the asserted C^2 estimate. $\square$

The mixed problem contains neither a full periodic inverse nor a remote growing mode as free initial data. This opposite-endpoint formulation makes the uniform passage estimate compatible with arbitrarily large winding.

4.2. Finite-dimensional matching equations select the winding strips. Let $y = (a, \theta)$ collect the finite nonangular matching variables and a residual angle. After decreasing the section width ν_N if necessary, let $\zeta = (\phi, \bar{\nu})$ range over the fixed rectangle $Z = I_\phi \times [-\nu_N, \nu_N]$. These are the source angle and the target action coordinate; the matching problem is therefore a two-parameter implicit equation, not a scalar pointwise equation. For each sign σ , suppose a limiting finite-dimensional matching equation $\mathcal{F}_{\sigma, \infty}(y, \zeta)$ has a solution $Y_{\sigma, \infty}(\zeta)$ for every $\zeta \in Z$. Assume this solution graph is C^2 and lies in the interior of the original matching block. Split the system into nonangular and angular components and assume

$$(4.41) \quad \|(D_a \mathcal{F}_{\sigma, \infty}^\perp)^{-1}\| \leq M_0, \quad |\Gamma_\sigma| \geq \gamma_0 > 0,$$

where

$$(4.42) \quad \Gamma_\sigma = \partial_\theta \mathcal{F}^\angle - D_a \mathcal{F}^\angle (D_a \mathcal{F}^\perp)^{-1} \partial_\theta \mathcal{F}^\perp.$$

The compact solution graph and the uniform inverse bound determine a smaller radius $r_* > 0$: in the closed fiber tube

$$\mathcal{N}_\sigma = \{(y, \zeta) : |y - Y_{\sigma, \infty}(\zeta)| \leq r_*\}$$

the limiting system has no zero other than the selected one, its derivative has one uniform inverse bound, and the centered Newton maps take every fiber tube strictly into itself. This is the “fixed tube” below; it is chosen before n . The exact matching map after n turns is $\mathcal{F}_{\sigma, n} = \mathcal{F}_{\sigma, \infty} + \mathcal{R}_{\sigma, n}$. Denote by $\Theta_\sigma(\zeta)$ the residual-angle component of $Y_{\sigma, \infty}(\zeta)$.

Proposition 4.5 (Uniform solution of the matching equation). *Suppose that for $0 \leq j \leq 2$*

$$(4.43) \quad \|D^j \mathcal{R}_{\sigma, n}\| \leq C_j (1 + n)^3 e^{-\kappa_+ n}$$

uniformly in both signs, for some $\kappa_+ > 0$. Then there is N such that, for every $n \geq N$, the exact matching map has a unique zero in the fixed tube. The zeros form C^2 graphs $Y_{\sigma, n}$ and

$$(4.44) \quad \|D^j (Y_{\sigma, n} - Y_{\sigma, \infty})\| \leq \tilde{C}_j (1 + n)^3 e^{-\kappa_+ n}, \quad 0 \leq j \leq 2.$$

Consequently their passage lengths satisfy

$$(4.45) \quad \beta T_{\sigma,n} = 2\pi n + \Theta_{\sigma}(\zeta) + O_{C^2}((1+n)^3 e^{-\kappa_+ n}).$$

Proof. Block elimination and (4.41) give a uniform inverse for $D_y \mathcal{F}_{\sigma,\infty}$ along the selected compact graph. The parameter-dependent inverse-function theorem and compactness give the radius r_* and centered Newton maps just described; this is precisely the isolation missing from a bare nonsingular zero. The C^1 part of (4.43) preserves this gap for all large n , while its order-zero part maps the tube into itself. Implicit differentiation once and twice gives (4.44); products of lower-order errors are absorbed by the order-two weight. $\square$

The scalar inversion of the marked clock, the exact marked action map, the endpoint trace estimate of Lemma 4.1, and composition with the fixed finite homoclinic matching equation supply (4.43). More precisely, on a fixed field ball put

$$(4.46) \quad \beta^* = \sup_{\mathbf{X} \in \mathcal{U}} \beta_{\mathbf{X}}, \quad \kappa_{\text{tr}} = \frac{2\pi\alpha}{\beta^*} \leq \inf_{\mathbf{X} \in \mathcal{U}} \frac{2\pi\alpha_{\mathbf{X}}}{\beta_{\mathbf{X}}}.$$

After shrinking the ball about $\mathbf{X}_*$, α may be chosen so that $\kappa < \kappa_+ < \kappa_{\text{tr}}$. The first gap proves the requested weighted estimate, and $\kappa_{\text{tr}} - \kappa_+$ absorbs every fixed polynomial caused by mixed state/parameter differentiation. The branchwise transverse scale of an individual edge remains the actual field-dependent quantity in (4.64). For the core, $\alpha_0 = \beta_0 = 2^{-1/2}$, so κ_{tr} can be made arbitrarily close to 2π and

$$(4.47) \quad T_{\sigma,n} = 2\pi\sqrt{2}n + O(1), \quad |\nu_{\sigma,n}| \asymp e^{-2\pi n}.$$

We now derive the hypotheses and the size of the completed cross forms from the marked local passage and the finite homoclinic matching equation, rather than infer them from transversality alone. Straighten the stable and unstable traces simultaneously in their Darboux section charts. If (ϕ, ν) are incoming action-angle coordinates and ψ is the outgoing angle, the critical local map is

$$(4.48) \quad \begin{aligned} \psi &= \phi + \Delta_{\mathbf{X},\sigma}(\nu), \\ \Delta_{\mathbf{X},\sigma}(\nu) &= \frac{\beta}{\alpha} \log |\nu| + b_{\sigma} + \tilde{\rho}_{\sigma}(\nu), \\ |D_{\log \nu}^k \tilde{\rho}_{\sigma}| &\leq C_k |\nu| (1 + |\log |\nu||) \quad (0 \leq k \leq 2). \end{aligned}$$

For the phase-normalized passage length $T_n = (2\pi n + \theta)/\beta$, the exact clock equation $T_{\mathbf{X},\sigma}(\nu) = T_n$ is equivalent to

$$(4.49) \quad \nu = \sigma c_{\mathbf{X},\sigma} e^{-\alpha T_n} \exp\{\alpha \tau_{\mathbf{X},\sigma}(\nu)\}.$$

On a fixed multiple of $e^{-\alpha T_n}$ the derivative of its right-hand side is $O((1 + T_n)e^{-\alpha T_n}) < 1/2$ after increasing N . It therefore has one small root on each signed side. Writing

$$r_{\sigma,n}^{\nu} = c_{\mathbf{X},\sigma} (\exp\{\alpha \tau_{\mathbf{X},\sigma}(\nu_{\sigma,n})\} - 1)$$

gives

$$(4.50) \quad \begin{aligned} \nu_{\sigma,n}(\theta) &= \sigma e^{-\alpha T_n} \{c_{\mathbf{X},\sigma} + r_{\sigma,n}^\nu(\theta)\}, \quad c_{\mathbf{X},\sigma} \geq c_* > 0, \\ \max_{|\gamma| \leq 2} \|D_{\theta,\varpi}^\gamma r_{\sigma,n}^\nu\| &\leq C(1+n)^3 e^{-\kappa+n}. \end{aligned}$$

Here ϖ is omitted when no field parameter is present. The inverse bound is at most two, so the same calculation gives the displayed mixed state/parameter two-jets. By (2.14), $c_{\mathbf{X},\sigma}$ is compatible with the actual clock. Put $\tilde{b}_\sigma = b_\sigma + \beta t_{\mathbf{X},\sigma}$. Substitution in (4.48) gives

$$(4.51) \quad \begin{aligned} \Delta_{\mathbf{X},\sigma}(\nu_{\sigma,n}(\theta)) &= -2\pi n - \theta + \tilde{b}_\sigma + \rho_{\sigma,n}(\theta), \\ \max_{0 \leq j \leq 2} \|D_\theta^j \rho_{\sigma,n}\| &\leq C(1+n)^3 e^{-\kappa+n}. \end{aligned}$$

The compact angular cell supplies the positive lower bound. Formula (4.50) proves the existence of both signs for every large n , rather than merely an unsigned sequence.

Let (ψ, ν_u) be coordinates before the finite homoclinic transition and write that transition as

$$(4.52) \quad (\bar{\phi}, \bar{\nu}) = (F^{\text{hom}}(\psi, \nu_u), G^{\text{hom}}(\psi, \nu_u)).$$

Thus G^{hom} is its target-action component, while F^{hom} is the target-angle component used below. With $(\phi, \bar{\nu})$ the prescribed source angle and target action, the exact transverse and lifted angular equations are

$$(4.53) \quad \mathcal{F}_{\sigma,n} = \begin{pmatrix} G^{\text{hom}}(\psi, \nu_{\sigma,n}(\theta)) - \bar{\nu} \\ \psi - \phi + \theta - \tilde{b}_\sigma - \rho_{\sigma,n}(\theta) \end{pmatrix}.$$

The second row is precisely the lifted identity $\psi = \phi + \Delta_{\mathbf{X},\sigma}(\nu_{\sigma,n}) + 2\pi n$. At the cut face the limiting matching rows are

$$(4.54) \quad \mathcal{F}_{\sigma,\infty} = \begin{pmatrix} G^{\text{hom}}(\psi, 0) - \bar{\nu} \\ \psi - \phi + \theta - \tilde{b}_\sigma \end{pmatrix}.$$

Lemma 2.10 gives the unique solution $\psi = \Psi_*(\bar{\nu})$ for every $|\bar{\nu}| \leq \nu_*$ and then determines θ uniquely from the second row of (4.54). Its inverse and boundary margins are uniform on Z for either source sign. After shrinking the coefficient ball, the two rows therefore supply the compact limiting graphs and the Schur-complement margins in (4.41).

For each (σ, n) the matching equation first produces the closure of a return strip $\tilde{\mathcal{V}}_{\sigma,n}$ over the whole rectangle Z . Inside the target-section collar, removing the target zero-action boundary gives the two regular return components

$$(4.55) \quad \mathcal{V}_{\sigma,n}^{\sigma'} = \{z \in \tilde{\mathcal{V}}_{\sigma,n} : 0 < \sigma' \bar{\nu}(z) < \nu_N\}, \quad \sigma' \in \{+, -\},$$

while $\mathcal{I}_{\sigma,n} = \{z \in \tilde{\mathcal{V}}_{\sigma,n} : \bar{\nu}(z) = 0\}$ is assigned to first exit. The preimages of $\bar{\nu} = \pm \nu_N$ are assigned to the fixed local-boundary first exits, not to return

edges. Because $\nu_N > 0$ and the solution is a graph over all of Z , both open components in (4.55) are nonempty for each source sign. Hence

$$(4.56) \quad \mathcal{G}_N = \{+, -\}^2.$$

The stable-manifold cut $\mathcal{I}_{\sigma,n}$ has type cut and is not part of either regular half. Compatibility at a shared vertex is enforced by using the same positive magnitude coordinate as in (3.2). The completed target-half embeddings are

$$(4.57) \quad \mathcal{I}_{\sigma'}(y') = \sigma' \nu_N y', \quad y' \in [0, 1], \quad \sigma' \in \{+1, -1\}.$$

Consequently the regular coding edges are

$$(4.58) \quad a = (\sigma, n, \sigma'), \quad n \geq N, \quad (\sigma, \sigma') \in \mathcal{G}_N, \quad s(a) = \sigma, \quad t(a) = \sigma'.$$

Finally solve the global normal equation as $\psi = R(\nu, \bar{\nu})$ and set $Q(\nu, \bar{\nu}) = F^{\text{hom}}(R(\nu, \bar{\nu}), \nu)$ for its angular output. A simultaneous symplectic shear straightens the image of the unstable axis and the stable axis without changing the local logarithmic map. To see this, take Darboux coordinates (ϕ_0, ω) with the stable trace $\{\omega = 0\}$. Transversality writes the image of the unstable trace as $\{\phi_0 = h(\omega)\}$. The shear

$$(4.59) \quad \bar{\phi} = \phi_0 - h(\omega), \quad \bar{\nu} = \omega, \quad \psi = \psi_0 - h(\nu)$$

is symplectic on both sections. Because the local map preserves the action ν , the two shear terms cancel in its angular equation. In these coordinates $F^{\text{hom}}(\psi, 0) = 0$, and therefore

$$(4.60) \quad Q(0, \bar{\nu}) \equiv 0.$$

Taylor's formula in ν , followed by (4.50), makes Q and its first two normalized state derivatives $O((1+n)^3 e^{-\alpha T_{\sigma,n}})$. The resulting return branch therefore has the following normalized cross form on the closed strip before the target cut:

$$(4.61) \quad \bar{\phi} = F_{\sigma,n}(\phi, \bar{\nu}) := Q(V_{\sigma,n}(\phi, \bar{\nu}), \bar{\nu}), \quad \nu = V_{\sigma,n}(\phi, \bar{\nu}) := \nu_{\sigma,n}(\theta_{\sigma,n}(\phi, \bar{\nu})),$$

with

$$(4.62) \quad \|F_{\sigma,n}\|_{C^j} + \|V_{\sigma,n}\|_{C^j} \leq C_j (1+n)^3 e^{-\kappa+n}, \quad 0 \leq j \leq 2.$$

For the core the unreduced exponent in this estimate is 2π . Increasing N makes the derivative bound smaller than one fixed $\vartheta < 1$. Restriction to either closed half in (4.55), followed by its vertex-compatible target chart, gives the completed edge cross form

$$(4.63) \quad f_{\sigma,n}^{\sigma'}(x, y') = F_{\sigma,n}(x, \mathcal{I}_{\sigma'}(y')), \quad g_{\sigma,n}^{\sigma'}(x, y') = \frac{\sigma}{\nu_N} V_{\sigma,n}(x, \mathcal{I}_{\sigma'}(y'))$$

for $(x, y') \in I \times [0, 1]$ and $a = (\sigma, n, \sigma')$. With

$$(4.64) \quad \epsilon_{\mathbf{X},a} = \nu_N^{-1} e^{-2\pi\alpha\mathbf{X}n/\beta\mathbf{X}}, \quad \tilde{g}_a = g_{\sigma,n}^{\sigma'}/\epsilon_{\mathbf{X},a},$$

the exact signed-normal law gives

$$(4.65) \quad \tilde{g}_a = e^{-(\alpha\mathbf{X}/\beta\mathbf{X})\theta_{\sigma,n}} \{c_{\mathbf{X},\sigma} + r_{\sigma,n}^\nu\}.$$

The compact residual-angle range and $c_{\mathbf{X},\sigma} \geq c_*$ give one $m_{\text{cf}} > 0$ with

$$(4.66) \quad m_{\text{cf}} \leq \tilde{g}_a \leq m_{\text{cf}}^{-1}, \quad \|\tilde{g}_a\|_{C^2} \leq C, \quad g_a \leq \frac{1}{2},$$

after increasing N . Moreover $Df_a, Dg_a = O(\epsilon_{\mathbf{X},a})$, and the matching estimate gives C^2 limits of f_a and $\tilde{g}_a$ for each fixed pair of source and target signs, with error $C(1+n)^3 e^{-\kappa+n}$. Thus the angular image has one uniform interior margin while the source-normal image has the rescaled transverse-action margin (4.66); no impossible absolute distance from $y = 0$ is asserted. The closure in the cut-open section is used for the estimate and is not treated as one edge across the removed zero-action boundary.

5. FIRST-EXIT GEOMETRY AND RENORMALIZED END DATA

5.1. Strict first events. We first isolate the global step which turns local rank bounds into stability of all connected components. This is the only place where the topology of the finite event arrangement, rather than a pointwise implicit-function theorem, is used.

The qualitative model is Thom's first isotopy lemma and its controlled vector fields [21, Sections 7–11]; Verona treats control data in the presence of faces [26, pp. 9–20 and 29–57]. Those general results give stratified topological trivializations. We need an ambient C^s diffeomorphism with quantitative control, so we record the elementary finite normal-crossing version used here.

Lemma 5.1 (Controlled lift of a moving neat arrangement). *Let $s \geq 1$, let M be a compact C^{s+1} manifold with corners, put $I = [0, 1]$, and let*

$$H = (H_1, \dots, H_q) \in C^{s+1}(M \times I; \mathbb{R}^q).$$

Fix a finite corner atlas, with boundary defining functions r_b in each chart. At (x, t) let

$$A(x, t) = \{j : H_j(x, t) = 0\}, \quad B(x) = \{b : r_b(x) = 0\}.$$

Suppose that the vertical conormal map

$$(5.1) \quad \mathcal{L}_{x,t} : T_x M \longrightarrow \mathbb{R}^{A(x,t)} \oplus \mathbb{R}^{B(x)}, \quad v \longmapsto ((d_x H_j v)_{j \in A(x,t)}, (dr_b v)_{b \in B(x)})$$

is onto at every (x, t) . Then there is a C^s time-dependent vector field V_t on M such that

$$(5.2) \quad d_x H_j(V_t) + \partial_t H_j = 0 \quad \text{on } H_j^{-1}(0),$$

$$(5.3) \quad dr_b(V_t) = 0 \quad \text{on } \{r_b = 0\}.$$

Its flow Φ_t exists on I , preserves every boundary face, and

$$(5.4) \quad \Phi_t(H_j(\cdot, 0)^{-1}(0)) = H_j(\cdot, t)^{-1}(0), \quad 1 \leq j \leq q.$$

Suppose in addition that the conormal maps have a common least singular value $\gamma > 0$, the H_j have a common C^{s+1} bound K , and the incidence types

lie in one fixed finite family of normal-crossing neighborhoods. Then the lift can be chosen with

$$(5.5) \quad \|V\|_{C^s(M \times I)} \leq C(s, q, \mathcal{U}, \gamma, K) \max_j \|\partial_t H_j\|_{C^s(M \times I)}.$$

Here $\mathcal{U}$ denotes the fixed finite atlas and active-star cover. The estimate is uniform for a class with common corner charts and cover, conormal bounds, and positive gaps for all empty face-incidence types.

Proof. Let

$$\mathcal{D} = \bigcup_{j=1}^q H_j^{-1}(0) \cup (\partial M \times I).$$

Normal-crossing coordinates and compactness give a finite *active-star cover* of a neighborhood of $\mathcal{D}$. Its members U_α have fixed index sets A_α, B_α such that

(1) the map

$$L_{\alpha,x,t}v = ((d_x H_j v)_{j \in A_\alpha}, (dr_b v)_{b \in B_\alpha})$$

is onto throughout U_α ;

(2) if $\bar{U}_\alpha$ meets $H_j^{-1}(0)$, then $j \in A_\alpha$;

(3) if $\bar{U}_\alpha$ meets $\{r_b = 0\}$, then $b \in B_\alpha$.

To construct the cover, start with product neighborhoods of the deepest nonempty intersections. After smaller inner tubes around the already treated strata are removed, the remaining part of each next stratum is compact. It has finitely many normal-crossing neighborhoods whose closures avoid every hypersurface and boundary face not containing that stratum. Descending induction on depth gives the asserted finite cover. Add one set U_0 whose closure is disjoint from $\mathcal{D}$, and choose a fixed C^s partition of unity $\{\chi_\alpha\}$ subordinate to these sets.

Choose auxiliary metrics and define on U_α

$$(5.6) \quad R_{\alpha,x,t} = L_{\alpha,x,t}^* (L_{\alpha,x,t} L_{\alpha,x,t}^*)^{-1}, \quad b_{\alpha,x,t} = ((\partial_t H_j)_{j \in A_\alpha}, (0)_{b \in B_\alpha}),$$

and $V_{\alpha,t} = -R_{\alpha,x,t} b_{\alpha,x,t}$; set $V_0 = 0$. Thus throughout U_α ,

$$d_x H_j(V_{\alpha,t}) = -\partial_t H_j \quad (j \in A_\alpha), \quad dr_b(V_{\alpha,t}) = 0 \quad (b \in B_\alpha).$$

Put $V = \sum_\alpha \chi_\alpha V_\alpha$. If $(x, t) \in H_j^{-1}(0)$ and $\chi_\alpha(x, t) \neq 0$, the active-star property gives $j \in A_\alpha$. Every nonzero summand satisfies the same affine equation for H_j , and their convex combination proves (5.2). The identical argument on a boundary face proves (5.3). No agreement of independently chosen local lifts is being assumed.

The right-inverse formula, the least-singular-value bound, the fixed partition, and the C^{s+1} coefficient bound give (5.5); taking s derivatives of R_α uses at most $s + 1$ derivatives of H . The vector field $\partial_t + V$ on $M \times I$ is tangent to every $H_j^{-1}(0)$ and every vertical boundary face. Compactness gives its flow on the whole interval. Uniqueness of solutions preserves each face and gives one inclusion in (5.4); the reverse flow gives equality. The

usual variational equations for the flow give the asserted C^s regularity and estimate. $\square$

Proposition 5.2 (Stability of a finite neat hypersurface arrangement). *Let $s \in \{1, 2\}$. Let M be a compact C^{s+1} manifold with corners and let $h^0 = (h_1^0, \dots, h_q^0)$ be a finite family of C^{s+1} functions on M . For $S \subset \{1, \dots, q\}$ put*

$$Z_S^0 = \{x \in M : h_j^0(x) = 0 \text{ for every } j \in S\}.$$

Suppose that every nonempty Z_S^0 is neat and that, in every corner chart, the covectors dh_j^0 , $j \in S$, together with the conormals of the boundary faces through x , are linearly independent at every $x \in Z_S^0$. Suppose also that every sign stratum

$$(5.7) \quad \{h_j^0 = 0 \ (j \in S), \quad \epsilon_j h_j^0 > 0 \ (j \notin S)\}, \quad \epsilon_j \in \{+1, -1\},$$

has finitely many connected components.

There is $\delta > 0$ such that, whenever a C^{s+1} family $h = (h_1, \dots, h_q)$ satisfies

$$(5.8) \quad \max_j \|h_j - h_j^0\|_{C^s(M)} < \delta,$$

there is a face-preserving C^s diffeomorphism $\Phi_h : M \rightarrow M$, joined to the identity by face-preserving C^s diffeomorphisms Φ_t which carry each h^0 -labelled sign stratum onto the corresponding h^t -labelled sign stratum, with

$$(5.9) \quad \Phi_h((h_j^0)^{-1}(0)) = h_j^{-1}(0), \quad 1 \leq j \leq q.$$

It maps each labelled sign stratum in (5.7) onto the corresponding sign stratum for h and hence gives a bijection of their connected components. In particular, no additional component can occur. The threshold is uniform for a class with a common finite corner atlas, a common C^{s+1} bound, a common positive conormal least-singular-value bound, and a common positive gap for every reference-empty face-incidence type. In such a class, Φ_h may be chosen so that

$$(5.10) \quad \|\Phi_h - \text{id}\|_{C^s(M)} \leq C \max_j \|h_j - h_j^0\|_{C^s(M)},$$

where C depends only on s, q , the fixed atlas and active-star cover, and the common conormal and C^{s+1} bounds. No differentiable dependence of Φ_h on an external parameter is asserted.

Proof. Let $\mathcal{F}$ be the finite family consisting of M and the closed coordinate-face pieces in the chosen finite corner atlas. For every subset S and every $F \in \mathcal{F}$, first restrict the incidence problem to F . If

$$\{x \in F : h_j^0(x) = 0 \text{ for every } j \in S\} = \emptyset,$$

compactness gives a positive lower bound on F for $(\sum_{j \in S} |h_j^0|^2)^{1/2}$; a sufficiently small C^0 perturbation therefore cannot create that face-incidence

type. Equivalently, the minimum

$$a_{\text{emp}} := \min_{\substack{F \in \mathcal{F}, S: \\ \{h_j^0=0 \mid (j \in S)\} \cap F = \emptyset}} \inf_{x \in F} \left(\sum_{j \in S} |h_j^0(x)|^2 \right)^{1/2}$$

is positive, with the minimum over an empty collection omitted. For every nonempty incidence, compactness and conormal independence give a positive least singular value on a fixed neighborhood. There are only finitely many subsets and corner faces. Hence the straight-line family

$$h^t = (1-t)h^0 + th, \quad 0 \leq t \leq 1,$$

has no reference-empty face incidence, and every incidence which occurs is neat with a common conormal bound, once (5.8) is small enough.

Apply Lemma 5.1 to $H_j(x, t) = h_j^t(x)$. Its hypotheses hold by the preceding incidence and rank checks. The resulting ambient C^s flow preserves every boundary face and transports every labelled zero hypersurface, which gives (5.9) at $t = 1$. Moreover, $\partial_t H_j = h_j - h_j^0$. Estimate (5.5) and the variational equations for the flow therefore prove the quantitative estimate (5.10). The flow equality in (5.4) now also proves that no reference incidence disappeared.

An inactive function cannot change sign along the image of a sign stratum without crossing its zero hypersurface. Hence the trivialization preserves all labels and signs. Its restriction is a diffeomorphism and therefore gives a bijection of connected components. The threshold uses precisely the common corner atlas, coefficient bound, conormal bound, and empty-incidence gaps stated in the proposition. This also explains why an empty-incidence margin is needed for a uniform family: the pair $h_1(x) = x$, $h_2(x) = x + \varepsilon$ loses its empty double-incidence gap as $\varepsilon \downarrow 0$. $\square$

We also use the following elementary first-hit statement repeatedly.

Lemma 5.3 (Compact first-hit stability). *Let K be compact and let $g = 0$ be a C^3 event face. Suppose every $z \in K$ has a first event at $\tau(z) \in [\tau_-, \tau_+]$, with*

$$(5.11) \quad Xg(\Phi^{\tau(z)}z) \geq m > 0,$$

and that every other face inequality has a common strict margin along the entire compact pre-event tube $\{\Phi^t z : z \in K, 0 \leq t \leq \tau(z)\}$, except for faces specified to meet simultaneously in the assigned first-exit component. Then a neighborhood of K has a unique C^3 first-hit time, the same event occurs first, and all event-image inequalities retain positive margins. The conclusion is uniform on compact subneighborhoods.

Proof. The implicit-function theorem gives a unique event near each event point. Before a common short final interval, compactness makes the maximum of $g(\Phi^t z)$ strictly negative. On the final interval, (5.11) retains a

positive lower bound and gives exactly one crossing. A finite subcover gives uniform constants and the C^3 dependence. $\square$

For later reference, here is the precise finite cell decomposition just described. For $\bullet \in \{u, r\}$, an active set S and a sign vector $\epsilon \in \{\pm 1\}^{J_\bullet \setminus S}$ define

$$(5.12) \quad C_{S,\epsilon}^\bullet = \{z \in B^\bullet : h_j(z) = 0 \ (j \in S), \ \epsilon_j h_j(z) > 0 \ (j \notin S)\},$$

with connected components named separately and empty components discarded. Thus $S = \emptyset$ gives two-dimensional interiors, $|S| = 1$ gives faces, and $|S| = 2$ gives corners. Transversality and the sign alternatives give the finite identity

$$(5.13) \quad B^\bullet = \bigsqcup_{S,\epsilon} C_{S,\epsilon}^\bullet.$$

Condition (H2) supplies the three pairwise disjoint aperture closures, the transverse sign stratifications, and the uniform event-order margins. Apply Lemma 5.3 to their compact pre-exit tubes. Pulling the outgoing, lateral, and return-face cells fixed in advance back by the local passage, then cutting by (3.3), produces the first-exit components and their event types.

Proposition 5.4 (Complete local return–exit decomposition). *Assume (H1)–(H2). After increasing N and decreasing ν_N , every point of Σ_N has exactly one of the following first itineraries:*

- (a) a return through one of the edges $\mathcal{V}_{\sigma,n}^{\sigma'}$, with current sign σ , next sign σ' , and $(\sigma, \sigma') \in \mathcal{G}_N$;
- (b) a first hit of the open pole aperture;
- (c) a hit of the codimension-one algebraic sheet;
- (d) a finite hit of the stable-manifold cut $\mathcal{I}_{\sigma,n}$; or
- (e) a hit of an outgoing or return-box boundary cell fixed in advance.

Writing ℓ for the finite cell labels, the relative partition is

$$(5.14) \quad \begin{aligned} \Sigma_N = & \bigsqcup_{n \geq N, (\sigma, \sigma') \in \mathcal{G}_N} \mathcal{V}_{\sigma,n}^{\sigma'} \sqcup \bigsqcup_{\sigma, n \geq N} \mathcal{I}_{\sigma,n} \\ & \sqcup \bigsqcup_{\substack{\mathbf{t} \in \{\mathbf{p}, \mathbf{a}, \mathbf{out}, \mathbf{rbox}\} \\ \sigma, n \geq N, \ell}} E_{\mathbf{t}, \sigma, n, \ell}. \end{aligned}$$

In the notation of the abstract coding theorem with exits,

$$(5.15) \quad E_{\text{cut}, \sigma, n, 1} := \mathcal{I}_{\sigma, n}.$$

Here and below ℓ ranges only over nonempty connected cell components; empty sign strata are omitted. In particular, the two components selected in (H2) persist for every $n \geq N$, whereas no assertion is made for an unselected algebraic or pole cell. Every first-exit component is a relatively open stratum of dimension zero, one, or two in its assigned cell. Residual angle, fixed cell coordinates, the vertex-compatible sign charts on recurrent sides, and finite

affine cell charts elsewhere identify all components with fixed n -independent reference cell charts. In those charts the active defining-function ranks, event speeds, and strict auxiliary first-hit-order inequalities have uniform positive margins, and the labelled sign pattern is preserved.

Proof. The local passage has a unique outgoing hit. The sign-vector identity (5.13) on B^u assigns exactly one pure-dimensional stratum there. A point in the homoclinic aperture has a unique finite hit on its fixed matching section, after which the same identity on B^r is exhaustive and disjoint. In particular, the equation $\bar{\nu} = 0$ defines the ideal stratum $\mathcal{I}_{\sigma,n}$, while its two open sides are the regular target-sign components in (4.55). Hence the zero-action boundary is counted exactly once and no connected edge crosses from Σ_+ to Σ_- . If several active faces vanish, their common intersection is its own lower-dimensional stratum; priority assigns only its first-exit type. Thus no geometric boundary is counted in an adjacent two-dimensional component.

Equation (3.4) and the sole half-open convention $\theta \in J_0$ make every sufficiently small radial component carry a unique source sign and label $n \geq N$. The target sign is then uniquely $\text{sgn } \bar{\nu}$ on the regular set. This proves local finiteness away from the two cut faces. The finite face family has only finitely many active-set types. For the two source signs, let $Z_{\text{exit}} = I_s \times [0, 1]$ be the fixed compact completion of a whole normalized winding cell, with its second coordinate the affine rescaling of $\theta \in \bar{J}_0$. Let $\Pi_{\sigma,n} : Z_{\text{exit}} \rightarrow B^u$ be the residual-phase-normalized local exit map and let $\Pi_{\sigma,\infty}$ denote the restriction of the marked cut-face template (2.59) to this reference cell. The exact scalar clock inversion (4.49), the lifted action law (4.51), and Lemma 4.1 give

$$(5.16) \quad \|\Pi_{\sigma,n} - \Pi_{\sigma,\infty}\|_{C_{z,\varpi}^2} \leq C(1+n)^3 e^{-\kappa+n}.$$

Thus (5.16) holds on the whole reference winding cell, independently of the matching graph over $Z = (\phi, \bar{\nu})$. Composition with each fixed finite flight, face defining function, and event-ordering function has the same bound. Transversality and compactness of the two source-sign limiting families give one positive rank, speed, unrelated inactive-face, and pre-event ordering margin m_0 . Increase N until the right-hand side of (5.16), after those finite compositions, is less than $m_0/2$. Lemma 5.3 then supplies the same first event and at least $m_0/2$ margin for every $n \geq N$. This convergence is used to prove the uniform first-exit margins; it does not add all first-exit defining functions to the weighted output claim. The uniform neat transversality in (H2), including the reference-cell boundary conormals, puts the pulled-back face functions in the setting of Proposition 5.2. Equation (5.16) gives the required C^2 closeness. First apply the proposition with $s = 2$ on Z_{exit} to the outgoing and aperture faces. This identifies every homoclinic cell with its finite-winding counterpart. Transport the latter back to the limiting cell, then apply the proposition with $s = 1$ a second time, relative to its boundary, to the return-band faces after the prescribed finite homoclinic flight. The fixed-neighborhood clause in (H2) makes all these compositions

well defined, while (5.10) and the composed-map part of (5.16) give the required C^1 closeness of the transported return-face functions. Since there are only finitely many limiting cells, one threshold works for both source signs and all $n \geq N$. The two component bijections together map each labelled limiting sign stratum onto its finite-winding counterpart. The strict event-order and unrelated inactive-face inequalities persist by the C^0 part of (5.16) and the margins in (H2); Lemma 5.3 preserves the assigned first hit. Thus no unlabelled component can be born, and the component labels used below are fixed independently of n .

Each limiting algebraic or pole anchor selected in (H2) lies a positive distance from the boundary of its reference cell. The implicit-function theorem for the active faces at that anchor, together with persistence of its strict inactive-face and event-order inequalities, therefore gives a nearby point in the corresponding finite-winding stratum for every $n \geq N$. The connected component containing that point is the asserted nonempty component. No assertion about the flow after a lateral hit enters the argument. $\square$

5.2. The algebraic reference. On the algebraic future graph, start the field-dependent reference at

$$(5.17) \quad z_{a,*}(\mathbf{X}) = (e_*, 0, 0, \Gamma_{\mathbf{X}}(e_*, 0, 0)).$$

In desingularized time s the reduced equations and the physical clock have the form

$$(5.18) \quad \begin{aligned} \frac{de}{ds} &= e^2 \widehat{v}_e(e, b), & \frac{db}{ds} &= e \widehat{f}_e(e, b) + b \widehat{v}_b(e, b), \\ \frac{dt}{ds} &= \chi_a(e, b) = e^{1/2} \widehat{\chi}_a(e, b), & \widehat{v}_e &\leq -a_*, \quad \widehat{v}_b \leq -\lambda_*. \end{aligned}$$

Writing $\mathbf{q} = e^{-1}$ puts the reference and every first-exit orbit on one fixed half-line. The two Volterra contractions in Lemma D.2 compare them directly at the same value of e and give, for every compact C^r coefficient family,

$$(5.19) \quad \|D_{b_0}^i D_{\varpi}^j \{b_{\varpi}(e; b_0) - b_{\text{ref}, \varpi}(e)\}\| \leq C_{i,j,m} e^m, \quad i+j \leq 2, \quad j \leq r, \quad m \geq 1.$$

Define the reference counterterms by the actual reference flight:

$$(5.20) \quad C_{\tau, a, \mathbf{X}}(e) = \int_{e_*}^e \frac{\chi_{a, \mathbf{X}}(u, b_{\text{ref}, \mathbf{X}}(u))}{u^2 \widehat{v}_e(u, b_{\text{ref}, \mathbf{X}}(u))} du,$$

$$(5.21) \quad C_{B, a, \mathbf{X}}(e) = \int_{e_*}^e \frac{\chi_{a, \mathbf{X}}(u, b_{\text{ref}, \mathbf{X}}(u)) \lambda_{\mathbf{X}}(X_{\mathbf{X}})(\mathcal{A}_{\mathbf{X}}(u))}{u^2 \widehat{v}_e(u, b_{\text{ref}, \mathbf{X}}(u))} du.$$

For $e < e_*$, the time integral is positive because the integration direction and $\widehat{v}_e < 0$ reverse together, while $\chi_a > 0$; the action integral uses the same physical forward orbit orientation. In particular, both counterterms vanish at the fixed cut $e = e_*$. Equation (5.19) supplies integrable majorants after any fixed singular power is removed, so the reference-subtracted time and action have uniform mixed state-parameter two-jet limits by Proposition D.3. The fixed cut is indispensable: a future-tempered condition at

$e = 0$ alone leaves a flat freedom of order $\exp\{-\lambda/(ae)\}$ and does not select a unique reference.

For the core, the exact algebraic orbit is

$$(5.22) \quad \gamma_0(t) = \left(-\frac{t^2}{12}, -\frac{t}{6}, \frac{1}{6} - \frac{t^4}{144}, -\frac{t^3}{36} \right).$$

With $e = (-U)^{-1}$ its exact counterterms are

$$(5.23) \quad \begin{aligned} C_{\tau,a}(e) &= 2\sqrt{3}(e^{-1/2} - e_*^{-1/2}), \\ C_{B,a}(e) &= -\frac{8\sqrt{3}}{21}e^{-7/2} + \frac{2\sqrt{3}}{9}e^{-3/2} \\ &\quad + \frac{8\sqrt{3}}{21}e_*^{-7/2} - \frac{2\sqrt{3}}{9}e_*^{-3/2}. \end{aligned}$$

5.3. The pole counterterm. Let $s = t_b - t$ be physical remaining time. For a general family in Definition 2.2, Proposition D.9 gives the finite Laurent–log singular density

$$\sum_{q=-m_p}^{-1} s^q A_{q,\mathbf{x}}(\log s, p)$$

and defines $C_{B,p,\mathbf{x}}$ by (D.47). This is the pole counterterm used in the main theorems. The coefficients depend on the pole-entry label and on the field; no lower-order jet is frozen. The pole time itself is finite.

For the normal-form system, most of the allowed singular terms vanish. In its pole chart the compactified equations give

$$(5.24) \quad W = s^4 w = -6s^4 \log s + \Omega(p)s^4 + O_{C_p^2}(s^{4+\nu}).$$

A finite dichotomy bootstrap yields

$$(5.25) \quad X = s^2 x = 6 + s^4 \left(-\frac{3}{5} \log s + \frac{\Omega}{10} - \frac{9}{50} \right) - \frac{C_p}{6} s^5 + O_{C_p^2}(s^{5+\nu}),$$

$$(5.26) \quad Y = s^3 y = 12 + s^4 \left(\frac{6}{5} \log s - \frac{\Omega}{5} + \frac{24}{25} \right) + \frac{C_p}{2} s^5 + O_{C_p^2}(s^{5+\nu}),$$

$$(5.27) \quad Z = sq = 6 + C_p s + O_{C_p^2}(s^4 |\log s|).$$

The potentially nonintegrable $C_p s^{-1}$ terms cancel in $\lambda(X) = y^2 - q^2$. The pole action counterterm in this fixed remaining-time normalization is therefore

$$(5.28) \quad C_{B,p}(s, p) = \frac{144}{5} s^{-5} + \frac{1}{s} \left(\frac{144}{5} \log s + \frac{396}{25} - \frac{24}{5} \Omega(p) \right).$$

Subtracting (5.28) leaves a C^2 first-exit limit. Proposition D.10 proves this explicit formula for the normal-form pole and order-eight-flat perturbations; Proposition D.9 proves the corresponding mixed state–parameter derivatives for the full varying-jet class.

5.4. Exact branch primitives and first-exit weights. For any finite first-hit map $P(z) = \Phi^{\tau(z)}z$ on $H = 0$, set

$$(5.29) \quad B_P(z) = \int_0^{\tau(z)} \lambda(X)(\Phi^t z) dt.$$

Cartan's formula gives the ambient variable-time identity

$$(5.30) \quad P^* \lambda - \lambda = dB_P + \tau dH.$$

If $\Sigma_-, \Sigma_+ \subset H^{-1}(0)$ are the source and target sections, restriction to their tangent spaces gives

$$(5.31) \quad P^*(\lambda|_{\Sigma_+}) - \lambda|_{\Sigma_-} = d_{\Sigma_-} B_P.$$

Thus B_P is a primitive function for the exact symplectic branch P . This is the standard exact-map meaning of “primitive”; the additive action co-cycle below is induced by composing these branch primitives. If $P_2 P_1$ is composable, splitting the same integral gives

$$(5.32) \quad B_{P_2 P_1} = B_{P_1} + B_{P_2} \circ P_1.$$

Under $\lambda \mapsto \lambda + d\chi$, the action changes by $\chi \circ P - \chi$. Admissible gauges are those respecting the marked reverser and chart overlaps and having uniformly controlled C^2 endpoint restrictions. These identities fix the branch constants and pass to the renormalized algebraic and pole limits.

Proposition 5.5 (Covariance under section changes and exact gauges). *Let two marked presentations describe the same Hamiltonian flow in the same physical return–exit tube. Suppose their sections are related by orbit-following C^3 diffeomorphisms*

$$(5.33) \quad S_v(z) = \Phi^{\delta_v(z)} z$$

whose joining orbit segments meet no event face. Suppose the two presentations otherwise use the same physical event hypersurface germs, end compactifications, cutoff coordinates, and algebraic and pole reference normalizations. Let their primitives satisfy $\lambda_1 = \lambda_0 + d\chi$ on the tube, where χ has controlled C^2 restrictions and limits on the section and end faces. If the reversible primitive convention is retained, assume also $\chi \circ \mathcal{R} = -\chi + \text{constant}$. Finally, assume that the slides carry each labeled component bijectively; this holds, for example, when the same angular cut is used and the residual-phase images have a uniform cut-avoidance margin. Then the physical first-return/first-exit components correspond bijectively. If a is a return component in the first presentation and $\rho(a)$ its counterpart in the second, then

$$(5.34) \quad P_{\rho(a)}^1 \circ S_{s(a)} = S_{t(a)} \circ P_a^0,$$

$$(5.35) \quad \tau_{\rho(a)}^1 \circ S_{s(a)} = \tau_a^0 - \delta_{s(a)} + \delta_{t(a)} \circ P_a^0.$$

Put

$$(5.36) \quad K_v(z) = \int_0^{\delta_v(z)} \lambda_0(X)(\Phi^t z) dt, \quad C_v = K_v + \chi \circ S_v.$$

Then

$$(5.37) \quad B_{\rho(a)}^1 \circ S_{s(a)} = B_a^0 - C_{s(a)} + C_{t(a)} \circ P_a^0.$$

The analogous formulas hold for a finite path ending at a corresponding first-exit event. The two recurrent symbolic systems are topologically conjugate through their coding maps. Hence closed-orbit periods and closed actions are unchanged. Here action invariance is for the fixed primitive and for exact gauge changes; it is not asserted for primitives differing by a non-exact closed one-form.

Proof. Because the orbit slides cross no event, they carry connected domains with the same first physical event to one another. The flow group law gives (5.34) and (5.35). Splitting the λ_0 integral at the two slide segments and adding the endpoint contribution of $d\chi$ gives (5.37); equivalently it follows from (5.30) applied to each S_v .

Let h_0, h_1 be the two coding maps and let S denote the piecewise section slide on the physical trapped set. Then $h_1^{-1} \circ S \circ h_0$ is a topological conjugacy of the two shifts. Equations (5.35) and (5.37) become endpoint coboundaries on code space and telescope on every periodic word. This proves the last claim. $\square$

Definition 5.6 (Admissible changes of coordinates and compactifications). Let $\mathcal{P}$ be compact, let $r \in \{0, 1, 2\}$, and let $\mathbf{X}_{\varpi}^0$ and $\mathbf{X}_{\varpi}^1$ be two C^r families of marked presentations in $\mathfrak{H}_{\text{adm}}^k$ of the same physical reversible exact Hamiltonian flows, return–exit tubes, and cooriented event hypersurface germs. The change of presentation is *admissible* when the following conditions hold.

- (a) The compact source and target sections are joined by event-preserving orbit slides

$$(5.38) \quad S_v(z) = \Phi_{\varpi}^{\delta_v(z)} z.$$

The functions δ_v , the slides, and their inverses have uniformly bounded mixed derivatives of total order at most three. On the physical tube,

$$(5.39) \quad \lambda_{\varpi}^1 = \lambda_{\varpi}^0 + d\chi_{\varpi}.$$

The restrictions of χ_{ϖ} to the sections, and its extensions to the two compactified terminal faces, have genuine limits with uniformly bounded mixed derivatives of total order at most three. The main output topology retains only total order two; the spare derivative is used in the fixed-cut and fixed-remaining-time finite-part arguments.

- (b) The transition between the saddle-focus charts extends to the real oriented blow-up of the stable and unstable axes as an exact symplectic C^3 diffeomorphism, with the same mixed bounds. It preserves the two axis faces, up to the permutation induced by a change of coorientation. In positive normal coordinates on a signed section it

has the form

$$(5.40) \quad \begin{aligned} \tilde{x} &= h_{\varpi, \sigma}(x) + y h_{\varpi, \sigma}^{(1)}(x, y), \\ \tilde{y} &= a_{\varpi, \sigma}(x)y + O_{C^3}(y^2), \quad 0 < a_* \leq a_{\varpi, \sigma} \leq a^*. \end{aligned}$$

On the angular boundary its lift has degree one,

$$(5.41) \quad \tilde{\theta} = \theta + d_{\varpi}(\theta, \zeta), \quad d_{\varpi}(\theta + 2\pi, \zeta) = d_{\varpi}(\theta, \zeta),$$

with d_{ϖ} uniformly bounded in C^3 .

On the compact parameter domains used for the residual-phase and first-event maps, the two angular cuts and the pulled-back event faces form a finite cleanly intersecting family, including at domain boundaries and over $\mathcal{P}$. The family admits parameter-uniform transverse collars and C^3 local trivializations over a finite stratification of $\mathcal{P}$; mixed derivatives on every total-space stratum extend uniformly to its closure. Thus cuts may cross return or first-exit families, but infinite-order contact with a limiting face is excluded.

- (c) The algebraic compactifications are related by a C^3 b -diffeomorphism preserving the energy face, limiting orbit, and end coorientation. Their boundary defining functions satisfy

$$(5.42) \quad \tilde{e} = u_{\varpi} e, \quad 0 < u_* \leq u_{\varpi} \leq u^*,$$

with uniform mixed bounds for the map and its inverse. The two reference cuts, and hence the two reference orbits, may differ.

- (d) The pole compactifications are related by a C^{m_p+3} b -diffeomorphism whose coordinate functions and inverse have uniformly bounded derivatives through order $m_p + 3$, with the corresponding mixed parameter bounds. It carries the compactified sink and its stable stratification to their counterparts and satisfies

$$(5.43) \quad \tilde{\rho} = v_{\varpi} \rho, \quad 0 < v_* \leq v_{\varpi} \leq v^*.$$

It transports the principal radial scaling and the normalized indicial operator (2.26), and both charts use the same compactified-action order m_p ; no equality of the finite boundary jets is required. Both presentations use physical remaining time $s = t_b - t$ and the normalization (D.46) to extract the singular pole action.

These conditions cover the standard choices used below—reversible symplectic eigenframes, exact normal-form charts extending to the real blow-up, smooth finite cut systems, algebraic boundary defining functions, and pole b -compactifications—provided their transitions satisfy the displayed bounds, clean-intersection condition, and fixed indicial type.

The proof has four steps. A bounded change of angular lift gives a bounded winding recoding; the two finite cut arrangements are then intersected on a common refinement. Section slides and exact gauges give

endpoint coboundaries, while aligned physical cutoffs give the end corrections. Finally, composition on the refined strata and comparability of the winding weights yield locally bi-Lipschitz equivalent weighted metrics.

Theorem 5.7 (Invariance under admissible changes of coordinates and compactifications). *Let the hypotheses of Definition 5.6 hold, and suppose that the zeroth presentation has compact image in a neighborhood in $\mathfrak{H}_{\text{adm}}^k$ to which Theorems 3.1–3.3 apply. Then its physical return/first-event relation and corrected roof/action data transport to the first presentation, and the weighted output topology is unchanged in the precise sense below. Full Markov coding transports only under the additional cut-avoidance condition in part (i); in general the symbolic comparison is the realized-itinerary factor statement there. After increasing the winding threshold if necessary, the following statements hold.*

- (i) *There are constants $K, Q < \infty$ and a common refinement $\mathcal{A}^\sharp$ of the two return/first-exit decompositions such that each old component is the union of at most Q refined components. If $c \in \mathcal{A}^\sharp$ projects to labels $a_j = \pi_j(c)$ and winding numbers $n_j(c)$, then*

$$(5.44) \quad |n_1(c) - n_0(c)| \leq K.$$

Let $\Sigma^\sharp$ be the set of bi-infinite itineraries of refined return components that are realized by trapped physical points. Label projection induces maps $\Pi_j : \Sigma^\sharp \rightarrow \Sigma_j$ that are one-block on this realized itinerary space and commute with the shift and the physical coding maps. No assertion is made that $\Sigma^\sharp$ is itself a full Markov edge shift. In particular, the two presentations determine the same trapped physical Poincaré system and exhaustive physical first-event relation. If neither cut crosses a residual-phase image, this refinement is the branchwise bijection in Proposition 5.5; in that case the full Markov coding and its summable-variation potentials transport by the induced conjugacy.

- (ii) *Put*

$$(5.45) \quad k_c = n_1(c) - n_0(c), \quad c_\tau = \frac{2\pi}{\beta_\varpi},$$

$$K_v(z) = \int_0^{\delta_v(z)} \lambda_\varpi^0(X_\varpi)(\Phi_\varpi^t z) dt, \quad C_v = K_v + \chi_\varpi \circ S_v.$$

On every refined return component,

$$(5.46) \quad P_{\pi_1(c)}^1 \circ S_{s(c)} = S_{t(c)} \circ P_{\pi_0(c)}^0,$$

$$(5.47) \quad \widehat{\tau}_{\pi_1(c)}^1 \circ S_{s(c)} = \widehat{\tau}_{\pi_0(c)}^0 - \delta_{s(c)} + \delta_{t(c)} \circ P_{\pi_0(c)}^0 - c_\tau k_c,$$

$$(5.48) \quad B_{\pi_1(c)}^1 \circ S_{s(c)} = B_{\pi_0(c)}^0 - C_{s(c)} + C_{t(c)} \circ P_{\pi_0(c)}^0.$$

Thus an angular-cut change contributes only the displayed deck term, and section and primitive changes contribute endpoint coboundaries. Closed periods and closed actions are unchanged.

- (iii) *At the algebraic end, align the two cutoff hypersurfaces along the same physical orbit and write $r_j(\epsilon, p)$ for their boundary defining-function values at the common cutoff $z_\epsilon(p)$. The limits*

$$(5.49) \quad \Gamma_{a,\tau}(p) = \lim_{\epsilon \downarrow 0} \{C_{a,\tau}^0(r_0(\epsilon, p)) - C_{a,\tau}^1(r_1(\epsilon, p))\},$$

$$(5.50) \quad \Gamma_{a,B}(p) = \lim_{\epsilon \downarrow 0} \{C_{a,B}^0(r_0(\epsilon, p)) - C_{a,B}^1(r_1(\epsilon, p)) + \chi_\varpi(z_\epsilon(p))\}$$

exist with all mixed state-parameter derivatives of total order at most two.

At the pole, use instead the common physical remaining-time cutoff $s = \epsilon$. If p_j denotes the label, in presentation j , of the same physical pole orbit and $z_\epsilon(p)$ is its point with $t_b - t = \epsilon$, set

$$(5.51) \quad \Gamma_{p,\tau}(p) = 0,$$

$$(5.52) \quad \Gamma_{p,B}(p) = \lim_{\epsilon \downarrow 0} \{C_{p,B}^0(\epsilon, p_0) - C_{p,B}^1(\epsilon, p_1) + \chi_\varpi(z_\epsilon(p))\}.$$

These limits have the same mixed- C^2 regularity; the gauge term has the genuine compactified limit required in Definition 5.6. For either $e \in \{a, p\}$, if E_c is the compact end-label map on a refined first-exit component, then

$$(5.53) \quad \widehat{\tau}_{e,c}^{\text{ren},1} \circ S_{s(c)} = \widehat{\tau}_{e,c}^{\text{ren},0} - \delta_{s(c)} - c_\tau k_c + \Gamma_{e,\tau} \circ E_c,$$

$$(5.54) \quad B_{e,c}^{\text{ren},1} \circ S_{s(c)} = B_{e,c}^{\text{ren},0} - C_{s(c)} + \Gamma_{e,B} \circ E_c.$$

If the primitive is fixed and the gauge representative is chosen as $\chi = 0$, the two charts have the same physical remaining-time singular part; the common zero finite-part normalization then gives $\Gamma_{p,B} = 0$.

- (iv) *Pull both labeled data sets to $\mathcal{A}^\sharp$, use the coordinate transitions of Definition 5.6, and make the corrections in (5.47)–(5.54). Let $\mathcal{T}_{10}$ denote this transition from presentation-0 data to presentation-1 data, and let $d_{\kappa,2}^{j,\sharp}$ be the metric obtained from (B.1)–(B.2) in the refined presentation- j chart. On every compact overlap there is $A \geq 1$ such that, for any two data sets $\mathcal{D}, \mathcal{E}$ in that overlap,*

$$(5.55) \quad A^{-1} d_{\kappa,2}^{0,\sharp}(\mathcal{D}, \mathcal{E}) \leq d_{\kappa,2}^{1,\sharp}(\mathcal{T}_{10}\mathcal{D}, \mathcal{T}_{10}\mathcal{E}) \leq A d_{\kappa,2}^{0,\sharp}(\mathcal{D}, \mathcal{E}).$$

The same local bi-Lipschitz estimate holds for the enhanced family metrics containing $D_z^i D_\varpi^j$, $i+j \leq 2$, on the total-space strata of Definition 5.6. Consequently the overlap maps define the same weighted output topology. The raw winding labels, limit representatives, and counterterms may still depend on the chosen presentation.

Proof. Two reversible symplectic eigenframes in which the stable linearization is $\alpha I + \beta \mathbb{J}$ differ by a matrix commuting with $\mathbb{J}$. It is a complex similarity $aI + b\mathbb{J}$, while reversibility and symplecticity determine the unstable frame. Hence every such eigenframe change induces a bounded radial rescaling and

bounded phase change on the real blow-up. Hadamard's lemma gives (5.40); the same argument applies to two exact smooth normal-form charts.

For the angular lifts, (5.41) compares the two degree-one deck decompositions of the same lifted physical phase. Thus $2\pi(n_1 - n_0)$ is a bounded periodic phase difference, up to the fixed endpoints of the two fundamental intervals. This proves (5.44). The transverse finite arrangement on the compact parameter domains has finitely many connected strata and uniform collars. Intersecting the two decompositions along these strata gives $\mathcal{A}^\sharp$ and the uniform multiplicity Q . For a trapped physical point, recording its intersection cell at every return defines an element of $\Sigma^\sharp$; projection at each coordinate gives its two old itineraries. This proves the stated one-block factor property on the actual itinerary set without asserting a new Markov crossing property for the refined pieces. Since the refinement only cuts domains belonging to the same physical orbit relation, both projected codes have the same physical image.

The flow group law gives (5.46) and the roof identity before renormalization. Substitution of $n_1 = n_0 + k_c$ gives the deck term in (5.47). Splitting the same physical orbit integral at the slide segments and using (5.39) gives (5.48). The endpoint terms telescope on a closed orbit.

At the algebraic end, choose one finite physical cross-section preceding both reference cuts and start both renormalizations there. Write each counterterm as the integral of the same physical time or action density along its reference orbit; the finite initial segments contribute smooth endpoint constants. On the common future tail, express both reference orbits in either b -chart. The flat shadowing estimate (5.19), with the second reference used as input to the first chart, makes the difference of the two physical densities absolutely integrable, together with its mixed two-jets. The relation $\tilde{e} = ue$ is now used only to change variables in these same physical integrals, not to infer cancellation of their singular parts. Adding the genuine limit of χ_ϖ proves (5.49)–(5.50) at the algebraic end.

At the pole, $s = t_b - t$ is physical. First keep the primitive fixed. Proposition D.9 shows directly that the finite Laurent–log singular density is determined by the physical action density and therefore agrees in the two charts after the end-label change. It also supplies a common integrable majorant after either counterterm is subtracted. Their difference therefore has the mixed- C^2 limit in (5.52); with the common zero finite-part normalization, the counterterm difference is zero.

For $\lambda^1 = \lambda^0 + d\chi_\varpi$, the physical tail identity (D.55) has a mixed- C^2 endpoint limit. Since both densities have finite Laurent–log expansions with integrable remainders, convergence of that identity forces the difference of their nonintegrable singular sums to vanish. Thus the same normalized pole counterterm is used after the gauge change, and the limit of $\chi_\varpi(z_\epsilon(p))$ is precisely the additional term in (5.52). Aligning the finite physical flights now proves (5.53)–(5.54).

Finally, C^3 composition and the implicit-function theorem control the transformed cross forms in C^2 . On a compact overlap, finite refinement and its inverse change coefficient differences by at most uniform constants. Moreover,

$$(5.56) \quad \frac{\epsilon_{\pi_1(c)}^1}{\epsilon_{\pi_0(c)}^0} = \frac{\nu_N^0}{\nu_N^1} \exp\left(-\frac{2\pi\alpha_\varpi}{\beta_\varpi} k_c\right),$$

which is bounded above and below by (5.44). The polynomial and exponential weights at n_0 and n_1 are comparable. The same chain-rule argument applies to mixed derivatives on each total-space stratum because the integer k_c is locally constant there; the uniform extensions to stratum closures give one constant on the finite arrangement. Applying the argument to two data sets proves the forward metric inequality in (5.55). Reversing the admissible change proves the opposite inequality. $\square$

Remark 5.8 (Why admissibility is needed). Theorem 5.7 is invariant under the class in Definition 5.6; no such statement can hold for all coordinate changes defined only on the punctured physical domain. For example, the exact symplectic shear

$$\tilde{\nu} = \nu, \quad \tilde{\phi} = \phi + e^{-1/\nu^2} \sin(1/\nu)$$

extends smoothly to the zero-action boundary, but its cut and the cut $\phi = \phi_0$ have infinitely many transverse intersections accumulating at an infinite-order boundary tangency. No uniformly finite common refinement exists. Likewise, treating the pole boundary jet as a list of scalar coefficients in one defining function makes even $\tilde{\rho} = (1+a)\rho$ appear to change its leading coefficient. Clean intersection of the cuts and the intrinsic boundary-jet interpretation are therefore necessary geometric restrictions, not proof artifacts.

Proof of Corollary 3.4. For each i , compactness of the image over $\bar{V}_i$ gives finitely many coefficient neighborhoods on which Theorems 3.1–3.3 have uniform constants. Take the maximum of the resulting winding and first-event thresholds over the finite atlas, increasing it by the finite overlap recoding bound in (5.44). The marked theorems then give an exhaustive local first-event partition in every chart.

On $V_i \cap V_j$, Theorem 5.7 supplies a finite common refinement and identifies the two partitions as the same physical first-event relation. Its section-transition formula conjugates the trapped Poincaré maps, while its endpoint-coboundary and end-transfer formulas identify the corrected weighted data. The endpoint terms cancel around a closed orbit, so its physical period and action for the fixed primitive are independent of the marking. Finally, each local full shift codes its local marked representatives; on a crossing overlap the common refinement records the same realized physical itinerary and gives the factor asserted in part (iii). This proves all three clauses without selecting a global lift. $\square$

Remark 5.9 (Change of angular lift). If a second angular lift is related on one connected branch by $\tilde{\theta} = \theta + \delta - 2\pi k_a$ and $\tilde{n} = n + k_a$, then

$$(5.57) \quad \widehat{\tau}_{\rho(a)} \circ S_{s(a)} = \widehat{\tau}_a - \delta_{s(a)} + \delta_{t(a)} \circ P_a^0 - \frac{2\pi}{\beta} k_a.$$

The integer k_a is constant on that branch provided the new cut does not cross its residual-phase image. If it does cross, use the common finite refinement of Theorem 5.7; on each refined piece the same formula holds and the resulting weighted tail norms are equivalent.

Combining Lemma 4.4, Proposition 4.5, the compact first-hit maps, and the two end estimates gives, for any $\kappa < \kappa_+$,

$$(5.58) \quad \widehat{\tau}_a = \widehat{\tau}_{s(a),t(a),\infty} + O_{C^2}((1+n(a))e^{-\kappa n(a)}),$$

$$(5.59) \quad B_a = B_{s(a),t(a),\infty} + O_{C^2}((1+n(a))e^{-\kappa n(a)}),$$

$$(5.60) \quad \widehat{\tau}_{t,\sigma,n,\ell}^{\text{ren}} = \widehat{\tau}_{t,\sigma,\infty,\ell}^{\text{ren}} + O_{C^2}((1+n)^3 e^{-\kappa n}),$$

$$(5.61) \quad B_{t,\sigma,n,\ell}^{\text{ren}} = B_{t,\sigma,\infty,\ell}^{\text{ren}} + O_{C^2}((1+n)^3 e^{-\kappa n}).$$

The first-exit normalization subtracts both the local dwell $(2\pi/\beta)n$ and the appropriate end counterterm.

6. COUNTABLE RETURN-EXIT CODING

We isolate the abstract step that turns the preceding return-exit data into dynamics. Let $I = [-1, 1]$, $J = [0, 1]$, let $\mathcal{G} = (\mathcal{V}, \mathcal{A}, s, t)$ be a directed graph with finite vertex set and countable edge set, and let $\Sigma = \bigsqcup_{v \in \mathcal{V}} \Sigma_v$ be the regular part of finitely many section rectangles. The closure of every vertex chart is $I \times J$, and the regular normal coordinate lies in $(0, 1)$; the face $y = 0$ is the removed zero-action boundary. For each edge there are closures of return strips $\overline{V}_a \subset \overline{\Sigma}_{s(a)}$ and $\overline{W}_a \subset \overline{\Sigma}_{t(a)}$. Their regular interiors, after the Markov and ideal faces fixed in advance are removed, are V_a and W_a . Each return branch $P_a : V_a \rightarrow W_a$ extends to the closed boxes and has cross form

$$(6.1) \quad x' = f_a(x, y'), \quad y = g_a(x, y'), \quad (x, y') \in I \times J.$$

We use the following four standard hypotheses.

- (i) *Markov strips.* For every edge a , V_a is the regular part of a completed vertical strip in $\Sigma_{s(a)}$, $W_a = P_a(V_a)$ is a horizontal strip in $\Sigma_{t(a)}$, and $P_a : V_a \rightarrow W_a$ is a C^2 diffeomorphism whose cross form on the closure is defined on all of $I \times J$. The nonempty domains V_a are pairwise disjoint, and the labeling is faithful: each connected return-domain component occurs as exactly one edge. The family $\{V_a\}$ is locally finite away from the two removed cut faces, and every accumulation at a cut face belongs to one of the two fixed-sign high-winding families. Vertex matching is the exact Markov incidence

rule: every pair with $t(a) = s(b)$ is represented by the full horizontal–vertical crossing, and no pair with unequal vertices is composable. A closed strip before cutting may cross a removed ideal face only before the edge set is defined: its regular sides are separate edges and the intervening face belongs to exactly one first-exit component.

- (ii) *Uniform cross-form estimates.* Every edge carries the branch-dependent exponential scale $\epsilon_a > 0$ and the scaled normal component $\tilde{g}_a = g_a/\epsilon_a$. There are $0 < \vartheta < 1$, $0 < m_{\text{cf}} < 1$, and $C < \infty$ such that

$$(6.2) \quad \max\{\|\partial_x f_a\| + \|\partial_{y'} f_a\|, \|\partial_x g_a\| + \|\partial_{y'} g_a\|\} \leq \vartheta,$$

the C^2 norms of f_a and $\tilde{g}_a$ are uniform, and

$$(6.3) \quad \begin{aligned} \text{dist}(f_a(I \times J), \partial I) &\geq m_{\text{cf}}, & m_{\text{cf}} &\leq \tilde{g}_a \leq m_{\text{cf}}^{-1}, \\ 0 < g_a &\leq 1 - m_{\text{cf}}, & \|Dg_a\|_{C^1} &\leq C\epsilon_a. \end{aligned}$$

Thus the angular and upper-normal faces have absolute margins, while the lower normal face has the normalized margin $g_a \geq m_{\text{cf}}\epsilon_a$; no uniform absolute distance from the zero-action boundary is required.

- (iii) *First-exit partition.* The complement of the regular return strips is a typed, locally finite, strict first-exit partition

$$(6.4) \quad \Sigma \setminus \bigcup_{a \in \mathcal{A}} V_a = \bigsqcup_{t, \ell} E_{t, \ell}.$$

Every removed Markov or cut-face piece of the closure of a return strip belongs to exactly one $E_{t, \ell}$; it is not restored to a neighboring return edge. All component charts, defining-function ranks, labelled sign patterns, event speeds, and renormalized first-exit data have uniform bounds. In a fixed-sign high-winding family the first-exit limits obey (5.61) with the fixed weight $p(n) = (1 + n)^3$.

- (iv) *Exact roof and action data.* Every branch has the exact flight data

$$(6.5) \quad P_a^* \lambda_\Sigma - \lambda_\Sigma = dB_a, \quad \tau_a = c_\tau n(a) + \hat{\tau}_a,$$

with uniformly bounded C^1 remainders—indeed C^2 in $\mathfrak{X}_\kappa^2$ —coherent action composition, and the weighted limits for each fixed source and target sign pair (5.59).

Give the countable edge alphabet the discrete topology and define

$$(6.6) \quad \Sigma_{\mathcal{G}} = \{(a_j)_{j \in \mathbb{Z}} \in \mathcal{A}^{\mathbb{Z}} : t(a_j) = s(a_{j+1}) \text{ for all } j\}.$$

This is the two-sided countable-state topological Markov shift associated with $\mathcal{G}$. Its topology is generated by cylinder sets obtained by fixing finitely many consecutive edges; equivalently, it is the subspace topology inherited from the product-discrete space $\mathcal{A}^{\mathbb{Z}}$.

Theorem 6.1 (Coding for a countable Markov family with first-exit sets). *Under the four hypotheses above:*

- (i) every word in $\Sigma_{\mathcal{G}}$ determines exactly one regular trapped point, and the resulting map $h : \Sigma_{\mathcal{G}} \rightarrow K$ is a homeomorphism satisfying $P \circ h = h \circ \sigma$;
- (ii) a one-sided future word determines a stable plaque with one free section coordinate;
- (iii) every point outside the forward trapped set has a unique finite return word followed by one first-exit component, and finite pullback preserves the dimension of that component;
- (iv) if two bi-infinite words agree on $[-m, m]$, their realized points differ by at most $C\vartheta^m$; consequently the renormalized roof and action have summable two-sided variations;
- (v) finite return words followed by a first exit carry the additive renormalized time and action, and admissible section or primitive changes produce only telescoping endpoint coboundaries.

All conclusions persist in the fixed weighted space of labeled return and first-exit maps as long as the contraction, angular, upper-normal, rescaled transverse-action, rank, and event margins remain strict and the labelled sign pattern is fixed.

Proof. For a bi-infinite word, its cross coordinates solve

$$(6.7) \quad x_{j+1} = f_{a_j}(x_j, y_{j+1}), \quad y_j = g_{a_j}(x_j, y_{j+1}).$$

On $\ell^\infty(\mathbb{Z}; I \times J)$ define

$$(6.8) \quad (\mathcal{T}x)_j = f_{a_{j-1}}(x_{j-1}, y_j), \quad (\mathcal{T}y)_j = g_{a_j}(x_j, y_{j+1}).$$

Conditions (6.2)–(6.3) make $\mathcal{T}$ a contraction of the closed product box with constant at most ϑ . At its fixed point the angular coordinate and the upper normal coordinate avoid their boundary faces uniformly. For every fixed index,

$$(6.9) \quad y_j = g_{a_j}(x_j, y_{j+1}) \geq m_{\text{cf}} \epsilon_{a_j} > 0,$$

so the zero-action boundary is excluded coordinatewise even though these distances may tend to zero along a sequence of winding labels. Hence the fixed point is in the regular strips. Shifting the word shifts this orbit. If two words agree on $[-m, m]$, m contraction iterates at index zero coincide, while both fixed-point tails are bounded by $C\vartheta^m$. This proves uniqueness, continuity and the variation estimate. The Markov-strip hypothesis, in particular disjointness and faithful edge labels, makes the itinerary of a regular point unique. Local finiteness away from the cut faces and the signed-tail charts at those faces make the itinerary map continuous. Thus the coding and its inverse are continuous.

On a half-line the same contraction leaves x_0 free and solves uniquely for the remaining coordinates. Varying x_0 gives the stable plaque and shows why a future word is not a point code. For a point outside the forward trapped set, take the first iterate outside every return strip; the strict partition (6.4) gives exactly one first-exit component. Finally, time additivity

and repeated use of (6.5) give the finite word formulas, while endpoint terms telescope. Strict inequalities and the uniform contraction inverse prove the persistence statement. $\square$

To compactify the two fixed-sign high-winding families, define

$$(6.10) \quad \widehat{B}_N = \{\pm n^{-1} : n \geq N\} \cup \{0^+, 0^-\},$$

with every regular symbol isolated and neighborhood bases

$$(6.11) \quad \widehat{U}_k^+ = \{0^+\} \cup \{n^{-1} : n > k\}, \quad \widehat{U}_k^- = \{0^-\} \cup \{-n^{-1} : n > k\}.$$

This alphabet is compact, metrizable, and Hausdorff. The closures of the uncut return strips extend in C^2 to the corresponding face of the cut-open section, so the alphabet times a compact free-coordinate box is homeomorphic to its geometric image. The actual regular edge alphabet retains the target sign:

$$(6.12) \quad \widehat{\mathcal{A}}_N = \{(b, \sigma') \in \widehat{B}_N \times \{+, -\} : (\text{sgn } b, \sigma') \in \mathcal{G}_N\},$$

where $\text{sgn}(0^\sigma) = \sigma$. It is a finite discrete extension of the two fixed-source-sign high-winding families. Its regular members are the triples (σ, n, σ') used in the graph shift; its boundary members only record the cut-open limiting correspondences and are not regular return edges. We do not compactify the first-exit symbols; their labels remain discrete.

7. ASSEMBLY OF THE MAIN THEOREMS AND AN OPTIONAL CONSEQUENCE

The three theorem proofs below use only the main dependency chain in Table 2. The final proof, for Corollary 3.9, is a logically independent selected-branch module and may be omitted on a first reading.

Proof of Theorem 3.1. Fix $\kappa \in (0, 2\pi\alpha_*/\beta_*)$. The proof chooses the field neighborhood and the winding threshold in the following order.

Step 1: continue finite geometry. Conditions (H1)–(H4) give a transverse homoclinic transition, a clean finite first-event arrangement, a forward-normally repelling algebraic future graph, and a strict pole entry into a hyperbolic sink basin. Proposition 2.9, the ordinary parameter-dependent implicit-function theorem, and Lemma 5.3 continue the homoclinic map, the algebraic entrance hypersurface, the pole map, and every finite cell face from the input coefficient data. The algebraic hypersurface persists because its local-maximality neighborhood, normal expansion, and order-three bunching margins are strict. When verified by an interval block, those margins follow from the strict face, cone, growth, and finite-time rate inequalities. The pole sink persists because its spectral inequalities are strict. The fixed-half-line comparison in Lemma D.2, recorded in (5.19), gives the same-e state- C^2 superalgebraic estimate in part (i).

Step 2: choose the two exponential rates. Shrink a relative coefficient ball and choose $\underline{\alpha} < \inf_{\mathbf{X} \in \mathcal{U}_\kappa} \alpha_{\mathbf{X}}$ until, with $\beta^* = \sup_{\mathbf{X} \in \mathcal{U}_\kappa} \beta_{\mathbf{X}}$,

$$(7.1) \quad \kappa < \kappa_+ < \frac{2\pi\underline{\alpha}}{\beta^*} \leq \inf_{\mathbf{X} \in \mathcal{U}_\kappa} \frac{2\pi\alpha_{\mathbf{X}}}{\beta_{\mathbf{X}}}.$$

This is possible because both rates in (7.1) approach $2\pi\alpha_*/\beta_*$ as the ball shrinks to $\mathbf{X}_*$. The parameter-uniform variation-of-constants estimate and the finite-dimensional matching equations then give (4.43) at rate κ_+ . Proposition 4.5 solves the matching equation on the closed strips for both source signs and all $n \geq N$ after one common increase of N . Cutting each closed strip along $\bar{\nu} = 0$ produces the regular edges (σ, n, σ') and assigns the intervening zero-action curve to the first-exit partition. Equations (4.62) and (4.66), with polynomial factors absorbed by the rate gap, give the contraction, angular, upper-normal, and rescaled transverse-action margins in (3.7).

Step 3: derive, rather than assume, the partition. Choose ν_N so that every point in (3.1) has winding at least N . Apply the actual outgoing and return first-hit maps to the fixed apertures and cell faces. Proposition 5.4 gives the exhaustive partition (5.14) and fixed component reference charts. Thus first-exit labels are outputs of first-hit equations; they have not entered the field norm.

This proves the continued geometry, the complete first-return/first-exit partition, and the uniform cross-form estimates asserted in Theorem 3.1. $\square$

Proof of Theorem 3.2. Removal of the end divergences. At the algebraic end, the fixed-cut Volterra equations and Proposition D.3 give uniform mixed-two-jet Cauchy limits. At the pole, the compact first-hit map and the event-speed margin in (H2) provide the mixed- C^3 entry maps (D.26). Hence Proposition D.9 leaves an integrable mixed-two-jet remainder after the field-dependent counterterm (D.47) is subtracted. Lemma 4.4 and smooth finite-flight composition now give the recurrent estimates (5.58)–(5.59). They also give the first-exit estimates (5.60)–(5.61). The gap $\kappa_+ - \kappa$ absorbs the polynomial two-jet factors. Equations (5.31)–(5.32) prove exactness and the coboundary laws.

These are precisely the two conclusions of Theorem 3.2. $\square$

Proof of Theorem 3.3. Coding and persistence. With vertices $\Sigma_\pm$, edge incidence $t(a) = s(b)$, and the zero-action boundary removed from every regular edge, the normalized output satisfies the four hypotheses of Theorem 6.1; hence Theorem 6.1 gives the two-sided coding, stable plaques, first-exit components, and summable variations. Fixed-point equations, finite first-hit maps, the normally expanding algebraic hypersurface, and end Cauchy limits all depend continuously on the coefficients. Their outputs therefore define a continuous map into one fixed weighted space of labeled return and first-exit data. The first step in the proof of Theorem 3.1 decreases $\mathcal{U}_\kappa$ so that the strict margins in (H1)–(H4) are uniform. Proposition B.3, stated for

an arbitrary base presentation satisfying those hypotheses, therefore applies the Volterra, clock-inversion, finite matching, first-hit, normally hyperbolic continuation, and end equations. It gives the stated compact-family mixed two-jets for the return, compact first-hit, and recurrent time/action data with the same tail weight. Because the marked chart is an input and the Banach-space family hypothesis controls parameter derivatives in the same state C^k norm, no state derivative is consumed. The theorem makes no additional regularity assumption at the algebraic end. The fixed-half-line arguments in Appendix D give the same mixed estimates for both noncompact end limits; at the pole they use the explicit family condition (2.30) and the finite-part construction of Proposition D.9. Section 8 is used only to verify the hypotheses for (1.1); no interval-validation or Krawczyk datum enters this abstract parameter-dependence argument. $\square$

Proof of Corollary 3.9. By the additional transverse-connection hypothesis and the parameter-dependent implicit-function theorem for transverse intersections, after decreasing $\mathcal{U}_\kappa$ the intersection $W^u(O_{\mathbf{X}}) \cap \mathcal{W}_{a,\mathbf{X}}$ contains one uniquely continued orbit $\Gamma_{a,\mathbf{X}}$ near $\Gamma_{a,*}$, with one uniform transversality margin. Proposition 4.3, applied to $\mathcal{W}_{a,\mathbf{X}}$ and this selected orbit, gives (3.39). The stable and unstable planes are Lagrangian. In the coordinates exchanged by the reverser, the linear symplectic off-diagonal matrix K is symmetric and satisfies

$$(7.2) \quad A_{\mathbf{X}}^T K = K A_{\mathbf{X}}.$$

The Hamiltonian Hessian at the saddle is nondegenerate. Since $H - H(0)$ vanishes on both invariant Lagrangian axes, its two diagonal Hessian blocks vanish; hence the remaining block K is invertible, in particular nonzero. Since $\beta_{\mathbf{X}} \neq 0$, this forces $\mathbb{J}K + K\mathbb{J} = 0$. Hence the quadratic Hamiltonian on $\text{Fix } \mathcal{R}$ is a nonzero traceless form. More explicitly,

$$(7.3) \quad K = \begin{pmatrix} a & b \\ b & -a \end{pmatrix}, \quad \sqrt{a^2 + b^2} \geq A_* > 0$$

after shrinking the coefficient neighborhood once. Put $x = \log r$ and $\hat{\gamma}_{\mathbf{X}}^a(x) = \gamma_{\mathbf{X}}^a(e^x)$. Then

$$(7.4) \quad H_{\mathbf{X}}(\hat{\gamma}_{\mathbf{X}}^a(x)) = A_{\mathbf{X}}^H e^{2x} \sin\left(2 \frac{\beta_{\mathbf{X}}}{\alpha_{\mathbf{X}}} x + \delta_{\mathbf{X}}^*\right) + O_{C_x^2}(e^{(2+\eta)x}).$$

Put

$$F_{\mathbf{X}}(x) = e^{-2x} \partial_x H_{\mathbf{X}}(\hat{\gamma}_{\mathbf{X}}^a(x)).$$

By (7.3), its leading periodic term is a nonzero linear combination of $\sin(2\beta_{\mathbf{X}}x/\alpha_{\mathbf{X}})$ and $\cos(2\beta_{\mathbf{X}}x/\alpha_{\mathbf{X}})$, with amplitude bounded below uniformly, while

$$F_{\mathbf{X}} - F_{\mathbf{X}}^{(0)} = O_{C_x^1}(e^{\eta x}).$$

Choose disjoint fixed neighborhoods of the simple zeros of $F_{\mathbf{X}}^{(0)}$ in one period. On their complement $|F_{\mathbf{X}}^{(0)}|$ has a uniform positive lower bound; for sufficiently negative x there is therefore no zero there, while the uniform implicit-function theorem gives exactly one zero in each chosen neighborhood. These are all sufficiently small critical points. They alternate between maxima and minima and have spacing $\pi\alpha_{\mathbf{X}}/(2\beta_{\mathbf{X}})$ in $-x$, which proves (3.40) and excludes additional critical points between them. $\square$

8. VERIFICATION FOR THE NORMAL-FORM SYSTEM

Fix throughout this section an integer $k \geq 5$.

8.1. Exact Hamiltonian structure. For (1.1), put

$$(8.1) \quad \lambda = P dU - Q dV, \quad \omega = d\lambda = -dU \wedge dP + dV \wedge dQ,$$

and

$$(8.2) \quad H = \frac{1}{2}(Q^2 - P^2) - \frac{1}{3}U^3 - UV.$$

Then $\iota_X \omega = dH$ and

$$(8.3) \quad \mathcal{R}(U, P, V, Q) = (U, -P, V, -Q)$$

satisfies (2.8). Moreover,

$$(8.4) \quad \lambda(X) = P^2 - Q^2.$$

The linearization gives $\alpha_0 = \beta_0 = 2^{-1/2}$, the pole compactification is (2.22), and the algebraic reference is (5.22). These statements are exact.

The core is analytic and its two expanding eigenvalues are nonresonant in the saddle-focus normal-form sense. The convergent Hamiltonian local normal form therefore supplies exact symplectic section coordinates in which the critical passage preserves ν and has

$$(8.5) \quad \Delta_{0,\sigma}(\nu) = \log |\nu| + b_{0,\sigma} + O_{D_{\log \nu}^{k-2}}(|\nu|(1 + |\log |\nu||)).$$

In the same convergent normal form, the longitudinal rate and the angular coefficient have the expansions

$$h_\xi(\nu) = \alpha_0 + O(\nu), \quad a_0(\nu) = -(\beta_0/\alpha_0)\nu + O(\nu^2).$$

The exact passage-time quadrature between the two fixed section radii then gives

$$(8.6) \quad T_{0,\sigma}(\nu) = -\alpha_0^{-1} \log |\nu| + t_{0,\sigma} + O_{D_{\log \nu}^{k-2}}(|\nu|(1 + |\log |\nu||)).$$

This is the critical local-map calculation used in [15]; it verifies (2.12)–(2.15), rather than deriving them from Lemma 4.1. In particular, $c_{0,\sigma} = e^{\alpha_0 t_{0,\sigma}} > 0$; orienting the two signed action charts outward fixes the displayed signs. Shrinking their fixed domains gives the common lower bound required in the chosen presentation.

Proposition 8.1 (Canonical algebraic future hypersurface). *For the core, put*

$$(8.7) \quad \begin{aligned} e &= (-U)^{-1}, & p &= Pe^{3/2}, & q &= Qe^{3/2}, \\ d &= q + 2/\sqrt{3}, & \omega &= 1 + Ve^2, & E &= 2H, \\ a_w &= d/e^3, & b_w &= (\omega - e^2/6)/e^4, & \zeta &= \{p - h_7(e, d, \omega)\}/e^8. \end{aligned}$$

Here h_7 is fixed algebraically, not fitted to a computed orbit. Start with

$$h^{(1)}(e, d, \omega) = -e/\sqrt{3} + \omega/\sqrt{2}$$

and, for $m = 2, \dots, 7$, add the unique homogeneous polynomial H_m for which the degree- m part of

$$(8.8) \quad \mathcal{R}_{\text{inv}}(h) = \frac{3}{2}h^2 - \omega - Dh \cdot \left(eh, \frac{3}{2}h(d - 2/\sqrt{3}) - e, e(d - 2/\sqrt{3}) + 2h(\omega - 1) \right)$$

vanishes. Thus $h_7 = h^{(1)} + \sum_{m=2}^7 H_m$ has coefficients in $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and every monomial in $\mathcal{R}_{\text{inv}}(h_7)$ has total degree at least eight. The exact recursion and complete coefficient table are stored at

`validation/future-target-fold/build_prototype.py`;

`validation/future-target-fold/tail_graph_generated.hpp`. Both hashes are pinned by the future-target and exact-source certificates. The weighted field extends polynomially to $e = 0$. In the physical corridor

$$(8.9) \quad \begin{aligned} 0 &\leq e \leq .06, & |a_w| &\leq .0065, & |b_w| &\leq .01, \\ |E| &\leq .012, & |\zeta| &\leq 2, \end{aligned}$$

its maximal forward-staying set is the graph of a unique C^4 function

$$(8.10) \quad \zeta = \Gamma(e, \mathcal{A}, b_w), \quad \mathcal{A} = -\frac{\sqrt{3}}{4}E.$$

Denote this corridor graph by $\mathcal{W}_a^{\text{tail}}$. The graph contains the exact orbit (5.22), and every one of its orbits with $e > 0$ reaches $e = 0$ only at infinite physical time. On the final shooting neighborhood the same hypersurface is

$$(8.11) \quad p = h_7(e, d, \omega) + \eta(e, c, \omega), \quad c = d - \frac{\sqrt{3}}{2}\omega,$$

where

$$(8.12) \quad \|\eta\|_{C^0} \leq 10^{-8}, \quad \|D\eta\|_2 \leq 10^{-5}, \quad \|D^2\eta\|_2 \leq 10^{-3}.$$

Uniqueness is relative to the fixed physical corridor.

Proof. In the variables (8.7),

$$(8.13) \quad E = e^{-3}(q^2 - p^2 + 2\omega - 4/3).$$

The interval enclosures on the two a_w faces and throughout the corridor are

$$(8.14) \quad \begin{aligned} E|_{a_w=-.0065} &\in [.013768862160662696, .016229671066303420], \\ E|_{a_w=.0065} &\in [-.0162295683074354, -.0137687595003354], \\ \partial_{a_w} E &\leq -2.3073894635767043. \end{aligned}$$

Thus $(e, \mathcal{A}, b_w, \zeta)$ are regular coordinates on the energy slab. The two b_w face margins are at least .012815359139113863 and .012859991059044963 inward, while the lower and upper ζ derivatives are at most -1.3523766005713447 and at least 4.287036728907498. The $e = .06$ face is inward and $e = 0$ is invariant. The action coordinate $\mathcal{A}$ is invariant. Thus all base and normal face hypotheses of Theorem A.1 hold.

Writing the derivative of the transformed field in the blocks (2.19), the validated bounds are

$$(8.15) \quad \begin{aligned} \mu_2(C) &\leq 0.022876876257191, & \|B\| &\leq 2.591943887076 \cdot 10^{-5}, \\ \|D\| &\leq 11.269134128863, & a &\geq 1.4142085902521. \end{aligned}$$

With $\alpha_g = 10$, these give

$$(8.16) \quad \begin{aligned} q_g &\geq 2.6415910672, & \lambda_g &\geq 0.2872951773659, \\ \gamma_{g,0}^{\text{chart}} &\geq 1.4139493958634. \end{aligned}$$

The normal-versus one-through-four base-rate gaps are, respectively,

$$(8.17) \quad \begin{aligned} &1.3908133252175, & 1.3676772545716, \\ &1.3445411839257, & 1.3214051132798. \end{aligned}$$

The transformed field is polynomial on a fixed collar of the corridor. By compactness and the strict displayed margins, after shrinking that collar the same inequalities hold there with slightly weakened positive constants. Theorem A.1, including its fourth-gap clause, therefore proves existence, uniqueness, and C^4 regularity of the physical staying graph. It also proves its mixed-total-three dependence for every compact coefficient family in the resulting open neighborhood. No boundaryless completion and no same-space C^3 Fréchet graph transform is used.

On $a_w = b_w = E = 0$, the exact polynomial

$$(8.18) \quad \zeta_{\text{alg}}(e) = \frac{-e/\sqrt{3} - h_7(e, 0, e^2/6)}{e^8}$$

satisfies the energy and tangency identities and obeys

$$(8.19) \quad \begin{aligned} &-1.0314380350086394 \leq \zeta_{\text{alg}}(e), \\ \zeta_{\text{alg}}(e) &\leq -1.0083599083642838, & \zeta_{\text{alg}}(e) &\in (-2, 2). \end{aligned}$$

Uniqueness places the exact algebraic orbit on the graph. The independently validated unweighted cone gives (8.12) for that same graph, not for an auxiliary completion.

Finally, along the graph

$$(8.20) \quad \begin{aligned} \dot{e} &= e^2(p/e), & dt/ds &= \sqrt{e}, \\ p/e &\in [-.5773524806450880, -.5773480533731503]. \end{aligned}$$

Hence $e(s) \geq (e(0)^{-1} + .5773524806450880s)^{-1}$, and $\int_0^s \sqrt{e(\sigma)} d\sigma \rightarrow \infty$. $\square$

The tail graph is not itself a positive- U object. Let $\mathcal{T}_a$ be the fixed union of the finite algebraic shooting and exact-source flow tubes used below. On every component of this tube the validated event equation gives a unique transverse first entry $\tau_a(z)$ into $e = .06$. Define the finite backward saturation

$$(8.21) \quad \mathcal{W}_a := \left\{ z \in \mathcal{T}_a : \Phi^{\tau_a(z)} z \in \mathcal{W}_a^{\text{tail}} \right\}.$$

It is a C^4 invariant hypersurface on this fixed tube, obtained by pulling back the tail graph through a finite regular flow. This notation does not assert a global maximal saturation or classify any component outside $\mathcal{T}_a$.

Proposition 8.2 (Invariant cone implies a finite-time pole). *In the core set*

$$(8.22) \quad x = -U, \quad y = -P, \quad q = -Q, \quad w = -V.$$

Then

$$(8.23) \quad x' = y, \quad y' = x^2 - w, \quad q' = x, \quad w' = q.$$

The region

$$(8.24) \quad \mathcal{K} = \{x \geq 10, y > 0, D := \tfrac{1}{2}x^2 - w \geq 0, H_{\text{cone}} := xy - q \geq 0\}$$

is forward invariant. Every orbit entering $\mathcal{K}$ has a finite maximal forward time and $x, y \rightarrow +\infty$ there. No sign assumption on q or w is required.

Proof. On $\mathcal{K}$,

$$(8.25) \quad x' = y > 0, \quad y' \geq \tfrac{1}{2}x^2, \quad D' = H_{\text{cone}},$$

and

$$(8.26) \quad H'_{\text{cone}} = y^2 + x(x^2 - w) - x \geq y^2 + \tfrac{1}{2}x^3 - x > 0.$$

These inward inequalities, including the simultaneous $D = H_{\text{cone}} = 0$ corner, give forward invariance by the first-exit argument. Since $x' > 0$,

$$(8.27) \quad y^2(x) \geq y^2(x_0) + \frac{x^3 - x_0^3}{3}.$$

If the orbit were global, x would tend to infinity, but then $t - t_0 = \int_{x_0}^{x(t)} d\xi/y(\xi)$ would have a finite limit by (8.27). If x stayed bounded at a finite maximal time, (8.23) would successively bound q, w, y and permit continuation. Hence the only alternative is the asserted finite-time pole. $\square$

Proposition 8.3 (The pole cone enters the compactified sink). *Every core orbit entering $\mathcal{K}$ in (8.24) satisfies, as $x \rightarrow +\infty$,*

$$(8.28) \quad \begin{aligned} y &= \sqrt{\frac{2}{3}} x^{3/2} \left(1 + O\left(\frac{\log x}{x^2}\right) \right), \\ q &= \sqrt{6} x^{1/2} + C_p + O(x^{-3/2} \log x), \\ w &= 3 \log x + C_w + O(x^{-1/2}). \end{aligned}$$

Consequently the compactified variables (2.21) converge to p_∞ in (2.23). A compact set of strict cone-entry points enters one fixed attracting neighborhood of p_∞ after a common finite continuation. The same conclusion holds for the continued entry set throughout a sufficiently small ball in the admissible coefficient class of Definition 2.2.

Proof. The entry value of q is finite and $q' = x > 0$. Since the remaining physical lifetime is finite by Proposition 8.2,

$$(8.29) \quad w(t) = w(t_0) + \int_{t_0}^t q(s) ds \geq w(t_0) + \min\{q(t_0), 0\}(t_b - t_0),$$

so w is bounded below. The lower growth estimate (8.27) and $y dy/dx = x^2 - w \leq x^2 + C$ give

$$(8.30) \quad cx^{3/2} \leq y(x) \leq Cx^{3/2}$$

for all sufficiently large x . Hence

$$(8.31) \quad \frac{dq}{dx} = \frac{x}{y} = O(x^{-1/2}), \quad \frac{dw}{dx} = \frac{q}{y} = O(x^{-1}),$$

and therefore $q = O(x^{1/2})$, $w = O(\log x)$.

The conserved energy in these variables is

$$(8.32) \quad E = q^2 - y^2 + \frac{2}{3}x^3 - 2xw.$$

Substitution of the coarse bounds gives $y^2 = \frac{2}{3}x^3 + O(x \log x)$ and the first line of (8.28). Substituting that line into the first equation of (8.31) gives

$$(8.33) \quad \frac{dq}{dx} = \sqrt{\frac{3}{2}} x^{-1/2} \left(1 + O\left(\frac{\log x}{x^2}\right) \right).$$

After subtraction of the derivative of $\sqrt{6}x^{1/2}$ the error is integrable, which proves the q expansion. Dividing that expansion by the first line of (8.28) yields

$$(8.34) \quad \frac{dw}{dx} = \frac{3}{x} + \frac{C_p}{\sqrt{2/3}} x^{-3/2} + O(x^{-3} \log x),$$

and proves the w expansion.

Now $\rho = x^{-1/2} \rightarrow 0$ and

$$(8.35) \quad A = \rho^3 y \rightarrow \sqrt{2/3}, \quad B = \rho q \rightarrow \sqrt{6}, \quad C = \rho^4 w \rightarrow 0.$$

Thus every cone orbit lies in the basin of p_∞ . For a compact strict-entry set, continuity of finite flight and a finite subcover put all orbits into one fixed local sink neighborhood. The sink, its spectral gap, and that local attracting neighborhood persist under perturbations in (2.29); this uses only the principal and indicial type, not equality of higher jets. Continuous dependence on the finite pre-entry segments then gives the final uniform assertion after shrinking the coefficient ball. $\square$

8.2. Validated finite margins. The computer-assisted part proves finite-dimensional roots, first-hit order, and strict inequalities. Each root is enclosed by an interval-Newton or parametric Krawczyk inclusion; continuation boxes are joined only by a common-root containment, not by overlap of scalar plotting parameters. The source programs use outward-rounded interval arithmetic and have package-level replay interfaces. Depending on the package, a replay either reconstructs the certificate or reruns every claim-bearing source check against its frozen hashes. The precise distinction, and the current fail-closed replay status, are recorded in the reproducibility statement below.

For the primary symmetric homoclinic, the local unstable source has radius .01 and phase ϕ . The first positive hit of $\text{Fix } \mathcal{R}$ satisfies

$$(8.36) \quad \phi \in [5.8615055856447817, 5.8615055856450482],$$

$$(8.37) \quad T \in [9.6374420678958099, 9.6374420678971511],$$

$$(8.38) \quad \det D_{(\phi,T)}(P(T), Q(T)) \in [149.56393055300413, 149.56404227745782].$$

The phase tangent has the independent nonzero rows

$$(8.39) \quad \partial_\phi U(T) \in [-10.889708535478462, -10.8897049543477],$$

$$(8.40) \quad \partial_\phi V(T) \in [35.417125972639965, 35.417127394127888].$$

Thus the symmetric homoclinic is transverse in the regular critical-energy hypersurface and its finite matching map has the nondegenerate component required by Proposition 4.5.

Proposition 8.4 (Certified origin-to-algebraic heteroclinic). *Let $\mathcal{W}_a$ be the finite saturation (8.21). Then $W^u(O) \cap \mathcal{W}_a \cap \{E = 0\}$ contains an orbit which is locally unique modulo time translation. The two invariant manifolds are transverse modulo their common flow direction. Its representative on the source circle $\|u\| = .01$ has phase*

$$(8.41) \quad \phi_a \in [5.7566913947049203, 5.7566913967948983].$$

Proof. In the exact linear saddle coordinates

$$(8.42) \quad \begin{aligned} U &= u_1 + s_1, & V &= u_2 + s_2, \\ P &= 2^{-1/2}(u_1 - s_1 - u_2 + s_2), & Q &= 2^{-1/2}(u_1 - s_1 + u_2 - s_2), \end{aligned}$$

write the true local unstable graph as $s = H^u(u)$. With $N = 36$, $T = 17.412624804302453$, $h = T/N$, and $Z = (z_0, \dots, z_N) \in (\mathbb{R}^4)^{37}$, the certified equation has the 148 rows

$$(8.43) \quad \begin{cases} z_{j+1} - \Phi_h(z_j) = 0 & (0 \leq j < 36), \\ \|u(z_0)\|^2 - .01^2 = 0, \\ s(z_0) - H^u(u(z_0)) = 0 & (\text{two rows}), \\ g_a(z_{36}) = 0, \end{cases}$$

where

$$(8.44) \quad g_a = p - h_7(e, d, \omega) - \eta(e, c, \omega).$$

The interval evaluation inserts the true source remainder budgets 10^{-20} in value and 10^{-18} in first derivative and the target budgets in (8.12). Its Krawczyk inclusion and weighted-max contraction ratios are

$$(8.45) \quad .18217055712874344, \quad .18216932777866471 < 1.$$

Thus the actual correlated source and target graphs have exactly one zero in the specified shooting box.

The complete final evaluation box, rather than only the root image, satisfies

$$(8.46) \quad \begin{aligned} e &\in [.057499998340262551, .057500001659737551], \\ a_w &\in [-.00040856190871630305, .00040780247295930716], \\ b_w &\in [-.0077092414472739561, .0076936304274827332], \end{aligned}$$

and, for all $|\zeta| \leq 2$, $E \in [-.001850408298773085, .0018504082847746351]$. It is strictly contained in (8.9), so (8.44) is the defining equation of the canonical physical hypersurface. The source lies on $W^u(O)$ and therefore has exact energy zero; the nonzero interval energy width is only dependency overestimation. Backward and forward invariance give the two asserted limits.

Finally eliminate the 36 continuity blocks in the differentiated shooting equation. The kernel of the radius and unstable-graph source rows is the phase tangent v_0 . Nonsingularity of the full bordered Jacobian is therefore equivalent to

$$(8.47) \quad Dg_a(z_N)D\Phi_T(z_0)v_0 \neq 0.$$

Since $\ker Dg_a = T\mathcal{W}_a$ and both invariant manifolds contain $X(z_N)$,

$$(8.48) \quad T_{z_N}W^u(O) + T_{z_N}\mathcal{W}_a = \mathbb{R}^4, \quad T_{z_N}W^u(O) \cap T_{z_N}\mathcal{W}_a = \text{span}\{X(z_N)\}.$$

The radius row fixes time translation, so Krawczyk uniqueness is precisely local uniqueness modulo flow. $\square$

For every true-unstable source phase $\phi \in [-.2, .2]$ modulo 2π , the orbit has a strict first hit of $x = 10$ with

$$(8.49) \quad T_{10} \in [10.885720013440157, 11.579632102474379],$$

$$(8.50) \quad y > 26.31, \quad D > 52.17, \quad H_{\text{cone}} > 262.34.$$

and

$$(8.51) \quad x' = y > 26.31, \quad y' > 102.17, \quad H'_{\text{cone}} > 1704.01.$$

Thus every certified endpoint lies strictly inside $\mathcal{K}$ in (8.24), and Proposition 8.2 proves a genuine finite-time pole rather than merely growth up to the validation section. In fact $y(T_{10})^2 > 10^3/3$, so (8.27) gives

$$(8.52) \quad t_b - T_{10} < \int_{10}^{\infty} \sqrt{3} x^{-3/2} dx = 2\sqrt{3/10}, \quad t_b < 12.676.$$

Lemma 5.3 thickens this source arc to an open two-dimensional pole window. The homoclinic phase, algebraic phase, and pole window have pairwise positive gaps; in particular, the algebraic–homoclinic gap exceeds .104814. This permits the three apertures and their finite boundary cells to be fixed simultaneously. The interval enclosures and the two strict phase gaps are displayed in Figure 6.

Here is the finite cell and incidence check used by the abstract theorem. Choose mutually disjoint compact subapertures inside those phase gaps and take their boundaries, together with transverse flowbox exit faces, as the functions h_j in (5.12). Name every connected stratum in the complement of the three outgoing subapertures as an outgoing-box lateral outcome. At the end of the homoclinic tube, name every complementary stratum outside the re-entry aperture and the stable cut as a return-box lateral outcome. Their closures, together with the three apertures, re-entry strata, and cut, cover the two chosen compact bands, so the return/first-exit assignment is total. On every nonempty active set their conormals are independent by the flowbox construction. The finite identities (5.13) on B^u and B^r then give all two-dimensional interiors, one-dimensional faces, and zero-dimensional corners. Choose the artificial faces of the signed reference cell in the certified phase gaps and transverse to the finitely many pulled-back active strata; a generic compact interior choice has this property. The certified homoclinic and algebraic phase-derivative bounds, the open pole window, and the flowbox faces then give the pullback rank and boundary-neatness margins required in (H2). The interval event bounds above, homoclinic transversality, and the algebraic graph margins remain strict on slightly smaller compact subapertures; the artificial lateral faces are transverse to the flow by construction. Compactness gives one positive minimum for event speed, competing-face sign, and event-order distance. The certified algebraic root and a point in the interior pole-entry interval lie a positive distance from the boundary of the chosen signed reference cell. Under the cut-face templates (2.59) they give the two interior anchors required in (H2). This verifies the exhaustive

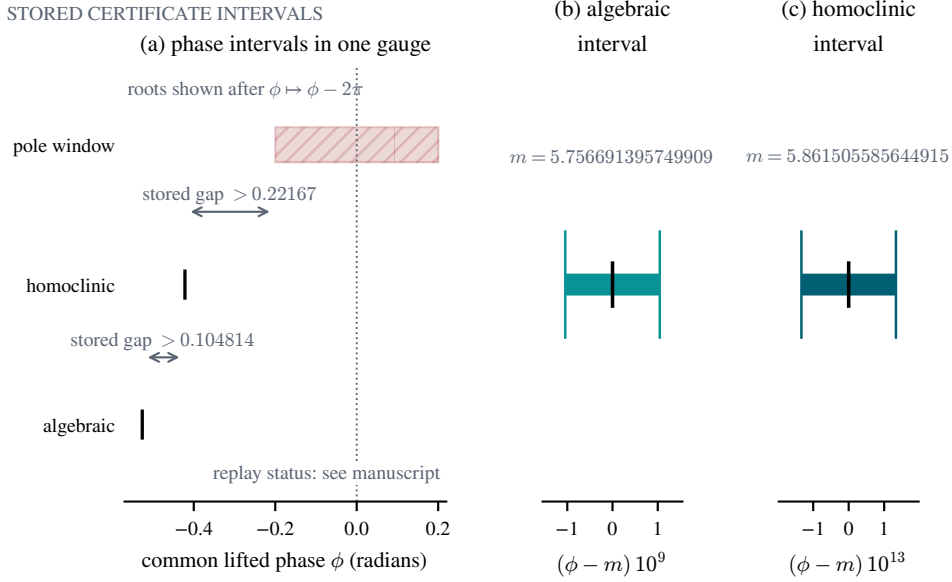


FIGURE 6. Validated separation of the source phases on the circle $\|u\| = 0.01$ in the common linearizing phase coordinate. (a) The interval-enclosed algebraic and symmetric-homoclinic roots are shown after subtracting 2π , together with the closed pole-entry cover $[-0.2, 0.2]$. The arrows give rigorous lower bounds for the consecutive gaps. (b,c) The complete algebraic and homoclinic root enclosures are magnified about their interval midpoints m . Thick segments are the full interval enclosures; black ticks are arithmetic midpoints used only for plotting and are not additional numerical evidence.

list of labelled connected strata, nonemptiness of both end channels, and all first-exit margins without inferring a global partition outside the fixed flow-tube domain.

For the Markov-strip hypothesis take the two vertices Σ_+ and Σ_- . For every (σ, n) , Proposition 4.5 gives one closed strip $\tilde{\mathcal{V}}_{\sigma, n}$ in its fixed matching tube before the target cut. The target set $\tilde{\nu} = 0$ cuts it exactly as in (4.55): the two nonempty regular components are the edges $a = (\sigma, n, \sigma')$, while $\mathcal{I}_{\sigma, n} = E_{\text{cut}, \sigma, n, 1}$ is a first-exit face. The two vertex charts (3.2) give every edge a completed cross form on $I \times J$. In particular, (4.63) uses the same positive magnitude coordinate at the source and target vertices, while (4.64)–(4.66) supply its field-dependent rescaled transverse-action margin. The first-hit Poincaré map sends that edge diffeomorphically to a horizontal strip in $\Sigma_{\sigma'}$, and the flowbox rectangles give one full horizontal–vertical crossing exactly when the preceding target sign is the following source sign. Unequal vertices are not composable.

The signed normal in (4.50), the strict phase monotonicity (3.4), and the half-open angular lift make the triples (σ, n, σ') faithful and the regular domains disjoint. They also prove local finiteness away from the two source cut faces and identify the two fixed-source-sign high-winding families, while retaining the target sign. Thus the return graph satisfies the Markov-strip hypothesis, and (5.14) places the zero-action boundary in exactly one first-exit component.

The rigorous numerical enclosures for the three source orbits and the independent finite collar are documented in Section C. Together with the analytic arguments of Sections 4 and 5, the first-table margins verify the common hypotheses of Theorems 3.1–3.3. That appendix also states the mathematical validation criteria and reproduction procedure. No low-winding or larger-section statement is imported into this verification.

8.3. Optional module: the selected reversible branch from the algebraic connection.

Proposition 8.5 (Continuation to the first nondegenerate critical point). *Let $\mathcal{W}_a$ be the canonical finite saturation (8.21). There is a connected C^2 arc*

$$(8.53) \quad c(U) = (U, 0, V(U), 0) \in \mathcal{W}_a \cap \text{Fix } \mathcal{R}, \quad 0 \leq U \leq U_f,$$

with

$$(8.54) \quad \begin{aligned} c(0) &= c_0 = (0, 0, 1/6, 0), \\ U_f &\in [.041527012176079285, .041527012805834346]. \end{aligned}$$

The restriction $E(c(U))$ has no critical point for $0 \leq U < U_f$ and has one nondegenerate critical point at U_f . Thus this is the first critical point of the restricted Hamiltonian on the selected arc followed from c_0 .

Proof. The proof has one analytic and one validated finite part. Appendix E constructs an exact Jost basis along the algebraic orbit, proves that the canonical tangent is

$$\text{span}\{(1, 0, -2k_0, 0)\}, \quad k_0 = \frac{\sqrt{\pi} \Gamma(1/4)}{4\sqrt{6} \Gamma(3/4)},$$

and excludes the growing Jost mode from the weighted tangent bundle. A fixed- U interval continuation starting from this analytic tangent then keeps $dE(c(U))/dU < 0$ up to the final cap, where an augmented Krawczyk argument proves one simple zero and identifies it with the first nondegenerate critical point on the connected selected arc. The exact representations, exclusion argument, continuation bridges, and numerical margins are recorded in that appendix. $\square$

A fixed critical-point/event continuation follows the other local side of this critical point into a 6,316-box intermediate collar. A further 9,725-box continuation, joined by 9,724 true common-root containments, reaches the circle $r = 2.4 \times 10^{-4}$. Fixed-radial uniqueness identifies its endpoint with the

local branch in (3.39). The full-state identification shared by the validated continuations therefore joins these pieces into one selected connected branch from c_0 through the first critical point to the local logarithmic spiral. No assertion is made about other components of $\mathcal{W}_a \cap \text{Fix } \mathcal{R}$.

8.4. Intrinsic realization and the normal-form corollary.

Proposition 8.6 (Intrinsic realization by the normal-form system). *For the one-point parameter space, the system (1.1), with the fixed local return–exit tube constructed in this section, is an intrinsic two-ended Hamiltonian configuration in the sense of Definition 2.1.*

Proof. We match the five clauses of the definition. For (I1), (8.1)–(8.3) are polynomial and satisfy $\mathcal{R}^*\lambda = -\lambda$; the origin has $\alpha_0 = \beta_0 = 2^{-1/2}$. The fixed local saddle block may be chosen inside the analytic normal-form ball, with auxiliary radial faces joined to its physical faces by compact event-free orbit slides.

For (I2), the determinant enclosure (8.38) and the independent tangent rows (8.39)–(8.40) prove transversality of the selected homoclinic in the regular critical-energy hypersurface. For (I3), the finite flowbox construction in lines (8.36)–(8.38) and the incidence check following Figure 6 give the clean physical event stratification, neat boundary faces, strict first-hit order, and the two interior anchors. The certified phase gaps allow the physical source cell and its two signed limiting traces to be placed in one proper arc with a complementary cut interval of positive length.

For (I4), Proposition 8.1 provides the algebraic compactification, the product interval block, and the physical finite saturation. The energy-face and flag ranks follow from the validated coordinate system and (8.14). The block bounds (8.15) and the positive gaps (8.16)–(8.17), inserted into (A.7), give the intrinsic finite-time secant-cone and order-three bunching inequalities (indeed also the optional fourth gap). Finally, (2.21)–(2.27) give the analytic pole chart, boundary sink, and simple indicial spectrum $\{1, 4, 6\}$ with $m_p = 6$; together with (8.4) they also show directly that $\rho^6\lambda(X)$ extends in $C^9 = C^{m_p+3}$. Propositions 8.2 and 8.3, with (8.49)–(8.51), supply the compact basin entry and strict radial margins required in (I5). Thus all intrinsic clauses hold for the same fixed physical tube. $\square$

Corollary 8.7 (Relative openness near the normal-form system). *Fix an integer $k \geq 5$, the physical saddle block, finite flow tubes, cooriented event hypersurface germs, and the two compactifications used above. Fix also $m_p = 6$ and the indicial conjugacy class (2.27). In the space of reversible exact Hamiltonians that are analytic on the saddle germ, C^k on the compact and algebraic tubes, and C^9 on the pole compactification, restrict to the subspace preserving the algebraic boundary factorizations, every designated structurally invariant face, and this pole indicial conjugacy class. There is a relatively open neighborhood $\mathcal{O}$ of the normal-form system such that every Hamiltonian in $\mathcal{O}$ satisfies the hypotheses of Theorem 1.1 on the continued*

physical tube. On each compact $K \subset \mathcal{O}$, the state-only strict margins and the threshold for time spent in the saddle block can be chosen uniformly. If a compact finite-dimensional C^r family, $r \leq 2$, has image in K , then after fixing

$$0 < \kappa < \inf_{H \in K} \frac{2\pi\alpha_H}{\beta_H},$$

the mixed state-parameter conclusions of Theorem 1.1 hold; constants involving parameter derivatives may depend on the C^r norm of that family.

Proof. The symmetric homoclinic persists by its transverse intersection, and the finite clean first-hit stratification, its exhaustive labelled strata, the proper-arc condition, and both end anchors persist because all defining rank, first-hit, event-order, and boundary margins are strict on compact sets. Proposition 2.9 continues the locally maximal normally expanding algebraic hypersurface and preserves the order-three bunching inequalities. The hyperbolic pole sink, its common basin-entry neighborhood, and the radial and spectral-gap margins persist inside the fixed indicial class. The local chart-production theorem then supplies the marked representations; no eigenframe or angular lift is included in the perturbation data. Compactness gives the uniform state-only margins and threshold on K . The parameter statement then follows from Theorem 3.3(ii) with the stated strict rate gap and its explicit dependence on the family norm. $\square$

Remark 8.8 (Why the openness is relative). Corollary 8.7 is not an assertion in an unrestricted compact-open C^k topology. Such a topology neither preserves state analyticity at the saddle nor controls extensions to the two chosen compactifications, and it allows the pole indicial roots to move through resonances. Removing those restrictions would require a finite-smooth saddle passage with the present weighted parameter bounds and a polyhomogeneous pole calculus with moving exponents. Neither is proved here.

Proof of Corollary 1.2. Proposition 8.6 shows that the core is an intrinsic two-ended Hamiltonian configuration, so Theorem 1.1 applies, after the fixed compact source-subcell restriction, on the region defined by sufficiently long physical saddle-block residence time within that subcell. In the locally produced marking, the exact calculations and interval enclosures in this section verify (H1)–(H4) of Definition 2.8. For every fixed $\kappa \in (0, 2\pi)$, the critical-energy flow of (1.1) has the high-winding return–exit decomposition, exact renormalized roof/action cocycle, two-sided countable coding, stable plaques, and the compactification by the two limit points $0^\pm$ from Theorems 3.1–3.3. Corollary 3.5 and Theorem 3.6 give the remaining conclusions stated in Corollary 1.2. $\square$

8.5. Optional selected-branch consequence for the normal-form system.

Corollary 8.9 (Optional selected branch for the normal-form system). *The algebraic connection in Proposition 8.4 selects a reversible branch whose successive critical points have physical Euclidean (U, V) radii satisfying*

$$(8.55) \quad \frac{R_{UV,m+1}}{R_{UV,m}} = e^{-\pi/2} (1 + O(R_{UV,m}^{\hat{\eta}})), \quad \hat{\eta} = \min\{\eta, 1\}.$$

The validated finite continuation extends this selected branch from the local annulus through its first nondegenerate critical point to the exact algebraic source c_0 .

Proof. Proposition 8.4 verifies the additional transverse algebraic connection required in Corollary 3.9. Choose the marked cross-form chart tangent to the linear coordinates (8.42). On $\text{Fix } \mathcal{R}$ its unstable coordinate obeys $u = \xi + O(|\xi|^2)$ through the required logarithmic jets, while $(U, V) = 2(u_1, u_2)$ in the linear chart. Consequently the physical radius satisfies $R_{UV} = 2r(1 + O(r))$; inserting $\alpha_0 = \beta_0$ into (3.40) gives (8.55). Proposition 8.5 and the subsequent continuation establish the final assertion. $\square$

9. OPTIONAL PERSPECTIVE: ORBIT-PRESERVING GENERICITY OF TRANSVERSE MATCHING

The nonzero derivative used in the finite homoclinic matching is not forced by the saddle-focus or by either compactified end. The following proposition shows that its failure is locally of codimension one under reversible Hamiltonian perturbations supported in the compact part of the orbit. The perturbations keep the selected connection itself fixed. This is an auxiliary transversality result; it is not used by the main theorem and does not provide a second system-level application.

Proposition 9.1 (Orbit-preserving genericity of transverse matching). *Let $k \geq 5$, let $\mathbf{X} \in \mathfrak{H}_{\text{adm}}^k$, and suppose the chosen compact Hamiltonian flow-box coordinates are C^{k+2} . Let Γ be a regular symmetric critical-energy homoclinic orbit in the fixed compact flight tube. Write $p = \Gamma \cap \text{Fix } \mathcal{R}$ and let ψ_* be its source phase in (4.52). Choose an open Hamiltonian flow box U on the unstable half of Γ , away from the saddle chart, all marked sections, all event faces, and both end charts, so small that*

$$\overline{U} \cap \mathcal{R}(\overline{U}) = \emptyset,$$

and such that $(\overline{U} \cup \mathcal{R}(\overline{U})) \cap \Gamma$ consists only of the chosen orbit segment and its reflected symmetric segment. Fix $U_0 \Subset U$.

Let $\mathcal{K}_{\Gamma}^{k+2}(U_0)$ be the closed Banach subspace of C^{k+2} functions K supported in $\overline{U}_0 \cup \mathcal{R}(\overline{U}_0)$ such that

$$(9.1) \quad K \circ \mathcal{R} = K, \quad K|_{\Gamma} = 0, \quad dK|_{\Gamma} = 0.$$

For K sufficiently small set

$$H^K = H + K, \quad \iota_{X^K} \omega = dH^K,$$

and retain the original primitive, reverser, sections, and marked charts. Then $\mathbf{X}^K \in \mathfrak{H}_{\text{adm}}^k$, the parametrized orbit Γ is unchanged, and the compact transition has the form

$$(\bar{\phi}, \bar{\nu}) = (f_K(\psi, \nu_u), g_K(\psi, \nu_u))$$

near $(\psi_*, 0)$. The matching determinant

$$(9.2) \quad \mathfrak{d}(K) := \partial_\psi g_K(\psi_*, 0)$$

is a C^1 function of K . At every sufficiently small K_0 for which $\mathfrak{d}(K_0) = 0$, its derivative

$$D\mathfrak{d}(K_0) : \mathcal{K}_\Gamma^{k+2}(U_0) \longrightarrow \mathbb{R}$$

is onto. Consequently, near each such K_0 , the set $\mathfrak{d}^{-1}(0)$ is a C^1 codimension-one Banach submanifold. Equivalently, there is a C^{k+2} reversible Hamiltonian K_* in this space for which

$$(9.3) \quad \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \mathfrak{d}(K_0 + \varepsilon K_*) \neq 0.$$

The same conclusion holds for the origin-to-algebraic connection if (9.2) is replaced by

$$(9.4) \quad \mathfrak{d}_a(K) = Dg_a(q_*) DP_a^K(z_*)v_*.$$

Here P_a^K is the compact first-hit map from the fixed unstable source trace to a section on which the algebraic future trace has defining function g_a , and v_* is the nonzero source-phase tangent. Assume that $Dg_a(q_*)$ restricts to a nonzero covector on the critical-energy target section, equivalently that its pullback by the finite flight is nonzero on the critical-energy symplectic normal plane. In this case define $\mathcal{K}_\Gamma^{k+2}(U_0)$ with Γ equal to the selected origin-to-algebraic orbit and choose $U \cup \mathcal{R}(U)$ also disjoint from the fixed future tail defining g_a , with $\mathcal{R}(\bar{U}) \cap \Gamma = \emptyset$ and $\bar{U} \cap \mathcal{R}(\Gamma) = \emptyset$. Formula (9.4) is the section form of (8.47).

In either case, within the orbit-preserving perturbation space the nondegenerate subset is locally open and dense. All other strict return–exit hypotheses persist on their relatively open subset, while the saddle passage, clock, primitive, algebraic coefficients, and pole boundary jet are unchanged.

Proof. An $\mathcal{R}$ -invariant Hamiltonian has a reversible Hamiltonian vector field because $\mathcal{R}$ is anti-symplectic. Moreover $K = dK = 0$ on Γ , and hence $X^K = X$ there. Thus the orbit, including its time parametrization, is fixed. The support condition leaves the local saddle-focus passage and clock, the algebraic factorization, and the pole boundary jet unchanged. It also leaves the primitive fixed. Therefore a small K gives a presentation in $\mathfrak{H}_{\text{adm}}^k$; the strict compact first-hit inequalities persist after decreasing its norm. Standard finite-time dependence of an ODE on its C^k vector field shows that (9.2) is C^1 in the C^{k+2} Hamiltonian.

We first treat the symmetric homoclinic. Replacing H by H^{K_0} , if necessary, reduces the submersion assertion at an arbitrary zero K_0 to the case

$K_0 = 0$; the orbit and every support condition are unchanged. Since p is a regular point of $H^{-1}(0)$, the quotient

$$N_p = \ker dH(p) / \text{span}\{X(p)\}$$

is a symplectic plane, with reduced form ω_p^{red} . The reverser induces an anti-symplectic involution R_p on N_p . If $v_K \neq 0$ spans the image of $T_p W^u(O)$ in N_p , then $R_p v_K$ spans the corresponding stable line. Hence

$$(9.5) \quad \delta(K) = \omega_p^{\text{red}}(v_K, R_p v_K)$$

vanishes exactly when the stable and unstable manifolds fail to be transverse modulo their common flow direction. Passing between the reduced plane and the fixed source and target section coordinates multiplies (9.5) by a nonzero C^1 factor. Thus

$$(9.6) \quad \mathfrak{d}(K) = a(K)\delta(K), \quad a(K) \neq 0.$$

It remains to vary δ while fixing Γ . By the Hamiltonian flow-box theorem, after shrinking U there are C^{k+2} coordinates (t, E, x, y) in which

$$\omega = dt \wedge dE + dx \wedge dy, \quad H = E, \quad \Gamma \cap U = \{E = x = y = 0\}.$$

The transverse coordinates may be chosen so that the transported unstable direction at the entrance to U is ∂_x . Choose $\chi \in C_c^\infty(\mathbb{R})$ supported in the t -interval of U_0 with $\int \chi(t) dt = 1$. With a transverse cutoff equal to one near $E = x = y = 0$, put

$$\varkappa(t, E, x, y) = \frac{1}{2}\chi(t)x^2$$

in U and extend it by zero. Its reversible extension is

$$(9.7) \quad K_* = \varkappa + \varkappa \circ \mathcal{R}.$$

It belongs to $\mathcal{K}_\Gamma^{k+2}(U_0)$. On the unstable half of Γ the second summand is absent, and the transverse variational equation for $H + \varepsilon K_*$ is

$$\dot{x} = 0, \quad \dot{y} = -\varepsilon\chi(t)x.$$

Thus passage through U changes the unstable direction by the symplectic shear

$$(9.8) \quad (x, y) \mapsto (x, y - \varepsilon x).$$

After the fixed transport from U to p , this realizes $v_\varepsilon = v_0 + \varepsilon w + O(\varepsilon^2)$ with $\omega_p^{\text{red}}(v_0, w) \neq 0$. The reflected bump lies on the stable half and ensures reversibility; the stable direction at p is $R_p v_\varepsilon$.

Suppose now that $\delta(0) = 0$. In a symplectic plane the lines spanned by v_0 and $R_p v_0$ then coincide. Differentiating (9.5) and using the anti-symplectic identity gives

$$\begin{aligned} \delta'(0) &= \omega_p^{\text{red}}(w, R_p v_0) + \omega_p^{\text{red}}(v_0, R_p w) \\ &= 2\omega_p^{\text{red}}(w, R_p v_0) \neq 0. \end{aligned}$$

Together with (9.6), this proves (9.3). The Banach-space implicit function theorem gives the codimension-one assertion.

For the algebraic connection, pull the target conormal $Dg_a(q_*)$ back to the symplectic normal plane at the exit of U . At a zero of (9.4) it annihilates the transported source tangent but is not the zero covector by the explicit target-section assumption. Choose the transverse coordinates so that the shear (9.8) moves that tangent in a direction on which the pulled-back conormal is nonzero. The same calculation then gives $\mathfrak{d}'_a(0) \neq 0$. Because $U \cup \mathcal{R}(U)$ is disjoint from the fixed algebraic tail, its defining function and all end coefficients are unchanged. This proves the last assertion. The local openness and density follow from the submersion result and the openness of the remaining strict margins. $\square$

Remark 9.2 (Scope of Proposition 9.1). The proposition concerns loss of transversality of a selected critical-energy connection. It neither asserts that such a connection exists for a generic Hamiltonian nor describes the boundary of the set on which all return, first-exit, and end hypotheses hold uniformly over a compact corridor. Its content is that, once the selected connection is present, exactness, reversibility, the fixed saddle passage, and the two end compactifications do not force its matching determinant to vanish: a one-parameter orbit-preserving reversible Hamiltonian family crosses the degeneracy set while preserving the orbit itself. This is a local genericity theorem for each selected connection separately; it does not provide a second system-level application.

10. SCOPE AND REMAINING QUESTIONS

The theorem has four precise scope limits. First, it concerns a fixed local flow tube and a compact source subcell: low windings are excluded, and no assertion is made about an orbit after it crosses a lateral face or about a global critical-energy partition. Second, the local coordinates are produced in the state-analytic saddle and geometric two-end class of Definition 2.1, not for an arbitrary unmarked C^k Hamiltonian family. Third, persistence is taken in the relative coefficient topology that fixes the physical block and tubes, the cooriented event germs, and the indicated end structures; unrestricted ambient openness is not claimed. Fourth, the pole theorem fixes the compactified-action order and the simple positive-integer indicial conjugacy class. The finite boundary jet may vary, but the indicial roots do not move or cross a resonance.

Two natural extensions are therefore clear. A larger dynamical result would validate the finitely many low windings and every new first-exit face in a larger flow tube. A broader pole theorem would allow moving indicial roots and resonance crossings; this requires parameter-dependent powers and uniform divided-difference coordinates rather than the fixed finite recursion used here.

Within these limits, an intrinsic saddle-focus configuration with a transverse homoclinic tube, clean physical event geometry, and the two end compactifications produces finitely many parameter-local reversible exact symplectic coordinate systems. Above one common residence-time cutoff, they determine the same exhaustive physical return/first-exit relation. The classical countable bifocal return dynamics then carries strict first-exit events, two compatible action finite parts, exact action composition, and mixed state–parameter control.

APPENDIX A. A COMPUTABLE CRITERION FOR THE ALGEBRAIC END

This appendix records the interval-block criterion used to verify the normally expanding algebraic hypersurface in the normal-form application. It is a sufficient test for the geometric hypothesis in Definition 2.1; its product coordinates and cone constants are not inputs to the return–exit theorem.

Theorem A.1 (Interval-block criterion for a normally expanding hypersurface). *Let $\mathcal{P}$ be a compact finite-dimensional parameter manifold, let $r \in \{0, 1, 2\}$, and let*

$$(A.1) \quad \varpi \mapsto Y_\varpi = (F_\varpi, G_\varpi)$$

be a C^r map into the C^5 vector fields on a fixed collar of an oriented C^5 interval bundle $\pi : Q_\Gamma \rightarrow B_\Gamma$, where B_Γ is a compact manifold with corners. Fix an oriented C^5 trivialization $Q_\Gamma \simeq B_\Gamma \times [-\zeta_\Gamma, \zeta_\Gamma]$, its product splitting, and adapted base and fiber metrics. Suppose every base face is strictly inward or is structurally invariant throughout the chosen coefficient class, and the two fiber faces are strict exits.

Assume that Q_Γ has a fixed ambient interval collar

$$\widehat{Q}_\Gamma = B_\Gamma \times [-\zeta_\Gamma - \varepsilon_\Gamma, \zeta_\Gamma + \varepsilon_\Gamma] \supset Q_\Gamma$$

on which the bundle projection, product splitting, adapted metrics, and the time- δ_g map extend for some $\delta_g > 0$. The normal component retains its strict outward sign beyond the two fiber faces. Fix a closed vertical cone and a closed horizontal Grassmann cone $\mathcal{H}_{\varpi, z}$ of base-dimensional graphs of slope at most α_g . We require the following two conditions.

First, the vertical cone is strictly forward invariant with a uniform angular margin. Whenever two orbit segments stay in the ambient collar, a separation initially in this cone remains in its interior and its vertical component grows at least as $e^{\lambda_g s}$ for one $\lambda_g > 0$. In addition, if $z' = \Phi_\varpi^{\delta_g} z$ and $P_+ \in \mathcal{H}_{\varpi, z'}$, then all backward planes

$$(D\Phi_\varpi^{\delta_g - s})^{-1} P_+, \quad 0 \leq s \leq \delta_g,$$

belong to the corresponding horizontal cones, and the initial plane lies in their relative interior. We refer to these requirements as the uniform secant-cone and backward projectivized-cone conditions.

Second, call a base-dimensional plane $P \subset T_z \widehat{Q}_\Gamma$, $z \in Q_\Gamma$, δ_g -admissible if the orbit segment stays in the ambient collar and every plane $P_s = D\Phi_\varpi^s P$ belongs to $\mathcal{H}_{\varpi, \Phi_\varpi^s z}$ for $0 \leq s \leq \delta_g$. Put $P' = P_{\delta_g}$ and $z' = \Phi_\varpi^{\delta_g} z$, and define

$$A_{\varpi, z}^\delta(P) = d\pi_{z'} \circ D\Phi_\varpi^{\delta_g}|_P \circ (d\pi_z|_P)^{-1},$$

$$N_{\varpi, z}^\delta(P) : T_z \widehat{Q}_\Gamma / P \longrightarrow T_{z'} \widehat{Q}_\Gamma / P'.$$

Use the base metric for A^δ . If $V_z = \ker d\pi_z$ and $q_P : T_z \widehat{Q}_\Gamma \rightarrow T_z \widehat{Q}_\Gamma / P$ is the quotient map, transport the fiber metric to the quotient through $q_P|_{V_z}$, and use the analogous norm at P' . Thus, for $P = \text{graph } S$, the quotient coordinate is $[(u, v)] \mapsto v - Su$. Assume the finite-time order-three inequalities (A.2)

$$\vartheta_{g,j} := \sup_{\substack{\varpi \in \mathcal{P}, z \in Q_\Gamma \\ P \text{ } \delta_g\text{-admissible}}} \|(N_{\varpi, z}^\delta(P))^{-1}\| \|A_{\varpi, z}^\delta(P)\|^j < 1, \quad j = 0, 1, 2, 3.$$

These hypotheses can be verified in one global adapted product chart by coefficient bounds. Write

$$DY_\varpi = \begin{pmatrix} C_\varpi & B_\varpi \\ D_\varpi & a_\varpi \end{pmatrix}$$

in base-fiber blocks. It is sufficient that, on a fixed neighborhood of Q_Γ , for some positive constants independent of ϖ ,

$$(A.3) \quad G_\varpi(x, -\zeta_\Gamma) \leq -m_g < 0 < m_g \leq G_\varpi(x, \zeta_\Gamma), \quad x \in B_\Gamma,$$

and

$$(A.4) \quad \mu_2(C_\varpi) \leq c_g, \quad \|B_\varpi\| \leq b_g, \quad \|D_\varpi\| \leq d_g, \quad a_\varpi \geq a_g.$$

Require the vertical-cone and normal-growth margins

$$(A.5) \quad q_g := \alpha_g(a_g - c_g) - d_g - b_g \alpha_g^2 > 0, \quad \lambda_g := a_g - d_g / \alpha_g > 0,$$

and the order-three rate gaps

$$(A.6) \quad \gamma_{g,j}^{\text{chart}} := a_g - jc_g - (j+1)b_g \alpha_g > 0, \quad j = 0, 1, 2, 3.$$

Then the cone conditions above hold and, along every admissible segment,

$$(A.7) \quad \|A_{\varpi, z}^\delta(P)\| \leq e^{(c_g + b_g \alpha_g) \delta_g},$$

$$m(N_{\varpi, z}^\delta(P)) \geq e^{(a_g - b_g \alpha_g) \delta_g},$$

$$\vartheta_{g,j} \leq e^{-\gamma_{g,j}^{\text{chart}} \delta_g} < 1.$$

One may instead verify the coordinate-independent finite-time conditions directly; bounds obtained by taking unrelated extrema in different charts are not used.

Under these hypotheses, the maximal forward-staying set

$$(A.8) \quad \mathcal{W}_\varpi^{\text{tail}} = \{z \in Q_\Gamma : \Phi_\varpi^s z \in Q_\Gamma \text{ for every } s \geq 0\}$$

is the image of a unique section $\Gamma_\varpi : B_\Gamma \rightarrow Q_\Gamma$. In the chosen adapted trivialization it is the graph

$$(A.9) \quad \zeta = \Gamma_\varpi(x), \quad x \in B_\Gamma, \quad \|D_x \Gamma_\varpi\| \leq \alpha_g.$$

It is normally expanding (hence locally repelling) in forward desingularized time. Equivalently, the local backward graph transforms contract its normal jets. For $P = T_z \mathcal{W}_\varpi^{\text{tail}}$, the tangent and normal quotient maps in the criterion coincide, under the base identification, with those in (2.5); hence $\vartheta_{a,j} \leq \vartheta_{g,j} < 1$ for $0 \leq j \leq 3$.

The graph has the scale regularity

$$(A.10) \quad \sup_{\substack{i+j \leq 3 \\ j \leq r}} \sup_{(x, \varpi) \in B_\Gamma \times \mathcal{P}} |D_x^i D_\varpi^j \Gamma_\varpi(x)| < \infty.$$

Thus, in parameter charts,

$$(A.11) \quad \Gamma \in C^0(\mathcal{P}; C^3(B_\Gamma)) \cap C^1(\mathcal{P}; C^2(B_\Gamma)) \cap C^2(\mathcal{P}; C^1(B_\Gamma)),$$

with the last one or two factors omitted when $r < 2$. If the additional intrinsic finite-time gap

$$(A.12) \quad \vartheta_{g,4} := \sup_{\varpi, z, P \text{ admissible}} \|(N_{\varpi, z}^\delta(P))^{-1}\| \|A_{\varpi, z}^\delta(P)\|^4 < 1$$

holds, then every state graph is C^4 , with a uniform C^4 bound. All these conclusions persist on a sufficiently small C^5 coefficient neighborhood relative to the subspace preserving every designated structurally invariant base face. If all base faces are strictly inward, this is an ordinary C^5 neighborhood. Constants involving parameter derivatives may depend on the C^r norm of the chosen family.

Proof. An oriented interval bundle is trivial because the orientation-preserving diffeomorphism group of an interval is contractible; the theorem fixes one such trivialization before stating the cone and bunching bounds. All differential estimates below use its adapted metrics and a finite corner atlas on the base. We separate existence and uniqueness from differentiability. Fix ϖ and $x \in B_\Gamma$. Among the initial values $\zeta \in [-\zeta_\Gamma, \zeta_\Gamma]$, let L_x and U_x consist of those whose first exit is through the lower and upper normal face, respectively. No orbit can leave through a base face. The strict normal face inequalities make L_x and U_x disjoint relatively open sets, and they contain the two endpoints. If no initial value stayed for all positive time, these two sets would disconnect the interval. Hence every fiber contains a forward-staying point.

For two orbit segments contained in one product/base chart write their separation as (u, v) in base and normal coordinates. Under the one-chart sufficient criterion, the mean-value formula applied to (2.19) gives

$$(A.13) \quad D^-|v| \geq a_g|v| - d_g|u|, \quad D^+|u| \leq c_g|u| + b_g|v|.$$

Still under that one-chart criterion, they imply

$$(A.14) \quad D^-(|v| - \alpha_g|u|) \geq q_g|u| > 0.$$

Inside the vertical cone they also imply

$$(A.15) \quad D^-|v| \geq \lambda_g|v|.$$

The same strict tangent-cone calculation, applied backward to a terminal horizontal graph plane, gives the backward projectivized-cone condition. In the general formulation, the uniform secant-cone hypothesis assumes the geometric consequences of these two inequalities—strict cone preservation and exponential vertical separation—along each common finite base-chart sequence; it does not assume the displayed coordinate Dini inequalities. Two different staying points in one fiber would therefore separate beyond the width of the normal strip. This proves uniqueness. The same argument applied to two arbitrary staying points shows $|\Gamma_\varpi(x_1) - \Gamma_\varpi(x_2)| \leq \alpha_g|x_1 - x_2|$ in every corridor chart. Thus the staying set is a uniformly Lipschitz graph, including at the corners.

We now bootstrap this graph by finite-horizon jet transforms. The fixed collar lets us pull local graphs backward up to every base face and then restrict them to B_Γ ; no completion of the physical staying set is introduced. In a product chart, if S is a graph slope with $\|S\| \leq \alpha_g$, the instantaneous tangent and normal quotient coefficients are

$$(A.16) \quad L = C + BS, \quad n = a - SB.$$

Consequently, under the one-chart sufficient criterion, the homogeneous part of the transform of an i th state jet has instantaneous normal-versus-base gap

$$(A.17) \quad n - i\mu(L) \geq \gamma_{g,i}^{\text{chart}} > 0.$$

This follows from $\mu_2(L) \leq c_g + b_g\alpha_g$ and $n \geq a_g - b_g\alpha_g$, and its lower bound is $\gamma_{g,i}^{\text{chart}}$ from (A.6). In the invariant formulation used by the theorem, integration over the fixed horizon δ_g is already encoded by $\vartheta_{g,i} < 1$. For $i = 0$ this controls a parameter variation at a fixed base point; for $i \geq 1$ it is the usual i -jet bunching inequality.

We now use one global finite-time graph transform. This is the finite-dimensional corridor version of the C'' section theorem and its Lipschitz-jet proof [11, Chapter 3]; we give the construction because the scale of the parameter derivatives is important here. The ambient collar in the theorem hypotheses retains the strict outward sign of the normal component outside the two normal faces and supports the time- δ_g map; put $\delta = \delta_g$. A segment satisfying the terminal graph equation below cannot cross a normal face and return before time δ : after its first such crossing the retained sign moves it farther into the corresponding outer collar. Let $\mathcal{L}_{\alpha_g}$ be the closed set of continuous graphs

$$h : B_\Gamma \longrightarrow [-\zeta_\Gamma, \zeta_\Gamma], \quad \text{Lip}(h) \leq \alpha_g.$$

For $h \in \mathcal{L}_{\alpha_g}$ and $x \in B_\Gamma$, follow (x, ζ_0) for time δ in the collar and impose the terminal condition

$$(A.18) \quad \pi_\zeta \Phi_\varpi^\delta(x, \zeta_0) = h \left(\pi_x \Phi_\varpi^\delta(x, \zeta_0) \right).$$

At the lower and upper initial normal faces the two sides have opposite strict signs. If $\zeta_0^2 > \zeta_0^1$, the difference of the two orbit segments begins in the vertical cone. The uniform secant-cone condition (or, under the one-chart test, (A.14)–(A.15)) and the Lipschitz bound on h show that the left side minus the right side in (A.18) is strictly increasing in ζ_0 . Hence there is one solution, denoted $(\mathcal{T}_\varpi h)(x)$. The same cone argument for two different base points proves $\mathcal{T}_\varpi h \in \mathcal{L}_{\alpha_g}$. At a base face the equation is solved in the ambient collar and restricted to the corresponding quadrant; strict inwardness or structural invariance keeps the resulting orbit in B_Γ .

The transform is a strict contraction in C^0 . To see the invariant estimate, join two terminal graphs by their affine path and differentiate (A.18) along that path. If P_- and P_+ are the initial and terminal tangent planes, respectively, the homogeneous variation is the terminal vertical displacement transported back by $(N^\delta(P_-))^{-1}$; the quotient norm was defined using the vertical identification. The $j = 0$ case of (A.2) therefore gives

$$(A.19) \quad \|\mathcal{T}_\varpi h_1 - \mathcal{T}_\varpi h_2\|_{C^0} \leq \vartheta_{g,0} \|h_1 - h_2\|_{C^0}.$$

The same inequality for Lipschitz terminal graphs follows directly from difference quotients (or by smooth approximation with the cone slope preserved). Thus $\mathcal{T}_\varpi$ has a unique fixed graph. Its forward time- δ image lies in the same graph, so it stays in the corridor for all positive time. Conversely the Lipschitz staying graph already constructed above satisfies (A.18). The two graphs are therefore the same Γ_ϖ .

It remains to record the jet contraction without losing compatibility in x . Start with the single global graph $h_0 = 0$ and put

$$(A.20) \quad h_{N,\varpi} = \mathcal{T}_\varpi^N h_0, \quad N = 0, 1, 2, \dots$$

Every $h_{N,\varpi}$ is one graph on all of B_Γ , is obtained from the same terminal graph equation, and is C^5 in the state variables. For a C^r coefficient family it is C^r in ϖ . Thus its local jets agree on overlaps and its compatible one-sided jets agree at every corner; this is precisely the compatibility absent from a pointwise choice of terminal germs.

We spell out the finite-order chain rule which is decisive for parameter regularity. In one parameter chart and one corner chart write

$$\Phi_\varpi^\delta(x, \zeta) = (X(\varpi, x, \zeta), Z(\varpi, x, \zeta)), \quad h = h_{N,\varpi}, \quad g = h_{N+1,\varpi},$$

and put

$$(A.21) \quad \beta(\varpi, x) = X(\varpi, x, g_\varpi(x)), \quad Q = Z_\zeta - D_y h_\varpi(\beta) X_\zeta.$$

The terminal equation is $Z(\varpi, x, g) = h_\varpi(\beta)$. In the quotient coordinates of the theorem, $Q = N^\delta(P_-)$, so Q^{-1} is exactly the inverse normal quotient

in (A.2). For a combined parameter–state direction $d = (v, u)$ set

$$\xi_d = (v, u, Dg[d]), \quad \tau_d = (v, D\beta[d]).$$

For directions d_a, d_b, d_c , define

$$\begin{aligned} \widehat{\beta}_{ab} &= D^2 X[\xi_a, \xi_b], & \widehat{Z}_{ab} &= D^2 Z[\xi_a, \xi_b], \\ \chi_{ab}^{(2)} &= (0, 0, D^2 g[d_a, d_b]), \\ \widehat{\beta}_{abc} &= D^3 X[\xi_a, \xi_b, \xi_c] + \sum_{\text{cyc}} D^2 X[\chi_{ab}^{(2)}, \xi_c], \\ \widehat{Z}_{abc} &= D^3 Z[\xi_a, \xi_b, \xi_c] + \sum_{\text{cyc}} D^2 Z[\chi_{ab}^{(2)}, \xi_c]. \end{aligned}$$

All derivatives of X, Z are evaluated at $(\varpi, x, g_\varpi(x))$, and derivatives of h at $(\varpi, \beta(\varpi, x))$. Twice and three times differentiating the terminal equation gives the exact identities

$$\begin{aligned} \text{(A.22)} \quad QD^2 g[d_1, d_2] &= D^2 h[\tau_1, \tau_2] + D_y h \widehat{\beta}_{12} - \widehat{Z}_{12}, \\ QD^3 g[d_1, d_2, d_3] &= D^3 h[\tau_1, \tau_2, \tau_3] + \sum_{\text{cyc}} D^2 h[(0, D^2 \beta[d_a, d_b]), \tau_c] \\ \text{(A.23)} \quad &+ D_y h \widehat{\beta}_{123} - \widehat{Z}_{123}. \end{aligned}$$

Thus the highest output jet is isolated on the left; the right side of (A.23) contains output jets only through order two.

To make the triangular order visible, put $A = D_x \beta$ and, when $r \geq 1$, $B = D_\varpi \beta$, and let $H_{i,j} = D_y^i D_\varpi^j h_\varpi(\beta)$. The first parameter derivative is the separate affine base recurrence

$$\text{(A.24)} \quad QD_\varpi g[v] = H_{0,1}[v] + D_y h X_\varpi[v] - Z_\varpi[v].$$

Indeed, $Bv = X_\varpi[v] + X_\zeta D_\varpi g[v]$, and the second term is already absorbed into Q on the left. Thus this recurrence contains no hidden current output jet. The order-two recurrences are

$$\begin{aligned} QD_x^2 g[u_1, u_2] &= H_{2,0}[Au_1, Au_2] + R_{2,0}, \\ \text{(A.25)} \quad QD_x D_\varpi g[u, v] &= H_{1,1}[Au; v] + H_{2,0}[Au, Bv] + R_{1,1}, \\ QD_\varpi^2 g[v_1, v_2] &= H_{0,2}[v_1, v_2] + \sum_{a \neq b} H_{1,1}[Bv_a; v_b] + H_{2,0}[Bv_1, Bv_2] + R_{0,2}. \end{aligned}$$

At total order three, the recurrences needed here are

$$\begin{aligned}
 QD_x^3 g[\mathbf{u}] &= H_{3,0}[\mathbf{Au}] + R_{3,0}, \\
 QD_x^2 D_\varpi g[u_1, u_2; v] &= H_{2,1}[Au_1, Au_2; v] + H_{3,0}[Au_1, Au_2, Bv] \\
 (A.26) \quad &+ R_{2,1}, \\
 QD_x D_\varpi^2 g[u; v_1, v_2] &= H_{1,2}[Au; v_1, v_2] + \sum_{a \neq b} H_{2,1}[Au, Bv_a; v_b] \\
 &+ H_{3,0}[Au, Bv_1, Bv_2] + R_{1,2}.
 \end{aligned}$$

For $2 \leq m = i + j \leq 3$, the same-total source terms are indexed by subsets S of the j parameter directions:

$$(A.27) \quad \sum_{S \subset \{1, \dots, j\}} H_{i+|S|, j-|S|}[\mathbf{Au}, (Bv_s)_{s \in S}; (v_t)_{t \notin S}].$$

The term $S = \emptyset$ is the homogeneous current jet; every nonempty S has larger state order. We therefore order jets first by total order and, at fixed total order, by decreasing state order:

$$\begin{aligned}
 \mathcal{I}_r &= \{(a, b) \in \mathbb{N}_0^2 : a + b \leq 3, b \leq r\}, \\
 (a, b) \triangleleft (i, j) &\iff a + b < i + j \text{ or } (a + b = i + j \text{ and } a > i), \\
 (A.28) \quad &(1, 0) \triangleleft (0, 1), \\
 &(2, 0) \triangleleft (1, 1) \triangleleft (0, 2), \\
 &(3, 0) \triangleleft (2, 1) \triangleleft (1, 2).
 \end{aligned}$$

After omitting pairs not in $\mathcal{I}_r$, the displayed chains give the order used below. Thus $\triangleleft$ also places every lower-total-order jet first. The next lemma records both the exact decomposition and the stability estimate used in the jet induction.

Lemma A.2 (Triangular remainder and adjacent-iterate estimate). *Work in one parameter chart and one closed-quadrant corner chart. Fix $(i, j) \in \mathcal{I}_r \setminus \{(0, 0), (1, 0)\}$, put $m = i + j$, and take $d_a = (0, u_a)$ for $1 \leq a \leq i$ and $d_{i+t} = (v_t, 0)$ for $1 \leq t \leq j$. In addition to the notation above, put*

$$G_{a,b} = D_x^a D_\varpi^b g, \quad H_{a,b} = D_y^a D_\varpi^b h_\varpi(\beta),$$

so that

$$A = X_x + X_\zeta G_{1,0}, \quad Q = Z_\zeta - H_{1,0} X_\zeta.$$

When $r \geq 1$, also put $B = X_\varpi + X_\zeta G_{0,1}$; no term involving B is present when $j = 0$. For an (i, j) -multilinear source jet K , define the homogeneous pullback operator by

$$(A.29) \quad (\mathcal{L}_i K)[\mathbf{u}; \mathbf{v}] = Q^{-1} K[\mathbf{Au}; \mathbf{v}].$$

(i) **Exact decomposition.** For $m \geq 2$, let $\mathcal{S}_{i,j}$ be the sum in (A.27)

restricted to nonempty subsets S , and set $\mathcal{S}_{0,1} = 0$. Define the lower outer-composition and inner-flow parts by

$$(A.30) \quad \begin{aligned} \mathcal{H}_{i,j} &= \begin{cases} \sum_{\text{cyc}} D^2 h[(0, D^2 \beta[d_a, d_b]), \tau_c], & m = 3, \\ 0, & m \leq 2, \end{cases} \\ \mathcal{F}_{i,j} &= \begin{cases} H_{1,0} X_{\varpi}[v] - Z_{\varpi}[v], & (i,j) = (0,1), \\ H_{1,0} \widehat{\beta}_{12} - \widehat{Z}_{12}, & m = 2, \\ H_{1,0} \widehat{\beta}_{123} - \widehat{Z}_{123}, & m = 3. \end{cases} \end{aligned}$$

Then the terminal equation has the exact mixed-jet form

$$(A.31) \quad Q D_x^i D_{\varpi}^j g[\mathbf{u}; \mathbf{v}] = H_{i,j}[\mathbf{A}\mathbf{u}; \mathbf{v}] + \mathcal{S}_{i,j} + \mathcal{H}_{i,j} + \mathcal{F}_{i,j}.$$

For $m = 2, 3$, the symbol in (A.25)–(A.26) is therefore $R_{i,j} = \mathcal{H}_{i,j} + \mathcal{F}_{i,j}$. The remainder in the solved graph-transform recurrence is exactly

$$(A.32) \quad \widetilde{R}_{i,j} = Q^{-1}(\mathcal{S}_{i,j} + \mathcal{H}_{i,j} + \mathcal{F}_{i,j}).$$

(ii) Triangular dependence. The three summands have the following exhaustive dependencies. The shifted-source part $\mathcal{S}_{i,j}$ contains only the same-total-order source jets $(i+s, j-s) \triangleleft (i,j)$, $s \geq 1$. The outer-composition part $\mathcal{H}_{i,j}$ contains source and current output jets of total order at most two. At total order m , the inner-flow part $\mathcal{F}_{i,j}$ contains derivatives of (X, Z) through order m , the source slope, and current output jets of total order at most $m-1$. In particular, the right side of (A.31) contains no unlisted current output jet of total order m , and every parameter derivative has order at most $j \leq 2$.

(iii) Adjacent-iterate estimate. Attach a superscript N to the objects in (A.32) when the source and output graphs are h_N and h_{N+1} . Suppose that the jets occurring above have a common $C^{0,\eta}$ bound, and write

$$e_N = \|h_{N+1} - h_N\|_{C^0}, \quad d_{a,b}^N = \|J_{a,b}^{N+1} - J_{a,b}^N\|_{C^0}.$$

On every bounded jet set on which Q^{-1} is uniformly bounded, there is one constant C such that, for $N \geq 1$,

$$(A.33) \quad \begin{aligned} \|\widetilde{R}_{i,j}^N - \widetilde{R}_{i,j}^{N-1}\|_{C^0} &\leq C(e_N^\eta + e_{N-1}^\eta + d_{1,0}^N + d_{1,0}^{N-1}) \\ &\quad + C \sum_{\substack{(a,b) \triangleleft (i,j) \\ (a,b) \in \mathcal{I}_r \\ (a,b) \notin \{(0,0), (1,0)\}}} (d_{a,b}^N + d_{a,b}^{N-1}), \end{aligned}$$

$$(A.34) \quad \|\mathcal{L}_i^N - \mathcal{L}_i^{N-1}\|_{C^0} + \|\beta_N - \beta_{N-1}\|_{C^0}^\eta \leq C(e_N^\eta + d_{1,0}^N + d_{1,0}^{N-1}).$$

The constants are uniform over the finite parameter and corner atlases.

Proof. Apply the set-partition form of the Faà di Bruno formula to $(X, Z) \circ (\varpi, x, g)$. The only partition containing the current m th derivative of g is the one-block partition. Its coefficients on the two sides of the terminal equation are Z_ζ and $H_{1,0}X_\zeta$; their difference is precisely Q , and hence that entire term is the left side of (A.31). Every other block has size at most $m-1$. Expanding the highest outer derivative $D^m h[\tau_1, \dots, \tau_m]$ amounts, for each parameter direction, to choosing either its explicit parameter component or its Bv state component. The all-explicit choice is $H_{i,j}[A\mathbf{u}; \mathbf{v}]$, and the other choices give exactly $\mathcal{S}_{i,j}$. Equivalently, the definitions give $D^2\beta = \hat{\beta}_{12} + X_\zeta D^2g$ and $D^3\beta = \hat{\beta}_{123} + X_\zeta D^3g$, which displays the same one-block cancellation at both orders. When $m = 3$, the lower outer partition uses $D^2\beta$, so $\mathcal{H}_{i,j}$ uses output jets only through order two. The exact second- and third-order formulas above now give (A.31) and all three dependency claims.

For the adjacent estimates, the $C^{0,\eta}$ bounds give

$$\|K \circ \beta_N - \overline{K} \circ \beta_{N-1}\|_{C^0} \leq \|K - \overline{K}\|_{C^0} + [\overline{K}]_\eta \|\beta_N - \beta_{N-1}\|_{C^0}^\eta.$$

The smooth finite-time flow and its derivatives are Lipschitz on the bounded jet set, while $\|\beta_N - \beta_{N-1}\|_{C^0} \leq Ce_N$. The differences of A and Q^{-1} are bounded by $C(e_N^\eta + d_{1,0}^N + d_{1,0}^{N-1})$; the difference of B has, in addition, the preceding term $d_{0,1}^N$ whenever that jet is present. Apply the finite-product difference identity to each of the finitely many contractions in $\mathcal{S}$, $\mathcal{H}$, and $\mathcal{F}$, and use $Q^{-1} - \overline{Q}^{-1} = Q^{-1}(\overline{Q} - Q)\overline{Q}^{-1}$. This proves (A.33) and (A.34). The identities first hold in the interiors of the corner charts and extend to their faces by continuity of the compatible one-sided jets. Equivalence of norms in the finite atlases makes C uniform. $\square$

This lemma explicitly accounts for every shifted-base term allowed by $r \leq 2$.

If K_+ is the highest i -state jet of the terminal graph, its homogeneous pullback is

$$K_- = (N^\delta(P_-))^{-1} \circ K_+ \circ (A^\delta(P_-))^{\otimes i}.$$

Hence, with $J_{i,j}^N = D_x^i D_\varpi^j h_{N,\varpi}$, the exact global form of the recurrence for $i+j \leq 3$, $(i,j) \notin \{(0,0), (1,0)\}$, is

$$(A.35) \quad J_{i,j}^{N+1}(x) = \mathcal{L}_i^N(x)[J_{i,j}^N(\beta_N(x))] + \tilde{R}_{i,j}^N(x), \quad \|\mathcal{L}_i^N\| \leq \vartheta_{g,i} < 1.$$

The graph $(0,0)$ is governed by the nonlinear contraction (A.19), and the slope $(1,0)$ by the projective transform treated below. For every remaining jet, the remainder contains only lower-total-order source or output jets and same-total-order source jets preceding (i,j) in (A.28). This is a finite triangular recursion, not a Fréchet differentiation of a map $C^3 \rightarrow C^3$.

We next control the shifted evaluation in (A.35). Compactness gives

$$M_g := \sup_{\varpi, h \in \mathcal{L}_{\alpha_g}} \max\{1, \text{Lip}(\beta_h)\} < \infty.$$

Choose $0 < \eta < 1$ so that

$$(A.36) \quad \vartheta_{g,i} M_g^\eta < 1, \quad 0 \leq i \leq 3,$$

and also for $i = 4$ for the optional C^4 conclusion. In a closed quadrant of a corner chart, the elementary difference-quotient estimate is

$$(A.37) \quad [\mathcal{L}_i^N(K \circ \beta_N)]_\eta \leq \vartheta_{g,i} M_g^\eta [K]_\eta + [\mathcal{L}_i^N]_\eta \|K\|_{C^0}.$$

The coefficient seminorm in the second term depends only on the bounded flow derivatives and preceding jets. Taking C^0 norms first in (A.35), and following (A.28), gives uniform sup-norm bounds for all target jets; the graph and slope bounds come respectively from (A.19) and the invariant cone.

The slope is the one nonlinear jet base case. The projective tangent transform and the same difference-quotient calculation give, for some $\theta_{1,\eta} < 1$,

$$(A.38) \quad [Dh_{N+1,\varpi}]_\eta \leq \theta_{1,\eta} [Dh_{N,\varpi}]_\eta + C.$$

For every remaining pair choose $\vartheta_{g,i} M_g^\eta < \theta_{i,\eta} < 1$. The recurrence (A.24), together with (A.25)–(A.27) and (A.37) then give

$$(A.39) \quad [J_{i,j}^{N+1}]_\eta \leq \theta_{i,\eta} [J_{i,j}^N]_\eta + C_{i,j} \left(1 + \sum_{\substack{(i',j') \triangleleft (i,j) \\ (i',j') \in \mathcal{I}_r}} ([J_{i',j'}^N]_\eta + [J_{i',j'}^{N+1}]_\eta) \right).$$

Every $N+1$ term on the right has already been obtained in the finite order (A.28). Starting from $h_0 = 0$ and using (A.38) for the base case therefore gives uniform Hölder bounds. This is the corridor version of the Hölder-jet step in the C^r section theorem [11, Chapter 3], with the decisive difference quotient displayed in (A.37).

For adjacent iterates, the induced projective tangent transform is a fiber contraction with constant $\theta_1 < 1$; the Hölder bound (A.38) controls its shifted terminal evaluation. Consequently

$$(A.40) \quad \|Dh_{N+1,\varpi} - Dh_{N,\varpi}\|_{C^0} \leq \theta_1 \|Dh_{N,\varpi} - Dh_{N-1,\varpi}\|_{C^0} + C \|h_{N,\varpi} - h_{N-1,\varpi}\|_{C^0}^\eta.$$

Together with (A.19), this proves uniform C^1 convergence.

For the remaining jets, fix numbers

$$(A.41) \quad \vartheta_{g,i} < \theta_i < 1, \quad 0 \leq i \leq 3,$$

enlarging the already chosen slope constant θ_1 , if necessary, without reaching one. Set

$$e_N = \|h_{N+1} - h_N\|_{C^0} \leq C \vartheta_{g,0}^N.$$

Since $\beta_N(x) = X(\varpi, x, h_{N+1, \varpi}(x))$, $\|\beta_N - \beta_{N-1}\|_{C^0} \leq Ce_N$. The uniform Hölder bounds give the explicit shifted-point estimate

$$(A.42) \quad \begin{aligned} & \|J_{i,j}^N \circ \beta_N - J_{i,j}^{N-1} \circ \beta_{N-1}\|_{C^0} \\ & \leq \|J_{i,j}^N - J_{i,j}^{N-1}\|_{C^0} + [J_{i,j}^{N-1}]_\eta \|\beta_N - \beta_{N-1}\|_{C^0}^\eta. \end{aligned}$$

Write $d_{i,j}^N = \|J_{i,j}^{N+1} - J_{i,j}^N\|_{C^0}$. Choose $\max\{\vartheta_{g,0}^\eta, \theta_1\} < \kappa_0 < 1$. The slope recurrence (A.40) gives $d_{1,0}^N \leq C\kappa_0^N$. Lemma A.2 now shows that coefficient changes, shifted evaluations, and every nonpreceding remainder term are bounded by $C\kappa_0^N$. Comparing adjacent instances of (A.35) therefore gives, for $(i, j) \notin \{(0, 0), (1, 0)\}$,

$$(A.43) \quad d_{i,j}^N \leq \theta_i d_{i,j}^{N-1} + C \sum_{\substack{(i', j') \triangleleft (i, j) \\ (i', j') \in \mathcal{I}_r \\ (i', j') \notin \{(0, 0), (1, 0)\}}} (d_{i', j'}^N + d_{i', j'}^{N-1}) + C\kappa_0^N.$$

Choose $\max\{\kappa_0, \theta_i : 0 \leq i \leq 3\} < \rho_g < 1$. At total order m , the only convolution required by the induction is bounded by

$$(A.44) \quad \sum_{k=0}^{N-1} \theta_i^{N-1-k} (1+k)^{m-1} \rho_g^k \leq C_m (1+N)^{m-1} \rho_g^N \sum_{\ell \geq 0} (\theta_i / \rho_g)^\ell.$$

Hence

$$(A.45) \quad \|J_{i,j}^{N+1} - J_{i,j}^N\|_{C^0} \leq C_{i,j} (1+N)^{i+j} \rho_g^N, \quad i+j \leq 3, \quad j \leq r.$$

The displayed polynomial is the finite triangular-convolution loss; after slightly enlarging ρ_g it is a deliberately loose bound.

The right side of (A.45) is summable. To identify the limiting jets at corners, represent B_Γ by closed quadrants in a finite compatible corner atlas. Derivatives are first computed in the interior and extended continuously to each face. Every h_N is one global function, so its jets obey the usual transition law on the dense interior of every chart overlap; continuity gives the same law for the one-sided jets at faces and corners. Applying the one-dimensional fundamental theorem of calculus successively in the quadrant coordinates, and then in parameter charts, identifies the uniform limits with the classical mixed derivatives. This proves (A.10) and (A.11). The C^5 coefficient budget covers the Hölder coefficient estimates through total order three; the same induction with $i = 4, j = 0$ proves the optional state- C^4 assertion under (A.12).

Finally, strict inward and exit inequalities persist in an ordinary C^1 neighborhood. A designated invariant base face persists only in the relative coefficient subspace that preserves it, as stated in the theorem. The rate gaps persist in C^1 , and the finite jet bounds depend continuously on the C^5 coefficients and on the C^r norm of the parameter family. This proves the asserted openness and family dependence. $\square$

APPENDIX B. WEIGHTED RETURN–EXIT DATA AND PARAMETER DEPENDENCE

The field topology of Definition 2.2 and the return–exit topology used after the first-return/first-exit construction are different objects. The former contains input coefficients but no orbit labels. The latter is a fixed-domain space for comparing the resulting families of Poincaré return and first-exit maps. We define the latter here and record the parameter estimates used in Theorem 3.3.

B.1. Coefficient topology. The product norm on $\mathfrak{H}_{\text{adm}}^k$ controls the C^k saddle Hadamard coefficients, the C^k coefficients of the marked local exact primitive $\tilde{\lambda}$ (and hence of $a = \tilde{\lambda}(X)$) on the fixed saddle bidisk, the C^k exact section charts, and the numbers $(\alpha_{\mathbf{X}}, \beta_{\mathbf{X}}, b_{\mathbf{X},\sigma}, t_{\mathbf{X},\sigma})$. It also controls the derived compatibility $c_{\mathbf{X},\sigma} = e^{\alpha_{\mathbf{X}} t_{\mathbf{X},\sigma}}$, the weighted local remainders in (2.13) and (2.15), and the $C^k(K_{\text{fin}})$ coefficients of (X, H, λ) .

At the algebraic end it controls $(Y_a, \hat{\chi}_a, a_a^c)$ in C^k and (v_e, f_e, f_E, v_b) in C^{k-1} . At the pole it controls the norm (2.28), the centering, and the linear normalization in (2.25)–(2.29). Finally, it controls the marked section embeddings and transitions among the saddle, finite-flight, and algebraic charts in C^k , and the pole charts in C^{m_p+3} . Event-face functions, the source, outgoing, and return cell charts, and transitions used only for the event stratification are controlled in C^3 . Parameter derivatives are taken componentwise in these respective Banach norms. This is an input-coefficient topology: it contains no return strip, orbit label, first-exit component, roof, or action potential.

B.2. Weighted C^2 topology on the branch data. Fix the two-vertex graph, the edge labels $a = (\sigma, n, \sigma')$, the winding and sign labels, the half-open boundary convention, and the finite first-exit cell types produced at the base field. Put $J = [0, 1]$. The closed extension of every return strip is identified with $I \times J$, with regular normal coordinate in $(0, 1)$. A first-exit component of dimension $d < 2$ is represented in a fixed cell $D_{\mathbf{t},\ell}$ as the zero set of

$$\mathfrak{d}_{\mathbf{t},\ell} : D_{\mathbf{t},\ell} \longrightarrow \mathbb{R}^{2-d},$$

with a uniform lower bound on its least singular value; a two-dimensional cell is represented by strict boundary inequalities. These representations include the finite first-hit map and the renormalized end label.

Each labeled datum carries its local exponent

$$\Lambda_{\mathcal{D}} = \frac{2\pi\alpha_{\mathcal{D}}}{\beta_{\mathcal{D}}} > \kappa, \quad \epsilon_{\mathcal{D},a} = \nu_N^{-1} e^{-\Lambda_{\mathcal{D}} n(a)}, \quad \tilde{g}_a = g_a / \epsilon_{\mathcal{D},a}.$$

This branchwise scale is not frozen at the core exponent. Put $p_1(n) = 1 + n$, $p_3(n) = (1 + n)^3$, and, for each fixed pair of source and target signs, let $h_{\sigma,\sigma',\infty}$ be the fixed half-chart restriction of the source-sign limit $h_{\sigma,\infty}$, for

$h \in \{f, \tilde{g}, \hat{\tau}, B\}$. Put $\Delta_a h = h_a - h_{\sigma(a), \sigma'(a), \infty}$. For a labeled datum $\mathcal{D}$, define its recurrent weighted size functional by

$$\begin{aligned}
 \|\mathcal{D}\|_{\text{reg}, \kappa} := & |c_{\mathcal{T}}| + |\Lambda_{\mathcal{D}}| + \sup_a (\|f_a\|_{C^2} + \|\tilde{g}_a\|_{C^2} + \|\hat{\tau}_a\|_{C^2} + \|B_a\|_{C^2}) \\
 & + \sum_{(\sigma, \sigma') \in \mathcal{G}_N} (\|f_{\sigma, \sigma', \infty}\|_{C^2} + \|\tilde{g}_{\sigma, \sigma', \infty}\|_{C^2}) \\
 & + \sum_{(\sigma, \sigma') \in \mathcal{G}_N} (\|\hat{\tau}_{\sigma, \sigma', \infty}\|_{C^2} + \|B_{\sigma, \sigma', \infty}\|_{C^2}) \\
 & + \sup_{a \text{ with fixed signs}} \left\{ \frac{e^{\kappa n(a)}}{p_3(n(a))} (\|\Delta_a f\|_{C^2} + \|\Delta_a \tilde{g}\|_{C^2}) \right. \\
 & \left. + \frac{e^{\kappa n(a)}}{p_1(n(a))} (\|\Delta_a \hat{\tau}\|_{C^2} + \|\Delta_a B\|_{C^2}) \right\}.
 \end{aligned}
 \tag{B.1}$$

For $h \in \{\hat{\tau}, B\}$ put

$$\Delta_{\mathbf{t}, \sigma, n, \ell} h = h_{\mathbf{t}, \sigma, n, \ell}^{\text{ren}} - h_{\mathbf{t}, \sigma, \infty, \ell}^{\text{ren}}.$$

Let $\mathcal{F}_{\text{hw}}$ be the set of quadruples $(\mathbf{t}, \sigma, n, \ell)$ in the fixed-sign high-winding first-exit families, and let $\mathcal{F}_{\infty}$ be its projection to $(\mathbf{t}, \sigma, \ell)$. Define the first-exit-potential weighted size functional by

$$\begin{aligned}
 \|\mathcal{D}\|_{\text{exit}, \kappa} := & \sup_{(\mathbf{t}, \sigma, n, \ell) \in \mathcal{F}_{\text{hw}}} (\|\hat{\tau}_{\mathbf{t}, \sigma, n, \ell}^{\text{ren}}\|_{C^2} + \|B_{\mathbf{t}, \sigma, n, \ell}^{\text{ren}}\|_{C^2}) \\
 & + \sum_{(\mathbf{t}, \sigma, \ell) \in \mathcal{F}_{\infty}} (\|\hat{\tau}_{\mathbf{t}, \sigma, \infty, \ell}^{\text{ren}}\|_{C^2} + \|B_{\mathbf{t}, \sigma, \infty, \ell}^{\text{ren}}\|_{C^2}) \\
 & + \sup_{(\mathbf{t}, \sigma, n, \ell) \in \mathcal{F}_{\text{hw}}} \frac{e^{\kappa n}}{p_3(n)} (\|\Delta_{\mathbf{t}, \sigma, n, \ell} \hat{\tau}\|_{C^2} + \|\Delta_{\mathbf{t}, \sigma, n, \ell} B\|_{C^2}).
 \end{aligned}
 \tag{B.2}$$

Use the first-exit defining functions, labelled sign patterns, cell maps, first-hit maps, and the reciprocals of the positive rank, event, cone, and face-ordering margins as additional chart coordinates. For two data sets, their distance is obtained by applying (B.1)–(B.2) to coefficient differences after the fixed chart identifications and adding the differences of these additional coordinates. The function $\tilde{g}$ is compared only after each datum is divided by its own branchwise $\epsilon_{\mathcal{D}, a}$, and $\Lambda_{\mathcal{D}}$ is compared separately. The resulting labeled space is denoted by $\mathfrak{X}_{\kappa}^2$; this notation is reserved here for the full- C^2 roof/action topology. A single expression is a fixed-presentation metric chart. For two admissibly related presentations, Theorem 5.7 pulls both charts to a common finite refinement, applies the deck and endpoint corrections, and proves that the overlap map is locally bi-Lipschitz. These overlap maps therefore define the same weighted topology on the physical output data.

For a finite-dimensional C^r family $\varpi \mapsto \mathcal{D}_\varpi$, $r \in \{1, 2\}$, the order- r enhanced norm means exactly that every displayed C^2 norm is replaced by

$$(B.3) \quad \|h\|_{C_{z,\varpi}^{2,r}} := \max_{\substack{i+j \leq 2 \\ j \leq r}} \sup_{(z,\varpi)} \|D_z^i D_\varpi^j h(z, \varpi)\|,$$

with the same tail weights, including the derivatives of $\Lambda_{\mathcal{D}_\varpi}$ and $\tilde{g}_{\varpi,a}$. Differentiating the physical factor $\epsilon_{\varpi,a}$ produces at most $n^j \epsilon_{\varpi,a}$ for $j \leq r$, which is absorbed by the gap between the two exponential rates. This definition controls mixed total two-jets for a family of data that has already been constructed. Under an admissible parameter-dependent change of presentation, the same definition is applied on the finitely many total-space strata, using the uniform extensions to their closures from Definition 5.6.

The unrenormalized first-exit time is intentionally absent from (B.2). Exactness, coherent orbit-integral normalization, the Markov incidence identities, and the first-hit covering identity define a closed subclass of $\mathfrak{X}_\kappa^2$. They are not manufactured by smallness of the metric.

Proposition B.1 (Openness in the weighted space of return and first-exit maps). *Suppose a labeled datum in $\mathfrak{X}_\kappa^2$ satisfies the four hypotheses of Theorem 6.1, with positive contraction, angular, upper-normal, rescaled transverse-action, rank, and event margins and a fixed labelled sign pattern. Every sufficiently close datum in the same admissible labeled class has the conclusions of Theorem 6.1, with uniform constants.*

Proof. Smallness in (B.1) preserves a contraction constant below one, the angular and upper-normal margins, and the positive lower bound for $\tilde{g}$. Thus each finite edge still has $g_a \geq m_{\text{cf}} \epsilon_{\mathcal{D},a} > 0$, although these physical distances correctly tend to zero with the winding. The fixed-point proof of Theorem 6.1 is therefore uniform. The weighted terms preserve the two source-sign limits, their fixed next-sign restrictions, and the typed first-exit limits.

On each fixed reference cell, closeness in the weighted metric gives the C^1 closeness of its finite defining-function family required by Proposition 5.2 with $s = 1$. For the return-band faces inside homoclinic cells, use the same two-stage relative application as in the proof of Proposition 5.4. The resulting component bijection preserves every dimension and label and excludes both an interior split and a component entering through a cell face. Event speed and ordering margins preserve the first-hit property. Exact action identities remain exact because closeness is taken inside the coherently normalized exact subclass. $\square$

B.3. Parameter derivatives and locality of variations. Let ϖ range over a compact finite-dimensional parameter manifold. For a family of recurrent fixed-point operators $\mathcal{T}_\varpi$, the derivative of its fixed point u_ϖ is

$$(B.4) \quad D_\varpi u_\varpi = (I - D_u \mathcal{T}_\varpi)^{-1} D_\varpi \mathcal{T}_\varpi(u_\varpi), \quad \|(I - D_u \mathcal{T}_\varpi)^{-1}\| \leq (1 - \vartheta)^{-1}.$$

A second differentiation gives the same inverse applied to the usual three terms containing $D_\varpi^2 \mathcal{T}$, $D_u D_\varpi \mathcal{T} D_\varpi u$, and $D_u^2 \mathcal{T} [D_\varpi u, D_\varpi u]$.

Proposition B.2 (Uniform parameter two-jets). *Let $\varpi \mapsto \mathcal{D}_\varpi$ be C^r , $r \in \{1, 2\}$, in the order- r enhanced version of $\mathfrak{X}_\kappa^2$ which controls all mixed state/parameter derivatives of total order at most two and parameter order at most r . Suppose all margins in the four hypotheses of Theorem 6.1 are uniform. Then:*

- (1) *every two-sided coded point and every one-sided stable-plaque graph depends C^r on ϖ , uniformly in the code;*
- (2) *all specified parameter derivatives of the coded renormalized roof and action have uniformly summable variations; and*
- (3) *every first-exit component is C^r in its fixed defining-function chart, and every fixed finite return word followed by a first exit has an action that may be differentiated r times.*

No estimate uniform in an unbounded finite-word length is asserted without an additional composition weight.

Proof. Formula (B.4) and its second derivative prove the uniform parameter bounds for coded points. The half-line operator has the same inverse and leaves the initial free coordinate untouched, proving the stable-plaque assertion. The parameter-dependent implicit-function theorem gives the first-exit components.

It remains to prove summable variations rather than mere uniform differentiability. The derivative $D_u \mathcal{T}_\varpi$ is nearest-neighbor in the sequence index. For an integer L split its inverse as

$$(B.5) \quad (I - D_u \mathcal{T}_\varpi)^{-1} = \sum_{j=0}^{L-1} (D_u \mathcal{T}_\varpi)^j + (D_u \mathcal{T}_\varpi)^L (I - D_u \mathcal{T}_\varpi)^{-1}.$$

The zeroth coordinate of the finite sum sees only indices at distance at most L . If two codes agree on $[-m, m]$, their fixed-point coordinates on the smaller block differ by at most $C\vartheta^{m-L}$. Hence the coefficient and inhomogeneous-term differences in the finite sum have that bound, while the two remainders in (B.5) are bounded by $C\vartheta^L$. Taking $L = \lfloor m/2 \rfloor$ gives

$$(B.6) \quad |D_\varpi u_\varpi(\omega)_0 - D_\varpi u_\varpi(\tilde{\omega})_0| \leq C\hat{\vartheta}^m$$

for some $\hat{\vartheta} < 1$. A second derivative produces only a fixed polynomial factor in m , which is absorbed after replacing $\hat{\vartheta}$ by a slightly larger number below one. Composition with the controlled mixed state/parameter jets of the roof and action proves a geometric, hence summable, variation bound. $\square$

For the coefficient-to-map application, the enhanced norm in Proposition B.2 is a conclusion only for those outputs whose defining operators have the required uniform inverses. The following proposition records exactly the part proved here.

Proposition B.3 (Mixed regularity of return, first-hit, and end data). *Let $\mathbf{X}_* \in \mathfrak{H}_{\text{adm}}^k$ satisfy Definition 2.8, and fix $\kappa < \kappa_+ < \kappa_{\text{tr}}$ as in (4.46). Decrease a relatively open coefficient neighborhood $\mathcal{U}$ of $\mathbf{X}_*$ so that the strict margins in (H1)–(H4) are retained uniformly: the homoclinic matching and interior-block margins in (H1); the rank, boundary-neatness, event-speed, and event-order margins and labelled sign pattern in (H2); the local-maximality neighborhood, normal-expansion, order-three bunching, cut, and saturation margins in (H3); and the common-basin, radial, spectral-gap, and entry margins in (H4).*

Let $\mathcal{P}$ be a compact finite-dimensional parameter manifold. Every C^r family $\varpi \mapsto \mathbf{X}_\varpi$ from $\mathcal{P}$ into $\mathcal{U}$, $r \in \{1, 2\}$, produces C^r families of return maps, compact first-hit maps, recurrent flight-time remainders, recurrent actions, and the two renormalized noncompact-end time/action limits. For each of these outputs and every $i + j \leq 2$ with $j \leq r$, the derivative $D_z^i D_\varpi^j$ has the same $e^{-\kappa n}$ weight as its state-only counterpart. The pole conclusion uses (2.30).

Proof. All equations are solved on domains fixed before the field is varied. We list the maps and their domains.

- (1) The marked time law is first inverted by the scalar fixed-point map (4.49). Its derivative is at most $1/2$ for all $n \geq N$, so its inverse is at most 2 in the normalized state/parameter two-jet space. The mixed passage is the contraction on the fixed Banach space $\mathcal{E}$ in (4.11); its inverse is bounded by $(1 - \vartheta)^{-1}$. Endpoint evaluation and the five-term functional (4.34) are bounded maps to the weighted trace and recurrent-action C^2 spaces.
- (2) The matching map is $\mathcal{F}_{\sigma,n} : \mathcal{N}_\sigma \rightarrow \mathbb{R}^{\dim y}$. The centered Newton tube supplies the finite-dimensional inverse M_{sel} uniformly in n . Composition with the fixed vertex charts maps its roots to $C^2(I \times J)$ cross forms on the closed strips, including the rescaled transverse-action component $\tilde{g}$.
- (3) Each finite event time solves $g(\Phi^\tau z) = 0$ in C^2 on a fixed compact cell; the scalar inverse is bounded by m_{hit}^{-1} . The sign-vector reference charts (5.12) use only fixed C^3 pullbacks.
- (4) The transverse homoclinic transition and the finitely many compact tube, section, and incidence maps continue from (H1)–(H2). The parameter-dependent implicit-function theorem and Lemma 5.3 give their uniform finite-dimensional and first-hit inverses from the stated matching, rank, and event-speed margins.
- (5) The algebraic future hypersurface and its mixed total-three jets are supplied by Proposition 2.9; the backward jet propagators contract by the factors $\vartheta_{a,i} < 1$ in (2.5). If (H3) is verified by an interval block, Theorem A.1 constructs the same hypersurface, and (A.7) bounds its factors by $e^{-\gamma_{g,i}^{\text{chart}} \delta_g}$ in the single product chart used for the concrete verification. On the fixed half-line, the reference and

aligned difference are the fixed points of

$$\mathcal{R} : \mathcal{P} \times \mathcal{Y}_1 \rightarrow \mathcal{Y}_1, \quad \mathcal{D} : I_{b_0} \times \mathcal{P} \times \mathcal{X}_\eta \rightarrow \mathcal{X}_\eta,$$

with inverse bounds $(1 - \theta_r)^{-1}$ and $(1 - \theta_d)^{-1}$ from (D.9) and (D.11). The cutoff density differences have one auxiliary mixed derivative beyond the C^2 output norm by (D.19).

- (6) At the pole, the compact first hit to $\rho = \rho_0$, the fixed- ρ phase integral (D.24), and the uniformly stable dichotomy act on fixed domains. Proposition D.9 gives a common integrable $s^{-\delta}$ majorant, $0 < \delta < 1$, through the auxiliary mixed three-jets.

The strict isolating-neighborhood, end-face, and ordering inequalities keep every domain fixed. None of these spaces or inverse bounds contains an orbit selected from the output maps.

The marked C^k input with $k \geq 5$ supplies all coefficient, flow, chart, and variational derivatives used in these six maps. For the scalar, Volterra, matching, and first-hit equations, differentiate each fixed-point or implicit equation once and twice; the same displayed inverse acts on the differentiated coefficient terms, and second derivatives add only products of already bounded first derivatives. For the algebraic staying graph use instead the scale estimates (A.35)–(A.45); this avoids differentiating a composition operator twice in the same C^3 space. The scalar clock inversion and Lemma 4.1 give only fixed polynomials in $1 + T$ as long-flight factors. Since $T = 2\pi n/\beta + O(1)$, the gap $\kappa_+ - \kappa$ absorbs every such polynomial. At the algebraic and pole ends, differentiation under the cutoff integral is justified by the common integrable majorants in Appendix D. At both ends the mean-value formula and chain rule use the auxiliary mixed third derivative before composition with the mixed C^3 first-hit labels; this preserves the displayed $e^{-\kappa n}$ rate in the output two-jets. Fixed Reference-domain pullbacks have bounded two-jets and introduce no additional inverse. This proves the asserted coefficient-to-map estimates and their tail weights. $\square$

B.4. Compactification by the two high-winding limit points. For clarity, we record a minor topological point used in the main text. The single-copy families printed in [20] have the form

$$\{0^+, 0^-\} \cup \{n^{-1} : n \geq k\}, \quad \{0^-, 0^+\} \cup \{-n^{-1} : n \geq k\}.$$

As written, they do not satisfy the basis-refinement axiom: the first set contains 0^- , but no set in the second family is contained in it, and conversely. We therefore use only the published set-level admissibility relation. The topology in (6.10)–(6.11) is defined independently from the two signed geometric limits. It is Hausdorff and metrizable; no one-sided point conjugacy or topology on the discrete first-exit outcomes is inferred from the older display.

APPENDIX C. VALIDATED NUMERICAL INGREDIENTS AND REPRODUCIBILITY

This appendix states exactly what is delegated to interval computation. The analytic arguments in the paper use the computer-assisted results only through finite roots, tangent equations, first-hit order, endpoint containment, and strict inequalities. No plotted continuation and no floating trajectory is a proof input.

C.1. Parametric roots and common-root bridges. Let $F : X \times I \rightarrow \mathbb{R}^d$ be a C^1 shooting equation, where X is an interval box and I a scalar parameter interval. Given a center $x_0 \in X$ and a point preconditioner C , its parametric Krawczyk image is

$$(C.1) \quad K(X, I) = x_0 - CF(x_0, I) + (I_d - CD_x F(X, I))(X - x_0).$$

Lemma C.1 (Interval continuation contract). *Suppose*

$$(C.2) \quad K(X, I) \subset \text{int } X, \quad \sup_{(x,p) \in X \times I} \|I_d - CD_x F(x, p)\| < 1.$$

Then, for every $p \in I$, $F(\cdot, p) = 0$ has a unique root in X , the Jacobian is nonsingular there, and the roots form a C^1 graph over I . If an interval Krawczyk solve for

$$(C.3) \quad D_x F(x(p), p)x'(p) = -D_p F(x(p), p)$$

is also strict, its enclosure contains the tangent of that graph.

For adjacent boxes (X_j, I_j) and (X_{j+1}, I_{j+1}) , suppose a common $p_ \in \text{int}(I_j \cap I_{j+1})$ has a point-parameter root enclosure strictly contained in both uniqueness boxes, with its next-chart parameter strictly inside I_{j+1} . Then the two local root graphs have an actual common root. A chain of such tests is one connected C^1 branch.*

Proof. The first assertion is the interval Newton–Krawczyk theorem [16], uniformly in the compact parameter interval. The contraction inequality gives nonsingularity and uniqueness, while the classical implicit-function theorem gives the C^1 graph. Applying the same theorem to the affine linear system (C.3) proves the tangent enclosure. At an adjacent bridge, uniqueness in both boxes identifies the two enclosed roots at p_* ; induction joins the entire chain. The interval inverse and tangent enclosures are instances of rigorous sensitivity control for nonlinear systems [23]; [25] gives broader background on the validated-numerics framework. $\square$

Every multiple-shooting problem below appends physical time as a state with $T' = 0$. Continuity equations join interval-enclosed flow segments, source equations impose the reversible plane or the true local unstable graph, and final equations impose an event fixed in advance and the physical future graph. The true local graphs are not replaced by their Taylor polynomials: certified C^0 and C^1 remainder budgets are inserted as interval parameters into every value and Jacobian evaluation.

Lemma C.2 (Validated first-hit criterion). *Let a shooting box enclose an orbit segment ending on $g = 0$. Suppose interval integration proves $g < 0$ on the entire segment before one final time slab, proves $Xg \geq m > 0$ on that slab, and proves strict inequalities for all competing faces. Then the enclosed event is the unique first $g = 0$ event for every orbit represented by the box. The conclusion is stable under every uncertainty included in the interval initial box and field coefficients.*

Proof. There is no earlier zero because the complete pre-event enclosure has strict sign. Strict monotonicity on the final slab gives at most one zero, and the final enclosure gives existence. Competing faces retain the wrong sign throughout, so none occurs first. $\square$

C.2. Validated computations. The tables retain the claim, the most constraining quantitative output, and the exact claim-to-certificate path. Complete box counts, full certificate, source, dependency, and bulk-output hashes, and toolchain pins remain in the machine-readable JSON and release metadata rather than in the mathematical narrative.

Main-theorem claim	Strict quantitative output	Code
Physical future graph and critical point	248 unknowns; root Krawczyk ratio .2675405 and contraction ratio .0520905	M1
Origin-to-algebraic connection	148 unknowns and 36 shooting segments; Krawczyk ratio .182170557 and contraction ratio .182169328	M2
Primary symmetric homoclinic	2 shooting variables with interval flow and first variation; Krawczyk ratio .000753174 and determinant bounds 149.563930553 and 149.564042277	M3
True-unstable pole entry	400 phase boxes of width .001; $y > 26.31$, $H_{\text{cone}} > 262.34$, and $H'_{\text{cone}} > 1704.01$ at the first $x = 10$ hit	M4

The certificate paths for this first table are, in order:

M1: `validation/future-target-fold/certificate.json`;

M2: `validation/origin-algebraic-heteroclinic/certificate.json`;

M3: `validation/universal-core-symmetric-homoclinic/certificate.json`;

M4: `validation/origin-unstable-pole-entry/certificate.json`. The first certificate supplies the physical target graph used by the second one. The homoclinic, algebraic, and pole certificates supply the finite margins used directly in Theorem 3.1.

Optional selected-branch claim	Strict quantitative output	Code
Selected branch to first critical point	17,345 main and 2,002 fine boxes; maximal Krawczyk ratios .934650, .740032, .912136; minimum cap margin $5.00006352 \cdot 10^{-10}$	S1
Fixed critical point to first event	interval flow through 37 shooting nodes; minimum state-containment margin $2.99988166 \cdot 10^{-8}$	S2
First finite collar	185 unknowns; 6,316 boxes; maximal root, tangent, and contraction ratios .8456984, .6733192, .7155494	S3
Finite collar to local zero-action boundary	545 unknowns; 9,725 boxes; maximal root, tangent, and contraction ratios .9492402115, .9991041442, .7733951226; minimum bridge margin $1.7765608 \cdot 10^{-9}$	S4
Fixed radial local zero-action boundary	fixed source annulus and exit-target enclosure; strict endpoint containment and local-arm uniqueness	S5

All adjacent interfaces in the two long covers pass true common-root tests. The complete interface margins and box ledgers are in the following exact certificate paths:

S1: `validation/exact-source-outer-fold/certificate.json`;

S2: `validation/fixed-fold-event-bridge/certificate.json`;

S3: `validation/finite-source-intermediate-collar/certificate.json`;

S4: `validation/finite-source-intermediate-collar/spiral_extension_certificate.json`;

S5: `validation/fundamental-annulus-overlap/certificate.json`. The first computation identifies its final augmented zero, in full state and tangent variables, with the nondegenerate critical point used by the next continuation. The five certificates therefore prove one connected selected branch from c_0 through its first outer critical point to the local spiral overlap. They do not classify any other component of $\mathcal{W}_a \cap \text{Fix } \mathcal{R}$.

C.3. Reproducibility and data availability. The validation directory contains one fail-closed entry point,

`validation/replay_all.py`,

together with the dependency graph `validation/replay_manifest.json` and the machine-readable environment contract `validation/environment.lock.json`. The manifest separates the four packages used by the normal-form realization of the main theorem, the five additional packages used only for the selected c_0 branch, and two ancillary packages. Among the eleven mathematical packages, four can reconstruct their certificate and seven rerun the claim-bearing source checks without reconstructing the historical JSON; none is classified as historical-JSON-only.

The driver first audits every declared source and dependency hash, then checks the pinned toolchain, and finally executes the package gates in dependency order in a temporary directory outside the repository. A dry run

or a toolchain preflight is labelled explicitly and never sets the full-replay flag. A full profile is labelled **PASS** only after all interval exit tests and the declared exact, canonical, or semantic certificate comparisons have passed. The environment contract pins CAPD 6.1.0 at commit 731079217a92, the FILIB backend, the static library hashes, the rounding-math marker, and the Python, MPFR, compiler, and CMake versions. The computations use CAPD [13] and FILIB [17]; for general validated-numerics background see [25].

The stale conditions disclosed in the preceding development version have been repaired. The future-target package was rebuilt after replacing its ordinary floating-point norm and rate reductions by CAPD interval reductions, and its source and certificate hashes were renewed. The exact-source package now resolves both archived analytic inputs at their repository paths and was regenerated. The pinned CAPD static library was reproduced byte for byte; the dependence of its byte hash on the recorded source path through `__FILE__` is stated explicitly in the environment and toolchain audit.

A manuscript build is *replay-release-eligible* only if the fail-closed driver runs the **all** profile from the exact fixed clean commit, all twelve manifest nodes pass, the clean preflight and postflight agree, and the machine-readable report sets both **profile_replay_pass** and **release_eligible** to true. This internal flag certifies the local replay contract only; it does not assert archival or journal readiness. A development run, a dry run, or an older clean report is not evidence for a revised PDF. Historical JSON is likewise not treated as replacement evidence.

To avoid a self-referential PDF/hash cycle, volatile identifiers are not embedded here. The external release record distributed with a candidate binds the exact Git commit, PDF SHA-256, replay-report SHA-256, deterministic archive SHA-256, and archive-metadata SHA-256. It also records the replay profile, environment manifest, and independent-machine status. Thus the release claim can be checked without interpreting prose from an earlier working build.

The source repository is available at

<https://github.com/h-lu/reversible-rfsn-ii-waves>.

The release tooling builds a deterministic payload containing the PDF, source, figures, certificates, replay driver, and manifests. Persistent identification and machine-readable provenance follow the stewardship principles in [29], but are requirements for the intended release and submission, not mathematical evidence. No immutable archival identifier or independent-machine replay is claimed in this working version; until those fields are populated in the external release record, the repository is the only data location claimed here.

APPENDIX D. UNIFORM LIMITS AT THE TWO NONCOMPACT ENDS

This appendix supplies the detailed estimates behind the end step in Section 5. The two ends require different arguments. The algebraic end has infinite physical time and a flat stable freedom, whereas the pole has finite physical time and a resonant logarithmic coefficient.

D.1. Fixed-cut asymptotic phase at the algebraic end. Restrict the future graph to $E = 0$ and use its desingularized time s . After substituting the graph, write

$$(D.1) \quad \frac{de}{ds} = e^2 v(e, b), \quad \frac{db}{ds} = ef(e, b) + b\mu(e, b), \quad \frac{dt}{ds} = \chi(e, b) = e^{1/2} \hat{\chi}(e, b),$$

where

$$(D.2) \quad v(e, b) \leq -a_* < 0, \quad \mu(e, b) \leq -\lambda_* < 0, \quad 0 < \chi_* \leq \hat{\chi}(e, b) \leq \chi^*.$$

All reduced coefficients have uniformly bounded state derivatives through order three, and $\hat{\chi}$ is controlled in C^k , throughout the coefficient neighborhood. The clock in (D.1) is part of the raw compactified coefficient data; it is not reconstructed from an unknown first-exit orbit.

Lemma D.1 (Mixed regularity of the algebraic future graph). *Let $\mathcal{P}$ be compact and let $\varpi \mapsto \mathbf{X}_\varpi$, $\varpi \in \mathcal{P}$, be a C^r family in $\mathfrak{H}_{\text{adm}}^k$, where $r \in \{1, 2\}$ and $k \geq 5$. After shrinking a common tubular neighborhood of the algebraic future-staying hypersurface, its graph over (e, E, b) ,*

$$\zeta = \Gamma_\varpi(e, E, b)$$

has uniformly bounded derivatives $D_{(e, E, b)}^i D_\varpi^j \Gamma_\varpi$ for $i + j \leq 3$ and $j \leq r$ on one fixed graph domain.

Proof. After shrinking the coefficient neighborhood, local maximality and all strict normal-expansion and bunching inequalities in (H3) hold uniformly. The coefficient topology gives a C^r map into the state- C^5 fields there. Proposition 2.9 gives (2.65), which is exactly the displayed mixed-total-three assertion. If (H3) is certified by an interval block, Theorem A.1 first constructs this same locally maximal hypersurface. In particular, $D_\varpi \Gamma_\varpi$ is controlled in C^2 and $D_\varpi^2 \Gamma_\varpi$ in C^1 ; two parameter differentiations are not claimed to retain three state derivatives. The graph is the maximal staying set in its fixed isolating neighborhood, not an orbit selected from the return or first-exit maps. $\square$

Put $a_\varpi = -v_\varpi > 0$, let $\mathbf{q} = e^{-1}$ and $\mathbf{q}_* = e_*^{-1}$, and substitute the future graph from Lemma D.1. At a common value of $\mathbf{q}$, the reduced equation is

$$(D.3) \quad \frac{db}{d\mathbf{q}} = G(\mathbf{q}, b, \varpi) := \frac{\mathbf{q}^{-1} f_\varpi(\mathbf{q}^{-1}, b) + b\mu_\varpi(\mathbf{q}^{-1}, b)}{a_\varpi(\mathbf{q}^{-1}, b)}.$$

Let r_ϖ and $u_{b_0, \varpi}$ solve

$$(D.4) \quad r' = G(\mathbf{q}, r, \varpi), \quad r(\mathbf{q}_*) = 0, \quad u' = G(\mathbf{q}, u, \varpi), \quad u(\mathbf{q}_*) = b_0,$$

and put $d = u - r$.

Lemma D.2 (Fixed- $\mathbf{q}$ algebraic passage). *After increasing $\mathbf{q}_*$ and decreasing the b_0 interval, there are $\eta > 0$ and uniform constants such that*

$$(D.5) \quad \max_{\substack{i+j \leq 3 \\ j \leq r}} \sup_{\mathbf{q} \geq \mathbf{q}_*} e^{\eta(\mathbf{q}-\mathbf{q}_*)} |D_{b_0}^i D_\varpi^j d(\mathbf{q})| \leq C, \quad \max_{j \leq r} \sup_{\mathbf{q} \geq \mathbf{q}_*} \mathbf{q} |D_\varpi^j r_\varpi(\mathbf{q})| \leq C.$$

Consequently, for every $m \geq 1$,

$$(D.6) \quad |D_{b_0}^i D_\varpi^j \{b_\varpi(e; b_0) - b_{\text{ref}, \varpi}(e)\}| \leq C_{i,j,m} e^m, \quad i + j \leq 3, \quad j \leq r.$$

The construction uses fixed half-line Banach spaces and is compatible with restriction to every finite $\mathbf{q}$ interval.

Proof. Use the fixed spaces

$$\begin{aligned} \mathcal{Y}_1 &= \{r \in C([\mathbf{q}_*, \infty)) : r(\mathbf{q}_*) = 0, \|r\|_1 = \sup_{\mathbf{q} \geq \mathbf{q}_*} \mathbf{q} |r(\mathbf{q})| < \infty\}, \\ \mathcal{X}_\eta &= \{d \in C([\mathbf{q}_*, \infty)) : \|d\|_\eta = \sup_{\mathbf{q} \geq \mathbf{q}_*} e^{\eta(\mathbf{q}-\mathbf{q}_*)} |d(\mathbf{q})| < \infty\}. \end{aligned}$$

Set

$$\ell_\infty(\varpi) = \partial_b G(\infty, 0, \varpi) = \frac{\mu_\varpi(0, 0)}{a_\varpi(0, 0)} \leq -c_0 < 0.$$

Compactified coefficient regularity gives, also through the required two parameter derivatives,

$$(D.7) \quad \begin{aligned} G(\mathbf{q}, b, \varpi) - \ell_\infty b &= O(\mathbf{q}^{-1}) + O(\mathbf{q}^{-1}|b| + |b|^2), \\ G_b(\mathbf{q}, b, \varpi) - \ell_\infty &= O(\mathbf{q}^{-1} + |b|). \end{aligned}$$

The reference is the fixed point in $\mathcal{Y}_1$ of

$$(D.8) \quad (\mathcal{R}_\varpi r)(\mathbf{q}) = \int_{\mathbf{q}_*}^{\mathbf{q}} e^{\ell_\infty(\varpi)(\mathbf{q}-\sigma)} \{G(\sigma, r(\sigma), \varpi) - \ell_\infty(\varpi)r(\sigma)\} d\sigma.$$

On a ball $\|r\|_1 \leq R$,

$$(D.9) \quad \|\mathcal{R}_\varpi r\|_1 \leq C_0 + \frac{C}{\mathbf{q}_*} (\|r\|_1 + \|r\|_1^2), \quad \|D_r \mathcal{R}_\varpi\| \leq \theta_r \leq \frac{C(1+R)}{\mathbf{q}_*} < 1.$$

Thus its inverse is bounded by $(1 - \theta_r)^{-1}$.

For $d = u - r$, the fixed-point operator on $\mathcal{X}_\eta$ is

$$(D.10) \quad \begin{aligned} (\mathcal{D}_{b_0, \varpi} d)(\mathbf{q}) &= e^{\ell_\infty(\varpi)(\mathbf{q}-\mathbf{q}_*)} b_0 \\ &+ \int_{\mathbf{q}_*}^{\mathbf{q}} e^{\ell_\infty(\varpi)(\mathbf{q}-\sigma)} \{G(\sigma, r + d, \varpi) - G(\sigma, r, \varpi) - \ell_\infty(\varpi)d\} d\sigma. \end{aligned}$$

Choose $0 < \eta < c_0$. On a sufficiently small ball,

$$(D.11) \quad \begin{aligned} \|\mathcal{D}_{b_0, \varpi} d\|_\eta &\leq |b_0| + \frac{C}{c_0 - \eta} (\mathbf{q}_*^{-1} \|d\|_\eta + \|d\|_\eta^2), \\ \|D_d \mathcal{D}_{b_0, \varpi}\| &\leq \theta_d \leq \frac{C}{c_0 - \eta} (\mathbf{q}_*^{-1} + R_d) < 1. \end{aligned}$$

The inverse bound is $(1 - \theta_d)^{-1}$. Parameter derivatives of the exponential kernel produce only $(\mathbf{q} - \sigma)^j e^{-c_0(\mathbf{q} - \sigma)}$, $j \leq 2$, which are bounded in the same η -weighted space.

For clarity, we write every variational equation used below. For parameter directions h, k , the reference variations satisfy

$$(D.12) \quad \begin{aligned} r'_h &= G_b r_h + G_h, & r_h(\mathbf{q}_*) &= 0, \\ r'_{hk} &= G_b r_{hk} + G_{bb}[r_h, r_k] + G_{bh} r_k + G_{bk} r_h + G_{hk}, & r_{hk}(\mathbf{q}_*) &= 0. \end{aligned}$$

Define $\hat{H}(\mathbf{q}, d, \varpi) = G(\mathbf{q}, r_\varpi + d, \varpi) - G(\mathbf{q}, r_\varpi, \varpi)$, with all terms below evaluated at the same $(\mathbf{q}, \varpi)$. Then

$$\begin{aligned} \hat{H}_d &= G_b(r + d), \\ \hat{H}_{dd} &= G_{bb}(r + d), \\ \hat{H}_h &= [G_b(r + d) - G_b(r)]r_h + G_h(r + d) - G_h(r), \\ \hat{H}_{dh} &= G_{bb}(r + d)r_h + G_{bh}(r + d), \end{aligned}$$

and

$$\begin{aligned} \hat{H}_{hk} &= [G_{bb}(r + d) - G_{bb}(r)][r_h, r_k] \\ &\quad + [G_b(r + d) - G_b(r)]r_{hk} \\ &\quad + [G_{bk}(r + d) - G_{bk}(r)]r_h \\ &\quad + [G_{bh}(r + d) - G_{bh}(r)]r_k \\ &\quad + G_{hk}(r + d) - G_{hk}(r). \end{aligned}$$

The difference variations are

$$(D.13) \quad \begin{aligned} d'_{b_0} &= \hat{H}_d d_{b_0}, & d_{b_0}(\mathbf{q}_*) &= 1, \\ d'_h &= \hat{H}_d d_h + \hat{H}_h, & d_h(\mathbf{q}_*) &= 0, \\ d'_{b_0 b_0} &= \hat{H}_d d_{b_0 b_0} + \hat{H}_{dd}[d_{b_0}, d_{b_0}], & d_{b_0 b_0}(\mathbf{q}_*) &= 0, \\ d'_{b_0 h} &= \hat{H}_d d_{b_0 h} + \hat{H}_{dh}[d_{b_0}, d_h] + \hat{H}_{dh} d_{b_0}, & d_{b_0 h}(\mathbf{q}_*) &= 0, \\ d'_{hk} &= \hat{H}_d d_{hk} + \hat{H}_{dd}[d_h, d_k] + \hat{H}_{dh} d_k + \hat{H}_{dk} d_h + \hat{H}_{hk}, & d_{hk}(\mathbf{q}_*) &= 0. \end{aligned}$$

After the preceding shrinkage, $\hat{H}_d \leq -c_1 < 0$. The zero-initial-value inverse of $w' - \hat{H}_d w = h$ satisfies

$$(D.14) \quad \|w\|_\eta \leq (c_1 - \eta)^{-1} \|h\|_\eta.$$

Equations (D.12)–(D.14), in increasing derivative order, prove the stated bounds through total order two. If $|\gamma| = 3$ and the parameter order in γ is at most r , a further differentiation has the form

$$(D.15) \quad (D^\gamma d)' - \widehat{H}_d D^\gamma d = K_\gamma,$$

where K_γ is a finite Faà di Bruno sum of coefficient derivatives, the reference derivatives in (D.12), and lower-order difference derivatives. Every summand contains at least one difference derivative, so $K_\gamma \in \mathcal{X}_\eta$. The inverse (D.14) proves the auxiliary total-three bound and hence (D.5). Since $e^{-\eta/e} \leq C_m e^m$, they also prove (D.6). Both Volterra equations restrict identically to every finite interval, proving restriction compatibility. $\square$

Proposition D.3 (Algebraic time and action mixed two-jets). *Let the time and action counterterms be (5.20)–(5.21). For each first-exit label b_0 , the differences between the actual first-exit flight and these reference flights have limits as $e \downarrow 0$ with every $D_{b_0}^i D_\varpi^j$, $i+j \leq 3$, $j \leq r$, uniformly on the compact coefficient family. The total-three estimate is auxiliary; the output topology and the main theorem retain only mixed total two-jets.*

Proof. On the future graph, the physical-time and action densities in the common $\mathbf{q}$ coordinate are

$$(D.16) \quad \mathcal{T}(\mathbf{q}, b, \varpi) = \mathbf{q}^{-1/2} \frac{\widehat{\chi}_{a, \varpi}}{a_\varpi}, \quad \mathcal{A}(\mathbf{q}, b, \varpi) = \mathbf{q}^{5/2} \frac{a_{a, \varpi}^c \widehat{\chi}_{a, \varpi}}{a_\varpi}.$$

For $Q = e^{-1}$, subtract at every finite cutoff before taking a limit:

$$(D.17) \quad R_\tau(Q, b_0, \varpi) = \int_{\mathbf{q}_*}^Q [\mathcal{T}(\mathbf{q}, r+d, \varpi) - \mathcal{T}(\mathbf{q}, r, \varpi)] d\mathbf{q},$$

$$(D.18) \quad R_B(Q, b_0, \varpi) = \int_{\mathbf{q}_*}^Q [\mathcal{A}(\mathbf{q}, r+d, \varpi) - \mathcal{A}(\mathbf{q}, r, \varpi)] d\mathbf{q}.$$

These are exactly the physical cutoff integrals minus the reference counterterms; no difference of two infinite quantities is used.

The chain rule, Lemma D.2, and the coefficient bounds give, for $i+j \leq 3$ and $j \leq r$,

$$(D.19) \quad \begin{aligned} |D_{b_0}^i D_\varpi^j [\mathcal{T}(\mathbf{q}, r+d, \varpi) - \mathcal{T}(\mathbf{q}, r, \varpi)]| &\leq C \mathbf{q}^{-1/2} e^{-\eta(\mathbf{q}-\mathbf{q}_*)}, \\ |D_{b_0}^i D_\varpi^j [\mathcal{A}(\mathbf{q}, r+d, \varpi) - \mathcal{A}(\mathbf{q}, r, \varpi)]| &\leq C \mathbf{q}^{5/2} e^{-\eta(\mathbf{q}-\mathbf{q}_*)}. \end{aligned}$$

Both majorants are integrable and independent of the family member. The Cauchy criterion and dominated convergence prove the mixed limits and justify interchanging parameter differentiation with $e \downarrow 0$. They also give the quantitative tail estimate

$$(D.20) \quad |D_{b_0}^i D_\varpi^j \{R_\bullet(\infty) - R_\bullet(Q)\}| \leq C(1+Q)^{5/2} e^{-\eta(Q-\mathbf{q}_*)}, \quad i+j \leq 3, \quad j \leq r.$$

For finite and limiting mixed C^3 first-hit labels, the mean-value formula and the chain rule use this one spare derivative to show that composition

retains both the mixed total-two bounds and the same exponential winding rate. $\square$

Remark D.4 (Why the cut is part of the normalization). The homogeneous stable displacement in the q variable has size $\exp(-\lambda q/a) = \exp(-\lambda/(ae))$. It is smooth and flat at $e = 0$. Thus boundedness, tempering, or agreement to every algebraic order at the end does not determine its coefficient. The condition $b_{\text{ref}}(e_*) = 0$ removes this one constant without putting an unknown full orbit into the field norm.

D.2. The pole phase and the action remainder. Let $\mathcal{P}$ be a compact parameter set and let $\varpi \mapsto \mathbf{X}_\varpi$ be a C^r coefficient family, $r \in \{1, 2\}$. For a function $h = h(s, z, \varpi)$ on a compact entry-label set K , write

$$(D.21) \quad \|h\|_{\text{mix}, m, r} = \max_{\substack{i+j \leq m \\ j \leq r}} \sup_{(s, z, \varpi)} |D_z^i D_\varpi^j h(s, z, \varpi)|.$$

The notation $O_{\text{mix}, m, r}(g(s))$ has the analogous meaning after division by $g(s)$. Denote by $\Psi_{\mathbf{p}, \varpi}^\tau$ the τ -flow of the compactified pole vector field $X_{\mathbf{p}, \varpi}$. For a pole-chart point z in the basin of its sink, define

$$(D.22) \quad \mathcal{S}_\varpi(z) = \int_0^\infty \rho(\Psi_{\mathbf{p}, \varpi}^\tau z) d\tau.$$

Since $dt/d\tau = \rho$, this is the physical time remaining to the pole.

Lemma D.5 (Parameter-dependent pole phase). *On one compact pole-entry window, (D.22) is finite and has all derivatives $D_z^i D_\varpi^j$ with $i + j \leq 3$ and $j \leq r$, uniformly on $K \times \mathcal{P}$. Its level $s = \mathcal{S}_\varpi(\Psi_{\mathbf{p}, \varpi}^\tau z)$ is a common remaining-time coordinate. Near the sink the change of variables $\rho \leftrightarrow s$ has the same mixed regularity and*

$$(D.23) \quad \frac{s}{\rho} \longrightarrow c_\varpi := -a_\varpi(0, 0)^{-1} > 0$$

with the corresponding mixed derivatives. For the normal-form system (1.1), $c_0 = \sqrt{6}$.

Proof. The boundary equilibrium is hyperbolic. The fixed indicial type and the strict radial inequality preserve a common spectral gap, while Proposition 8.3 puts the compact core entry image in one fixed basin neighborhood and proves that this remains true on the continued coefficient ball. More generally, this common-basin property is part of (H4). There, $|D_z^i D_\varpi^j \rho(\Psi_{\mathbf{p}, \varpi}^\tau z)| \leq C_{ij} e^{-c\tau}$ for $i + j \leq 3$, $j \leq r$. These estimates follow by differentiating the flow equation: every variational equation has the same stable linear part, and its inhomogeneous term is a finite sum of lower-order variations and the coefficient derivatives controlled by (2.30). Before entry into the attracting neighborhood, the flight is finite. The fixed event-speed bound and $k \geq 5$ give its mixed derivatives through total order three directly from the finite-time variational equations and the implicit-function theorem.

Differentiation under (D.22) is therefore justified by a common integrable majorant.

For the phase change itself, choose a fixed section $\rho = \rho_0$ inside the common sink neighborhood. The event speed satisfies $X_{p,\varpi}^\rho \leq -c\rho_0$, so the entry map to this section has the same mixed derivatives. Parametrize the tail orbit by its ρ -coordinate, writing it $\Gamma_\varpi(v; z)$. Then

$$(D.24) \quad s_\varpi(\rho, z) = \int_0^\rho \frac{-v}{X_{p,\varpi}^\rho(\Gamma_\varpi(v; z))} dv.$$

Along Γ_ϖ let $\hat{z}$ denote its u -coordinate. The stable dichotomy gives $\hat{z} \rightarrow 0$, together with its fixed- ρ mixed label derivatives through total order three. Equation (2.25) gives $X_{p,\varpi}^\rho = v\{a_\varpi(0, 0) + O(v + |\hat{z}|)\}$. The integrand in (D.24) and its required mixed derivatives therefore extend to $v = 0$, with limiting value $-a_\varpi(0, 0)^{-1}$. This proves (D.23). A parameter-uniform inverse-function theorem, applied while $\partial_\rho s$ stays away from zero, gives the asserted mixed regularity of $\rho = \rho(s, z, \varpi)$ without a loss of a power of s . $\square$

Classical Frobenius theory explains the appearance of resonant logarithms at a regular singular point; see, for example, [28]. The next lemma gives the finite nonlinear, parameter-dependent recursion and mixed regularity required here rather than importing those conclusions from the linear theory.

Lemma D.6 (Finite power-log expansion at a regular singular point). *Let K be a compact C^3 manifold, let $r \in \{1, 2\}$, and consider a C^r parameter family*

$$(D.25) \quad \rho u_\rho = \mathcal{I}u + \mathcal{G}_\varpi(\rho, u), \quad \mathcal{G}_\varpi(\rho, u) = O(\rho + |u|^2),$$

on $[0, \rho_0] \times U_p$, where $m_p \geq 2$ is fixed, $\mathcal{I}$ is the fixed matrix in (2.26) whose spectrum Λ_p is simple and contained in the positive integers, and the coefficients are C^{m_p+3} in (ρ, u) . Suppose a family of entry maps

$$(D.26) \quad \iota_\varpi : K \longrightarrow \{\rho = \rho_0\}$$

has uniformly bounded derivatives $D_p^i D_\varpi^j$, $i + j \leq 3$, $j \leq r$, and that the corresponding solutions converge to $u = 0$ as $\rho \downarrow 0$.

There is an integer D , depending only on m_p and the fixed indicial matrix, and polynomials $U_{m,\varpi}(\ell, p)$ of degree at most D in ℓ , for $1 \leq m \leq m_p - 1$, whose coefficients have the same mixed derivatives, such that for every $\delta \in (0, 1)$,

$$(D.27) \quad u_\varpi(\rho, p) = \sum_{m=1}^{m_p-1} \rho^m U_{m,\varpi}(\log \rho, p) + O_{\text{mix},3,r}(\rho^{m_p-\delta}).$$

The polynomials U_m are independent of δ .

Proof. Put $t = -\log \rho$ and let P_λ be the spectral projector of $\mathcal{I}$ associated with $\lambda \in \Lambda_p$. These projectors are fixed because the full normalized indicial

operator, not merely its spectrum, is fixed. Variation of constants, first on a common finite interval and then on the attracting tail, gives

$$(D.28) \quad D_p^i D_\varpi^j u = O(e^{-(1-\varepsilon)t}) \quad (i+j \leq 3, j \leq r)$$

for every $\varepsilon > 0$. The estimate follows by applying Gronwall's inequality to the three projected equations; the forcing $\mathcal{G}(e^{-t}, u)$ is $O(e^{-t} + |u|^2)$, and the entry maps supply the mixed initial data.

We now proceed by finite induction. After the terms of orders less than m have been subtracted, let v_{m-1} denote the residual. The induction statement is

$$(D.29) \quad v_{m-1} = O_{\text{mix},3,r}(\rho^{m-\varepsilon}),$$

$$(D.30) \quad \rho \partial_\rho v_{m-1} - \mathcal{I} v_{m-1} = \rho^m \mathcal{Q}_{m,\varpi}(\log \rho, p) + O_{\text{mix},3,r}(\rho^{m+1-\varepsilon}),$$

for every fixed $\varepsilon > 0$. The case $m = 1$ follows from (D.28). Taylor expansion of $\mathcal{G}$ shows that $\mathcal{Q}_{m,\varpi}$ is a polynomial in $\log \rho$ determined by the previous coefficients, and that subtracting the next coefficient improves (D.29) by one power. That coefficient must solve

$$(D.31) \quad U'_{m,\varpi} + (mI - \mathcal{I})U_{m,\varpi} = \mathcal{Q}_{m,\varpi}.$$

On the sum of the nonresonant eigenspaces this has a unique polynomial solution. If $m \in \Lambda_p$, its projection onto $\ker(mI - \mathcal{I})$ is polynomial integration and hence may contain one free constant. After the polynomial particular solution has been subtracted, write w_m for the new residual. In the resonant case its projection satisfies

$$(\partial_t + m)P_m w_m = O_{\text{mix},3,r}(e^{-(m+1-\varepsilon)t}), \quad t = -\log \rho.$$

Consequently that constant is defined from the actual orbit by

$$(D.32) \quad c_{m,\varpi}(p) = \lim_{t \rightarrow \infty} e^{mt} P_m w_m(t, p, \varpi), \quad m \in \Lambda_p \cap \{1, \dots, m_p - 1\},$$

where P_m denotes the spectral projector for the root m . Here w_m is the residual after the polynomial particular solution at order m has been removed. The projected variation-of-constants formula shows that the limit equals its value at a fixed tail section plus an absolutely convergent integral. After the resonant constant, when present, is also removed, use the terminal variation-of-constants formula in every eigendirection $\lambda \leq m$ whose homogeneous coefficient has already been removed, and the initial-value formula in every direction $\lambda > m$. The source in (D.30) is $O(e^{-(m+1-\varepsilon)t})$; the remaining linear term is $O(e^{-(1-\varepsilon)t}|w_m|)$ and is absorbed after increasing the common tail time. Since all indicial roots are positive integers, the initial-value homogeneous modes with $\lambda > m$ decay at least as fast as $e^{-(m+1)t}$. These projected formulas prove

$$w_m = O_{\text{mix},3,r}(e^{-(m+1-\varepsilon)t}),$$

which closes the induction. Differentiating the same projected equations proves both this estimate and the asserted mixed three-jets of $c_{m,\varpi}$. This

constructs $U_1, \dots, U_{m_p-1}$ and accounts for every indicial resonance below order m_p .

For completeness, write $u^{[m_p-1]}$ for the constructed finite sum, put $v = u - u^{[m_p-1]}$, and choose t_0 in the common tail neighborhood. In the variable $t = -\log \rho$, direct substitution gives

$$(D.33) \quad v_t = -\mathcal{I}v + F(t, p, \varpi) + N(t, v, p, \varpi).$$

Finite Taylor expansion and $|t|^M = O(e^{\varepsilon t})$ give $F = O_{\text{mix},3,r}(e^{-(m_p-\delta/2)t})$, while

$$N(t, v) = O(e^{-(1-\varepsilon)t}|v| + |v|^2)$$

with the analogous mixed derivative bounds. The extracted coefficients impose zero terminal coefficient in every eigendirection $\lambda < m_p$. Thus (D.33) is equivalent to the projected Lyapunov–Perron equations

$$(D.34)$$

$$P_\lambda v(t) = -e^{-\lambda t} \int_t^\infty e^{\lambda \tau} P_\lambda \{F(\tau) + N(\tau, v(\tau))\} d\tau, \quad \lambda \in \Lambda_p, \lambda < m_p,$$

$$P_\lambda v(t) = e^{-\lambda(t-t_0)} P_\lambda v(t_0)$$

$$(D.35) \quad + \int_{t_0}^t e^{-\lambda(t-\tau)} P_\lambda \{F(\tau) + N(\tau, v(\tau))\} d\tau, \quad \lambda \in \Lambda_p, \lambda \geq m_p.$$

On the Banach space with norm

$$\|v\|_{m_p-\delta} := \sup_{t \geq t_0} e^{(m_p-\delta)t} |v(t)|,$$

the right-hand side is a contraction after increasing t_0 . Equations (D.34)–(D.35) therefore give $v = O(e^{-(m_p-\delta)t})$. Differentiating these same equations in (p, ϖ) gives, inductively in total derivative order, the identical weighted bound. Every free homogeneous mode with $\lambda \geq m_p$ is retained in (D.35); its order ρ^λ already lies in the claimed remainder. This proves (D.27). $\square$

Lemma D.7 (Finite power–log inversion). *Suppose, with uniformly bounded mixed derivatives through total order three,*

$$(D.36)$$

$$s = c_\varpi \rho + \sum_{m=2}^{m_p} \rho^m S_{m,\varpi}(\log \rho, p) + O_{\text{mix},3,r}(\rho^{m_p+1-\varepsilon}), \quad c_\varpi \geq c_* > 0.$$

Then, after decreasing s_0 , the inverse has a finite expansion in powers of s and $\log s$ through order m_p , and its remainder is $O_{\text{mix},3,r}(s^{m_p+1-\delta})$ for every $\delta > \varepsilon$.

Proof. Starting with $\rho_1 = s/c_\varpi$, substitute recursively in (D.36) and cancel one power at a time. This constructs a power–log polynomial $\rho^{[m_p]}(s, p, \varpi)$ whose substitution defect is $O_{\text{mix},3,r}(s^{m_p+1-\delta})$; finite powers of $|\log s|$ are absorbed by the gap $\delta - \varepsilon$. Since $\partial_\rho s \geq c_*/2$, the mean-value theorem gives the same bound for $\rho - \rho^{[m_p]}$. Differentiating the identity $s(\rho(s, p, \varpi), p, \varpi) = s$ and using induction in total derivative order proves the mixed bounds. $\square$

Lemma D.8 (Finite pole index set and uniqueness). *Let $m_p \geq 2$ and let $\Lambda_p \subset \mathbb{N}$ be fixed. Define integers d_m recursively by*

$$(D.37) \quad d_0 = 0, \quad d_m = m \left(1 + \max_{0 \leq j < m} d_j \right) + \mathbf{1}_{\{m \in \Lambda_p\}}, \quad 1 \leq m \leq m_p,$$

and put

$$(D.38) \quad d_u = \max_{0 \leq m \leq m_p} d_m, \quad d_p = 1 + m_p^2(1 + d_u).$$

The regular-singular recursion (D.31), finite power-log inversion, and composition with a Taylor polynomial of degree m_p use only the finite index sets

$$(D.39) \quad \mathcal{E}_u = \{(m, d) : 1 \leq m \leq m_p - 1, 0 \leq d \leq d_m\},$$

$$(D.40) \quad \mathcal{E}_{\text{sing}} = \{(q, d) : -m_p \leq q \leq -1, 0 \leq d \leq d_p\}.$$

Thus the orbit coefficients at order ρ^m have logarithmic degree at most d_m , and every singular density and counterterm coefficient has logarithmic degree at most d_p . These bounds are deliberately nonoptimal and depend only on m_p and the fixed indicial data.

Moreover, let

$$(D.41) \quad L(s) = \sum_{q=-m_p}^{-1} s^q P_q(\log s)$$

where every P_q is a polynomial. If either $L \in L^1((0, s_0))$ or the improper primitive

$$(D.42) \quad \lim_{s \downarrow 0} \int_s^{s_0} L(\sigma) d\sigma$$

exists and is finite, then $P_q = 0$ for every q . Consequently two finite Laurent-log sums have identical singular coefficients whenever their difference is absolutely integrable at zero or has a convergent improper primitive there.

Proof. At step m of (D.31), the coefficient $\mathcal{Q}_m$ is obtained by collecting the order- m part of a Taylor polynomial in ρ and the already constructed U_j . An order- m monomial contains at most m previous factors, so its logarithmic degree is at most $m \max_{j < m} d_j$. Solving the nonresonant rows does not change that degree, while polynomial integration in a resonant row increases it by one. The larger recursive bound in (D.37) therefore closes the induction. Power-log inversion through order m_p consists of finitely many additions, products, differentiations, and substitutions of these polynomials. Taylor composition of degree m_p and the final integration fit inside the deliberately larger budget d_p in (D.38). This proves the index-set assertions.

For uniqueness, put $t = -\log s$. After multiplication by $ds = e^{-t} dt$, a term with exponent $q < -1$ becomes an exponentially growing function $e^{-(q+1)t} P_q(-t)$. Choose the smallest q for which $P_q \neq 0$; terms with larger exponents cannot cancel its exponential rate, so neither its absolute integral

nor its improper primitive can converge. Once all $q < -1$ terms vanish, the $q = -1$ term has a divergent primitive unless $P_{-1} = 0$. This proves both alternatives and the final assertion. $\square$

Proposition D.9 (Pole finite part for a fixed simple positive-integer indicial spectrum). *Let $\mathcal{P}$ be compact, let $r \in \{1, 2\}$, and let $\varpi \mapsto \mathbf{X}_\varpi$ be a C^r family satisfying (2.30) and the common-basin condition (H4) of Definition 2.8. The compactified-action order m_p and the normalized indicial matrix $\mathcal{I}$, with simple positive-integer spectrum, are fixed across the family. No equality with the normal-form system is imposed on the nonlinear boundary jet of $X_{p,\varpi}$, on any positive normal jet of $X_{p,\varpi}$, or on the boundary jet of $a_{p,\varpi}^c$.*

Let K be a compact C^3 manifold of pole-entry labels. Assume that finite flight to a fixed tail section $\rho = \rho_0$ gives entry maps $\iota_\varpi : K \rightarrow \{\rho = \rho_0\}$ satisfying (D.26), with a uniform event-speed bound and uniformly bounded mixed derivatives through total order three. Let $s = t_b - t$ be physical remaining time; all label and parameter derivatives below are taken at fixed s . There are polynomials in ℓ ,

$$A_{q,\varpi}(\ell, p), \quad -m_p \leq q \leq -1,$$

of degree at most d_p from Lemma D.8, whose coefficients have all mixed derivatives $D_p^i D_\varpi^j$, $i + j \leq 3$, $j \leq r$. These polynomials do not depend on δ , and for every $\delta \in (0, 1)$,

$$(D.43) \quad \lambda_\varpi(X_\varpi)(s, p) = \sum_{q=-m_p}^{-1} s^q A_{q,\varpi}(\log s, p) + R_{p,\varpi}(s, p),$$

where

$$(D.44) \quad R_{p,\varpi} = O_{\text{mix},3,r}(s^{-\delta}).$$

In particular, the remainder is integrable at $s = 0$.

For $-m_p \leq q \leq -2$, let $Q_{q,\varpi}(\ell, p)$ be the unique polynomial solution of

$$(D.45) \quad Q'_{q,\varpi} + (q+1)Q_{q,\varpi} = -A_{q,\varpi},$$

and determine $Q_{-1,\varpi}$ by

$$(D.46) \quad Q'_{-1,\varpi} = -A_{-1,\varpi}, \quad Q_{-1,\varpi}(0, p) = 0.$$

Then

$$(D.47) \quad C_{B,p,\varpi}(s, p) = \sum_{q=-m_p}^{-2} s^{q+1} Q_{q,\varpi}(\log s, p) + Q_{-1,\varpi}(\log s, p)$$

satisfies

$$(D.48) \quad -\partial_s C_{B,p,\varpi} = \sum_{q=-m_p}^{-1} s^q A_{q,\varpi}(\log s, p).$$

Consequently the limit

$$(D.49) \quad B_{p,\varpi}^{\text{ren}}(p) = \lim_{s \downarrow 0} \left\{ \int_{t_{\text{in}}}^{t_b - s} \lambda_{\varpi}(X_{\varpi})(\Phi_{\varpi}^t z_{\text{in}}(p)) dt - C_{B,p,\varpi}(s, p) \right\}$$

exists with all mixed derivatives $D_p^i D_{\varpi}^j$, $i + j \leq 3$, $j \leq r$.

The singular density and counterterm are intrinsic to physical remaining time. Suppose two admissible C^{m_p+3} pole charts in the sense of Definition 5.6 use the same physical time scale and label the same physical orbit. Under their coordinate change $\tilde{p} = v_{\varpi} \rho$, $v_{\varpi} > 0$, and the induced end-label change $\tilde{p} = \psi_{\varpi}(p)$,

$$(D.50) \quad \tilde{C}_{B,p,\varpi}(s, \psi_{\varpi}(p)) = C_{B,p,\varpi}(s, p).$$

An event-preserving orbit slide adds only the action of the finite segment. If $\tilde{\lambda}_{\varpi} = \lambda_{\varpi} + d\chi_{\varpi}$ and χ_{ϖ} has a mixed- C^3 limit $\chi_{\infty,\varpi}$ at the pole, then, with the same source section,

$$(D.51) \quad \tilde{B}_{p,\varpi}^{\text{ren}}(p) = B_{p,\varpi}^{\text{ren}}(p) + \chi_{\infty,\varpi}(p) - \chi_{\varpi}(z_{\text{in}}(p)).$$

Proof. Use ρ as the independent variable in (2.25). Along a pole orbit, its u -coordinate satisfies the regular-singular equation

$$(D.52) \quad \rho \frac{du}{d\rho} = \mathcal{I}u + \mathcal{G}_{\varpi}(\rho, u), \quad \mathcal{G}_{\varpi}(\rho, u) = O(\rho + |u|^2).$$

Since a_{ϖ} is uniformly bounded away from zero, division by a_{ϖ} preserves C^{m_p+3} regularity. The entry hypothesis and Lemma D.6 give

$$(D.53) \quad u_{\varpi}(\rho, p) = \sum_{m=1}^{m_p-1} \rho^m U_{m,\varpi}(\log \rho, p) + O_{\text{mix},3,r}(\rho^{m_p-\varepsilon})$$

for every $\varepsilon \in (0, 1)$, with one set of coefficient polynomials.

Physical remaining time satisfies

$$(D.54) \quad \frac{ds}{d\rho} = -\frac{1}{a_{\varpi}(\rho, u_{\varpi}(\rho, p))}, \quad s(0, p) = 0.$$

Choose $0 < \varepsilon < \delta$. Substitution of (D.53) and termwise integration give

$$s = c_{\varpi} \rho + \sum_{m=2}^{m_p} \rho^m S_{m,\varpi}(\log \rho, p) + O_{\text{mix},3,r}(\rho^{m_p+1-\varepsilon}),$$

$$c_{\varpi} = -a_{\varpi}(0, 0)^{-1} > 0.$$

Lemma D.7 gives the corresponding expansions of ρ and u in powers of s and $\log s$, with a $O_{\text{mix},3,r}(s^{m_p-\delta})$ remainder for u .

Taylor-expand the smooth compactified action coefficient $a_{p,\varpi}^c(\rho, u)$ through total degree m_p and substitute the preceding expansions into

$$\lambda_{\varpi}(X_{\varpi}) = \rho^{-m_p} a_{p,\varpi}^c.$$

The finite sum in (D.53) is $O_{\text{mix},3,r}(\rho(1 + |\log \rho|)^D)$. Hence the degree- m_p Taylor remainder after composition with the orbit is $O_{\text{mix},3,r}(\rho^{m_p+1}(1 +$

$|\log \rho|)^{D'})$; after multiplication by ρ^{-m_p} it is integrable. The error caused by the orbit remainder is $O_{\text{mix},3,r}(\rho^{-\delta})$. Since s/ρ stays bounded above and below, these errors are $O_{\text{mix},3,r}(s^{-\delta})$. The terms with powers $s^{-m_p}, \dots, s^{-1}$ form the finite sum in (D.43). This proves (D.43)–(D.44). Equations (D.45)–(D.46) integrate the singular sum exactly. Dominated convergence proves (D.49) and its mixed derivatives.

For reference, the finite recursion has the following regularity ledger. All derivatives in the last column are taken at fixed ρ before time inversion and at fixed physical remaining time s afterwards.

Object	Determination	Uniform regularity
$U_{m,\varpi}$	homological recursion (D.31)	polynomial on $\mathcal{E}_u$; coefficientwise $D_p^i D_\varpi^j$, $i + j \leq 3, j \leq r$
$c_{m,\varpi}$ time and inverse- time coefficients	projected limit (D.32) (D.54) and Lemma D.7	$D_p^i D_\varpi^j, i + j \leq 3, j \leq r$ same mixed derivatives; finite power–log index set
$A_{q,\varpi}$	Taylor composition of $a_{p,\varpi}^c$	polynomial on $\mathcal{E}_{\text{sing}}$ with the same mixed derivatives
$Q_{q,\varpi}$	(D.45)–(D.46)	same mixed derivatives and log-degree bound d_p
$R_{p,\varpi}$	Taylor and orbit remain- ders	$O_{\text{mix},3,r}(s^{-\delta})$ for every $\delta \in (0, 1)$; locally integrable

First keep the primitive fixed. Both pole charts then use the same physical remaining time and the same physical action density. The difference of their two finite Laurent–log singular sums is integrable. A finite linear combination of $s^q(\log s)^m$, $q \leq -1$, is covered by Lemma D.8. Hence the singular density is unique after the end-label change, and the normalization in (D.46) gives (D.50). Orbit slides add the action of a finite segment.

Now let $\tilde{\lambda} = \lambda + d\chi$. Along the same physical pole orbit,

$$(D.55) \quad \int_{t_{\text{in}}}^{t_b-s} \{\tilde{\lambda}(X) - \lambda(X)\}(\Phi^t z_{\text{in}}) dt = \chi(z_s) - \chi(z_{\text{in}}), \quad t_b - t(z_s) = s.$$

The right-hand side has a mixed- C^3 limit as $s \downarrow 0$. Hence the improper integral of the density difference converges. Both density expansions already have integrable remainders, so the difference of their finite Laurent–log singular sums must have a convergent primitive. Uniqueness forces that difference to vanish and therefore makes the normalized counterterms agree. Taking the limit in (D.55) proves exactly (D.51). $\square$

Proposition D.10 (Explicit counterterm for the normal-form pole). *Let $r \in \{1, 2\}$ and let $\varpi \mapsto \mathbf{X}_\varpi$ be a C^r family whose pole coefficients agree with those of (2.22) through order seven, in the precise sense that for $0 \leq j \leq r$,*

$$(D.56) \quad D_\varpi^j(X_{p,\varpi} - X_{p,0}), \quad D_\varpi^j(a_{p,\varpi}^c - a_{p,0}^c) \in \rho^8 C_b^4.$$

Here C_b^4 means that derivatives of order at most four generated by $\rho\partial_\rho$ and ∂_u remain bounded. On a compact pole-entry label set K , assume the entry maps have the mixed three-jets in (D.26), and take all derivatives at fixed remaining-time phase s . There are C_p, Ω with all mixed derivatives of total order at most three and $\nu \in (0, 1)$ such that

$$(D.57) \quad \begin{aligned} W &= s^4\{-6\log s + \Omega - C_p s\} + O_{\text{mix},3,r}(s^8|\log s|), \\ X &= 6 + s^4\left(-\frac{3}{5}\log s + \frac{\Omega}{10} - \frac{9}{50}\right) - \frac{C_p}{6}s^5 + O_{\text{mix},3,r}(s^{5+\nu}), \\ (D.58) \quad Y &= 12 + s^4\left(\frac{6}{5}\log s - \frac{\Omega}{5} + \frac{24}{25}\right) + \frac{C_p}{2}s^5 + O_{\text{mix},3,r}(s^{5+\nu}), \\ Z &= 6 + C_p s + O_{\text{mix},3,r}(s^4|\log s|). \end{aligned}$$

Moreover,

$$(D.59) \quad \lambda_\varpi(X_\varpi) = 144s^{-6} + \left(\frac{144}{5}\log s - \frac{24}{5}\Omega - \frac{324}{25}\right)s^{-2} + O_{\text{mix},3,r}(s^{-1+\nu}).$$

Thus (5.28), with $\Omega = \Omega_\varpi(p)$, removes every nonintegrable term, and the renormalized pole action has every derivative $D_z^i D_\varpi^j$, $i + j \leq 3$, $j \leq r$, uniformly on $K \times \mathcal{P}$. Total order three is used only as the spare derivative before composition; the main theorem claims total order two.

Proof. Proposition 8.3 places the core pole-entry image compactly in the basin of the sink (2.23). Finite-time continuous dependence, followed by the uniform local sink estimate, puts all continued entry images in one common attracting neighborhood after shrinking $\mathcal{U}_\kappa$. The order-eight condition (D.56) fixes the boundary equilibrium and its spectral gap. The flow and its mixed variational equations through total order three therefore decay exponentially. Lemma D.5 puts them in the common fixed- s phase and gives $s/\rho \rightarrow \sqrt{6}$ with the same mixed regularity.

Use logarithmic time $\hat{\tau} = -\log s$. Phase alignment and the conormal hypothesis give

$$(D.60) \quad \begin{aligned} \dot{X} &= Y - 2X + \mathcal{E}_X, & \dot{Y} &= X^2 - W - 3Y + \mathcal{E}_Y, \\ \dot{Z} &= X - Z + \mathcal{E}_Z, & \dot{W} &= -4W + s^4 Z + \mathcal{E}_W, \end{aligned}$$

where, for some $\nu_0 \in (0, 1)$,

$$(D.61) \quad (\mathcal{E}_X, \mathcal{E}_Y, \mathcal{E}_Z) = O_{\text{mix},3,r}(s^{6+\nu_0}), \quad \mathcal{E}_W = O_{\text{mix},3,r}(s^8).$$

Indeed, the only division by a boundary variable is $R_\rho/\rho = O(\rho^7)$; all other perturbation terms remain $O(\rho^8)$, and $s \asymp \rho$. Hence no perturbation changes a coefficient through order five.

For $\xi = (X - 6, Y - 12)^\top$,

$$(D.62) \quad \dot{\xi} = \begin{pmatrix} -2 & 1 \\ 12 & -3 \end{pmatrix} \xi + \begin{pmatrix} 0 \\ (X - 6)^2 - W \end{pmatrix} + \mathcal{E}_\xi, \quad \text{spec} = \{1, -6\}.$$

Endpoint convergence excludes the growing mode. The uniform dichotomy, also for every mixed variational equation through total order three, gives first $\xi = O_{\text{mix},3,r}(s^4 |\log s|)$. Explicitly, each such variation V_γ satisfies the same dichotomy equation with a forcing formed from lower-order variations and a coefficient derivative; the growing coefficient vanishes because $D^\gamma \xi \rightarrow 0$ at the sink. With $h = (Z - 6)/s$ and $f = W/s^4$,

$$(D.63) \quad \dot{h} = s^{-1}(X - 6) + O_{\text{mix},3,r}(s^{5+\nu_0}), \quad \dot{f} = Z + O_{\text{mix},3,r}(s^4).$$

After subtraction of the displayed leading terms, the right sides are integrable. Thus $h \rightarrow C_p$ and $f = -6 \log s + \Omega - C_p s + O_{\text{mix},3,r}(s^4 |\log s|)$. Solving (D.62) at orders four and five gives (D.58); the remaining decaying homogeneous mode is s^6 . Equivalently, the two coefficient functions are the convergent mixed limits

$$(D.64) \quad C_{p,\varpi}(z) = \lim_{s \downarrow 0} \frac{Z - 6}{s}, \quad \Omega_\varpi(z) = \lim_{s \downarrow 0} \left(\frac{W}{s^4} + 6 \log s + C_{p,\varpi}(z)s \right).$$

The integrable remainders above justify differentiation of these limits through total order three.

For the core compactified action coefficient,

$$(D.65) \quad \rho^{-6} a_{p,0}^c = s^{-6} Y^2 - s^{-2} Z^2,$$

whereas (D.56) gives

$$(D.66) \quad \rho^{-6} (a_{p,\varpi}^c - a_{p,0}^c) = O_{\text{mix},3,r}(\rho^2) = O_{\text{mix},3,r}(s^2).$$

Substitution gives (D.59); the $C_p s^{-1}$ terms cancel between the two squares. Finally,

$$(D.67) \quad -\partial_s C_{B,p} = 144s^{-6} + \left(\frac{144}{5} \log s - \frac{24}{5} \Omega - \frac{324}{25} \right) s^{-2}.$$

The $C_p s^{-1}$ terms cancel identically before any derivative is taken. The remaining mixed three-jet is bounded by $Cs^{-1+\nu}$ and is integrable. Differentiation under the cutoff integral followed by dominated convergence therefore proves the auxiliary mixed three-jets, and hence the asserted mixed parameter two-jets, of the renormalized pole action. $\square$

APPENDIX E. OPTIONAL MODULE: JOST IDENTIFICATION AND VALIDATION OF THE SELECTED REVERSIBLE BRANCH

This appendix contains the application-specific bridge from the canonical algebraic tangent to the first outer critical point. It is not used in the construction of the countable return–exit decomposition; it supplies the analytic starting tangent for Proposition 8.5 and thereby the finite-continuation assertion in Corollary 8.9.

Proof of Proposition 8.5. For completeness, we identify the tangent at the algebraic source analytically. Along γ_0 , the variational equation is $z' = A_0(t)z$, where

$$(E.1) \quad A_0(t) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ t^2/6 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

If

$$(E.2) \quad \mathcal{W}[a] = \left(a, a', \frac{t^2}{6}a - a'', \left(\frac{t^2}{6}a - a'' \right)' \right),$$

then $\mathcal{W}[a]$ solves this system exactly when

$$(E.3) \quad a^{(4)} - \frac{t^2}{6}a'' - \frac{2t}{3}a' + \frac{2}{3}a = 0.$$

Put

$$(E.4) \quad \begin{aligned} Y_0(t) &= {}_2F_3 \left(-\frac{1}{4}, 1; \frac{1}{4}, \frac{1}{2}, \frac{3}{4}; \frac{t^4}{96} \right), \\ Y_2(t) &= t^2 {}_1F_2 \left(\frac{1}{4}; \frac{3}{4}, \frac{5}{4}; \frac{t^4}{96} \right), \\ Y_3(t) &= t^3 {}_1F_2 \left(\frac{1}{2}; \frac{5}{4}, \frac{3}{2}; \frac{t^4}{96} \right), \\ B_2 &= \frac{\Gamma(3/4)96^{5/8}}{8\sqrt{\pi}}, \quad B_3 = \frac{\Gamma(1/4)96^{7/8}}{16\sqrt{\pi}}. \end{aligned}$$

Coefficient recurrence in (E.3) verifies these entire solutions. With k_0 as in (E.17), set

$$(E.5) \quad \mathcal{T} = \mathcal{W}[t], \quad \mathcal{Z} = \mathcal{W}[Y_0 + k_0 Y_2], \quad \mathbf{s} = \mathcal{W}[B_3 Y_2 - B_2(Y_3 + 12k_0 t)].$$

The positive-axis identity

$$(E.6) \quad B_3 Y_2 - B_2(Y_3 + 12k_0 t) = -\frac{96^{3/4}}{16\sqrt{\pi}} t^{5/2} \int_1^\infty r^{-1/4} K_{1/4} \left(\frac{t^2 r}{2\sqrt{6}} \right) dr$$

proves that $\mathbf{s}$ is recessive. Replacing the minus sign between the two terms by a plus sign gives the growing scalar $u_{\text{sc}} = B_3 Y_2 + B_2(Y_3 + 12k_0 t)$. More explicitly, with $x = t^2/(2\sqrt{6})$,

$$(E.7) \quad \begin{aligned} u_{\text{sc}}(t) &= 12B_2 k_0 t \\ &+ \frac{\sqrt{2\pi} 96^{3/4}}{32} t^{5/2} \int_0^1 r^{-1/4} \{I_{-1/4}(xr) + I_{1/4}(xr)\} dr. \end{aligned}$$

Indeed, termwise integration of the entire $I_{\pm 1/4}$ series gives Y_2 and Y_3 , respectively, and $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ gives the displayed common coefficient.

Thus the integrand in (E.7) is positive for $t > 0$. Set $\mathbf{u} = \mathcal{W}[u_{\text{sc}}]$. Their Cauchy data, obtained directly from (E.4), are

$$(E.8) \quad \begin{aligned} \mathcal{T}(0) &= (0, 1, 0, 0), & \mathcal{Z}(0) &= (1, 0, -2k_0, 0), \\ \mathbf{s}(0) &= (0, -12B_2k_0, -2B_3, 6B_2), & \mathbf{u}(0) &= (0, 12B_2k_0, -2B_3, -6B_2). \end{aligned}$$

Since $B_2B_3 = 6\sqrt{3} > 0$, these four solutions form a fundamental splitting. We next check, rather than assume, that $\mathcal{Z}$ is an algebraic staying direction. The Cauchy data give the conserved symplectic pairings

$$(E.9) \quad \omega(\mathbf{s}, \mathcal{T}) = \omega(\mathbf{u}, \mathcal{T}) = \omega(\mathbf{s}, \mathcal{Z}) = \omega(\mathbf{u}, \mathcal{Z}) = 0, \quad \omega(\mathcal{T}, \mathcal{Z}) = 1.$$

Put $c = Y_0 + k_0Y_2$. There is a unique vector $\mathcal{E} = (0, p_E, w_E, q_E)$ in $\text{span}\{\mathcal{T}, \mathcal{Z}\}$ with $\omega(\mathcal{T}, \mathcal{E}) = 1$. Solving the three pairing rows gives the following symbol expansion. Here and below $R(t) = O_{\text{sym}}(t^{-m})$ means $|\partial_t^j R(t)| \leq C_j t^{-m-j}$ for $0 \leq j \leq 3$; these are the only symbol derivatives used in the argument.

$$(E.10) \quad p_E(t) = \frac{\mathbf{s}_Q(t)\mathbf{u}_V(t) - \mathbf{s}_V(t)\mathbf{u}_Q(t)}{t\omega(\mathbf{s}, \mathbf{u})} = 36t^{-5}(1 + 252t^{-4} + O_{\text{sym}}(t^{-8})).$$

Here the second equality follows by Watson endpoint expansion of the two positive Bessel representations (E.6) and (E.7); applying the derivative recurrences before division gives the displayed symbol remainder, so no cancellation of unvalidated compound asymptotic series is used. The variational equation gives the exact relations

$$(E.11) \quad w_E = -p'_E - p_E/t, \quad q_E = w'_E + t^2 p_E/6, \quad q'_E = -t p_E/2.$$

In particular, $w_E = 144t^{-6} + O(t^{-10})$ and $q_E = 6t^{-3} + O(t^{-7})$. Since both $\mathcal{Z}$ and $\mathcal{E}$ pair to one with $\mathcal{T}$ and have no $\mathbf{s}$ or $\mathbf{u}$ component,

$$(E.12) \quad \mathcal{Z}(t) = \frac{c(t)}{t}\mathcal{T}(t) + \mathcal{E}(t), \quad c(t) = C_*t - \frac{36}{5}t^{-4}(1 + 140t^{-4} + O_{\text{sym}}(t^{-8}))$$

for a finite $C_* > 0$; indeed $(c/t)' = p_E/t$, and positivity follows from the exact identity

$$(E.13) \quad \int_0^\infty \left[(q_*)^2 + \frac{t^2}{6}(q'_*)^2 + \frac{5}{6}q_*^2 \right] dt = 2k_0C_* > 0,$$

where $q_* = t^2(c - C_*t)/6 - (c - C_*t)''$.

Thus $\mathcal{Z}$ is a flow multiple plus an algebraically decaying center variation, not a hidden growing Jost mode. We now identify the tangent space without using the numerical endpoint slope. The four-rate graph transform in Proposition 8.1 gives, along γ_0 in the weighted corridor, an invariant rank-three horizontal bundle

$$(E.14) \quad T\mathcal{W}_a^{\text{tail}} = \{(\delta e, \delta \mathcal{A}, \delta b_w, \delta \zeta) : \delta \zeta = L(t)(\delta e, \delta \mathcal{A}, \delta b_w)\}, \quad \sup_t \|L(t)\| < \infty.$$

Every solution of the variational equation has constant coefficients in the fundamental basis $\{\mathbf{s}, \mathcal{T}, \mathcal{Z}, \mathbf{u}\}$. The first is recessive, $\mathcal{T}$ is polynomial, and (E.12) shows that $\mathcal{Z}$ remains at most polynomial after the algebraic coordinate change.

It remains only to exclude a growing coefficient. Put $x = t^2/(2\sqrt{6})$. Watson endpoint expansion of (E.7) gives

$$(E.15) \quad \mathbf{u}(t) = e^x \begin{pmatrix} 12\sqrt{3}t^{-1/2}(1 + O_{\text{sym}}(t^{-2})) \\ 6\sqrt{2}t^{1/2}(1 + O_{\text{sym}}(t^{-2})) \\ 72\sqrt{3}t^{-5/2}(1 + O_{\text{sym}}(t^{-2})) \\ 36\sqrt{2}t^{-3/2}(1 + O_{\text{sym}}(t^{-2})) \end{pmatrix}.$$

Along the exact orbit $e = 12t^{-2}$, $d = 0$, and $\omega = e^2/6$. Differentiating (8.7), using $h_7 = -e/\sqrt{3} + \omega/\sqrt{2}$ plus terms of total degree at least two, gives for this solution

$$(E.16) \quad \begin{aligned} \delta e_{\mathbf{u}} &= 1728\sqrt{3}e^x t^{-9/2}(1 + O(t^{-2})), \\ \delta \mathcal{A}_{\mathbf{u}} &= 0, \\ \delta b_{\mathbf{w}, \mathbf{u}} &= -\frac{\sqrt{3}}{72}e^x t^{11/2}(1 + O(t^{-2})), \\ \delta \zeta_{\mathbf{u}} &= \frac{288\sqrt{6}}{128}e^x t^{27/2}(1 + O(t^{-2})). \end{aligned}$$

Here $\delta \mathcal{A}_{\mathbf{u}} = 0$ because its conserved energy variation vanishes by (E.9). Consequently the ratio of the normal component to the weighted base norm is asymptotic to a positive constant times t^8 . Any nonzero $\mathbf{u}$ coefficient eventually dominates the polynomial/recessive contributions of the other three basis vectors and enters the validated vertical cone, contradicting (E.14). Every tangent solution therefore has zero $\mathbf{u}$ coefficient. Since the horizontal tangent bundle has rank three, this proves

$$T_{\gamma_0(t)}\mathcal{W}_a = \text{span}\{\mathcal{T}(t), \mathcal{Z}(t), \mathbf{s}(t)\}.$$

This equality is first identified in the weighted tail and is then transported by the variational flow through the finite saturation (8.21), including to $t = 0$. At $t = 0$, a vector $a\mathcal{T}(0) + b\mathcal{Z}(0) + d\mathbf{s}(0)$ belongs to $T_{c_0} \text{Fix } \mathcal{R} = \{P = Q = 0\}$ only if the Q row gives $d = 0$ and the P row then gives $a = 0$. Therefore

$$(E.17) \quad T_{c_0}\mathcal{W}_a \cap T_{c_0} \text{Fix } \mathcal{R} = \text{span}\{(1, 0, -2k_0, 0)\}, \quad k_0 = \frac{\sqrt{\pi}\Gamma(1/4)}{4\sqrt{6}\Gamma(3/4)}.$$

A directed 256-bit evaluation gives

$$(E.18) \quad k_0 \in [.53522525701880774, .53522525701880797], \quad V'(0) = -2k_0,$$

and the first tangent Krawczyk image strictly contains this analytic interval. Hence the numerical branch starts on the canonical Jost line, not on a separately selected slope.

The fixed- U continuation has 17,345 base-and-tangent uniqueness boxes and 17,344 true adjacent common-root bridges. Every box uses the canonical target budgets and proves a unique first entry into $e = .06$. The energy derivative is strictly negative on every main box for which its sign is required. A true common-root bridge then enters a 2,002-box fine cover with 2,001 adjacent bridges. On its first 1,867 boxes,

$$(E.19) \quad \frac{d}{dU} E(c(U)) \leq -2.9561716261977677 \cdot 10^{-5}.$$

The final 140 complete base-and-tangent boxes, five of which overlap the strict-negative subcover, lie strictly inside one 248-dimensional augmented critical-point domain, with minimum containment margin $1.0001554357993005 \cdot 10^{-9}$. The negative-sign and cap parameter covers overlap by $9.000000178449596 \cdot 10^{-8}$.

The augmented Krawczyk ratio is $.912135743122722 < 1$, so the cap contains one critical point. The full Krawczyk image of the independently verified nondegenerate critical point lies inside that cap with margin $1.947670835849842 \cdot 10^{-9}$ and inside the final fixed- U uniqueness box in both state and parameter. Therefore the unique augmented zero is the endpoint of the connected regular arc, is simple, and is its first critical point. $\square$

REFERENCES

1. Inmaculada Baldomá, Mar Giral, and Marcel Guardia, *Breakdown of homoclinic orbits to L_3 in the RPC3BP (II): An asymptotic formula*, Advances in Mathematics **430** (2023), 109218, Author manuscript: arXiv:2107.09941.
2. Luis Barreira and Godofredo Iommi, *Suspension flows over countable Markov shifts*, Journal of Statistical Physics **124** (2006), no. 1, 207–230.
3. Pablo G. Barrientos, Artem Raibekas, and Alexandre A. P. Rodrigues, *Chaos near a reversible homoclinic bifocus*, Dynamical Systems **34** (2019), no. 3, 504–516, Author manuscript: arXiv:1810.06359.
4. Sergey V. Bolotin, *Dynamics of slow-fast Hamiltonian systems: The saddle-focus case*, Regular and Chaotic Dynamics **30** (2025), no. 1, 76–92.
5. Amadeu Delshams, Rafael de la Llave, and Tere M. Seara, *Geometric properties of the scattering map of a normally hyperbolic invariant manifold*, Advances in Mathematics **217** (2008), no. 3, 1096–1153.
6. Robert L. Devaney, *Homoclinic orbits in Hamiltonian systems*, Journal of Differential Equations **21** (1976), no. 2, 431–438.
7. ———, *Blue Sky catastrophes in reversible and Hamiltonian systems*, Indiana University Mathematics Journal **26** (1977), no. 2, 247–263.
8. Neil Fenichel, *Persistence and smoothness of invariant manifolds for flows*, Indiana University Mathematics Journal **21** (1971), no. 3, 193–226.
9. Antonio Giorgilli, *Unstable equilibria of Hamiltonian systems*, Discrete and Continuous Dynamical Systems **7** (2001), no. 4, 855–871.
10. Jörg Härterich, *Cascades of reversible homoclinic orbits to a saddle-focus equilibrium*, Physica D: Nonlinear Phenomena **112** (1998), no. 1–2, 187–200.
11. Morris W. Hirsch, Charles C. Pugh, and Michael Shub, *Invariant manifolds*, Lecture Notes in Mathematics, vol. 583, Springer-Verlag, Berlin, 1977.

12. A. J. Homburg, J. S. W. Lamb, and D. V. Turaev, *Symmetric homoclinic tangles in reversible dynamical systems have positive topological entropy*, *Advances in Mathematics* **464** (2025), 110131.
13. Tomasz Kapela, Marian Mrozek, Daniel Wilczak, and Piotr Zgliczyński, *CAPD::DynSys: A flexible C++ toolbox for rigorous numerical analysis of dynamical systems*, *Communications in Nonlinear Science and Numerical Simulation* **101** (2021), 105578, Article 105578; author manuscript: arXiv:2010.07097.
14. Shane Kepley and J. D. Mireles James, *Chaotic motions in the restricted four body problem via Devaney's saddle-focus homoclinic tangle theorem*, *Journal of Differential Equations* **266** (2019), no. 4, 1709–1755, Author manuscript: arXiv:1711.06932.
15. Oksana Koltsova and Lev Lerman, *Hamiltonian dynamics near nontransverse homoclinic orbit to saddle-focus equilibrium*, *Discrete and Continuous Dynamical Systems* **25** (2009), no. 3, 883–913.
16. R. Krawczyk, *Interval extensions and interval iterations*, *Computing* **24** (1980), no. 2–3, 119–129.
17. Michael Lerch, German Tischler, Jürgen Wolff von Gudenberg, Werner Hofschuster, and Walter Krämer, *FILIB++, a fast interval library supporting containment computations*, *ACM Transactions on Mathematical Software* **32** (2006), no. 2, 299–324.
18. L. M. Lerman, *Complex dynamics and bifurcations in a Hamiltonian system having a transversal homoclinic orbit to a saddle focus*, *Chaos: An Interdisciplinary Journal of Nonlinear Science* **1** (1991), no. 2, 174–180.
19. ———, *Homo- and heteroclinic orbits, hyperbolic subsets in a one-parameter unfolding of a Hamiltonian system with heteroclinic contour with two saddle-foci*, *Regular and Chaotic Dynamics* **2** (1997), no. 3–4, 139–155.
20. ———, *Dynamical phenomena near a saddle-focus homoclinic connection in a Hamiltonian system*, *Journal of Statistical Physics* **101** (2000), no. 1–2, 357–372.
21. John N. Mather, *Notes on topological stability*, *Bulletin of the American Mathematical Society* **49** (2012), no. 4, 475–506.
22. Jürgen Moser, *On the generalization of a theorem of A. Liapounoff*, *Communications on Pure and Applied Mathematics* **11** (1958), no. 2, 257–271.
23. Siegfried M. Rump, *Rigorous sensitivity analysis for systems of linear and nonlinear equations*, *Mathematics of Computation* **54** (1990), no. 190, 721–736.
24. Omri M. Sarig, *Thermodynamic formalism for countable Markov shifts*, *Ergodic Theory and Dynamical Systems* **19** (1999), no. 6, 1565–1593.
25. Warwick Tucker, *Validated numerics: A short introduction to rigorous computations*, Princeton University Press, Princeton, NJ, 2011.
26. Andrei Verona, *Stratified mappings—structure and triangulability*, *Lecture Notes in Mathematics*, vol. 1102, Springer-Verlag, Berlin, 1984.
27. Theodore Vo, Arjen Doelman, and Tasso J. Kaper, *Les canards de Turing*, *SIAM Journal on Applied Dynamical Systems* **24** (2025), no. 4, 2618–2684, Author manuscript: arXiv:2409.02400.
28. Wolfgang Wasow, *Asymptotic expansions for ordinary differential equations*, *Pure and Applied Mathematics*, vol. 14, Interscience Publishers, New York, 1965.
29. Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, et al., *The FAIR guiding principles for scientific data management and stewardship*, *Scientific Data* **3** (2016), 160018.

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